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The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

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Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

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Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today -3 September with 942 score.

# Hashicorp

(Terraform Associate 004)

HashiCorp Certified

Total: **331 Questions**

Link: <https://certyiq.com/papers/hashicorp/terraform-associate-004>

### Question: 1

laC (Infrastructure as Code) can be stored in a version control system along with application code.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Answer: A – True**

Infrastructure as Code is commonly version-controlled just like application source code. Storing Terraform scripts, Ansible playbooks, CloudFormation templates, or similar artifacts in a Git repository (or other VCS) enables:

**Auditability & traceability** of infrastructure changes.

**Collaboration** among developers, ops, and security teams.

**Consistent, repeatable deployments** through CI/CD pipelines.

**Branching strategies** that mirror software release flows (feature branches, pull-request reviews, versioned releases).

Keeping IaC alongside application code in the same repository promotes a unified source of truth, simplifies dependency management, and supports the “treat-infrastructure-as-code” mindset required for modern DevOps practices.

**Why the other option is less suitable**

B. False – This statement contradicts the core principle of IaC. Treating infrastructure as immutable, versioned artifacts is a best practice; treating it as an after-thought or external artifact undermines traceability, reproducibility, and the ability to automate provisioning through CI/CD pipelines.

**References**

HashiCorp Learn – Terraform on GitHub: <https://learn.hashicorp.com/terraform/cloud-get-started/github-actions>

Terraform Registry Documentation – Version Control Best Practices:  
<https://developer.hashicorp.com/terraform/language/v1.5/backends/gitlab> (valid at time of writing)

### Question: 2

It is best practice to store secret data in the same version control repository as your Terraform configuration.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Answer: B – False**

**Security posture:** Storing secrets (e.g., database passwords, API keys) alongside Terraform files risks accidental exposure in version-control history, pull-requests, or public forks. Secrets should be kept in dedicated secret-management systems or encrypted stores, not in plain-text source control.

**Least-privilege access:** By separating secrets from infrastructure code, you can grant Terraform execution roles only the permissions needed to read the secret at runtime, rather than exposing the entire configuration to any user who can clone the repo.

**Operational hygiene:** Version-controlled Terraform configurations should be immutable in terms of infrastructure definition; injecting secrets makes drift harder to detect and can inadvertently cause drift when secret values are rotated.

**Compliance & auditability:** Best-practice frameworks (e.g., SOC2, ISO27001) require separation between code and secret material to maintain clear audit trails and to support automated secret-scan policies.

**Terraform provider behavior:** The AWS provider will reject or mask secret values when they are sourced from variables that are not marked as sensitive and stored in version control, encouraging the use of external secret providers or encrypted backends.

### Why the other option is less suitable

**Option A (True)** would imply that putting secrets in the same repo is the recommended approach. This contradicts industry security standards, increases the attack surface, and violates principles of secret isolation that the AWS provider and broader Terraform ecosystem promote.

### References

Terraform AWS Provider Documentation – Secure Credential Management:

<https://registry.terraform.io/providers/hashicorp/aws/latest/docs/guides/credential-management>

HashiCorp Best Practices for Secrets in Terraform:

<https://developer.hashicorp.com/terraform/language/modules/develops/using-modules#using-modules>  
(section “Storing Secrets”).

### Question: 3

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Which is an advantage of using IaC (Infrastructure as Code) that is not possible when provisioning with a GUI (Graphical User Interface)?

- A. Let's you version, reuse, and share infrastructure configuration.
- B. Secures your credentials.
- C. Provisions the same resources at a lower cost.
- D. Prevents manual modifications to your resources.

**Answer: A**

**Explanation:**

**Why option A is the correct advantage of IaC:**

IaC treats infrastructure as version-controlled source code, enabling **history tracking, branching, pull-request reviews, and rollbacks** of configuration changes.

The same configuration can be **reused across environments** (dev, staging, prod) and **shared across teams** through repositories and registries, fostering collaboration and reducing drift.

This capability is fundamentally a **code-centric workflow** that GUIs cannot provide; a GUI interacts only with the current state and offers no native versioning or sharing mechanisms.

### Why the other options are less appropriate:

**B – Secures your credentials:** Security of credentials depends on how secrets are stored and accessed; both IaC tools and some GUIs can be configured securely. The advantage lies in the automation and consistency of secret handling, not in the mere existence of a GUI.

**C – Provisions the same resources at a lower cost:** Cost is determined by the underlying cloud/services and design decisions, not by the provisioning method (IaC vs. GUI).

**D – Prevents manual modifications to your resources:** While IaC encourages declarative control and can enforce changes via CI/CD pipelines, a GUI can also enforce policies or lock resources; however, it does not offer the same systematic enforcement or auditability that version-controlled IaC provides.

### References

Terraform documentation on state management and version control:

<https://developer.hashicorp.com/terraform/language>

AWS CloudFormation documentation on stacks and versioning: <https://docs.aws.amazon.com/cloudformation/>

These sources illustrate how IaC's declarative, versionable nature enables capabilities that pure GUI-based provisioning lacks.

### Question: 4

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What is an advantage of immutable infrastructure?

- A. Automatic infrastructure upgrades
- B. In-place infrastructure upgrades
- C. Quicker infrastructure upgrades
- D. Less complex infrastructure upgrades

### Answer: D

#### Explanation:

**Immutable infrastructure replaces resources rather than modifying them in-place**, so an upgrade is performed by creating a new set of infrastructure and retiring the old one.

This approach produces **repeatable, auditable changes** that are easier to plan, test, and roll back, reducing the operational overhead associated with large, complex upgrade tasks.

The declarative state stored in a configuration management system (e.g., Terraform) remains simple — upgrades are expressed as a new desired state, not as a series of incremental patches — hence they are less complex to execute.

**Option D – “Quicker infrastructure upgrades”** is not the primary advantage; the speed gain may be incidental, whereas the reduction in complexity is consistent and measurable.

**Option A – “Automatic infrastructure upgrades”** misleads because the upgrade still requires explicit definitions and orchestration; it is not “automatic” in the sense of zero-effort.

**Option B – “In-place infrastructure upgrades”** directly contradicts the immutable model, which deliberately avoids modifying existing resources.

**Option C – “Quicker infrastructure upgrades”** is not a guaranteed benefit; performance gains depend on provisioning pipelines and can be offset by new resource creation latency.

**Therefore, the correct answer is D – it highlights the central benefit of immutable infrastructure: simpler, less complex upgrade processes.**

## References

HashiCorp Learn – Immutable Infrastructure: <https://learn.hashicorp.com/terraform/cloud-infra/terraform-state>

Terraform Documentation – Immutable Resources:

<https://developer.hashicorp.com/terraform/language/resources#mutable-vs-immutable-resources>

### Question: 5

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What is the primary purpose of IaC (Infrastructure as Code)?

- A. To define a pipeline to test and deliver software.
- B. To provision infrastructure cheaply.
- C. To programmatically create and configure resources.
- D. To define a vendor-agnostic API.

**Answer: C**

**Explanation:**

#### Primary purpose of IaC

IaC enables the **programmatic creation and configuration of infrastructure resources** (servers, networks, storage, etc.) through declarative code rather than manual processes. This makes provisioning repeatable, version-controlled, and easily replicable across environments.

#### Why option C is correct

It directly describes IaC's core capability: programmatically (i.e., via code/write-ups) defining the desired state of infrastructure and creating and configuring the underlying resources to match that state.

#### Why the other options are less suitable

**A. To define a pipeline to test and deliver software** – This describes CI/CD pipelines, not IaC. IaC may be integrated into pipelines, but its primary focus is infrastructure provisioning, not software testing or delivery.

**B. To provision infrastructure cheaply** – Cost efficiency can be a side effect of IaC, but “cheap” is a business outcome, not the defining purpose. IaC can be used for costly architectures as well.

**D. To define a vendor-agnostic API** – IaC tools often support multiple clouds, yet they are not primarily about exposing an API; they are about declaratively describing infrastructure, often tied to a specific provider or tooling.

#### Conclusion

Option **C** best captures the essence of IaC: using code to programmatically create and configure resources.

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## References

HashiCorp Learn – What is Infrastructure as Code?

<https://learn.hashicorp.com/tutorials/terraform/install-cli>

Terraform Documentation – Introduction to Terraform

<https://developer.hashicorp.com/terraform/tutorials/basics>

### Question: 6

Your team adopts an AWS CloudFormation as the standardized method for provisioning public cloud resources. Which scenario presents a challenge for your team?

- A. Deploying new infrastructure into Microsoft Azure.
- B. Automating a manual, web console-based provisioning process.
- C. Building a reusable code base that can deploy resources into any AWS region.
- D. Managing a new application stack built on AWS-native services.

**Answer: A**

**Explanation:**

#### Why option A is the challenge

AWS CloudFormation only provisions resources that exist within the AWS ecosystem; it cannot directly manage Azure services.

Introducing Azure requires a different IaC engine (e.g., Azure Resource Manager templates or Terraform), adding cross-cloud complexity, extra toolchains, and operational overhead.

The team's standardized workflow is built around CloudFormation's native features, so extending it to a non-AWS platform breaks that standardization.

#### Why the other options are not challenges

**B – Automating a manual, web-console process** – This is precisely the purpose of CloudFormation; it reduces manual effort rather than creating difficulty.

**C – Building a reusable code base for any AWS region** – CloudFormation already supports regional deployment through parameters, mappings, and stacksets, turning this into a capability, not a hurdle.

**D – Managing a new AWS-native application stack** – This aligns with the core use case of CloudFormation; it leverages existing constructs and best practices without introducing new constraints.

#### References

AWS CloudFormation User Guide – Regions and Endpoints:

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/using-regions.html>

Overview of AWS CloudFormation Stack Sets:

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/stacksets.html>

### Question: 7

Which is not a benefit of adopting IaC (Infrastructure as Code)?

- A. Reusability of code
- B. Automation
- C. A GUI (Graphical User Interface)
- D. Versioning

**Answer: C**

**Explanation:**

## Why option C is not a benefit

**Graphical User Interface (GUI) is not inherent to IaC** – IaC models infrastructure as declarative code, which is meant to be stored, reviewed, and executed programmatically. While some tools provide GUIs for convenience, the core benefit of IaC does not depend on a graphical interface; it could be fully managed through version-controlled scripts alone.

## Why the other options are genuine benefits

**A. Reusability of code** – Modules, templates, and libraries can be shared across projects, enabling consistent environments and reducing duplication of effort.

**B. Automation** – IaC eliminates manual provisioning steps; the same code can be applied repeatedly to spin up, modify, or destroy resources automatically.

**D. Versioning** – Infrastructure definitions are treated as source code, so they can be tracked, rolled back, reviewed, and tested in the same way as application code.

## Conclusion

Option **C (A GUI)** is not a fundamental advantage of Infrastructure as Code; it is merely an optional convenience layer, whereas reusability, automation, and versioning are intrinsic capabilities of an IaC approach.

## References

HashiCorp Documentation – Terraform Overview: <https://developer.hashicorp.com/terraform/tutorials/aws-get-started/intro>

Kubernetes Hardening Guide – Infrastructure as Code Best Practices:

<https://kubernetes.io/docs/concepts/cluster-administration/manage-automation/#tools-for-automation>

## Question: 8

CertyIQ

How does the use of Infrastructure as Code (IaC) enhance the reliability of your infrastructure? (Choose two.)

- A. Configuration drift is reduced with declarative definitions.
- B. Updates are deployed with zero downtime.
- C. Proposed changes can be reviewed before being applied.
- D. Incorrect configurations cannot be deployed.
- E. Infrastructure is automatically scaled to meet demand.

**Answer: AC**

**Explanation:**

**Why options A and C are correct**

**A – Configuration drift is reduced with declarative definitions** –

Infrastructure as Code stores the desired state in version-controlled files (e.g., HCL, JSON, YAML). Terraform treats these files as the single source of truth, constantly reconciling the actual provisioned resources with that declared state. When the real environment diverges, a terraform plan detects the mismatch and flags the required remediation, preventing unsolicited drift.

**C – Proposed changes can be reviewed before being applied** –

terraform plan generates an execution plan that lists every create, modify, or destroy action that would occur. This preview lets operators audit the impact, verify dependencies, and confirm that the changes meet security

or compliance policies before they are persisted with terraform apply.

### Why the other options are less suitable

#### B – Updates are deployed with zero downtime –

Terraform itself does not guarantee zero-downtime deployment; achieving that requires additional patterns (e.g., rolling updates, load-balancer shifting). Terraform's role is to provision the desired resources, not to manage runtime availability during the update.

#### D – Incorrect configurations cannot be deployed –

While a plan can catch syntax errors or dependency issues, Terraform cannot prevent logical or business-logic mistakes that are not manifest in the configuration language. Incorrect resource properties may still be applied if they are syntactically valid.

#### E – Infrastructure is automatically scaled to meet demand –

Autoscaling is a runtime capability of cloud services (e.g., AWS Auto Scaling, Azure Scale Sets). Terraform can provision scaling resources, but the scaling logic must be defined separately; Terraform does not automatically adjust capacity in response to demand.

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## References

**Terraform State & Drift Detection** – <https://developer.hashicorp.com/terraform/language/state/drift>

**Terraform Plan & Apply Workflow** – <https://developer.hashicorp.com/terraform/language/works/plans>

These documents detail how declarative state management mitigates drift and how Terraform plans enable pre-apply reviews, directly supporting the rationale above.

## Question: 9

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Your team often uses API calls to create and manage cloud infrastructure.

In what ways does Terraform differ from conventional infrastructure management approaches?

- A. Terraform replaces cloud provider APIs with its own protocols, enabling automated deployments.
- B. Terraform describes infrastructure with version-controlled, repeatable configurations that specify the desired state.
- C. Terraform is merely a wrapper for cloud provider APIs, so there is little to no difference in calling the API directly.
- D. Terraform enforces infrastructure through imperative scripts to ensure tasks are completed in the proper order.

### Answer: B

#### Explanation:

#### Why option B is correct

Terraform uses a **declarative configuration language** (HCL) where the desired infrastructure state is expressed in plain text files.

These files are **version-controlled**, enabling reproducible builds, audit trails, and collaborative workflows.

The Terraform engine compares the declared state with the actual provider state and

**creates/updates/destroys resources only as needed** to achieve convergence, abstracting away the details of individual provider APIs.

## Why the other options are less suitable

**OptionA** – Terraform does **not replace** provider APIs; it uses them under the hood. Its power comes from the declarative model, not from inventing new protocols.

**OptionC** – While Terraform ultimately calls provider APIs, it adds a **layer of abstraction** (state management, plan/apply workflow, dependency graph) that shields users from direct, imperative API calls.

**OptionD** – Terraform is **not imperative**; it does not rely on scripts that manually sequence actions. Instead, it calculates a plan and applies changes in a **declarative, idempotent** manner, ensuring the correct order only when required by dependencies.

## References

### Terraform Documentation – Overview of the Planning and Applying Process

<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/install-cli>

### Terraform Documentation – State Management and Remote State

<https://developer.hashicorp.com/terraform/language/state>

These links provide the official, up-to-date explanations of Terraform's declarative approach and its state-driven behavior, supporting the rationale above.

## Question: 10

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Which of these workflows is only enabled by the use of Infrastructure as Code?

- A. Automatic scaling of resources based on application load.
- B. Cost optimization of infrastructure deployment.
- C. Role-based access control of cloud resources.
- D. Reviewing the proposed changes for potential security issues.

## Answer: D

### Explanation:

**Why D is correct** – Only Infrastructure asCode makes every change **explicit, version-controlled, and previewable**. Terraformplan generates a detailed execution plan that can be reviewed (by humans or automated policy engines) before any resource is created or modified, allowing security teams to spot unsafe configurations, privilege escalations, or mis-aligned network rules early. This pre-execution security review is a direct consequence of IaC tooling.

### Why A, B, C are not exclusive to IaC –

Automatic scaling (A) can be achieved with native services (e.g., Auto Scaling groups, Lambda triggers) without IaC; IaC merely provisions the scaling rules.

Cost optimization (B) is a business/ops practice that may leverage IaC for repeatability but is not inherent to IaC itself.

RBAC (C) can be applied manually or via IAM policies outside of code; IaC only provides a way to express those policies programmatically.

**Why the IaC-enabled review is unique** – The plan-stage output is deterministic and can be integrated with policy-as-code (e.g., Sentinel, OPA) to automatically flag violations, something that cannot be done reliably without a declarative, version-controlled representation of the infrastructure.

## References

### Question: 11

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What does Terraform use to deploy infrastructure for different cloud providers?

- A. Custom APIs developed by HashiCorp
- B. Vendors' CLI tools
- C. Vendors' UI
- D. Vendor-specific providers

**Answer: D**

**Explanation:**

#### Technical justification

Terraform extends its core engine with **provider plugins** that contain the API calls and logic required to interact with each cloud platform.

Each provider is **vendor-specific**, implementing the resources, data sources, and authentication methods unique to that cloud (e.g., aws, azure, google, oci).

The provider framework abstracts the underlying **REST/HTTP APIs** of the cloud services, allowing Terraform to manage resources uniformly across providers without custom Terraform code.

Options A, B, and C describe indirect or unrelated mechanisms:

**A** – HashiCorp does not expose custom APIs for every cloud; it relies on third-party provider plugins.

**B** – Vendors' CLI tools are for interactive administration, not for programmatic IaC orchestration by Terraform.

**C** – Cloud UI interfaces are GUI-based and not used by Terraform, which operates headlessly.

Therefore, deploying infrastructure across different clouds is achieved through **vendor-specific providers** that translate Terraform's resource definitions into the appropriate cloud API calls.

#### References

Terraform Providers Documentation – <https://developer.hashicorp.com/terraform/language/providers>

Provider Registry – <https://registry.terraform.io/providers>

### Question: 12

CertyIQ

How can you enable verbose logging to troubleshoot?

- A. Set the log level command line flag.
- B. Set the TF\_LOG environment variable.
- C. Set the log level in your terraform block.

**Answer: B**

**Explanation:**

### Answer: B

Terraform's built-in logging is controlled by the **TF\_LOG** environment variable; setting it to DEBUG or TRACE produces the most detailed output.

The command-line flag `--log-level` does not exist for Terraform, so optionA is invalid.

The `log_level` directive belongs inside a provider or backend block, not a generic terraform block, and it does not affect CLI-level output (optionC is misleading).

Using **TF\_LOG** overrides any configuration-level logging settings and is the standard, documented method for enabling verbose logs during troubleshooting.

### References

<https://developer.hashicorp.com/terraform/tutorials/cli/config-file#setting-configuration-options>

[https://developer.hashicorp.com/terraform/cli/config/environments#tf\\_log](https://developer.hashicorp.com/terraform/cli/config/environments#tf_log)

### Question: 13

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Which command lets you experiment with Terraform expressions?

- A. terraform console
- B. terraform env
- C. terraform validate
- D. terraform test

### Answer: A

#### Explanation:

#### Why optionA is the correct choice

**terraform console** launches an interactive REPL where you can input and evaluate any Terraform expression (including function calls, data lookups, and module arguments). This is specifically designed for “playing with” expressions before incorporating them into real configurations.

It provides immediate feedback on expression results, helps debug complex expressions, and allows you to test variable interpolation without needing a full build plan.

Because it operates on the current working directory's configuration, it respects providers, data sources, and local values, making the evaluations realistic and context-aware.

#### Why the other options are unsuitable

**terraform env** – historically used for setting environment variables in early Terraform versions; it does not provide an expression evaluator and is unrelated to testing expressions.

**terraform validate** – checks the syntactic and semantic validity of the configuration files. It does **not** execute or evaluate expressions; it only reports errors related to configuration structure and schema compliance.

**terraform test** – runs external test suites that invoke Terraform commands in a controlled environment; it is meant for integration/behavior testing, not for ad-hoc expression experimentation.

#### Key takeaway

For rapid, interactive experimentation with Terraform expressions, the official command is **terraform console**, as it directly offers an REPL dedicated to that purpose.

## References

**terraform console** – Official CLI documentation:

<https://developer.hashicorp.com/terraform/cli/commands/console>

**Expressions** – Terraform language reference on functions and expressions:

<https://developer.hashicorp.com/terraform/language/expressions>

### Question: 14

CertyIQ

You can install Community/Partner plugins using terraform init.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Justification**

The statement is **True**.

terraform init automatically downloads and installs any declared provider plugins that are not already present in the working directory. This includes providers sourced from the public Terraform Registry, which hosts Community and Partner providers. When a provider block references a registry entry, terraform init resolves the provider address, fetches the appropriate version, and installs it under .terraform/providers.

Option **B (False)** is unsuitable because it contradicts the documented behavior of terraform init. The command is explicitly designed to handle provider installations from the Registry, encompassing both community-maintained and HashiCorp-partner providers. Claiming it cannot do so ignores the core purpose of the init step and would lead to errors such as “Could not find provider configuration” when a provider is missing.

**References**

Terraform Registry Documentation: <https://developer.hashicorp.com/terraform/tutorials/providers/how-to-use-registry>

Official terraform init Command: <https://developer.hashicorp.com/terraform/cli/commands/init>

### Question: 15

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What functionality do providers offer in Terraform? (Choose three.)

- A. Group a collection of Terraform configuration files that map to a single state file.
- B. Provision resources for public cloud infrastructure services.
- C. Interact with cloud provider APIs.
- D. Enforce security and compliance policies.
- E. Provision resources for on-premises infrastructure services.

**Answer: BCE**

**Explanation:**

**Technical justification**

**B – Provision resources for public cloud infrastructure services** – Terraform providers expose resources that map to the services of public-cloud platforms (e.g., `aws_instance`, `azurerm_virtual_network`). When a provider is configured for a public cloud, Terraform can create, update, and destroy those cloud resources, making provider support essential for public-cloud provisioning.

**C – Interact with cloud provider APIs** – By definition, a Terraform provider is a plugin that implements the HTTP or RPC interface of a remote service. It translates Terraform's declarative actions into API calls defined by the provider, enabling any external system – not just clouds – to be managed through Terraform.

**E – Provision resources for on-premises infrastructure services** – Providers are not limited to public clouds; they can also represent APIs for on-premises systems (e.g., VMware, Kubernetes, HashiCorp Vault). When a provider for such a platform is used, Terraform can manage the corresponding resources, fulfilling the on-premises use-case.

**Why the other options are not correct**

**A – Group a collection of Terraform configuration files that map to a single state file** – This describes a backend configuration or a module organization, not a provider. Providers do not bundle configuration files; they expose typed resources to Terraform.

**D – Enforce security and compliance policies** – Policy enforcement is performed by tools like Sentinel, OPA, or Terraform Cloud policy sets. Providers solely interact with external APIs; they do not contain built-in enforcement logic for security/compliance controls.

**Conclusion** – Providers are the mechanism through which Terraform consumes external APIs, enabling resource management for both public-cloud and on-premises environments. Therefore, the three correct statements are **B, C, and E**.

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## References

Terraform Language Documentation – Providers:

<https://developer.hashicorp.com/terraform/language/providers>

Terraform Registry – Providers Overview: <https://developer.hashicorp.com/terraform/registry/providers>

## Question: 16

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Terraform can only manage resource dependencies if you set them explicitly with the `depends_on` argument.

- A. True
- B. False

**Answer: B**

**Explanation:**

Terraform automatically infers resource ordering from the configuration language; it determines execution order based on **implicit dependencies** (e.g., when a resource references the output or attribute of another resource).

The `depends_on` attribute is only needed when the dependency is **not expressible** through configuration

syntax, such as when using external services, provisioners, or third-party providers that have no natural attribute references.

Without an explicit **depends\_on**, Terraform still respects both **implicit** and any **output-driven** dependencies, so it can manage most dependencies without additional declarative ordering.

### Why Option A (True) Is Incorrect

It overgeneralizes Terraform's dependency resolution. The engine does not strictly require an explicit `depends_on` clause; it can infer and satisfy dependencies through resource arguments and output references, making implicit ordering the default behavior.

### Why Option B (False) Is Correct

**False** accurately reflects that Terraform can manage resource dependencies **without** an explicit `depends_on` argument, as long as the dependency is represented semantically within the configuration (e.g., referencing a resource's address). `depends_on` is reserved for edge cases where Terraform cannot infer the order automatically.

## References

1. Terraform documentation – Dependency graph and ordering:  
<https://developer.hashicorp.com/terraform/language/resources/dependencies>
2. Terraform documentation – `depends_on` argument:  
[https://developer.hashicorp.com/terraform/language/resources/dependencies#the-depends\\_on-argument](https://developer.hashicorp.com/terraform/language/resources/dependencies#the-depends_on-argument)

### Question: 17

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Which of the following is not a valid Terraform collection type?

- A. set
- B. list
- C. map
- D. tree

**Answer: D**

**Explanation:**

The correct answer is **D. tree** because Terraform does not define "tree" as a built-in collection type. Terraform's core collection types are **set** (unordered unique elements), **list** (ordered sequence), and **map** (key-value pairs), all explicitly supported in documentation and syntax.

"Tree" is a conceptual data structure (e.g., nested maps/lists), but it is **not a native Terraform type**; complex structures must be built using combinations of existing types (e.g., `map(string)` with nested maps), not as a standalone "tree" type.

Options A (set), B (list), and C (map) are all valid Terraform collection types, as confirmed by HashiCorp's official documentation.

Option D is a distractor that mimics the structure of valid types but lacks any basis in Terraform's type system.

References:

**Question: 18**

A resource block is shown in the Exhibit space of this page.  
resource "aws\_vpc" "main"  
name = "test"

What is the provider for this resource?

- A. main
- B. vpc
- C. aws
- D. test

**Answer: C**

**Explanation:**

**Key concepts, documents or landing pages related to the question**

<https://registry.terraform.io/providers/hashicorp/aws/latest/docs> – official AWS provider documentation, which defines the provider name used in Terraform configuration.

<https://developer.hashicorp.com/terraform/language/resources> – guide to Terraform resource blocks and the syntax for specifying a provider.

**Technical justification (exam-style explanation)**

The block shown is a resource declaration that targets the AWS provider (resource "aws\_vpc" ...).

Terraform resolves the provider for a resource by reading the provider argument that appears in the resource label. In aws\_vpc "main" the identifier aws is the type of provider, and Terraform automatically selects the AWS provider when such a type is used.

Therefore, the provider referenced by this resource block is **aws** (optionC).

The other options are not valid providers: main and test are resource names, while vpc is not a registered provider type. Consequently, they would cause a configuration error if used as the provider name.

**References**

<https://registry.terraform.io/providers/hashicorp/aws/latest/docs>

<https://developer.hashicorp.com/terraform/language/resources>

**Question: 19**

You must use different Terraform commands depending on the cloud provider you use.

- A. True
- B. False

**Answer: B**

### Explanation:

### Justification

Terraform's CLI commands (terraform init, terraform plan, terraform apply, terraform destroy, etc.) are **provider-agnostic**; they operate on the underlying state and configuration regardless of whether the target is AWS, Azure, GCP, or any other supported service.

What varies across providers is **only the provider block and its arguments** (e.g., region, credentials, API endpoints). The same set of commands is used after the configuration is written.

Therefore, the statement "You must use different Terraform commands depending on the cloud provider you use" is **incorrect**, making option **B (False)** the correct choice.

Option **A (True)** would only be true if Terraform exposed provider-specific CLI sub-commands, which it does not; such a design would fragment the workflow and violate Terraform's universal command model.

### References

Terraform Core CLI Overview: <https://developer.hashicorp.com/terraform/cli>

Provider Configuration Documentation: <https://developer.hashicorp.com/terraform/language/providers>

### Question: 20

CertyIQ

Terraform providers are always installed from the internet.

- A. True
- B. False

### Answer: B

### Explanation:

### Justification

Terraform can install providers from multiple sources:

Direct download from the public registry (registry.terraform.io) over the internet.

Local files placed in a working directory or in a custom directory (~/.terraform.d/plugins/).

Private package registries or Artifactory instances that are accessed via configuration (cli\_config\_credentials).

Because providers can be sourced locally or from private repositories, the statement "Terraform providers are always installed from the internet" is inaccurate. Hence the correct answer is **B. False**.

Option **A. True** would only apply when a provider is retrieved from the public registry, but it does not cover scenarios where the provider is pre-downloaded, cached, or served from an internal registry. Therefore it is not universally correct and is the less suitable choice.

### References

**Terraform Provider Installation** – Official HashiCorp documentation describing the various installation methods, including local files and private registries:

<https://developer.hashicorp.com/terraform/language/providers/installation>

**Terraform Registry** – Overview of public and private registry capabilities, confirming that providers are not

limited to internet downloads:

<https://registry.terraform.io/browse/providers>

### Question: 21

CertyIQ

You need to determine from which paths Terraform is loading the providers referenced in your \*.tf files. How can you enable additional logging to see this information?

- A. Set the environment variable TF\_VAR\_log=TRACE.
- B. Set the environment variable TF\_LOG=TRACE.
- C. Set the environment variable TF\_LOG=PATH.
- D. Set verbose logging for each provider in your Terraform configuration.

**Answer: B**

**Explanation:**

**Correct option: B – Set the environment variable TF\_LOG=TRACE.**

TF\_LOG=TRACE forces Terraform to emit the full trace-level log, which includes the exact file paths and the order in which providers are located and loaded from the registry, the working directory, and any custom source paths.

This log output is written to stdout/stderr and can be inspected with any text viewer, making it the standard way to debug provider resolution.

The other options do not achieve this purpose:

**A. TF\_VAR\_log=TRACE** – TF\_VAR\_\* prefixes are used only to set input variable values; they are not recognized by Terraform for controlling logging.

**C. TF\_LOG=PATH** – There is no such environment variable in Terraform; setting TF\_LOG to “PATH” would not enable any logging and would be ignored.

**D. “Set verbose logging for each provider in your Terraform configuration.”** – Terraform does not expose a per-provider logging flag in the configuration language; provider-level verbosity must be enabled globally via the log level, which is controlled by TF\_LOG.

Therefore, only option **B** directly activates the detailed provider-loading information needed for troubleshooting.

### References

Terraform CLI configuration – Environment variables:

<https://developer.hashicorp.com/terraform/cli/config/environment>

Logging and debugging: <https://developer.hashicorp.com/terraform/cli/logs>

### Question: 22

CertyIQ

Terraform requires using a different provider for each cloud provider where you want to deploy resources.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Justification**

Terraform's architecture is built around provider plugins; each plugin implements the API of a specific service (e.g., AWS, Azure, GCP).

To target a different cloud platform you must declare a separate provider configuration block (or an aliased one when multiple instances of the same provider are used).

Terraform does **not** automatically infer the provider from the resource type – the target cloud is defined explicitly in the provider block, so one provider block is required per distinct cloud provider.

Using a single provider for multiple clouds would leave the other clouds unreachable, making the statement “Terraform requires using a different provider for each cloud provider where you want to deploy resources” **true**.

If the statement were false, it would imply that a single provider block could manage resources across unrelated cloud APIs, which contradicts how Terraform's provider model operates.

**References**

Provider Configuration: <https://developer.hashicorp.com/terraform/language/providers>

AWS Provider Documentation: <https://registry.terraform.io/providers/hashicorp/aws/latest/docs>

### Question: 23

CertyIQ

Terraform cannot use a newly-defined cloud backend until it has been initialized with terraform init.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Answer: True**

When a cloud backend (e.g., S3, GCS, Azure Blob) is added or changed, Terraform treats the configuration as a new backend block.

Terraform only recognizes a backend after the **terraform init** (or **terraform init -backend-config=...**) command has run, which performs the backend migration and verifies the new backend's connectivity and credentials. Until this initialization step completes, Terraform continues using the previously configured backend (or the local state) and will display an error if a backend switch is attempted without initialization.

OptionB (“False”) is incorrect because the backend cannot be used immediately after definition; the explicit init step is required to switch backends safely.

**Why this is the best answer:**

The statement directly reflects Terraform's design: backend configuration changes are not applied until the working directory is re-initialized, ensuring that all backend-specific operations (state locking, data encryption, remote operations, etc.) are correctly established.

**Why the other option is less suitable:**

OptionB ignores the mandatory initialization phase and would mislead someone into thinking they can switch

backends ad-hoc, which would cause runtime errors and potential state corruption.

## References

### Terraform Documentation – Backend Configuration

<https://developer.hashicorp.com/terraform/language/settings/backends>

### Terraform CLI – Working with Backends

<https://developer.hashicorp.com/terraform/cli/commands/init#initializing-a-backend>

## Question: 24

CertyIQ

You have a list of numbers that represents the number of free CPU cores on each virtual cluster: numcpus = [ 18, 3, 7, 11, 2]

What Terraform built-in function would you use to select the largest number from the list?

- A. top(numcpus)
- B. max(numcpus)
- C. ceil(numcpus)
- D. high[numcpus]

**Answer: B**

**Explanation:**

**Justification**

**B. max(numcpus)** – This built-in Terraform function iterates over all elements of the input list and returns the highest numeric value. It is the directly appropriate way to obtain the largest core count (25 in the example).

**A. top(numcpus)** – top converts a set or tuple to a list and returns the first element(s) in sorted order, but it does not perform a numeric comparison; it cannot be used to extract the maximum from a plain list.

**C. ceil(numcpus)** – ceil rounds each number up to the nearest integer; it does not compare values, so it cannot determine the maximum.

**D. high[numcpus]** – high is not a recognized Terraform function; it will cause a validation error.

Therefore, max() is the only Terraform function designed specifically to retrieve the greatest element from a collection.

**References**

Terraform Functions – max: <https://developer.hashicorp.com/terraform/language/functions/math#max>

Terraform Functions – ceil: <https://developer.hashicorp.com/terraform/language/functions/math#ceil>

## Question: 25

CertyIQ

Which of these is stored in the .terraform directory?

- A. Providers and modules

- B. Lock file
- C. State file
- D. Configuration files

**Answer: A**

**Explanation:**

The **.terraform** directory is created by the Terraform CLI to keep all downloaded **provider binaries** and the **module files** that are referenced in the configuration. These artifacts are cached there so they can be reused on subsequent runs.

**OptionA** directly describes what the directory contains, making it the most accurate answer.

**OptionB** – The lock file (.terraform.lock.hcl) is stored in .terraform, but its primary purpose is to record version hashes; it is not the primary thing the directory is known for, and many exam questions treat it as a secondary detail.

**OptionC** – The state file (terraform.tfstate) resides in the working directory or a remote backend, not in .terraform.

**OptionD** – Configuration files (\*.tf) are kept alongside the configuration files themselves, outside of .terraform.

**References**

Terraform Registry & Plugin Installation:

<https://developer.hashicorp.com/terraform/language/providers/installation>

Working Directory & .terraform Directory:

<https://developer.hashicorp.com/terraform/cli/config/versionshcl#working-directory>

**Question: 26**

**CertyIQ**

A variable block is shown in the Exhibit space on this page:

```
variable "tags" {  
  description = "Metadata tags for resources"  
  type = _____  
}
```

You will use this variable as the value for the tags argument in several resources. The data format must be a set of key value pairs.

Which type argument would you use?

- A. type = map(string)
- B. type = any
- C. type = list(string)
- D. type = object( tags = string )

**Answer: A**

**Explanation:**

A. type = map(string)

In Terraform, `map(string)` is used when you want to define a variable that holds key-value pairs where both keys and values are strings – this is exactly how things like tags are typically structured.

### Question: 27

CertyIQ

Which of the following is allowed as a Terraform variable name?

- A. name
- B. source
- C. version
- D. count

**Answer: A**

**Explanation:**

**Why option A is the only valid Terraform variable name**

Terraform variables are identifiers that must follow HCL identifier rules: they can contain letters, numbers and underscores, must start with a letter, and cannot be a reserved Terraform keyword.

**name** complies with these rules – it begins with a letter, uses only alphanumeric characters, and is not a reserved keyword.

**source**, **version**, and **count** are each reserved identifiers in Terraform's syntax (e.g., **source** is a required attribute in module blocks, **version** is used in provider and module constraints, and **count** is a meta-argument for resource instantiation). Using any of these as a plain variable name would clash with the language's built-in semantics and is therefore prohibited.

**Validation logic applied**

1. **Starts with a letter?** – Yes for name; No violation for others.
2. **Contains only allowed characters?** – All options satisfy this, so it does not discriminate.
3. **Is the identifier a reserved keyword or built-in block argument?** – **source**, **version**, and **count** are reserved, making them invalid as free-standing variable names.
4. **Does it conflict with Terraform's internal attributes?** – Yes for the three reserved words, but not for **name**.

Hence, among the given choices, **name** is the only allowed variable name.

**References**

Terraform Language – Variables – <https://developer.hashicorp.com/terraform/language/expressions/variables>

Terraform Language – Reserved Words –

<https://developer.hashicorp.com/terraform/language/syntax#comments-and-reserved-words>

### Question: 28

CertyIQ

Terraform providers are part of the Terraform core binary.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. False**

The statement "Terraform providers are part of the Terraform core binary" is **False** for the following reasons:

**Modular Architecture:** Terraform is designed with a modular architecture. This means the core binary is intentionally kept lightweight and focused on the core functionality of infrastructure provisioning and state management.

**External Plugins:** Providers in Terraform are implemented as external plugins. This design allows for several benefits, including:

**Easier Maintenance and Updates:** Providers can be updated independently of the Terraform core, enabling more agile development and deployment of new features and fixes for cloud and infrastructure services.

**Customizability and Extensibility:** The plugin architecture makes it easier for the community and HashiCorp to develop and integrate new providers without altering the core binary.

**Reduced Binary Size:** By not bundling all providers within the core binary, the overall size of the Terraform download is significantly reduced, improving user experience, especially for those only needing a subset of providers.

**Deployment and Usage:** When using Terraform, providers are typically installed alongside the Terraform core through the terraform init command, which downloads and configures the required providers for a specific configuration. This step is necessary because the providers are not inherently part of the core binary.

**Why Other Options are Less Suitable (in this case, only one incorrect option exists, so focusing on why it's incorrect):**

**A. True:** This option is incorrect because it contradicts the modular, plugin-based design of Terraform. If providers were part of the core binary, it would lead to a bloated core, make updates more cumbersome, and contradict the observed behavior of needing to initialize providers separately.

**References**

[HashiCorp Terraform Documentation - Plugins](#)

[HashiCorp Terraform Documentation - Getting Started with Plugins \(Implicit in terraform init Explanation\)](#)

**Question: 29**

**CertyIQ**

Where can Terraform not load a provider from?

- A. Provider plugin cache
- B. Source code
- C. Official HashiCorp distribution on releases.hashicorp.com
- D. Plugins directory

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. Source code**

The correct answer is **B. Source code** because Terraform is designed to load providers from specific locations, but not directly from provider source code. Here's a breakdown of why:

#### Why B (Source code) is NOT a load location:

Terraform expects providers to be compiled binaries, not raw source code. Loading a provider from source code would require compilation on the fly, which is not a supported workflow in Terraform's architecture. Terraform's provider loading mechanism is designed to work with pre-compiled, distributable artifacts, not source code repositories.

#### Why other options are more suitable (making B the correct answer by elimination):

**A. Provider plugin cache:** This is a valid location. Terraform caches downloaded provider plugins in its cache directory (~/.terraform/d/plugins on Linux/Mac or C:\Users\<Username>\.terraform\d\plugins on Windows) to reuse them across runs, improving efficiency.

**C. Official HashiCorp distribution on releases.hashicorp.com:** This is the primary source for official Terraform providers. Terraform can download providers from this official repository when specified in the provider block with the correct version.

**D. Plugins directory:** While the specifics of the "Plugins directory" might vary (as it could refer to a custom directory configured for Terraform), in general, Terraform supports loading providers from a local directory if the provider binary is placed there and Terraform is configured to look for providers in that custom location (though this is less common for production workflows).

**Conclusion:** Given Terraform's design and workflow, loading a provider directly from **Source code (B)** is not supported, as it does not align with the expected binary distribution and loading process. The other options (A, C, and D, with proper configuration for D) are all viable ways to load providers into Terraform, albeit with differing use cases and frequencies of application.

#### References

[Terraform Providers Documentation](#) - Overview of working with providers in Terraform.

[Terraform Provider Installation](#) - Details on how Terraform installs and loads providers.

#### Question: 30

CertyIQ

You are creating a Terraform configuration which needs to make use of multiple providers, one for AWS and one for Datadog.

Which provider blocks would allow you to do this?

```
provider "aws" {
  profile = var.aws_profile
  region  = var.aws_region
}

provider "datadog" {
  api_key = var.datadog_api_key
  app_key = var.datadog_app_key
}
```

A.

```
provider {  
  "aws" {  
    profile = var.aws_profile  
    region  = var.aws_region  
  }  
  
  "datadog" {  
    api_key = var.datadog_api_key  
    app_key = var.datadog_app_key  
  }  
}
```

B.

```
terraform {  
  provider "aws" {  
    profile = var.aws_profile  
    region  = var.aws_region  
  }  
}
```

C.

```
terraform {  
  provider "aws" {  
    profile = var.aws_profile  
    region  = var.aws_region  
  }  
  
  provider "datadog" {  
    api_key = var.datadog_api_key  
    app_key = var.datadog_app_key  
  }  
}
```

D.

Answer: A

Explanation:

```
provider "aws" {  
  profile = var.aws_profile  
  region  = var.aws_region  
}  
  
provider "datadog" {  
  api_key = var.datadog_api_key  
  app_key = var.datadog_app_key  
}
```

### Question: 31

You have developed a new service that runs on Azure and uses proprietary APIs. You want to use Terraform to create, manage, and delete users from the service.  
How can Terraform interact with the service?

- A. Terraform can manage users for any service that is hosted on a public cloud provider.
- B. Terraform cannot use proprietary APIs.
- C. Develop and publish a custom provider to interact with the service using its proprietary APIs.
- D. Use the azurerm provider to leverage existing resource types.

**Answer: C**

**Explanation:**

#### Technical Justification

The correct answer is **C. Develop and publish a custom provider to interact with the service using its proprietary APIs**. Here's why:

#### Why C is the best option:

Terraform relies on providers to interact with cloud services, on-premises systems, and custom applications. Since the service in question uses **proprietary APIs**, a standard, existing provider (like azurerm for general Azure resources) won't support these custom APIs out of the box. Developing a **custom provider** allows for the direct integration of these proprietary APIs, enabling Terraform to create, manage, and delete users for the specific service.

#### Why other options are less suitable:

##### A. Terraform can manage users for any service that is hosted on a public cloud provider:

**Incorrect** because this statement overgeneralizes. Terraform's capability is provider-specific. While it can manage many resources, it requires a provider that supports the service's API, which is proprietary in this case.

##### B. Terraform cannot use proprietary APIs:

**Incorrect** as this is a misconception. Terraform can indeed interact with proprietary APIs, but it requires a custom provider to do so, as these APIs are not part of the standard, publicly supported providers.

##### D. Use the azurerm provider to leverage existing resource types:

**Incorrect for this specific task** because:

1. The azurerm provider manages Azure resources (e.g., VMs, Storage, DNS) but not custom, proprietary services unless they expose their APIs through Azure Resource Manager (ARM) in a way that's supported by the azurerm provider, which is not indicated here.
2. Managing users of a custom service with proprietary APIs falls outside the scope of standard Azure resource management.

#### References

[HashiCorp Terraform: Custom Providers](#)

[Azure Provider for Terraform \(for understanding azurerm limitations in this context\)](#)

### Question: 32

terraform init creates an example main.tf file in the current directory.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Answer: B –False**

**Why B is correct:** terraform init only prepares the working directory for subsequent commands by initializing the backend, downloading required provider plugins, and setting up the state file configuration. It never generates example configuration files such as main.tf; those must be created manually or via other scaffolding tools.

**Why A is not suitable:** The statement incorrectly assumes that terraform init adds an example main.tf. In reality, the command's purpose is initialization and dependency resolution, not file creation. Generating a sample file would be unexpected behavior and is not part of the official CLI semantics.

#### References

Terraform CLI init documentation – <https://developer.hashicorp.com/terraform/cli/init>

Terraform CLI command reference – <https://developer.hashicorp.com/terraform/cli/commands/init>

### Question: 33

Which command must you run before you run a plan or apply for the first time?

- A. terraform validate
- B. terraform workspace
- C. terraform import
- D. terraform init

**Answer: D**

**Explanation:**

#### Technical justification

**terraform init is required before any plan or apply on a fresh repository** because it:

Sets up the working directory (.terraform) where provider plugins, module templates and configuration state artifacts are stored.

Downloads the configured provider binaries and module files needed for the configuration language.

Initializes the backend (local or remote) and validates that the required configuration (e.g., backend "azurerm" or backend "s3" blocks) is present.

Without this step Terraform cannot resolve providers or modules, resulting in errors such as "Terraform has not been initialized. Run terraform init".

### Why the other options are not the mandatory first step

**terraform validate** – Checks syntactic correctness and internal consistency of the configuration, but it can be run later or even omitted on the very first execution; Terraform will still initialize and locate providers before validation is needed.

**terraform workspace** – Manages separate state files for the same backend; it is unrelated to the initialization of the working directory and is only relevant after the project has been set up.

**terraform import** – Brings existing external resources under Terraform management; it is optional and only needed when you want to bring pre-existing infrastructure into the state file, not for the initial plan/apply cycle.

Hence, terraform init is the prerequisite command that must be executed before any planning or applying operation for the first time.

---

## References

**Terraform init – Initialise a configuration directory –**

<https://developer.hashicorp.com/terraform/cli/commands/init>

**Lifecycle of a Terraform configuration –**

<https://developer.hashicorp.com/terraform/language/modules/develop#lifecycle>

These links are from the official Terraform documentation and provide detailed guidance on initialization and the overall workflow.

### Question: 34

CertyIQ

As a developer, you want to ensure your plugins are up-to-date with the latest versions. Which Terraform command should you use?

- A. terraform apply -upgrade
- B. terraform providers -upgrade
- C. terraform init -upgrade
- D. terraform refresh -upgrade

**Answer: C**

**Explanation:**

#### Technical justification

**terraform init -upgrade** is the official command that re-initializes the working directory and forces Terraform to download the latest compatible provider plugins, even if the current versions appear to be satisfied. This guarantees that the plugin binaries used during subsequent apply or plan runs are up-to-date with the newest releases available in the registry.

The other answer choices are invalid or do not achieve the goal:

**terraform apply -upgrade** – the -upgrade flag is not defined for apply; applying provisions resources and does not refresh provider plugins.

**terraform providers -upgrade** – there is no such command; terraform providers only lists versions and does not download or upgrade them unless invoked with terraform init -upgrade or terraform providers install.

**terraform refresh -upgrade** – refresh is used to update the state snapshot; it has no -upgrade flag and does not impact provider version installation.

Therefore, the only command that explicitly triggers a provider-plugin upgrade during initialization is **C. terraform init -upgrade**.

## References

**Terraform CLI Install and Upgrade** – Provides details on the -upgrade flag for terraform init.

<https://developer.hashicorp.com/terraform/cli/commands/init#upgrade>

**Terraform Provider Install & Management** – Explains how Terraform resolves and installs provider plugins, including the role of terraform init -upgrade.

<https://developer.hashicorp.com/terraform/cli/plugins#installing-providers>

## Question: 35

CertyIQ

What does terraform apply change after you approve the execution plan? (Choose two.)

- A. The state file
- B. The cloud infrastructure
- C. The execution plan
- D. The Terraform code
- E. The .terraform directory

**Answer: AB**

**Explanation:**

**Terraform apply updates the persisted state file** – After the plan is approved, apply calculates the actions required to achieve the desired state and writes that state back to the state file, reflecting the new resources and attributes.

**Terraform apply creates or modifies real cloud infrastructure** – The execution phase performs the actual API calls that provision, update, or destroy resources in the target provider (e.g., AWS, Azure, GCP), making the changes live in the environment.

**Options C and D are not modified by apply** – The execution plan is a read-only artifact generated by terraform plan; it is displayed but not changed by apply. The Terraform source code resides on disk and is never edited by the command.

**Option E is a runtime data directory** – .terraform holds internal working files (e.g., provider plugins, cached modules). While apply may update caches, these files are not considered “changed” in the sense of user-visible state; the command does not alter the directory structure itself.

**Bottom line:** terraform apply persists the new state to the state file and, through that, orchestrates the actual creation/modification of cloud resources, which is why **A and B** are the correct choices.

## References

1. Terraform Docs –apply command:<https://developer.hashicorp.com/terraform/cli/commands/apply>

### Question: 36

You modified your local Terraform configuration and ran terraform plan to review the changes. Simultaneously, your teammate manually modified the infrastructure component you are working on. Since you already ran terraform plan locally, the execution plan for terraform apply will be the same.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Explanation**

**Why the correct answer is “False” (B)**

When a teammate makes manual changes to the target infrastructure, the actual state stored in the remote backend is altered.

Your locally-generated Terraform plan was created based on the **original** remote state, not on the new state that now exists in the provider.

Consequently, the plans differ: yours reflects only your local changes, while the provider sees modifications you did not account for. Running terraform apply after this point will apply a plan that includes both your changes **and** whatever the provider now thinks needs to be reconciled, which is generally not the same as your original plan.

**Why the other option (“True”) is unsuitable**

The premise assumes that a locally-executed terraform plan captures the full desired state regardless of external drift.

In reality, Terraform plans are **state-driven**; they compare the desired configuration to the **current remote state**. If the remote state diverges — even without you making additional changes — the plan will be out-of-sync with what the provider expects.

Therefore, it is unsafe to assume that the execution plan will remain identical after concurrent manual modifications.

**Result:** The statement is false; you must refresh/pull the latest state before generating a new plan if any external modifications may have occurred.

---

### References

Terraform documentation: Backend Options –

<https://developer.hashicorp.com/terraform/language/settings/backends>

Terraform documentation: State management –<https://developer.hashicorp.com/terraform/language/state>

### Question: 37

Which is the correct workflow for deploying new infrastructure with Terraform?

- A. 1. Write Terraform configuration.  
2. Run terraform apply to create infrastructure.  
3. Use terraform validate to confirm Terraform deployed resources correctly.
- B. 1. Write Terraform configuration.  
2. Run terraform init to initialize the working directory or workspace.  
3. Run terraform apply.
- C. 1. Write Terraform configuration.  
2. Run terraform plan to initialize the working directory or workspace.  
3. Run terraform apply to create the infrastructure.
- D. 1. Write Terraform configuration.  
2. Run terraform show to view proposed changes.  
3. Run terraform apply to create new infrastructure.

**Answer: B**

**Explanation:**

**Why optionB is the correct workflow**

- 1. **terraform init must be executed first** – it sets up the working directory (downloads provider plugins, configures any backend, creates the .terraform directory). Without this step Terraform cannot locate the necessary providers or configure the execution environment, so any subsequent command would fail.
- 2. **terraform apply follows init** – after the environment is initialized, apply reads the configuration, builds the execution plan, prompts for approval (or uses -auto-approve), and then creates or updates the real infrastructure.
- 3. This sequence (write→init→apply) is the minimal, deterministic order required when introducing **new** infrastructure. It respects the dependency graph of resources while ensuring the provider state is ready.

**Why the other options are unsuitable**

**OptionA** – terraform validate is a syntax-and-consistency check; it does **not** confirm that resources were deployed correctly and should be run **before** apply. Using it after apply adds no value and interrupts the expected workflow.

**OptionC** – terraform plan does not initialize the working directory; it only generates a preview of changes. Running it before init will fail because Terraform cannot locate provider binaries or backend settings.

**OptionD** – terraform show merely displays the state or plan output; it is not a required step in the deployment pipeline and should not be placed before apply. Using it as a prerequisite step misrepresents the standard process.

**Reference Implementation**

Official Terraform CLI installation & command

overview:<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/install-cli>

Core workflow documentation (init → init →

apply):<https://developer.hashicorp.com/terraform/language/workspace/workflow>

These links detail the proper initialization and application sequence that underpins optionB.

### Question: 38

After creating a new Terraform configuration, your config passes terraform validate but gives an “Access Denied” error from the cloud provider when running terraform plan.

Why didn't validate catch this issue?

- A. The working directory was not initialized, so the cloud provider plugin wasn't available to use when running the terraform validate command.
- B. The remote backend wasn't configured, so terraform validate couldn't load the state and detect the missing credentials.
- C. terraform validate only checks if a configuration is syntactically correct and internally consistent, and does not communicate with providers.
- D. Variables are only applied and validated during a terraform plan, so validate assumed defaults and returned the success message.

**Answer: C**

**Explanation:**

**Why option C is the correct answer**

terraform validate only performs local checks: syntax correctness, required arguments are present, and internal consistency of expressions.

It never contacts any provider, reads state, or evaluates variable values that are only resolved during a plan/apply.

Consequently, an “AccessDenied” error caused by missing provider credentials is invisible to validate; the command considers the configuration valid and exits with a success status.

**Why the other options are not appropriate**

**A.** The provider plugins are loaded based on the backend configuration, not on whether the working directory has been initialized. validate can run without a prior terraform init and still perform its checks.

**B.** validate does not depend on a remote backend or state file; it operates entirely on the configuration files present in the working directory.

**D.** Variables are not “applied” during validate; they are parsed and type-checked, but no provider interaction occurs. The error is therefore unrelated to the phase of execution.

**Key takeaway**

terraform validate is deliberately limited to static analysis. Provider authentication failures are only detected when Terraform attempts to communicate with a provider, which happens during plan, apply, or other commands that require provider access.

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### References

**Terraform CLI Documentation – validate:** <https://developer.hashicorp.com/terraform/cli/commands/validate>

**Terraform Errors – Access Denied:**

<https://developer.hashicorp.com/terraform/language/expressions/functions#access-denied-errors-during-provider-interaction>

### Question: 39

Only the user that generated a terraform plan may apply it.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Correct answer: B – False**

Terraform executes plans and applies them using the credentials of the user who runs the command. The state file records the user who performed each apply, and applying a plan created by another user requires explicit permissions on the backend or remote state store.

By default, any user with access to the state can run terraform apply, regardless of who generated the plan. Restricting apply rights solely to the plan creator would require additional tooling (e.g., a custom script, policy-enforcement layer, or CI/CD approval step) and is not a built-in behavior of Terraform.

**Why the other option is less suitable**

**True** would imply that Terraform enforces per-user apply restrictions automatically, which it does not. Therefore, the correct logical statement is **False**, reflecting Terraform's default operation.

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## References

**Terraform documentation – Working with State:** <https://developer.hashicorp.com/terraform/language/state>

**Terraform documentation – CLI commands – apply:**

<https://developer.hashicorp.com/terraform/cli/commands/apply>

## Question: 40

**CertyIQ**

terraform apply will fail if you have not run terraform plan first to update the plan output.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Technical Justification**

The terraform plan command generates an execution plan that describes the actions Terraform will take, but it does **not** modify any real infrastructure.

terraform apply can be run directly; it will create a new plan internally if one does not already exist or if the underlying state has changed. Therefore, the requirement to first execute terraform plan is optional, not mandatory.

Skipping terraform plan is perfectly valid when you are confident in the configuration or when you intentionally want to apply changes without previewing them.

**Why OptionB (False) Is Correct**

OptionB states that the assertion “terraform apply will fail if you have not run terraform plan first” is false. This aligns with the behavior described above: Terraform will not abort; it will generate its own plan internally and proceed with the apply.

### Why OptionA (True) Is Incorrect

OptionA claims the statement is true. That would only be correct if Terraform enforced a hard rule that an external plan file must be provided before any apply. Terraform does **not** enforce such a rule, so the statement is inaccurate.

### Conclusion

The correct answer is **B. False**, because terraform apply can execute without a prior terraform plan run.

### References

Terraform Documentation – CLI Commands > *apply*:

<https://developer.hashicorp.com/terraform/cli/commands/apply>

Terraform Documentation – CLI Commands > *plan*:

<https://developer.hashicorp.com/terraform/cli/commands/plan>

## Question: 41

CertyIQ

What's the proper syntax for the plan command?

- A. terraform plan -generate-config-out=tfplan
- B. terraform plan -target=tfplan
- C. terraform plan -out=tfplan
- D. terraform plan -var-file=tfplan

### Answer: C

#### Explanation:

**Correct syntax:** terraform plan -out=tfplan

The -out flag directs Terraform to write the generated plan to a binary file (tfplan), which can later be applied with terraform apply. This is the standard way to persist a plan for audited or automated workflows.

#### Why the other options are incorrect:

**A. terraform plan -generate-config-out=tfplan** – -generate-config-out writes the configuration files, not a plan; it never creates a tfplan file.

**B. terraform plan -target=tfplan** – -target limits the scope of the plan to specific resources; it does not output a plan file.

**D. terraform plan -var-file=tfplan** – -var-file specifies a file containing variable definitions; it does not control the output of the plan.

### References

1. Terraform CLI documentation – Plan command options:  
<https://developer.hashicorp.com/terraform/cli/commands/plan#-out>
2. Using saved plans for reliable deployments:

### Question: 42

Which is true about terraform apply? (Choose two.)

- A. It only operates on infrastructure defined in the current working directory or workspace.
- B. You cannot target specific resources for the operation.
- C. Depending on provider specification, Terraform may need to destroy and recreate your infrastructure resources.
- D. You must pass the output of a terraform plan command to it.
- E. By default, it does not refresh your state file to reflect the current infrastructure configuration.

**Answer: AC**

**Explanation:**

**Why A and C are correct**

**A. Scope of operation** – terraform apply only manages resources that are declared in the current working directory (or an associated workspace). It cannot ingest files from outside that directory hierarchy unless they are referenced through a relative path or a module source that resolves inside the directory. This limitation is documented and reinforces the importance of running the command from the appropriate directory.

**C. Replacement behavior** – Many cloud providers require an in-place change to be implemented as a destroy-and-recreate cycle when the desired state cannot be achieved by updating the existing resource (e.g., changing resource instance types, modifying immutable fields, or updating certain networking configurations). terraform apply evaluates the plan and, when a replacement is required, issues a destroy of the old resource followed by a create of the new one, respecting the provider's semantics.

**Why the other options are not correct**

**B. Targeting resources** – Terraform does support targeting specific resources (e.g., terraform apply -target=module.web, terraform apply -target=aws\_s3\_bucket.my-bucket). Therefore, the statement “You cannot target specific resources for the operation” is false.

**D. Requirement of a plan output** – terraform apply can be invoked without any prior terraform plan output. When a plan file is not supplied, apply will internally generate a new plan and then apply it. Hence, passing the output of terraform plan is optional, not mandatory.

**E. State refresh behavior** – terraform apply always refreshes the state to reflect the real infrastructure before applying changes (unless -refresh=false is explicitly set). The default behavior is to refresh, making the claim “it does not refresh your state file” incorrect.

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### References

**Terraform Apply Documentation** – <https://developer.hashicorp.com/terraform/cli/commands/apply>

**Terraform State Management Overview** –

<https://developer.hashicorp.com/terraform/language/state/statefile> (includes discussion of refresh and replacement behavior)

### Question: 43

Which syntax check returns an error when you run terraform validate?

- A. None of these will return an error.
- B. There is a missing variable block.
- C. The state file does not match the current infrastructure.
- D. The code contains tabs for indentation instead of spaces.

**Answer: A**

**Explanation:**

**Technical justification (exam-style)**

**Purpose of terraform validate** – The command checks only the structural and syntactic correctness of the HCL configuration files (e.g., correct block delimiters, allowed characters, well-formed expressions). It does **not** perform any semantic analysis such as evaluating missing variables, comparing the state file with the live infrastructure, or enforcing code-style rules like indentation.

**Option A – “None of these will return an error.”** – Correct. Because terraform validate only validates syntax, none of the listed conditions (missing variable block, state-configuration drift, or tab-based indentation) will cause a validation error; they either are ignored or are caught later by other commands (e.g., terraform plan/apply or terraform fmt).

**Why option B is unsuitable** – A missing variable block is a semantic omission. While it may lead to runtime errors when the variable is referenced, terraform validate does not detect this omission; it only ensures that the block syntax itself is correct.

**Why option C is unsuitable** – State file consistency is verified by commands like terraform plan -refresh=false or the backend health checks, not by validate. validate never reads or compares the state, so a mismatch will not surface as a validation error.

**Why option D is unsuitable** – Although tabs violate the Terraform style guide, the HCL parser accepts them and does not raise a validation error. Formatting concerns are addressed by terraform fmt or third-party linting tools, not by validate.

**Conclusion:** terraform validate will only fail on syntax errors; therefore, none of the listed scenarios will cause a validation error, making **Option A** the definitive answer.

#### References

Terraform CLI Documentation – validate command:

<https://developer.hashicorp.com/terraform/cli/commands/validate>

Terraform Configuration Language – HCL syntax rules:

<https://developer.hashicorp.com/terraform/language/syntax>

### Question: 44

What is the purpose of the .terraform directory in a Terraform workspace?

- A. The directory contains plugins and modules that Terraform downloads during initialization, along with other important information.
- B. The directory contains the provider credentials and the .tfvars files to prevent them from being committed to version control by accident.

- C. The directory is where Terraform creates and maintains the state file to track the underlying resources it creates and manages.
- D. The directory is used to convert and store Terraform configuration files into API calls to communicate with the targeted platform.

**Answer: A**

**Explanation:**

**Why option A is correct**

The .terraform folder is created automatically after terraform init. Its primary role is to hold everything Terraform needs to manage the working directory independently of the configuration files. This includes:

Downloaded provider plugins and their cached binaries (/terraform/providers/...).

The module installation cache (/terraform/modules/).

The lock file that records provider and module versions (/terraform.lock.hcl).

Internal state-metadata files used by Terraform's debugging and introspection mechanisms.

Because these artifacts are generated on-the-fly, they must be kept separate from source code to allow version-controlled configurations and reproducible builds.

**Why the other options are not suitable**

**B.** Provider credentials and .tfvars files belong in the root of the configuration or in environment-specific variable files; they are **not** stored inside .terraform. The directory does not manage secret handling.

**C.** The actual persistent state file (terraform.tfstate) resides in the configuration directory (or a backend) and is **not** managed by .terraform. The folder contains only temporary/internal data, not the canonical state.

**D.** Terraform translates HCL into API calls internally; the conversion and request generation happen in memory while executing a plan/apply. There is no conversion or storage step that results in API-call artifacts being saved under .terraform.

**References**

Terraform CLI Working Directories – <https://developer.hashicorp.com/terraform/cli/workspaces#working-directories>

terraform init documentation – <https://developer.hashicorp.com/terraform/cli/commands/init#plugins-and-modules>

**Question: 45**

**CertyIQ**

How can terraform plan aid in the development process?

- A. Validates your expectations against the execution plan without permanently modifying state.
- B. Reconciles Terraform's state against deployed resources and permanently modifies state using the current status of deployed resources.
- C. Initializes your working directory containing your Terraform configuration files.
- D. Formats your Terraform configuration files.

**Answer: A**

**Explanation:**

## Technical justification

### A – Validates expectations against the execution plan without permanently modifying state

*terraform plan* generates a preview of the actions Terraform will take to achieve the desired configuration. It compares the current state with the desired state, lists additions, modifications, and destructions, and presents this as a read-only plan. This lets developers verify that the upcoming changes align with design goals, catch misconfigurations, and avoid accidental resource drift – all without touching the persisted state file.

### B – Reconciles state against deployed resources and permanently modifies state

This describes *terraform apply*, not *plan*. *apply* executes the plan and updates the state. *plan* only creates a tentative execution plan; it never writes state changes.

### C – Initializes the working directory

Initialization is performed by *terraform init*, which downloads provider plugins and sets up the backend. *plan* assumes the directory is already initialized.

### D – Formats configuration files

Formatting is handled by *terraform fmt*. *plan* does not perform any formatting operations.

Therefore, option **A** correctly captures the purpose of *terraform plan* in the development workflow.

## References

Terraform Documentation – [terraform plan](#)

Terraform Documentation – [terraform init](#)

## Question: 46

CertyIQ

You have a saved execution plan containing desired changes for infrastructure managed by Terraform. After running the command `terraform apply my.tfplan`, you receive the error shown in the Exhibit space on this page.

| Error: Saved plan is stale

|

| The given plan file can no longer be applied because the state was changed by another operation after the plan was created.

How can you apply the desired changes? (Choose two.)

- A. Generate a new execution plan file with `terraform plan`, and apply the new plan.
- B. Run `terraform apply` without the saved execution plan.
- C. Refresh the current state data using the `-refresh-only` flag.
- D. Force the `apply` command by adding the flag `-lock=false`.
- E. Update the current plan file using the `terraform state push` command.

## Answer: AB

### Explanation:

Provide a short, structured technical justification suitable for a certification exam review. do not use table structure, use paragraph or bullet point format. Explain why the correct option is best and why other options are less suitable. Be precise, technical, and professional. optional - Put 2 valid and working documentation links at the very end under a section titled References.

The correct options are A and B because they address the root cause of the “Saved plan is stale” error: the state has diverged from the plan, so the existing plan can no longer be safely applied. Generating a new

execution plan with `terraform plan` produces a fresh plan that reflects the current state, and then applying it with `terraform apply` updates the infrastructure according to the latest desired state. Simply running `terraform apply` without the saved plan also works, but it implicitly triggers a new plan within the `apply` command, effectively achieving the same outcome without needing a separate plan file.

Option C (refresh-only) only updates the state metadata; it does not create a new plan and therefore cannot resolve a stale plan error on its own. Option D (`-lock=false`) bypasses state locking but does not fix the staleness condition and can lead to race conditions or corruption. Option E (terraform state push) is unrelated to plan application and does not affect the plan file's validity. Hence, only A and B correctly resolve the issue by ensuring a fresh, consistent plan is applied.

References:

1. Terraform Documentation – Working with Plans:  
<https://developer.hashicorp.com/terraform/language/plan>
2. Terraform Documentation – apply Command:  
<https://developer.hashicorp.com/terraform/cli/commands/apply>

### Question: 47

CertyIQ

Which of these can you do with `terraform plan`? (Choose two.)

- A. Schedule Terraform to run at a planned time in the future.
- B. View the execution plan and check if the changes match your expectations.
- C. Execute a plan in a different workspace.
- D. Save a generated execution plan to apply later.

**Answer: BD**

**Explanation:**

**Why B and D are correct**

#### **B – View the execution plan and verify changes**

`terraform plan` generates a detailed proposal of the actions Terraform will take to reach the desired state. It lists resource additions, modifications, and destructions, allowing you to review and confirm that the upcoming changes align with expectations before any modifications are persisted.

#### **D – Save a generated execution plan for later application**

The `-out` (or `-out=plan.out`) flag lets you serialize the plan into a binary file. This saved plan can be reapplied later with `terraform apply plan.out`, enabling controlled, repeatable deployments (e.g., in CI pipelines or manual approvals).

**Why A and C are not appropriate**

#### **A – Schedule Terraform to run at a future time**

Scheduling execution is outside the scope of `terraform plan`. Scheduling is handled by external orchestration tools (e.g., Jenkins, GitHub Actions, Terraform Cloud) and is not a feature of the `plan` command itself.

#### **C – Execute a plan in a different workspace**

While workspaces isolate state, the `plan` command does not execute changes; it merely produces a plan. To apply a plan in another workspace you would need to switch workspaces (e.g., `terraform workspace select`

other) or copy the plan file, but the command does not perform execution across workspaces.

## References

### Terraform documentation – terraform plan

<https://developer.hashicorp.com/terraform/cli/commands/plan>

### Saving and loading plans

<https://developer.hashicorp.com/terraform/cli/commands/plan#save-the-plan>

These links provide the official specifications for viewing plans and persisting them for later use.

## Question: 48

CertyIQ

Which task does terraform init not perform?

- A. Sources any remote modules and downloads them.
- B. Connects to the configured backend.
- C. Validates that values are set for all required input variables.
- D. Sources all providers used in the configuration and downloads them.

**Answer: C**

### Explanation:

**Why the correct option (C) is the only one that does not describe a task performed by terraform init**

**A. Sources any remote modules and downloads them** – terraform init resolves and fetches module sources defined with source arguments, placing them in the module cache. This is a core responsibility of the initialization command.

**B. Connects to the configured backend** – During initialization Terraform reads the backend configuration (e.g., S3, Azure Blob, Terraform Cloud) and establishes the connection needed for state storage.

**D. Sources all providers used in the configuration and downloads them** – Provider plugins are discovered from the required\_providers block and from any required\_providers entries, then installed into the working directory's .terraform folder.

**C. Validates that values are set for all required input variables** – Variable validation and requirement checking are performed by terraform validate and terraform plan, not by terraform init. Initialization only prepares the environment; it does **not** evaluate input variable definitions.

Therefore, option **C** correctly identifies a task that terraform init does not execute.

### Why the other options are unsuitable

**A, B, and D** each describe a concrete step that terraform init must carry out to make the working directory usable: module resolution, backend connection, and provider installation.

**C** confuses initialization with later validation phases; Terraform never checks variable completeness during init.

## References

Terraform CLI Documentation – terraform init: <https://developer.hashicorp.com/terraform/cli/commands/init>  
Remote Module Execution – Modules: <https://developer.hashicorp.com/terraform/language/modules/development> (module source resolution details)

### Question: 49

CertyIQ

terraform init retrieves and caches the configuration for all remote modules.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Justification**

The terraform init command initializes a working directory that contains Terraform configuration files. During initialization it downloads and installs the required provider plugins and, if the configuration references any **remote modules**, it fetches those modules from their source locations (e.g., Terraform Registry, Git, S3, etc.) and stores them in the .terraform directory.

After the retrieval, each module's source files become available locally, and Terraform can later create resources based on those modules without needing external network calls.

Therefore, the statement “retrieves and caches the configuration for all remote modules” accurately describes the purpose of terraform init.

**Why the other option is less suitable**

Selecting “False” would contradict the documented behavior of terraform init, which explicitly includes downloading remote modules as part of its initialization steps. Choosing “False” ignores the core responsibility of the command and would be technically incorrect.

**References**

Terraform Documentation – Initialization: <https://developer.hashicorp.com/terraform/cli/init>  
Terraform Registry – Using Remote Modules: <https://developer.hashicorp.com/terraform/language/modules/sources#source-addresses>

### Question: 50

CertyIQ

You just upgraded the version of a provider in an existing Terraform configuration. What will install the new provider?

- A. Run terraform plan.
- B. Run terraform apply.
- C. Run terraform init -upgrade.
- D. Run terraform refresh.

**Answer: C**

**Explanation:**

**Why option C is the correct choice**

**terraform init -upgrade** explicitly tells Terraform to re-download and install any newer versions of referenced providers that satisfy the version constraints in the configuration. This command is designed for exactly the scenario of upgrading a provider version in an already-existing codebase without destroying resources. It performs a **safe upgrade**: providers are fetched into the .terraform directory, the backend is re-initialized if needed, and the existing state and configuration remain untouched until a subsequent terraform apply is run. Other commands address different phases of the workflow:

**A. terraform plan** – only analyzes the configuration and produces a preview; it does not download or install new provider binaries.

**B. terraform apply** – creates or modifies infrastructure but will fail if the required provider version is not present; it does not install anything by itself.

**D. terraform refresh** – updates the state file to reflect the real infrastructure but never touches provider binaries.

Therefore, to actually install the new provider version after changing its constraints, you must run **terraform init -upgrade**.

**Why the other options are unsuitable**

They either miss the step of downloading the updated provider binaries or they only operate on the state/plans, not on provider installation.

Using any of them alone would not guarantee the new provider code is available for subsequent operations, potentially leading to errors or unexpected behavior.

**References**

Terraform Documentation – Initialization: [https://developer.hashicorp.com/terraform/cli/commands/init# - upgrade](https://developer.hashicorp.com/terraform/cli/commands/init#-upgrade)

Terraform Registry – Provider Installation:

<https://developer.hashicorp.com/terraform/language/providers/installation#installing-providers>

**Question: 51**

**CertyIQ**

terraform validate confirms the syntax of Terraform files.

A. True

B. False

**Answer: A**

**Explanation:**

**Answer: A – True**

**Technical justification**

terraform validate performs a static analysis of all .tf files. It checks for syntax errors, provider configuration

issues, and correct usage of Terraform language constructs (blocks, arguments, expressions). If the validation succeeds, Terraform reports “Success! The given configuration is valid.” indicating the syntax is correct; otherwise it returns detailed error messages describing the specific problems. The command does **not** perform state checks, plan previews, or resource creation — it only verifies structural and lexical correctness of the HCL code. Therefore the statement “terraform validate confirms the syntax of Terraform files” is accurate. Option **B** is incorrect because terraform validate does confirm syntax; it does not skip that step.

#### Why the other option is less suitable

B (“False”) would be chosen only if the command performed validation beyond syntax (e.g., checking resource dependencies). Since its primary purpose is syntax checking, labeling it as “False” mischaracterizes its function.

#### References

Terraform documentation – Validate: <https://developer.hashicorp.com/terraform/cli/commands/validate>  
Terraform language documentation – Configuration syntax: <https://developer.hashicorp.com/terraform/language/syntax>

### Question: 52

CertyIQ

A Terraform configuration is returning errors when running terraform plan. How would you enable debug logging to investigate further?

- A. Rerun terraform plan with the correct flag.
- B. Edit the client configuration file.
- C. Update the backend configuration.
- D. Set the correct environment variable.

**Answer: D**

#### Explanation:

The correct choice is **D. Set the correct environment variable.**

Enabling debug-level logging in Terraform is done by setting the TF\_LOG environment variable to DEBUG (or a specific level such as INFO, WARN, ERROR). When this variable is exported before running terraform plan, Terraform prints detailed internal logs that help diagnose configuration-related errors. This method directly influences the logging behavior without modifying files or infrastructure settings.

#### Why the other options are unsuitable

**A – Rerun terraform plan with the correct flag:** Terraform does not have a built-in “flag” for increasing log output; the only supported way is via TF\_LOG.

**B – Edit the client configuration file:** Terraform’s client configuration (e.g., ~/.terraformrc) controls provider credentials and backend settings, not log verbosity.

**C – Update the backend configuration:** Backend settings govern where state is stored and how it is accessed; they do not affect logging levels.

Therefore, setting TF\_LOG=DEBUG (or another appropriate level) is the precise, officially supported mechanism for obtaining debug output during a plan run.

## References

Set environment variables for logging –

<https://developer.hashicorp.com/terraform/language/values/functions/log>

Debugging and logging in Terraform – <https://developer.hashicorp.com/terraform/tutorials/cli/config-file#logging>

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### Question: 53

CertyIQ

Why would you use the -replace flag for terraform apply?

- A. You want to force Terraform to destroy a resource on the next apply.
- B. You want Terraform to ignore a resource on the next apply.
- C. You want Terraform to destroy all the infrastructure in your workspace.
- D. You want to force Terraform to destroy and recreate a resource on the next apply.

Answer: D

Explanation:

#### Why the -replace flag is the correct choice

The -replace flag tells Terraform to treat a specified resource address as if it were newly created, forcing a **destructive destroy followed by a fresh apply** of that resource.

This is required when the configuration of a resource must be changed in an **immutable** way (e.g., altering a calculated attribute, changing a lifecycle rule, or updating a field that cannot be drifted).

By using -replace, you can keep the rest of the state intact while ensuring the targeted resource is recreated on the next terraform apply, avoiding plan failures caused by configuration drift.

The operation is **explicit and surgical** — only the addressed resource is affected, unlike a full-workspace destroy.

#### Why the other options are incorrect

##### A. “Force Terraform to destroy a resource on the next apply.”

Destroying a resource alone can be done with terraform destroy or by removing it from the configuration; -replace does more than merely destroy — it recreates the resource after the destroy.

##### B. “Ignore a resource on the next apply.”

Ignoring resources is achieved via the -ignore-apply-errors flag or by using lifecycle directives, not by -replace.

##### C. “Destroy all the infrastructure in your workspace.”

-replace targets a **specific resource address**; it does not automatically affect every resource in the workspace. A full-workspace destroy requires different commands.

**Key takeaway:** -replace is the precise mechanism to **force a destroy-and-recreate cycle for a single resource**, making option **D** the only technically accurate answer.

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## References

Terraform CLI -replace flag: <https://developer.hashicorp.com/terraform/cli/commands/apply#replace>

### Question: 54

CertyIQ

How does Terraform determine dependencies between resources when it creates an execution plan?

- A. Terraform builds a resource graph based on your configuration and your state file (if present).
- B. Terraform requires resources in your configuration be listed in the order they will be created to determine dependencies.
- C. Terraform requires all dependencies between resources be specified using the `depends_on` parameter.
- D. Terraform requires resource dependencies be defined as modules and sourced in order.

**Answer: A**

**Explanation:**

**Why option A is correct**

Terraform constructs an internal resource graph by parsing every module and root module in the configuration.

If a **state file** is present, Terraform also reads the current state to identify any implicit dependencies (e.g., a resource that references an output exported by another).

The graph is traversed to create an execution plan that respects all prerequisite relationships before the plan is applied.

**Why the other options are not suitable**

**B.** Terraform does **not** require the user to list resources in the order they will be created; the engine determines the order automatically from the dependency graph.

**C.** While `depends_on` can force a dependency, it is **not required**; most dependencies are inferred automatically, and omitting it does not prevent Terraform from building a valid plan.

**D.** Dependencies do not need to be expressed as ordered modules or sourced sequentially; modules are just reusable containers and the engine resolves intra-module dependencies using the same graph-building logic.

**References**

Terraform Documentation – Resources and Dependencies:

<https://developer.hashicorp.com/terraform/language/resources-deps>

Terraform Documentation – Resource Graph: <https://developer.hashicorp.com/terraform/language/resource-graph>

### Question: 55

CertyIQ

You create a `main.tf` Terraform configuration consisting of an application server, a database, and a load balancer. You run `terraform apply` and Terraform created all the resources successfully. Now you realize that you do not actually need the load balancer, so you run `terraform destroy` without any flags. What will happen?

- A. Terraform will prompt you to confirm that you want to destroy all the infrastructure.
- B. Terraform will prompt you to pick which resource you want to destroy.

- C. Terraform will immediately destroy all the infrastructure.
- D. Terraform will destroy the application server because it is listed first in the code.
- E. Terraform will destroy the main.tf file.

**Answer: A**

**Explanation:**

#### **Technical Justification for Correct Answer: A**

When running terraform destroy without any flags, Terraform's behavior is to terminate all resources managed in the current working directory's configuration (in this case, defined in main.tf). Here's why the correct answer is A and why other options are less suitable:

**Correct Answer: A**

**Terraform will prompt you to confirm that you want to destroy all the infrastructure.**

**Reason:** By default, terraform destroy will display a preview of the destruction plan and then prompt for confirmation before proceeding. This is a safety feature to prevent accidental destruction of entire infrastructures. Since the command lacks flags that might specify a target resource (e.g., -target) or auto-confirm destruction (e.g., -auto-approve), Terraform seeks user confirmation to ensure the action is intentional.

#### **Incorrect Answers with Justifications**

**B. Terraform will prompt you to pick which resource you want to destroy.**

**Why Incorrect:** This behavior would require a specific interactive mode or flag not mentioned in the question. Terraform does not natively offer a "pick which resource" prompt during destroy without additional configuration or flags like -target.

**C. Terraform will immediately destroy all the infrastructure.**

**Why Incorrect:** Without the -auto-approve flag, Terraform will not proceed with destruction without user confirmation, making this outcome incorrect for the given command.

**D. Terraform will destroy the application server because it is listed first in the code.**

**Why Incorrect:** The order of resource definition in Terraform configuration files does not dictate the order of destruction in a blanket terraform destroy command. Terraform determines the destruction order based on dependency graphs, not the order in the config file.

**E. Terraform will destroy the main.tf file.**

**Why Incorrect:** terraform destroy targets the infrastructure resources defined in the configuration files, not the configuration files themselves. The file main.tf would remain intact.

#### **References**

[Terraform Destroy Command Documentation](#)

[Terraform Configuration and State Documentation](#)

#### **Question: 56**

**CertyIQ**

You developed a new Terraform configuration for two virtual machines with a cloud provider. You're ready to create the infrastructure for the first time. Which Terraform command should you run first?

- A. terraform apply
- B. terraform plan
- C. terraform init
- D. terraform show

**Answer: C**

**Explanation:**

#### **Technical Justification for Correct Answer: C (terraform init)**

When creating infrastructure for the first time with a new Terraform configuration, the initial step is crucial for setting up the environment correctly. Here's why terraform init is the best first choice, followed by why other options are less suitable for the initial step:

#### **Why C (terraform init) is Correct**

**Initialization Requirement:** Before any operations (plan, apply, or show), Terraform requires initialization to set up the working directory, download and install the required providers (in this case, the cloud provider), and configure the backend (if any).

**Prerequisite for Other Commands:** Running terraform init ensures the environment is properly configured for subsequent commands like terraform plan or terraform apply. Without initialization, these commands cannot execute as they rely on the setup provided by terraform init.

**Safety and Configuration Check:** It performs a safety check on the configuration, ensuring that the prerequisites for deployment are met before attempting to plan or apply the changes.

#### **Why Other Options are Less Suitable as the First Step**

##### **A. terraform apply:**

**Risk of Unprepared Environment:** Directly applying changes without ensuring the environment is set up correctly can lead to errors or unexpected behavior.

**Lacks Preliminary Check:** Skipping initialization and planning steps can lead to unforeseen infrastructure provisions without a prior review of the changes.

##### **B. terraform plan:**

**Dependent on Initialization:** Like apply, plan requires the environment to be initialized first. Running it without terraform init will fail.

**Useful After Initialization:** However, it's highly recommended to run terraform plan after terraform init and before terraform apply to review the planned changes.

##### **D. terraform show:**

**Irrelevant to New Infrastructure Creation:** terraform show is used to display the current infrastructure's state. Since no infrastructure exists yet, this command does not contribute to the setup process.

**Utility in Existing Environments:** Useful for understanding the current state of already provisioned infrastructure.

#### **Recommended Initial Workflow for New Infrastructure**

1. **terraform init** (Correct Answer for the first step)
2. **terraform plan** (To review planned changes)
3. **terraform apply** (To provision the infrastructure)

#### **References**

[HashiCorp Terraform Documentation - terraform init](#)

**Question: 57**

When should you run terraform init?

- A. After you start coding a new Terraform project, and before you run terraform plan for the first time.
- B. Every time you run terraform apply.
- C. Before you start coding a new Terraform project.
- D. After you run terraform plan for the first time in a new Terraform project, and before you run terraform apply.

**Answer: A**

**Explanation:**

**Technical Justification for Correct Answer: A**

The correct answer is **A. After you start coding a new Terraform project, and before you run terraform plan for the first time.** Here's why:

**Initialization Necessity:** terraform init is used to initialize a Terraform working directory. This command prepares the working directory by installing and configuring required providers, setting up backend settings (if configured), and managing the Terraform state (if a backend is specified). It's essential to run this after starting to code to ensure the infrastructure as code (IaC) configuration is properly set up with the required dependencies.

**Sequence Logic:**

Coding the project first allows for defining the infrastructure requirements.

Running terraform init afterward ensures the project directory is properly initialized with the necessary providers and settings before any planning or application of changes.

Running terraform plan subsequently allows for reviewing the planned changes without applying them, which is a best practice for verifying the correctness of the infrastructure changes.

**Why Other Options are Less Suitable:**

**B. Every time you run terraform apply:**

Unnecessary Overhead: Running terraform init every time before terraform apply introduces unnecessary overhead, as the initialization only needs to occur once per project (or when the backend/provider versions change).

Potential for Unintended Provider Updates: Frequent re-initialization could lead to unintended provider version updates if not carefully managed.

**C. Before you start coding a new Terraform project:**

Premature Initialization: Initializing before coding doesn't make sense because there's no configuration for Terraform to initialize with. The command would essentially do nothing useful in this context.

**D. After you run terraform plan for the first time in a new Terraform project, and before you run terraform apply:**

Incorrect Sequence for Initialization: terraform plan can technically run without a full terraform init (though init is usually implied in the first run of any command in a directory), but the key point is that initialization should happen before planning to ensure everything is set up correctly from the start.

## References

[HashiCorp Terraform Documentation - terraform init](#)

[HashiCorp Terraform Getting Started Guide](#) (See the "Initialize the Working Directory" section)

### Question: 58

CertyIQ

Which configuration consistency error does terraform validate report?

- A. Declaring a resource identifier more than once.
- B. Differences between local and remote state.
- C. An out-of-date module.
- D. A mix of spaces and tabs in configuration files.

**Answer: A**

**Explanation:**

#### Technical Justification for Correct Answer: A

The Terraform validate command is primarily designed to check the syntax, format, and certain aspects of configuration consistency within Terraform configuration files. Here's why option A is the correct answer and why the others are less suitable:

#### Why A is Correct: Declaring a resource identifier more than once.

**Reason:** Terraform's validate command checks for duplicate resource identifiers because having two resources with the same identifier in the same configuration would lead to a conflict during the apply phase, as Terraform would not know how to reconcile the two declarations.

**Technical Impact:** Duplicate resource identifiers violate the fundamental principle of resource uniqueness in Terraform, which is crucial for maintaining a predictable infrastructure state.

**Validation Scope:** This check falls under the configuration consistency errors that terraform validate can detect through static analysis of the configuration files.

#### Why Other Options are Less Suitable:

##### B. Differences between local and remote state.

**Reason for Exclusion:** While crucial for Terraform's operational integrity, differences between local and remote states are typically identified during terraform plan or terraform apply (when Terraform interacts with the backend/state store), not during terraform validate. validate focuses on configuration syntax and consistency, not state synchronization.

##### C. An out-of-date module.

**Reason for Exclusion:** Module version checks (e.g., ensuring a module is up-to-date) are not within the scope of terraform validate. This would more likely be addressed through other practices (e.g., regular dependency updates, using specific version pins in module declarations).

##### D. A mix of spaces and tabs in configuration files.

**Reason for Exclusion:** Although a mix of spaces and tabs can lead to configuration readability and potential merge conflicts in version control, Terraform's validate command does not check for this as it's more of a

coding style issue rather than a configuration consistency error that would prevent Terraform from functioning correctly. Tools like linters (e.g., Terraform Lint, terraform fmt for formatting) are more suited for addressing such issues.

## References

[Terraform Validate Command Documentation](#)

[Terraform Resource Documentation \(emphasizing unique identifiers\)](#)

## Question: 59

CertyIQ

Which Terraform command checks that your configuration syntax is correct?

- A. terraform init
- B. terraform show
- C. terraform fmt
- D. terraform validate

**Answer: D**

**Explanation:**

**Technical Justification for Correct Answer: D. terraform validate**

The correct option for checking the syntax of your Terraform configuration is **D. terraform validate**. Here's why:

**Reason for Choosing D:** terraform validate is specifically designed to check the configuration files for syntax errors and ensure that the configuration is valid according to the current Terraform version's schema. It verifies that the Terraform file format is correct and that there are no obvious errors in the configuration that would prevent terraform apply from executing.

**Why Other Options are Less Suitable:**

**A. terraform init:** This command initializes a Terraform working directory, configuring the backend, copying modules, and installing providers. While it may indirectly identify some configuration issues by failing if critical errors exist, its primary purpose is not syntax validation. **Key Difference:** Initialization vs. Syntax Validation.

**B. terraform show:** This command displays a human-readable representation of the infrastructure configuration and state. It does not perform syntax checking on the configuration files. **Primary Use:** Visualization of Current State.

**C. terraform fmt:** Although crucial for maintaining consistency in formatting, terraform fmt auto-formats the Terraform configuration files based on a standard style to make them more readable. It does not check for syntax errors. **Focus:** Code Formatting vs. Syntax Validation.

**Key Points Reinforcing D as the Best Choice:**

**Specificity:** terraform validate is specifically designed for syntax and configuration validation.

**Accuracy:** It provides direct feedback on syntax errors without the overhead of other operations.

**Best Practice:** Running terraform validate is a recommended step in the Terraform workflow for ensuring config correctness before initiation or application.

## References

[HashiCorp Terraform Command Reference - Validate Terraform Workflow - Validation Step](#)

### Question: 60

CertyIQ

Which of the following is not an action performed by terraform init?

- A. Create template configuration files.
- B. Retrieve the source code for all referenced modules.
- C. Initialize a configured backend.
- D. Load required provider plugins.

**Answer: A**

**Explanation:**

#### Technical Justification for Correct Answer: A

The correct answer is A, "Create template configuration files", as it is **not** an action performed by terraform init. Here's a breakdown justifying why A is incorrect and why the other options are correct actions of terraform init:

**A. Create template configuration files:** This is **not** an action of terraform init. The creation of template configuration files (e.g., main.tf, variables.tf, etc.) is a manual process performed by the user before running terraform init. Terraform init expects these files to already exist. Instead, it processes and sets up the environment based on the configurations defined in these files.

**B. Retrieve the source code for all referenced modules:** This is an action of terraform init. When you run terraform init, if your configuration references external modules (e.g., from a Git repository, AWS S3, etc.), Terraform will retrieve the source code for these modules to make them available for the Terraform workflow.

**C. Initialize a configured backend:** This is an action of terraform init. The command initializes the backend as configured in your Terraform settings (e.g., setting up the backend for state management with AWS S3, GCP Cloud Storage, etc.). This step is crucial for managing the Terraform state.

**D. Load required provider plugins:** This is an action of terraform init. Running terraform init downloads and installs the provider plugins (e.g., AWS, Azure, GCP, etc.) specified in your provider blocks or inferred from your resource types, making them available for use in your Terraform execution.

#### Why A is the Correct Answer (i.e., Not an Action of terraform init):

**Pre-Existence Requirement:** Terraform requires configuration files to pre-exist because terraform init is designed to process and set up the Terraform working directory based on the configurations you've manually created.

**Tool's Design Purpose:** The primary design of terraform init is to prepare the Terraform working directory for execution (e.g., downloading modules and providers, initializing the backend), not to generate the initial configuration templates.

## References

[Terraform Documentation - terraform init](#)

[Terraform Getting Started - Configuring State](#) (Implicitly covers the initialization aspect of backend configuration)

### Question: 61

Which of the following arguments are required when declaring a Terraform output?

- A. default
- B. value
- C. description
- D. sensitive

**Answer: B**

**Explanation:**

#### Technical justification

**value** – This argument is mandatory in an output block because it tells Terraform what piece of data the output should expose; without it the block would be syntactically invalid and the configuration would fail to plan/apply.

**default** – Only used to provide a fallback value when the output is not set; it is optional and does not affect the block's validity.

**description** – A human-readable annotation that improves documentation and IDE hints; it is purely informational and not required for execution.

**sensitive** – Controls whether the output value is masked in logs and UI; it is a security flag and can be omitted when the output is not sensitive.

Therefore, among the listed arguments, **only value is required** when declaring a Terraform output.

#### References

Terraform Documentation – Output Values: <https://developer.hashicorp.com/terraform/language/values/outputs>

Terraform CLI – output Block Details: <https://developer.hashicorp.com/terraform/cli/outputs>

### Question: 62

Outside of the required\_providers block, Terraform configurations always refer to providers by their local names.

- A. True
- B. False

**Answer: A**

**Explanation:**

#### Justification

-In a Terraform configuration, provider references outside of the required\_providers block are made by the alias that you assign in that block (e.g., aws, azurearm).

- The alias is defined once and can then be used throughout the rest of the configuration (module sources, provider arguments, etc.). This is why the statement is **True**.
- OptionB (“False”) would imply that you must use the full provider address (hashicorp/aws) outside the block, which contradicts the way Terraform resolves providers internally and would break the configuration.

#### Why the other option is not suitable

-Terraform resolves providers only by the alias supplied in required\_providers. Using the address syntax elsewhere is not required and is discouraged; it would defeat the purpose of the alias-based local naming scheme. Hence “False” does not correctly describe the behavior described in the question.

#### References

- [Terraform Registry – Provider Requirement Syntax]  
(<https://developer.hashicorp.com/terraform/language/providers/require>)
- Provider Configuration – Using Aliases

### Question: 63

CertyIQ

The Exhibit section of this page shows part of a configuration you’ve been asked to update. data “azurerm\_resource\_group” “example” name = var.resource\_group\_name

```
resource “azurerm_virtual_network” “example”  
name = _____
```

The name of the Azure Virtual Network should be set to the name of the resource group followed by a dash and the word “vnet”.

Which expression fulfills this requirement?

- A. “\$ azurerm\_resource\_group.example.name -vnet”
- B. join("-",var.resource\_group\_name,"vnet")
- C. concat(data.azurerm\_resource\_group.example.name, "-", "vnet")
- D. “\$ data.azurerm\_resource\_group.example.name -vnet”

**Answer: D**

**Explanation:**

#### Why optionD is the correct expression

It uses the proper **data source reference** (data.azurerm\_resource\_group.example.name) that matches the declared data block in the example.

It encloses the interpolation in double-quotes, allowing Terraform to resolve the expression as a quoted string.

The resulting value is exactly <resource-group-name>-vnet, satisfying the requirement to append “-vnet” to the resource-group name.

#### Why the other options are less suitable

**OptionA** – Uses \$ azurerm\_resource\_group.example.name which references a **resource** block name instead of the declared **data source**; the name azurerm\_resource\_group.example does not refer to the data block defined above, so Terraform cannot resolve it.

**OptionB** – Calls join("-", var.resource\_group\_name, "vnet"). join expects a list as its second argument, and introducing a variable that holds only the raw name bypasses the data source altogether. The expression therefore does not guarantee it reads the value from the configured data source.

**OptionC** – Although `concat()` would produce a valid string, it must be used on a list and returns a list; the syntax shown (`concat(data.azure_rm_resource_group.example.name, "-", "vnet")`) does not wrap the result in parentheses or explicitly convert it to a string, which can lead to type mismatches in newer Terraform versions.

Hence, optionD is the most precise, idiomatic and reliable way to construct the required VNet name.

---

## References

**Terraform Language – Expressions:** <https://developer.hashicorp.com/terraform/language/expressions>

**Data Sources – Azure Resource Group:**

[https://registry.terraform.io/providers/hashicorp/azurerm/latest/docs/data-sources/resource\\_group](https://registry.terraform.io/providers/hashicorp/azurerm/latest/docs/data-sources/resource_group)

These links provide the official documentation supporting the correct use of data source interpolation and expression syntax in Terraform.

### Question: 64

CertyIQ

Your team is collaborating on infrastructure using Terraform and are formatting their code to follow Terraform language style conventions.

How can you update your code to meet these requirements?

- A. Run `terraform validate` prior to executing `terraform plan` or `terraform apply`.
- B. Replace all tabs with spaces within your Terraform configuration files.
- C. Terraform automatically formats configuration on `terraform apply`.
- D. Run `terraform fmt` to update your Terraform configurations.

**Answer: D**

**Explanation:**

**Technical justification**

**Option D – terraform fmt**

`terraform fmt` rewrites configuration files in the canonical Terraform style, normalizing spacing, line breaks, and indentation while preserving content.

It can be run automatically (e.g., in CI pipelines) to enforce a consistent formatting convention across all files. The command is idempotent; running it on already-formatted code produces no changes, making it safe for iterative development and code-review workflows.

**Why the other options are less suitable**

**A. terraform validate** – Checks syntactic correctness and internal consistency but does **not** modify formatting or style.

**B. Replacing tabs with spaces** – Manual text substitution does not address the full set of style conventions (e.g., line length, quote usage, ordering of blocks) and is error-prone; Terraform's formatter handles these rules automatically.

**C. Terraform automatically formats configuration on terraform apply** – Terraform never rewrites files during `apply`; it only validates and applies changes. Formatting must be performed explicitly via `terraform fmt`.

Therefore, the recommended approach is to run terraform fmt on the codebase.

## References

Terraform CLI – terraform fmt documentation:

<https://developer.hashicorp.com/terraform/cli/commands/formatted>

Terraform style and formatting best practices: <https://developer.hashicorp.com/terraform/language/style-guide#formatting-files>

### Question: 65

CertyIQ

A resource block is shown in the Exhibit space of this page.

```
resource "aws_instance" "web"
```

```
count = 2
```

```
name = "terraform-$ count.index "
```

How do you reference the name value of the second instance of this resource?

- A. `aws_instance.web.*.name`
- B. `aws_instance.web[2].name`
- C. `element(aws_instance.web, 2)`
- D. `aws_instance.web[1].name`
- E. `aws_instance.web[1]`

**Answer: D**

**Explanation:**

**Correct answer – D. `aws_instance.web[1].name`**

The resource `aws_instance.web` is defined with `count = 2`, creating two instances indexed from 0.

Index 1 refers to the second instance, and accessing the name attribute (`aws_instance.web[1].name`) returns exactly the name value of that second instance.

**Why the other options are incorrect**

**A. `aws_instance.web.*.name`** – The `.` syntax expands to a set of values, returning a collection rather than a single string; it cannot be used directly to obtain a scalar value.

**B. `aws_instance.web[2].name`** – Indexing starts at 0, so `[2]` attempts to access a non-existent third instance, causing a runtime error.

**C. `element(aws_instance.web, 2)`** – The `element` function expects an index; using 2 again points to a non-existent element, whereas the correct index is 1.

**E. `aws_instance.web[1]`** – This returns the entire resource object for the second instance, not just its name attribute.

## References

Terraform CLI Documentation – Resource Addressing:

<https://developer.hashicorp.com/terraform/cli/syntax/resources>

Terraform Functions – `element`: <https://developer.hashicorp.com/terraform/language/functions/element>

### Question: 66

When using multiple configurations of the same Terraform provider, what meta-argument must you include in any non-default provider configurations?

- A. alias
- B. name
- C. depends\_on
- D. id

**Answer: A**

**Explanation:**

#### Technical Justification

When Terraform needs to instantiate more than one instance of a single provider (e.g., two distinct AWS accounts), every non-default provider configuration must be given a unique **alias** meta-argument.

The alias serves as a distinct provider name that Terraform uses internally to reference that specific configuration instance. Without an alias, Terraform cannot differentiate the multiple providers and will raise an error at plan/apply time.

The alias is required only for the **non-default** (i.e., explicitly aliased) providers; the default provider block does **not** need an alias.

#### Why the other choices are incorrect

**B. name** – Terraform does not expose a name meta-argument for providers; the concept of naming a provider block is handled solely by the alias attribute.

**C. depends\_on** – This meta-argument controls ordering of resources or modules, not provider instance identification, and it cannot be used to differentiate provider configurations.

**D. id** – id is the computed resource address of a provisioned resource after apply; it is an output value, not a configuration attribute that can be set on a provider block.

Therefore, **alias** is the only meta-argument that must be included when referencing a non-default provider configuration multiple times within the same Terraform module.

#### References

Terraform Provider Configuration: <https://developer.hashicorp.com/terraform/language/providers>

Using Provider Aliases:

[https://developer.hashicorp.com/terraform/language/providers#configuring\\_multiple\\_instances\\_of\\_one\\_provider](https://developer.hashicorp.com/terraform/language/providers#configuring_multiple_instances_of_one_provider)

### Question: 67

You want to create a string that is a combination of a generated random\_id and a variable and reuse that string several times in your configuration.

What is the simplest correct way to implement this without repeating the random\_id and variable?

- A. Add an output value.
- B. Add a local value.
- C. Use a data source.

D. Use a module.

**Answer: B**

**Explanation:**

**Answer:**

**B. Add a local value** – Correct

Locals are designed for storing intermediate results (e.g., a concatenated string of a random ID and another variable) that can be referenced multiple times throughout the same configuration via `local.<name>`.

Using a local eliminates repetition of the random ID and the variable and keeps the expression evaluation clear and manageable.

The value can be defined once and reused anywhere in the configuration without needing to re-evaluate the expression.

**A. Add an output value** – Incorrect

Outputs expose values to the caller (other configurations) but do not provide an in-module variable that can be reused locally; they are intended for external consumption, not internal reuse.

Overkill when the goal is simply to avoid repeated calculations inside the same configuration.

**C. Use a data source** – Incorrect

Data sources fetch external data (e.g., from other providers) and are not meant for storing computed strings based only on internal inputs.

They introduce unnecessary complexity and cannot be used as a persistent, reusable variable within the same module.

**D. Use a module** – Incorrect

Modules encapsulate sets of resources and configurations, typically for re-usability across separate projects or environments.

Creating a module solely to reuse a concatenated string would violate the principle of “doing the simplest thing that works” and adds an unnecessary abstraction layer.

---

## References

HashiCorp Terraform Docs – Locals: <https://developer.hashicorp.com/terraform/language/expressions/locals>

HashiCorp Terraform Docs – Expressions – Concatenation:

[https://developer.hashicorp.com/terraform/language/expressions#string\\_literal\\_concatenation](https://developer.hashicorp.com/terraform/language/expressions#string_literal_concatenation)

## Question: 68

**CertyIQ**

Your Terraform configuration manages a resource that requires maximum uptime. You need to update the resource, and when you run `terraform plan`, Terraform notes that the update requires the resource to be destroyed and recreated.

Which lifecycle rule can you add to the resource to reduce the downtime of the resource while still applying the update?

A. `prevent_destroy = true`

B. `destroy = false`

C. `create_before_destroy = true`

D. ignore\_changes = all

**Answer: C**

**Explanation:**

**Why option C is correct**

Adding create\_before\_destroy = true tells Terraform to provision a new resource first, then switch traffic to it, and finally destroy the old resource. This sequencing eliminates any gap where the resource is absent, thus minimizing downtime while still allowing the replacement of the resource that must be recreated.

The other options do not address the recreate-only situation:

prevent\_destroy = true stops Terraform from destroying the resource at all, which would block the required update.

destroy = false merely postpones the destroy step but still destroys the resource later without mitigating the outage caused by the recreation.

ignore\_changes = all tells Terraform to ignore all changes to the resource, which would also prevent you from applying the necessary update and therefore is not a viable solution.

**Why the remaining options are unsuitable**

**A – prevent\_destroy = true:** Useful when you never want to delete the resource, but it would stop Terraform from ever recreating the resource that needs replacement.

**B – destroy = false:** Can delay a destroy, yet it still results in a period where the resource is missing while Terraform attempts to rebuild it, leading to the same downtime issue.

**D – ignore\_changes = all:** By ignoring all attribute changes, Terraform will not notice any modification you intend to make, making it impossible to trigger the required recreation and update.

**References**

Terraform Documentation – Lifecycle Configuration:

<https://developer.hashicorp.com/terraform/language/resources/lifecycle>

Create Before Destroy – How to use it safely:

[https://developer.hashicorp.com/terraform/language/resources/lifecycle#create\\_before\\_destroy](https://developer.hashicorp.com/terraform/language/resources/lifecycle#create_before_destroy)

### Question: 69

CertyIQ

The Terraform configuration shown in the Exhibit space on this page will create a new AWS instance. data “aws\_instance” “web” filter name = “tag:Name” values = [“web”]

A. True

B. False

**Answer: B**

**Explanation:**

**Justification**

The data "aws\_instance" "web" block is used to **read** existing EC2 instances that already have a tag Name equal to "web"; it does **not** create a new instance.

Creation of an EC2 instance in Terraform is performed with the aws\_instance resource block, not with a data block.

Therefore the configuration described will **only search** for matching instances, and if none are found it will simply return an empty result – it will not provision a new instance.

Consequently, the statement “the configuration will create a new AWS instance” is **false**.

### Why option A is not suitable

Option A claims the configuration creates a new instance, which contradicts how Terraform’s data sources work.

Data sources are read-only; they cannot modify infrastructure, so they cannot be used to launch resources.

### Result

The correct answer is **B – False**.

### References

Terraform AWS Provider – Data Sources:

<https://registry.terraform.io/providers/hashicorp/aws/latest/docs/data-sources>

aws\_instance Resource – Creation Example:

<https://registry.terraform.io/providers/hashicorp/aws/latest/docs/resources/instance#example-usage>

## Question: 70

CertyIQ

What kind of configuration block will manage an infrastructure object with settings specified within the block?

- A. data
- B. locals
- C. provider
- D. resource

### Answer: D

### Explanation:

### Justification

**Resource block (D)** – This is the Terraform configuration element that creates and manages an infrastructure object (e.g., aws\_instance, google\_storage\_bucket). All settings that define the desired state of that object are written inside its body.

**Data block (A)** – Intended for reading existing external data or gathering information at plan time; it does not create or modify resources.

**Locals block (B)** – Used to define internal, reusable values for the current configuration. It does not interact with the provider or affect any real infrastructure.

**Provider block (C)** – Configures authentication and feature enablement for a specific service (e.g., AWS, GCP). It does not itself manage an infrastructure object; it only tells Terraform how to talk to the service.

Therefore, the block type that directly manages an infrastructure object with its settings defined inside the

block is the **resource** block.

#### References

[Terraform Language – Resources](#)

[Terraform Language – Providers](#)

#### Question: 71

CertyIQ

You can execute `terraform fmt` to standardize all Terraform configurations within the current working directory to Terraform's canonical format and style.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Justification**

`terraform fmt` rewrites every supported configuration file under the current working directory (and its sub-directories) to HashiCorp's canonical formatting rules.

It normalizes whitespace, ordering of arguments, line breaks, and other stylistic elements, ensuring a consistent code base without altering the underlying configuration semantics.

The command exits with status 0 when changes are applied, making it safe to run as part of CI pipelines or pre-commit hooks.

Because the tool's primary purpose is exactly to standardize Terraform code within the directory you invoke it from, option **A – True** correctly describes its behavior.

Option **B – False** would imply the command does not perform formatting or that it operates differently; however, the official documentation confirms that `terraform fmt` precisely applies canonical formatting to all files in the target directory, so the false statement does not align with Terraform's documented functionality.

**References**

Official CLI documentation: <https://developer.hashicorp.com/terraform/cli/commands/fmt>

Terraform tutorial on formatting: <https://developer.hashicorp.com/terraform/tutorials/cli/fmt>

#### Question: 73

CertyIQ

A resource block is shown in the Exhibit space of this page.  
resource "kubernetes\_namespace" "example"  
name = "test"

How would you reference the attribute name of this resource in HCL?

- A. `resource.kubernetes_namespace.example.name`
- B. `data.kubernetes_namespace.name`
- C. `kubernetes_namespace.example.name`
- D. `kubernetes_namespace.test.name`

**Answer: C**

**Explanation:**

**Technical justification (bullet-point format)**

The HCL syntax for referencing a resource attribute is `<resource_type>.<resource_name>.<attribute>`.

In the excerpt the resource block defines a resource of type **kubernetes\_namespace** with the local name **example**.

Therefore the attribute **name** is accessed as **kubernetes\_namespace.example.name** – option **C**.

**OptionA** (`resource.kubernetes_namespace.example.name`) is invalid because `resource` is not a valid prefix for attribute lookup; only the resource type and local name are used.

**OptionB** (`data.kubernetes_namespace.name`) references a data source, which does not exist in the given snippet and thus cannot resolve the attribute.

**OptionD** (`kubernetes_namespace.test.name`) uses the wrong local name (**test** vs **example**) that was declared in the block.

**References**

HashiCorp Configuration Language (HCL) Resource addressing:

[https://developer.hashicorp.com/terraform/language/syntax#resource\\_addresses](https://developer.hashicorp.com/terraform/language/syntax#resource_addresses)

Terraform Namespacing and naming conventions:

<https://developer.hashicorp.com/terraform/language/modules/developing#naming-scope>

## Question: 74

CertyIQ

You need to deploy resources into two different regions in the same Terraform configuration using the block in the Exhibit space on this page. provider “aws” region = “us-east-1”

```
provider “aws”  
region = “us-west-2”
```

What do you need to add to the provider configuration to deploy the resource to the us-west-2 AWS region?

A. resource “aws\_instance” “example-us-west-2”  
ami = data.aws\_ami.ubuntu.id  
instance\_type = “t3.micro”

B. provider “aws”  
region = “us-east-1”

provider “aws” “west”  
region = “us-west-2”

C. provider “aws\_west”  
region = “us-west-2”

D. provider “aws”  
region = “us-east-1”

```
provider “aws”  
alias = “west”  
region = “us-west-2”
```

**Answer: D**

**Explanation:**

## Technical justification

To reference multiple AWS providers within a single configuration, each provider must be given a unique **alias** and its own region setting.

The original default provider (provider "aws" region = "us-east-1" ) continues to serve the us-east-1 region.

Adding a second provider with an alias (e.g., "west") allows Terraform to switch to a different region (us-west-2) when that alias is used in resource blocks, without affecting the original provider.

**Option D** correctly implements this pattern:

```
provider "aws" alias = "west" region = "us-west-2"
```

The alias can then be applied to a resource, e.g.,

```
resource "aws_instance" "example-west" provider = aws.west ...
```

## Why the other options are unsuitable

- A** – Defines a resource block that references data.aws\_ami.ubuntu.id but does not specify a provider at all; Terraform would default to the first (us-east-1) provider, so the resource would not be deployed to us-west-2.
- B** – Attempts to add a new provider block without an alias, which would create a duplicate default provider and cause a configuration error (multiple providers with the same name).
- C** – Uses an invalid provider type (aws\_west) that does not match the required syntax (aws), leading to a parse error.

## References

Provider Configuration Syntax – <https://developer.hashicorp.com/terraform/language/providers/config>

Using Provider Aliases – <https://developer.hashicorp.com/terraform/language/resources/providers#using-aliases-for-multiple-providers-of-the-same-type>

## Question: 75

CertyIQ

Two resources blocks are shown in the Exhibit space on this page: azurerm\_linux\_web\_app, and azurerm\_role\_assignment. When provisioned, the web app will use the role assignment during creation, so the role assignment must be created first.

```
resource "azurerm_linux_web_app" "app" {
  name = "example-app"
  resource_group_name = azurerm_resource_group.rg.name
  location = azurerm_resource_group.rg.location
  service_plan_id = azurerm_service_plan.plan.id

  identity {
    type = "UserAssigned"
    identity_ids = [azurerm_user_assigned_identity.app.id]
  }
}
```

```
resource "azurerm_role_assignment" "kv_access" {
  scope = azurerm_key_vault.kv.id
  role_definition_name = "Key Vault Secrets User"
  principal_id = azurerm_user_assigned_identity.app.principal_id
}
```

How do you ensure the azurerm\_role\_assignment resource is created first?

- A. Add a depends\_on argument to the azurerm\_linux\_web\_app.
- B. Change the order of the azurerm\_linux\_web\_app and azurerm\_role\_assignment blocks.
- C. Add a create\_before\_destroy argument to the azurerm\_role\_assignment.
- D. Add a count argument to both resources.

**Answer: A**

### Question: 76

CertyIQ

A Terraform local value can reference other Terraform local values.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Answer: A.True**

#### Technical justification

Terraform's locals block defines a set of key-value pairs that act as a namespace for intermediate values. These values are evaluated before any resource or provider configuration is processed, and they follow the same reference rules as variables: a local can read the value of another local that has been declared earlier in the same block, or that depends on it.

The Terraform engine builds a dependency graph, so circular references that are not self-contained are rejected, but straightforward cross-references (e.g., local.instance\_name = "app-\$ local.environment ") are fully supported.

This capability is deliberately designed to let you compute derived values once and reuse them throughout the configuration, making the configuration more maintainable and less error-prone.

#### Why the alternative (B) is less suitable

OptionB ("False") incorrectly asserts that locals cannot reference one another. In reality, Terraform explicitly allows such references, and the language reference explicitly describes this behavior.

Choosing B would contradict both the language semantics and the official documentation, leading to an incorrect answer on the exam.

#### References

Terraform Documentation – Locals: <https://developer.hashicorp.com/terraform/language/expressions/locals>

### Question: 77

CertyIQ

You want to define multiple data disks as nested blocks inside the resource block for a virtual machine. What Terraform feature would help you define the blocks using the values in a variable?

- A. Count arguments
- B. Collection functions
- C. Local values
- D. Dynamic blocks

**Answer: D**

**Explanation:**

**Justification**

**Dynamic blocks** allow a resource argument (such as disk) to be generated repeatedly from a variable list, enabling each disk to be expressed as a separate nested block without manually writing repetitive code. By iterating over a collection (e.g., `var.disks`), Terraform produces a block instance for each element, preserving the hierarchical structure expected by the provider.

This approach integrates cleanly with `for_each` or indexed collections, keeping configuration DRY and making the VM definition scalable when the number of disks varies.

**Why the other choices are less appropriate**

**Count arguments** – only support a single numeric count and create multiple resource instances, not nested blocks with arbitrary attributes; they cannot dynamically vary block parameters per element.

**Collection functions** – are utility functions for manipulating collections; they help generate data but do not provide a Terraform mechanism for rendering nested blocks.

**Local values** – store computed values internally for reuse within a module but cannot influence resource configuration structures such as nested blocks.

**References**

Dynamic Blocks: <https://developer.hashicorp.com/terraform/language/resources/dynamic-blocks>  
for\_each and nested resources: [https://developer.hashicorp.com/terraform/language/count-arguments#for\\_each](https://developer.hashicorp.com/terraform/language/count-arguments#for_each)

### Question: 78

CertyIQ

You are creating a reusable Terraform configuration and want to include an optional `billing_dept` tag so your Finance team can track team-specific spending on resources. Which of the following `billing_dept` variable declarations will achieve this?

- A. variable "billing\_dept"  
default = ""
- B. variable "billing\_dept"

type = optional(string)  
C. variable "billing\_dept"  
optional = true  
D. variable "billing\_dept"  
type = default

#### Answer: A

#### Explanation:

The correct declaration is **A – variable "billing\_dept" default = ""** because it defines a string-typed variable and assigns an empty string as the default value. An empty default makes the argument optional (i.e., it may be omitted when the variable is used), while still preserving the declared type. This is the idiomatic Terraform pattern for optional string variables.

#### Why A is correct:

- Uses the standard default attribute to provide a fallback value.
- The empty string ("" ) signals that the variable is optional but always returns a string.
- Terraform treats any variable without a supplied value as having the default value, fulfilling the Finance team's requirement to tag resources optionally.

#### Why the other options fail:

- **B – type = optional(string)** – The optional keyword is not a valid argument for the type attribute; Terraform does not support a type = optional(...) syntax.
- **C – optional = true** – optional is not a recognized argument within a variable block; only default, type, and a few other meta-arguments are allowed.
- **D – type = default** – type must be a specific type such as string, number, or any, not the keyword default; placing default here is syntactically invalid.

Thus, only option A follows the correct Terraform syntax and semantics for an optional billing department tag.

#### References

<https://developer.hashicorp.com/terraform/language/variables/definition#default>  
<https://developer.hashicorp.com/terraform/language/variables/definition#type>

#### Question: 79

CertyIQ

Part of a configuration is shown in the Exhibit space of this page. data "vsphere\_datacenter" "dc" resource "vsphere\_folder" "parent" path = "Production" type = "vm" datacenter\_id = \_\_\_\_\_

You want to pass the id of the vsphere\_datacenter data source to the datacenter\_id argument of the vsphere\_folder resource.

Which reference would you use?

- A. data.vsphere\_datacenter.dc
- B. data.vsphere\_datacenter.dc.id
- C. vsphere\_datacenter.dc.id
- D. data.dc.id

#### Answer: B

## Explanation:

### Justification

#### Option B – `data.vsphere_datacenter.dc.id`

The `vsphere_datacenter` data source returns a map that contains the attribute `id`, which uniquely identifies the datacenter object in vSphere.

To satisfy the `datacenter_id` argument of the `vsphere_folder` resource, you must supply the ID (a string/number) rather than the whole data source reference.

Therefore, the expression `data.vsphere_datacenter.dc.id` correctly extracts that identifier and passes it to the resource.

#### Option A – `data.vsphere_datacenter.dc`

This returns the entire data object, which includes numerous attributes (e.g., `mo_id`, `name`, `parent`).

The `vsphere_folder` data source expects a **scalar** value for `datacenter_id`; passing a complex map causes a type-mismatch error.

#### Option C – `vsphere_datacenter.dc.id`

This refers to a local variable or resource name named `vsphere_datacenter`.

No such variable is declared in the snippet; only a **data source** named `vsphere_datacenter` exists. Using the raw type name without the `data.` prefix would result in an undefined reference.

#### Option D – `data.dc.id`

The identifier `dc` is the label (or alias) of the data source, not its type name.

The correct syntax to reference the data source is `data.vsphere_datacenter.dc`. Without the leading `data.` the reference is ambiguous and will be ignored by the HCL parser.

Hence, the only expression that both accesses the data source correctly and extracts its `id` attribute is **`data.vsphere_datacenter.dc.id`**.

---

## References

Terraform vSphere Provider – Data Sources: `vsphere_datacenter`

<https://registry.terraform.io/providers/hashicorp/vsphere/latest/docs/data-sources/datacenter>

Terraform `vsphere_folder` Resource – `datacenter_id` argument

[https://registry.terraform.io/providers/hashicorp/vsphere/latest/docs/resources/folder#datacenter\\_id](https://registry.terraform.io/providers/hashicorp/vsphere/latest/docs/resources/folder#datacenter_id)

## Question: 80

CertyIQ

Your Terraform configuration declares a variable, you want to enforce that its value meets your specific requirements, and you want to block the Terraform operation if it does not.

What should you add to your configuration?

- A. Add a top-level check block.
- B. Add a validation block to the variable block.
- C. Add a top-level validation block.
- D. Add a check block to the variable block.

**Answer: B**

**Explanation:**

**Justification**

**Option B – Add a validation block to the variable declaration**

The validation meta-argument (available from Terraform 0.12+) lets you define a custom validation rule using condition and an informative error\_message.

When the condition evaluates to false, Terraform aborts the plan/apply with the supplied error, effectively “blocking” the operation.

The block is placed **inside** the variable resource, making the rule scoped precisely to that variable.

**Why the other options are unsuitable**

**A. Top-level check block** – check is a generic testing construct used in provisioners or in the Terraform Cloud/Enterprise “policy as code” feature; it cannot be used directly in a variable block, nor does it provide a standard validation message.

**C. Top-level validation block** – A top-level validation block is not a supported syntax; validation blocks must be nested under specific resources (e.g., variable, resource, locals). Placing it at the root would cause a parsing error.

**D. check block to the variable block** – check is not a valid attribute for a variable; it belongs to resource-level testing or policy checks, and Terraform will reject it as an illegal nested block.

**Conclusion**

Only a validation block **inside** the variable declaration satisfies the requirement of enforcing a specific value and halting execution when the condition is not met.

**References**

[Variable Validation – HashiCorp Documentation](#)

[Validation Blocks – HashiCorp Documentation](#)

**Question: 81**

**CertyIQ**

Running terraform fmt without any flags in a directory with Terraform configuration files will check the formatting of those files, but will never change them.

A. True

B. False

**Answer: B**

**Explanation:**

**Key Points**

terraform fmt without flags **re-writes** the files to apply the recommended formatting.

The command is designed to modify the files in-place; it does not perform a read-only check.

If you want only a diagnostic check (no changes), you must add the -check flag, which makes the operation

return a non-zero exit status when files need formatting but leaves them untouched.

#### Why OptionB (False) is correct

The statement claims that terraform fmt will never change files. This is inaccurate because the default behavior is to **format and save** the files, thereby altering them.

#### Why OptionA (True) is not suitable

It misrepresents the command's effect. While terraform fmt -check can be used to verify formatting without modification, the plain terraform fmt always applies formatting changes when needed.

#### Technical Summary

Default execution: reads each .tf file, rewrites it with canonical formatting, and writes the changes back to disk.

Exit code: 0 on success, non-zero only when files are unchanged.

Without -check, Terraform treats the operation as a transformation step, making it effectively a **mutating** command.

#### References

Terraform CLI – fmt command: <https://developer.hashicorp.com/terraform/cli/commands/fmt>

Formatting and checking files: <https://developer.hashicorp.com/terraform/language/formatting>

### Question: 82

CertyIQ

You want to retrieve the ID of an existing cloud resource for use in your Terraform configuration. Which block type would you use?

- A. resource
- B. import
- C. data
- D. output

#### Answer: C

#### Explanation:

##### Explanation

To reference an existing cloud resource without creating it, Terraform uses **data sources**, which are declared with the **data** block. This block retrieves the resource's attributes (such as its identifier) so they can be used elsewhere in the configuration, for example in output variables or other resources.

**resource block** creates a new resource; it cannot retrieve an already-existing one.

**import** is a command-line operation that brings an existing resource under Terraform management but is not a configuration block.

**output** only exposes values after they have been computed; it does not fetch external resource data.

Therefore, the appropriate block to **retrieve the ID of an existing cloud resource** is the data block.

#### Why the other options are unsuitable

**A. resource** – creates new resources; it does not read existing ones.

**B. import** – a CLI command, not a block used inside the configuration file.

**D. output** – only publishes values generated by Terraform; it cannot query external resources directly.

## References

Terraform Data Sources: <https://developer.hashicorp.com/terraform/language/data-sources>

Data Blocks Overview: <https://developer.hashicorp.com/terraform/language/resources/data>

### Question: 83

CertyIQ

Which of these commands makes your code more human-readable?

- A. terraform show
- B. terraform fmt
- C. terraform output
- D. terraform validate

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. terraform fmt**

The correct option for making your code more human-readable is **B. terraform fmt**. Here's why:

**B. terraform fmt:** This command automatically formats your Terraform configuration files to adhere to the standard Terraform style guide. It ensures consistent indentation, spacing, and formatting, making the code significantly more readable for humans. By enforcing a uniform style, terraform fmt enhances code readability, maintainability, and collaboration among team members.

**Why Other Options Are Less Suitable:**

**A. terraform show:** This command displays the current state of your infrastructure resources in a human-readable format, but it does **not** modify your configuration files to improve their readability. Its purpose is more about inspecting the current state of resources rather than enhancing the readability of your Terraform code.

**C. terraform output:** Similar to terraform show, this command is used to display values from infrastructure after apply, based on output blocks in your configuration. It has no impact on the readability of your Terraform configuration files themselves.

**D. terraform validate:** Although crucial for ensuring your Terraform configuration is valid and follows the language rules, terraform validate (or the newer terraform plan with validation capabilities) does **not** alter the formatting or readability of your code. Its primary function is to check for errors, not to improve code aesthetics or readability.

**References:**

[HashiCorp Terraform Documentation - terraform fmt](#)

[HashiCorp Terraform Documentation - Commands Overview](#) (for context on other commands)

## Question: 84

You want to split a large main.tf file into multiple files for better organization within the same working directory. What impact should you expect when you run a terraform plan?

- A. The terraform plan will fail since variable and provider blocks must appear before resource blocks.
- B. Significant changes should be expected since moving resources between files can trigger a destroy/replacement.
- C. No impact since Terraform reads all .tf files as a single configuration when executing a terraform plan.
- D. Only minor impacts since Terraform parses the main.tf file first, which could change the order in which resources are created.

**Answer: C**

**Explanation:**

**Technical Justification for Correct Answer C**

The correct answer is **C. No impact since Terraform reads all .tf files as a single configuration when executing a terraform plan.** Here's why:

**Terraform Configuration Aggregation:** When running terraform plan (or any Terraform command), Terraform aggregates all files with the .tf extension in the working directory into a single configuration set. This means that regardless of how you split your configuration across multiple .tf files, Terraform treats them as a unified set of configurations.

**Order of Evaluation, Not File Structure:** While it's true that certain blocks (like provider and variable definitions) need to be evaluated before resources can be planned, this is based on the logical dependency graph within the aggregated configuration, not the file structure. As long as the split files maintain the necessary logical order of definitions (e.g., providers and variables before resources that depend on them), there will be no issue.

**Resource Tracking by ID, Not File Location:** Terraform tracks resources by their ID (typically a combination of the resource type and a unique identifier within that type) in the state file, not by the file in which they were defined. Therefore, moving a resource block from one file to another within the same directory does not change its identity in Terraform's state, leading to **no expectation of destroy/replacement triggers solely due to file reorganization.**

**Why Other Options Are Less Suitable:**

**A. Terraform plan will fail since variable and provider blocks must appear before resource blocks.**

**Incorrect:** While the statement about the order of blocks is technically correct (providers and variables should be defined before resources that use them), Terraform's aggregation of all .tf files into one configuration set means this order is considered across all files, not per file. A fail would occur if the aggregated config violates this rule, but not merely from splitting files.

**B. Significant changes should be expected since moving resources between files can trigger a destroy/replacement.**

**Incorrect:** The trigger for destroy/replacement is a change in the resource's configuration or its dependencies, not the physical location of the resource block within the configuration files.

**D. Only minor impacts since Terraform parses the main.tf file first, which could change the order in which resources are created.**

**Incorrect:** The parsing order (if any slight preference exists in practice) does not significantly impact the

planning outcome, as the final plan is based on the dependency graph, not the parsing order of files. The term "minor impacts" is misleading in this context, as the expected outcome is actually **no impact** on the plan due to file splitting.

## References

HashiCorp Terraform Documentation - Configuration Files

[HashiCorp Terraform Documentation - Resource Dependencies]

([https://www.terraform.io/docs/language/meta-arguments\\_deprecation.html#depends\\_on](https://www.terraform.io/docs/language/meta-arguments_deprecation.html#depends_on))

## Question: 85

CertyIQ

You need to deploy resources into two different regions in the same Terraform configuration. To do this, you declare multiple provider configurations as shown in the Exhibit space on this page.

```
provider "aws"
```

```
region = "us-east-1"
```

```
provider "aws"
```

```
alias = "west"
```

```
region = "us-west-2"
```

What meta-argument do you need to configure in a resource block to deploy the resource to the us-west-2 AWS region?

- A. provider = west
- B. provider = aws.west
- C. alias = aws.west
- D. alias = west

## Answer: B

### Explanation:

#### Technical Justification for Correct Answer: B

To deploy resources across multiple regions in the same Terraform configuration, declaring multiple provider configurations with aliases (as shown) is the correct approach. The key step is correctly referencing the aliased provider in the resource block to specify the target region.

#### Why B (provider = aws.west) is Correct:

When using multiple provider configurations with aliases, Terraform requires the provider meta-argument in the resource block to specify which provider configuration to use.

The syntax provider = aws.west directly references the named provider configuration by its alias (west) within the aws provider namespace, ensuring the resource is deployed to the us-west-2 region as configured.

This syntax is in line with Terraform's documentation for using multiple providers with aliases.

#### Why Other Options are Less Suitable:

**A (provider = west):** Incorrect syntax. Does not specify the provider namespace (aws), which could lead to ambiguity if multiple providers (e.g., AWS, Azure) have an alias named west.

**C (alias = aws.west):** Misuse of the alias meta-argument. The alias is used in the provider configuration to define an alias, not in the resource block to select a provider.

**D (alias = west):** Similar to C, this misuses the alias meta-argument in the resource block. Additionally, it lacks the provider namespace specification.

### Correct Configuration Example for a Resource in us-west-2:

```
# Provider configurations (as given)provider "aws"    region = "us-east-1" provider "aws"    alias = "west" region = "us-west-2"
```

#### References:

[Terraform Documentation: Multiple Provider Configurations](#)

[Terraform Documentation: Aliases with Multiple Provider Configurations](#)

### Question: 86

CertyIQ

A resource block is shown in the Exhibit space on this page.  
resource "google\_compute\_instance" "main"  
name = "test"

What is the Terraform resource name of the resource block?

- A. test
- B. compute\_instance
- C. google
- D. main

#### Answer: D

#### Explanation:

##### Technical Justification for Correct Answer: D. main

The correct Terraform resource name of the given resource block is **D. main**. Here's why:

**Resource Block Syntax:** In Terraform, a resource block is defined with the syntax resource "TYPE" "NAME" ... . In this case, the syntax breaks down into:

resource: Keyword indicating a resource block.

"google\_compute\_instance": The **TYPE** of the resource (a Google Compute Instance).

"main": The **NAME** of the resource (assigned by the user for reference within the Terraform configuration).

**Resource Name vs. Attribute Name:** The **resource name** (in this context) refers to the user-assigned name within the resource block ("main"), not to be confused with the name attribute ("test") which is a property of the resource that sets the actual name of the created Google Compute Instance.

#### Why Other Options are Less Suitable:

**A. test:** This is the value of the name attribute within the resource block, which sets the name of the Google Compute Instance itself, not the Terraform resource name.

**B. compute\_instance:** Part of the **TYPE** (google\_compute\_instance), indicating the type of resource, not the name of the resource block.

**C. google:** Also part of the **TYPE** (google\_compute\_instance), specifying the provider, not the resource block's name.

#### References

For deeper understanding of Terraform's resource block syntax and configuration:

1. [Terraform Documentation - Resource Syntax](#)

### Question: 87

A variable declaration is shown in the Exhibit space on this page.

```
variable "sizes"
```

```
type = list(string)
```

```
description = "Valid server sizes"
```

```
default = ["small", "medium", "large"]
```

Which is the correct way to get the value of medium from this variable?

- A. `var.sizes[0]`
- B. `var.sizes[1]`
- C. `var.sizes[2]`
- D. `var.sizes[3]`

**Answer: B**

**Explanation:**

#### Technical Justification for Correct Answer (B)

The variable "sizes" is declared with the type `list(string)` and has a default value set to `["small", "medium", "large"]`. To access the value "medium" from this list, one must understand how indexing works in Terraform (the HashiCorp tool implied in the context of this question). In Terraform, like in many programming languages, list indexing starts at **0**. This means the first element of the list is at index 0, the second at index 1, and so on.

**Correct Answer (B) - `var.sizes[1]`:** Since "medium" is the second element in the list `["small", "medium", "large"]`, it is correctly accessed using the index **1**. This aligns with zero-based indexing principles.

#### Why Other Options Are Less Suitable:

**A. `var.sizes[0]`:** This would return "small", the first element of the list, not "medium". Incorrect for the purpose of retrieving "medium".

**C. `var.sizes[2]`:** This accesses "large", the third element in the list. While this is a valid operation, it does not retrieve "medium" as required.

**D. `var.sizes[3]`:** This option attempts to access an index that is out of range for the given list (since the list only has three elements, indexed 0 through 2). In Terraform, this would result in an error, as the index 3 does not exist for this variable.

#### References

For further understanding of Terraform's data types and how to work with lists, refer to the official HashiCorp Terraform documentation:

[Terraform Configuration Basics: Data Types](#)

[Terraform Variables: Using Lists](#) (Specifically, look for examples on list handling)

### Question: 88

You have declared a variable called `var.list`, which is a list of objects that all have an attribute `id`. Which expressions will evaluate to a list of the objects' `id` attributes? (Choose two.)

- A. `[ var.list[*].id ]`
- B. `[ for o in var.list : o.id ]`
- C. `var.list[*].id`
- D. `for o in var.list : o => o.id`

**Answer: BC**

**Explanation:**

**Technical Justification for Correct Options: B and C**

**Correct Option B: `[ for o in var.list : o.id ]`**

**Expression Type:** This is a **For-Expression** in Terraform's configuration language (HCL). It iterates over a collection (in this case, `var.list`) and applies an operation to each element.

**Evaluation:** For each object `o` in `var.list`, it extracts the `id` attribute and collects these into a new list.

**Result:** Directly evaluates to a **list** of the objects' `id` attributes, making it a straightforward and efficient choice for this task.

**Why It's Best for This Scenario:** Explicitly designed for transforming collections, making the intent clear and the code readable.

**Correct Option C: `var.list[*].id`**

**Expression Type:** This utilizes **Element Access** with a **Splat** (`*`) in Terraform's HCL. The splat operator is used to access all elements of a list (or map) at once.

**Evaluation:** The `*` splat operator iterates over the list, and for each element, it accesses the `id` attribute, implicitly collecting these into a list.

**Result:** Also evaluates to a **list** of the objects' `id` attributes, providing a concise way to achieve the goal.

**Why It's Suitable:** Offers a very concise syntax for simple attribute extraction from all elements in a list, perfect when the operation is straightforward.

**Why Other Options Are Less Suitable**

**Option A: `[ var.list[*].id ]`**

**Issue:** This is essentially the same as Option C but unnecessarily wrapped in an additional list literal. It still evaluates to a list of IDs but is less concise than necessary.

**Result Type:** List of IDs (same as C, but less idiomatic)

**Option D: `for o in var.list : o => o.id`**

**Issue:** This is a **Map Comprehension** (denoted by the curly braces), which evaluates to a **map** (with the object itself as the key and the `id` as the value), not a list.

**Result Type:** Map (incorrect type for the question's requirement)

**References**

[Terraform Documentation - For Expressions](#)

[Terraform Documentation - Value Expressions: Element Access](#)

## Question: 89

CertyIQ

What type of block is used to construct a collection of nested configuration blocks?

- A. dynamic
- B. nesting
- C. for\_each
- D. repeated

**Answer: A**

**Explanation:**

#### Technical Justification

When constructing a collection of nested configuration blocks in Terraform (a HashiCorp product), the most appropriate type of block to use is the **dynamic block**. Here's why:

**A. Dynamic:** Dynamic blocks are designed to generate nested blocks dynamically based on a local variable or an expression. This makes them highly flexible for creating collections of nested configurations where the structure or number of blocks might vary. Dynamic blocks allow for the iteration over a collection (like a list or map) to produce multiple blocks of the same type, fitting the requirement perfectly.

**B. Nesting:** While nesting is a fundamental concept in Terraform for organizing resources and modules, it simply refers to the practice of placing one block inside another. Nesting itself does not inherently facilitate the dynamic creation of a collection of blocks based on variable input.

**C. for\_each:** The for\_each argument is used to create multiple instances of a resource or a module based on a map or a set. However, its primary use is for creating multiple top-level resources or module instances, not for generating nested blocks within a single resource or module configuration.

**D. Repeated:** "Repeated" is not a recognized block type in Terraform for dynamically generating nested configuration blocks. While you can repeat blocks manually, this approach is static, error-prone for large collections, and does not support dynamic generation based on variables.

**Conclusion:** The dynamic block (A) is the best choice for constructing a collection of nested configuration blocks due to its flexibility and dynamic nature, tailored for variable and potentially large sets of nested configurations. The other options either serve different purposes (B, C) or are not valid block types for this specific use case (D).

#### References

[HashiCorp Terraform Documentation - Dynamic Blocks](#)

[HashiCorp Terraform Documentation - for\\_each](#)

#### Question: 90

CertyIQ

Which type of block fetches information from outside of Terraform, or computes data for use in a Terraform configuration?

- A. resource
- B. provider
- C. data
- D. output

**Answer: C**

**Explanation:**

**Technical Justification for Correct Answer: C. data**

The correct answer is **C. data** because, in Terraform, a data block is specifically designed to fetch information from external sources or compute data that can then be used within a Terraform configuration. This is crucial for dynamically retrieving values that are not managed by Terraform itself, such as the latest AMI ID for an AWS instance, user input, or data from other infrastructure systems.

**Why C. data is the Best Choice:**

**External Information Retrieval:** data blocks can fetch data from sources outside the Terraform configuration, such as cloud provider APIs, files, or other external data sources.

**Computed Data:** They can also compute data based on predefined functions or logic, tailored for configuration use.

**Read-Only Nature:** Aligns with the intent of fetching/computing data without modifying external states, distinguishing it from resource creation/modification.

**Why Other Options are Less Suitable:**

**A. resource:**

**Purpose:** Resources are for managing infrastructure (e.g., creating an AWS EC2 instance).

**Misalignment:** Using a resource to merely fetch data would be a misuse, as it implies state management of that resource.

**B. provider:**

**Purpose:** Providers are the bridge between Terraform and the infrastructure/services being managed. They define capabilities and authentication but do not fetch or compute data for configurations.

**Misalignment:** Providers are about enabling resource management, not data fetching/computation for config use.

**D. output:**

**Purpose:** Outputs are used to display values after applying a configuration, making selected data available for external scripts/users.

**Misalignment:** Outputs are for exporting computed or known values post-apply, not for importing or computing data for the configuration itself.

**References:**

[HashiCorp Terraform Documentation - Data Sources](#)

[Terraform by HashiCorp - Data Blocks](#)

## Question: 91

CertyIQ

Which option does not keep secret variable values out of Terraform configuration files?

- A. Pass the secret values as environment variables.
- B. Configure the variable as sensitive.
- C. Pass the secret values via a -var flag.
- D. Store the secrets in a shared credential file.

**Answer: B**

**Explanation:**

#### Technical Justification

The question asks which option does not keep secret variable values out of Terraform configuration files. The correct answer is **B. Configure the variable as sensitive**. Here's why each option fares in terms of keeping secrets out of Terraform configuration files:

**A. Pass the secret values as environment variables:** This approach keeps secret values out of Terraform configuration files by storing them outside of the Terraform codebase in the environment. Environment variables are not committed to version control along with Terraform files, making this a secure method for handling secrets.

**B. Configure the variable as sensitive: This is the correct answer** because marking a variable as sensitive in Terraform (e.g., `sensitive = true` in a variable block) only controls the output of the variable in Terraform's console output and state file, but **it does not prevent the secret from being stored in the Terraform configuration files themselves**. If the secret is directly written in the Terraform config file (which should never happen), marking it as sensitive does nothing to remove it from the file.

**C. Pass the secret values via a -var flag:** Similar to environment variables, passing secrets via the `-var` or `-var-file` command line flags keeps them out of the Terraform configuration files. The secrets are not hardcoded in the files but are provided at runtime.

**D. Store the secrets in a shared credential file:** While this might seem less secure than some other methods due to the "shared" aspect, if properly implemented (e.g., not committed to version control, access controlled), it keeps secrets out of the primary Terraform configuration files. The security of this method heavily depends on how the credential file is managed.

#### Why B is the Least Suitable for Keeping Secrets Out of Configuration Files:

Unlike the other options, configuring a variable as sensitive does not inherently remove or prevent the storage of secret values within Terraform configuration files.

The primary benefit of `sensitive = true` is obscuring the value in output and state, not in securing the input/configuration files themselves.

**Recommendation for Secure Practice:** For keeping secret values out of Terraform configuration files, options A, C, and properly managed D are more secure. Always avoid hardcoding secrets directly in Terraform files.

#### References

1. HashiCorp Terraform Documentation - Sensitive Values  
<https://www.terraform.io/docs/configuration/values.html#sensitive-values>
2. HashiCorp Terraform Documentation - Input Variables  
<https://www.terraform.io/docs/configuration/inputs.html#passing-input-variable-values>

#### Question: 92

CertyIQ

Which statements are true? (Choose two.)

- A. When a resource has a count meta-argument, Terraform evaluates the precondition for each instance.

- B. A precondition in a resource block must pass for Terraform to create the resource.
- C. A precondition cannot reference attributes of another resource that are only known after apply.
- D. A resource can either have a precondition or a postcondition. block, but not both.
- E. If a resource's postcondition block fails, Terraform still creates the resources that depend on it.

**Answer: AB**

**Explanation:**

**Justification for Correct Answer: AB**

**Why A is Correct:**Option A states that "When a resource has a count meta-argument, Terraform evaluates the precondition for each instance." This is true. Terraform's count meta-argument is used to create multiple instances of a resource. When a precondition (defined using precondition block with a check argument in Terraform 1.x, or more commonly via validation blocks in resource definitions for schema validation, though dynamic preconditions as described here are more aligned with null\_resource and custom scripts) is present on such a resource, Terraform indeed evaluates this precondition for each instance. This ensures that each instance of the resource meets the predefined conditions before creation, which is crucial for managing multiple resources with potentially varying configurations.

**Why B is Correct:**Option B, "A precondition in a resource block must pass for Terraform to create the resource," is accurate. The fundamental purpose of a precondition is to define a condition that must be met before Terraform can proceed with creating the resource. If the precondition fails (i.e., the condition is not met), Terraform will not create the resource, thus preventing potentially erroneous or undesired configurations from being deployed. This aligns with Terraform's design to enforce consistency and safety in infrastructure provisioning.

**Why C, D, and E are Less Suitable or Incorrect:**

**C: Incorrect** - A precondition **can** technically reference attributes of another resource that are known before the apply phase, thanks to Terraform's planning phase where dependencies are resolved. However, attributes that are **only** known after apply (e.g., randomly generated IDs) cannot be referenced in preconditions because these values are not available during the planning phase when preconditions are evaluated. This makes the statement somewhat misleading but more incorrect in the context of how Terraform's lifecycle works.

**D: Incorrect** - There is no strict either/or limitation imposed by Terraform regarding having **both** a precondition and a postcondition on a resource. While less common due to the differing natures of these checks (pre-creation vs. post-creation), there's no technical barrier to defining both, though in practice, precondition checks are more about input(validation) and postcondition about output verification, and both can coexist for comprehensive validation.

**E: Incorrect** - If a resource's postcondition block fails, Terraform **will not** consider the resource as successfully created, and by default, resources dependent on it **will not** be created. Postconditions are checks that run after a resource is created, and failure here indicates the resource is not in a desired state, triggering an error and preventing further dependent creations.

**References**

[Terraform Documentation - Resource Meta-Arguments \(Count\)](#)

[Terraform Documentation - Validations \(closest to precondition concept for resource attributes\)](#)

Your configuration contains a module block that references a module from the Terraform registry, and sets the version argument to 1.0. You just published a new version of the module and updated your configuration to point to version 1.1.

Which command must be run to install the new version?

- A. terraform apply
- B. terraform init
- C. terraform plan
- D. terraform modules

**Answer: B**

**Explanation:**

**Justification**

**terraform init** is the command that resolves and installs module versions from the registry. When the version argument in a module block is changed (e.g., from 1.0 to 1.1), the new version is fetched only after a re-initialization of the working directory. Running terraform init (optionally with -upgrade) tells Terraform to download the new module version that satisfies the updated constraint.

**Why the other options are incorrect:**

**A – terraform apply** provisions or updates infrastructure resources; it does not download or install new module code.

**C – terraform plan** generates an execution plan but does not interact with the module source registry.

**D – terraform modules** is not a valid Terraform CLI command.

Therefore, the correct command to fetch and install the newly released module version is **B – terraform init** (with -upgrade recommended to ensure the new version is retrieved).

**References**

**Terraform CLI Init Command** – <https://developer.hashicorp.com/terraform/cli/init>

**Modules and Version Constraints** –

<https://developer.hashicorp.com/terraform/language/modules/develop/src>

## Question: 94

CertyIQ

If one of your modules uses a local value, you can expose that value to callers of the module by defining a terraform output in the module's configuration.

- A. True
- B. False

**Answer: A**

**Explanation:**

**Correct answer: A – True**

A module can reference its own local variables only inside the module; to make those values usable by its

callers, the module must declare a corresponding output attribute that returns the value.

When the calling configuration references the module, the output is resolved and can be assigned to a variable, used in other resources, or passed to other modules.

This mechanism is the standard way to expose module-scoped information; without an output, a local value remains internal and invisible to the caller.

### Why other options are unsuitable

The question only provides a binary choice; “False” would imply you cannot expose local values via outputs, which contradicts both the design of Terraform modules and the official documentation.

Any justification that ignores the requirement to declare an output attribute would be incomplete and inaccurate for a certification context.

### References

Exposing module outputs: <https://developer.hashicorp.com/terraform/language/modules/develop/outputs>

Using outputs in calling configurations:

<https://developer.hashicorp.com/terraform/language/modules/usage#exposing-module-outputs>

## Question: 95

CertyIQ

If a module declares a variable without a default value, you must pass the value of the variable within the module block when you call the module in your configuration.

- A. True
- B. False

### Answer: A

### Explanation:

#### Answer: A – True

Terraform requires all input variables to be defined with a value before the module is instantiated.

If a variable is declared without a default value, the configuration will not compile unless the caller explicitly passes a value for that variable in the module block.

Omitting the required argument leads to a validation error (Variable "xxx" is required), preventing the plan/apply from proceeding.

### Why B is incorrect:

Statement B claims that passing the value is optional; this contradicts Terraform’s strict variable-requirement semantics, which mandate that any non-default variable must be supplied.

### References

[Variables in Terraform](#)

[Module Call Syntax](#)

## Question: 96

CertyIQ

When you run terraform apply, the Terraform CLI will print output values from both the root module and any child

modules.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Answer: B.False**

#### Technical justification

The terraform apply command executes a plan and then provisions any resources that differ from the desired state. It prints a summary of the actions it will take (e.g., "Created instance ...") and any error or warning messages, but it **does not automatically render output values** (output blocks) from the root or any child modules.

Output values are only emitted when you explicitly invoke the **terraform output** command after a successful apply (or terraform apply -json for machine-readable output).

To make module outputs visible in the root module, they must be re-exposed through root-module output declarations, which is an explicit step and not the default behavior of apply.

Therefore, the statement that terraform apply prints output values from both the root and child modules by itself is inaccurate.

---

#### References

##### Terraform CLI documentation – apply

<https://developer.hashicorp.com/terraform/cli/commands/apply>

##### Terraform CLI documentation – output

<https://developer.hashicorp.com/terraform/cli/commands/output>

These links confirm that apply does not display output values automatically and that separate commands or explicit output definitions are required to view them.

#### Question: 97

CertyIQ

Which type of information does the Terraform Registry provide about the modules it hosts?

- A. Required input variables
- B. Outputs
- C. Optional input variables and default values
- D. All of these are provided

**Answer: D**

**Explanation:**

The Terraform Registry publishes a **module page** that enumerates all the metadata required for consumption:

- **Required input variables** – listed to indicate mandatory parameters.
- **Optional input variables and their default values** – shown with defaulting behavior.
- **Outputs** – listed to give users a clear view of what the module returns.

Because these three pieces of information are all presented together on the same page, the only answer that captures the complete set is **“All of these are provided.”**

Options A and B isolate only a subset (required variables only, or outputs only), which omits the other equally important details; therefore they are incomplete descriptions of what the Registry supplies.

## References

[Terraform Registry Module Page Documentation](#) – explains the fields displayed for each module, including required/optional variables and outputs.

[Module Requirements & Variables Reference](#) – details the variable definitions and output contracts published by modules in the Registry.

## Question: 98

CertyIQ

When you include a module block in your configuration that references a module from the Terraform Registry, the version attribute is required.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Justification**

The module block may contain a version argument, but it is not mandatory – Terraform will accept the block without a version specifier and will use the latest published version of the module.

Providing a version constraint is only a best-practice measure to lock the module to a specific release or to guard against breaking changes; the Terraform language grammar does not enforce it.

The Registry automatically resolves the most recent stable release when no version is declared, so excluding version is syntactically valid and operationally functional.

However, omitting the version makes the configuration less deterministic and can lead to unexpected upgrades in future runs, which is why the guidance recommends specifying a version, not that it is required.

**Why the other option is less suitable**

Option A claims the version attribute is required; this misrepresents the Terraform configuration syntax and would cause an unnecessary validation error if a user simply omitted it.

In exam contexts, precision about optional versus mandatory fields is critical; recognizing that version is optional aligns with the official documentation and prevents misunderstandings about Terraform’s parsing rules.

## References

Modules – Using Modules from the Terraform Registry:

<https://developer.hashicorp.com/terraform/language/modules/develop/source>

Version constraints – Terraform Registry module version constraints:

<https://developer.hashicorp.com/terraform/language/modules/develop/source#version-constraints>

### Question: 99

CertyIQ

A child module can always access variables declared in its parent module.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Justification**

The statement is **False**.

Terraform modules have their own scope; a child module does **not** automatically inherit variables defined in its parent.

Variables are only accessible to a child when the parent explicitly **passes** them via **output values** that are then declared as **input variables** in the child, or when the child receives them as arguments in its module block.

Unpassed variables remain private to the parent and cannot be read by the child, regardless of the hierarchical relationship.

This design enforces encapsulation and prevents unintended coupling between modules, which is a core principle of Terraform's module system.

---

### References

Terraform Modules Documentation – Passing Input Variables:

<https://developer.hashicorp.com/terraform/language/modules/develop/input-variables>

Terraform Modules Documentation – Output Variables:

<https://developer.hashicorp.com/terraform/language/modules/develop/outputs>

**Correct Option:** B.False

### Question: 100

CertyIQ

You are updating a child module with the resource block shown in the Exhibit space on this page. The `public_ip` attribute of the resource needs to be accessible to the parent module. resource "aws\_instance" "example" ami = "ami-0a123456789abcdef" instance\_type = "t3.micro"

How do you meet this requirement?

- A. Add a data source to the parent module.
- B. Create a local value in the child module.
- C. Add an import block to the parent module.
- D. Create an output in the child module.

**Answer: D**

**Explanation:**

## Why option D is correct

**Expose the attribute via an output** – The child module can declare a named output that returns `public_ip`. The parent module can then reference that output directly, making the value available as a first-class variable.

**Decouples modules** – Using an output keeps the parent module agnostic of the child's internal resources, preserving encapsulation while still sharing needed data.

**Terraform semantics** – Outputs are the official mechanism for exposing a module's results to its callers; they are resolved after the child plan is applied and can be consumed in any expression within the parent.

## Why the other options are not suitable

**Add a data source to the parent** – A data source reads external data, but it cannot directly expose a resource attribute defined inside another child module without importing state, which adds unnecessary complexity and breaks module boundaries.

**Create a local value in the child** – A `locals` block is scoped to the module where it is defined; it does not automatically become visible to the parent. The parent would still need a mechanism to retrieve the value, which an output fulfills more naturally.

**Add an import block to the parent** – Import blocks are used to bring resources created outside the current state into the current configuration. They do not provide a clean, declarative way to expose the child's `public_ip` attribute.

## Conclusion

Declaring an output in the child module that returns `public_ip` and referencing that output in the parent is the idiomatic, maintainable way to share the attribute across modules.

## References

Modules: <https://developer.hashicorp.com/terraform/language/modules/develop>

Outputs: <https://developer.hashicorp.com/terraform/language/modules/develop#outputs>

## Question: 101

CertyIQ

Your configuration defines the module block shown in the Exhibit space on this page. The `web_stack` module accepts an input variable named `servers`.  
module "web\_stack" source = "./modules/web\_stack"

Which of the following changes to the module block sets the `servers` variable to the value of "3"?

A. module "web\_stack"  
source = "./modules/web\_stack"  
servers = 3

B. module "web\_stack"  
source = "./modules/web\_stack"  
inputs.servers = 3

C. module "web\_stack"  
source = "./modules/web\_stack"  
inputs =  
servers = 3

D. module "web\_stack"  
source = "./modules/web\_stack"  
var.servers = 3

**Answer: A**

**Explanation:**

## Justification

**OptionA** – `servers = 3` is a valid attribute directly on the module block. When a module declares an input variable named `servers`, assigning `servers = 3` in the module call passes the value 3 to that variable. This syntax follows the exact form required by Terraform.

**OptionB** – `inputs.servers = 3` is invalid because the module block does not accept a top-level `inputs` attribute; module arguments are specified as separate attribute-value pairs, not as a dotted-notation map.

**OptionC** – `inputs = { servers = 3 }` incorrectly uses `inputs` as the argument name and wraps the assignment in braces. Terraform expects each argument to be a simple attribute, not a map keyed by `inputs`.

**OptionD** – `var.servers = 3` references a `var` namespace that does not exist in a module block. Variables are passed in, not defined or mutated via `var.` inside the block.

Therefore, only **OptionA** correctly sets the `servers` input to 3.

## References

### References

Module block syntax – Terraform Language:

<https://developer.hashicorp.com/terraform/language/modules/requirements>

Passing input variables to modules – Terraform Documentation:

<https://developer.hashicorp.com/terraform/language/modules/sources>

## Question: 102

CertyIQ

Your team is using a version 3.1.4 of a module from the public Terraform Registry, and they're worried about possible breaking changes in future versions of the module.

Which version argument should you add to the module block to prevent newer versions from being used?

- A. `version = "3.1.4"`
- B. `version = "< 3.2"`
- C. `version = "!=3.1.5"`
- D. `version = "~>3.1.4"`

**Answer: D**

**Explanation:**

**Why optionD is the correct choice**

The constraint `~>3.1.4` expands to “greater than or equal to 3.1.4, but less than 3.2.0”.

This means the module can only use patch releases (e.g., 3.1.5) until the next minor version (3.2.x) is reached, effectively **preventing any newer minor or major version from being pulled** while still allowing safe patch upgrades.

It matches the team's intent to “lock the module to the current 3.1.x line” without unnecessarily restricting older versions.

**Why the other options are unsuitable**

**A – version = "3.1.4"** pins the module to that exact string. Any later patch or minor release (e.g., 3.1.5) would be rejected, which may be overly restrictive if patch upgrades are desired. It also does not express the typical

semantic-versioning intent of “use this series only”.

**B – version = "< 3.2"** allows any version prior to 3.2, including older major releases such as 2.x. This could bring in unexpected breaking changes from previous major versions, contradicting the goal of staying on a known 3.1.x level.

**C – version = "!=3.1.5"** only excludes a single patch version; it does **not** stop newer minor or major releases like 3.2.0 from being used, so it fails to limit future breaking changes.

Hence, `~>3.1.4` is the precise, industry-standard way to restrict usage to the current minor series while still permitting safe patch updates.

---

## References

**Module version constraints** – HashiCorp Terraform documentation:

<https://developer.hashicorp.com/terraform/language/modules/sources#version-constraints>

**Semantic versioning constraints (~> notation)** – HashiCorp Terraform documentation:

<https://developer.hashicorp.com/terraform/language/version-constraints>

### Question: 103

CertyIQ

Which of the following locations can Terraform use as a private source for modules? (Choose two.)

- A. Public repository on GitHub.
- B. Internally hosted VCS (Version Control System) platform.
- C. Public Terraform Registry.
- D. Private repository on GitHub.

**Answer: BD**

**Explanation:**

#### Technical justification

**Internally hosted VCS platform (B)** – Terraform can reference a module stored in any Git-compatible VCS that is reachable only inside the organization’s network (e.g., an internal Git server). Because the repository is not publicly exposed, it meets the definition of a private source. Access is typically provided via SSH or HTTPS with credentials, allowing Terraform to clone the exact commit or tag required for the configuration.

**Private repository on GitHub (D)** – A GitHub repository that is owned by an organization or user and has its visibility set to private can also be used as a module source. Terraform fetches the module using the same Git URL scheme, but the request includes authentication (personal access token, SSH key, or OAuth) that proves the caller is authorized. Since the repo is not publicly reachable, it qualifies as a private location for modules.

**Public repository on GitHub (A) – not private** – Although a public GitHub repo can be used as a module source, its contents are openly accessible on the internet. The defining characteristic of a private source is that it is reachable only by authorized entities; a publicly reachable repository fails this requirement.

**Public Terraform Registry (C) – not private** – The public registry hosts community-maintained modules that anyone can publish and consume. By definition it is a public source; Terraform treats it as an open registry and does not consider it a private location for module storage.

References

Module source types and private source authentication:

<https://developer.hashicorp.com/terraform/language/modules/sources>

Managing private modules in the Terraform Registry:

<https://developer.hashicorp.com/terraform/registry/private-modules>

### Question: 104

CertyIQ

Your configuration defines the module block shown in the Exhibit space on this page.

```
module "production" {  
  source = "../modules/web_stack"  
}
```

This module declares an output named hostnames.  
How do you access the value of this output?

- A. production.hostnames
- B. module.production.hostnames
- C. web\_stack.hostnames
- D. module.web\_stack.hostnames

**Answer: B**

**Explanation:**

**B. module.production.hostnames**

In **Terraform**, outputs from a module are accessed using the syntax:

module.<module\_name>.<output\_name>

If the module block is defined as:

```
module "production"  source = "../web_stack"
```

and the module contains:

```
output "hostnames"  value = ...
```

then you reference the output as:

```
module.production.hostnames
```

### Question: 105

CertyIQ

Which argument can you set on a module block to prevent Terraform from updating the module's configuration during an init or get operation?

- A. count
- B. lifecycle

- C. source
- D. version

**Answer: D**

**Explanation:**

**Why the correct answer is D – version**

The version argument on a module block pins the module to a specific released version (e.g., =1.2.3 or ">=1.0.0, <2.0.0").

When Terraform runs **terraform init** or **terraform get**, it resolves and downloads only that exact version, never automatically upgrading to a newer one. This stops Terraform from silently updating the module configuration during those operations.

Because the version is locked, Terraform treats the module as immutable with respect to version selection, preventing unintended updates.

**Why the other options are unsuitable**

**A. count** – count is used on resource definitions to create multiple instances; it is **not** a module-level argument.

**B. lifecycle** – A lifecycle block controls resource-level behavior (e.g., create\_before\_destroy, prevent\_destroy) but cannot be applied to a module block, so it has no effect on version resolution during init/get.

**C. source** – source points to the module's address (repository or registry) but does **not** constrain version selection; Terraform will still fetch the latest matching version unless a specific version constraint is given.

Thus, only the version argument can directly prevent Terraform from updating a module's configuration during init or get.

---

## References

**Terraform Module Blocks – Version Constraints:**

<https://developer.hashicorp.com/terraform/language/modules/develop/src#version-constraints>

**Module Versioning and Dependency Management:**

<https://developer.hashicorp.com/terraform/language/modules/develop/sources#versioning>

## Question: 106

**CertyIQ**

You are writing Terraform configuration that calls a child module. This child module declares an output value. How can you ensure that Terraform will print out this value when you run Terraform CLI commands such as terraform apply?

- A. Nothing. Terraform will automatically print out all output values from child modules.
- B. Nothing. Terraform cannot use output values from child modules.
- C. Declare a new output in your root configuration that references the module's output.
- D. None of these will print the value.

**Answer: C**

**Explanation:**

**Technical Justification for Correct Answer: C**

The correct option to ensure Terraform prints out the output value from a child module when running Terraform CLI commands (like terraform apply) is **C. Declare a new output in your root configuration that references the module's output**. Here's why:

### Why C is Correct

**Explicit Output Declaration Requirement:** Terraform requires explicit declaration of outputs in the root module configuration to display values after an operation like apply. Child module outputs are not automatically exposed to the root level for display.

**Reference and Re-export:** By declaring a new output in the root configuration that references the child module's output (e.g., `value = module.child_module.output_name`), you effectively "re-export" the value, making it visible at the root level and thus printed by Terraform after applying changes.

### Why Other Options are Less Suitable

**A. Nothing. Terraform will automatically print out all output values from child modules.**

**Incorrect:** Terraform does not auto-print child module outputs at the root level without explicit configuration. This would lead to unexpected behavior if relied upon.

**B. Nothing. Terraform cannot use output values from child modules.**

**Misleading:** While the statement about printing is incorrect (as it implies a capability issue rather than a configuration one), the core issue is about visibility, not usability. Outputs from child modules can be used (e.g., for resource configuration), just not automatically displayed.

**D. None of these will print the value.**

**Incorrect in Context of Available Choices:** Given that option C provides a valid method to achieve the desired outcome, this statement is false in the context of the question.

### Example for Clarity (Implicit in Justification but Explicit for Understanding)

In your **root module configuration**:

```
module "child_module" # ... module source and variables ... output "child_module_output" value = module.child_module.output_value
```

In your **child\_module/main.tf**:

```
# ... resource definitions ... output "child_output_value" value = "Value to be printed"
```

### References

[HashiCorp Terraform Documentation - Output Values](#)

[HashiCorp Terraform Documentation - Modules](#)

## Question: 107

CertyIQ

All modules published on the official Terraform Registry are maintained and tested by HashiCorp.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. False**

The correct answer is **B. False**, and here's why:

**Maintenance and Testing Responsibility:** Modules published on the official Terraform Registry are not all maintained and tested by HashiCorp. While HashiCorp does maintain and test some core and widely used modules, the Terraform Registry also hosts a vast number of community-driven modules. The maintenance and testing of these community modules fall on their respective authors or contributor teams, not necessarily HashiCorp.

**HashiCorp's Role:** HashiCorp's primary role regarding the Terraform Registry includes:

Providing the platform for module discovery and sharing.

Establishing guidelines for module quality and compatibility.

Maintaining and testing **official** HashiCorp-supported modules (e.g., those for their own products like Vault, Consul, etc., and some core infrastructure providers).

**Why Other Options Are Less Suitable:**

**A. True:** Incorrect because it implies a blanket responsibility by HashiCorp for all modules, which is not the case due to the community-driven nature of many listings.

**Key Points for Certification Exam Review:**

Understand the distinction between HashiCorp-maintained modules and community-driven modules on the Terraform Registry.

Recognize HashiCorp's role in the ecosystem versus direct maintenance responsibility.

Be able to identify the implications of community maintenance on module reliability and support.

## References

[Terraform Registry Documentation - Contributing Modules](#) gösterihow HashiCorp encourages community contributions, indicating not all are maintained by them.

[HashiCorp's Official Modules on Terraform Registry](#) illustrates HashiCorp's direct involvement with specific, officially supported modules.

## Question: 108

CertyIQ

You root module contains a variable named num\_servers.

Which is the correct way to pass its value to a child module with an input named servers?

- A. servers = var(num\_servers)
- B. servers = num\_servers
- C. \$ var.num\_servers
- D. servers = var.num\_servers

**Answer: D**

**Explanation:**

**Technical Justification for Correct Answer (D)**

The correct way to pass the value of the num\_servers variable from the root module to a child module's input named servers is option D: servers = var.num\_servers`. Here's a breakdown of why this option is best and why the others are less suitable:

**D: servers = var.num\_servers**

**Correctness:** This is the appropriate syntax for referencing a variable in Terraform (as of the latest versions up to my last update). `var.` is the prefix used to access variables declared in the root module from within a child module when passing values, or in the root when referencing the variable directly.

**Clarity and Convention:** It clearly indicates that `num_servers` is a variable, enhancing readability and adhering to Terraform's naming conventions and best practices.

**A: `servers = var(num_servers)`**

**Incorrect Syntax for Variable Passing:** While `var()` syntax is used for accessing variables, it's more commonly seen in expressions or when the variable name is dynamically determined. For static variable names, `var.name` is preferred for clarity and simplicity in this context.

**Less Suitable for Direct Variable Reference:** Not the idiomatic way to pass a simple variable value to a module input.

**B: `servers = num_servers`**

**Ambiguity and Potential Error:** Without the `var.` prefix, Terraform would look for a local variable, input variable, or a predefined value named `num_servers` in the current scope (e.g., within the child module or the block where this assignment is made), which is not the intention here. This could lead to errors if `num_servers` is not defined in this scope.

**Lacks Explicitness:** Does not clearly communicate that `num_servers` is a variable from the root module.

**C: `$ var.num_servers`**

**Unnecessary String Interpolation:** The syntax `$ ...` is used for string interpolation in Terraform. Since `servers` expects a value (not necessarily a string), and `var.num_servers` already evaluates to a value, this interpolation is unnecessary and could lead to type mismatches if not carefully managed.

**Redundant for Non-String Contexts:** In a non-string assignment, directly referencing the variable is more appropriate.

## References

For further reading on Terraform variables and module inputs:

1. **Terraform Documentation - Variables:**

<https://www.terraform.io/docs/configuration/basics/variables.html>

2. **Terraform Documentation - Module Inputs:**

<https://www.terraform.io/docs/modules/modules.html#input-variables>

## Question: 109

CertyIQ

You're using a module block to reference a module from the Terraform Registry. How do you specify version 1.0.0 of the module?

- A. Add a `version = "1.0.0"` argument to the module block.
- B. Nothing. Modules stored on the public Terraform module Registry always default to version 1.0.0.
- C. You cannot. Modules stored on the public Terraform Registry do not support versioning.
- D. Append a `?ref=v1.0.0` argument to the source path.

**Answer: A**

**Explanation:**

### Technical Justification for Correct Answer: A

The correct option to specify version 1.0.0 of a module from the Terraform Registry within a module block is **A. Add a version = "1.0.0" argument to the module block**. Here's why:

**Precision and Explicitness:** By adding version = "1.0.0" directly to the module block, you explicitly define the desired version, ensuring clarity and precision in your Terraform configuration. This approach avoids potential version ambiguities.

**Terraform Best Practices and Documentation:** Terraform's official documentation recommends specifying the version attribute for modules to ensure reproducibility and control over the version used. This aligns with best practices for infrastructure as code (IaC), emphasizing version control for dependency management.

#### Why Other Options are Less Suitable:

**B. Nothing (Default to 1.0.0):** This option is incorrect because modules on the Terraform Registry do not default to a specific version (like 1.0.0) if not specified. Without a version attribute, Terraform would either use the latest version (which might not be 1.0.0) or fail if a version range is not satisfied.

**C. No Support for Versioning:** This is factually incorrect. Modules on the Terraform Registry fully support versioning, which is a key feature for managing dependencies effectively.

**D. Append ?ref=v1.0.0 to the Source Path:** While appending a ?ref=<version> (or more commonly, ?ref=<tag> or ?ref=<commit\_hash>) to the source path in the module block can specify a version (note the format should typically match the repository's version tagging, e.g., v1.0.0 for a Git repository), this approach is less direct and less recommended for specifying a known, tagged version from the Terraform Registry compared to using the version attribute. The ?ref= parameter is more flexible (supporting tags, branches, commits) but less concise for specifying a known, published module version.

#### Correct Implementation (Based on A):

```
module "example" source = "registry.example/module/name" version = "1.0.0" # ... other arguments ...
```

#### References:

[Terraform Documentation - Module Versioning](#)

[Terraform Documentation - Specifying Module Sources](#)

### Question: 110

CertyIQ

You need to destroy all of the resources in your Terraform workspace, except for aws\_instance.ubuntu[1], which you want to keep.

How can you tell Terraform to stop managing that specific resource without destroying it?

- A. Change the value of the count argument on the resource.
- B. Remove the resource block from your configuration.
- C. Run terraform state rm aws\_instance.ubuntu[1].
- D. Use a moved block.

**Answer: C**

**Explanation:**

**Correct answer: C –Run terraform state rm aws\_instance.ubuntu[1]**

**Why this is the best choice**

terraform state rm detaches a resource instance from the state file without deleting it from the underlying infrastructure. The resource stays provisioned, but Terraform treats it as “orphaned” and will no longer try to manage or destroy it.

This is the only option that implicitly removes the dependency while preserving the actual AWS EC2 instance, allowing you to retain `aws_instance.ubuntu[1]` for future manual operations.

### Why the other options are unsuitable

**A – Modify the count argument:** Changing count only affects how many instances Terraform expects to create; it does not remove a specific indexed resource from the state, so the instance would either be recreated or left with an inconsistent index.

**B – Remove the resource block:** Deleting the block from the configuration and running terraform apply would cause Terraform to destroy the resource because it would think the resource is missing. The instance would be removed from the infrastructure.

**D – Use a move block:** The move argument is used to tell Terraform to treat a resource as a new identity; it does not stop management, it merely renames/moves the address in state. It does not prevent future destruction or updates.

### References

[Terraform State Command – rm](#)

[Resource Lifecycle – delete vs orphaned](#)

### Question: 111

CertyIQ

When you run `terraform apply -refresh-only`, which of the following is not consulted by Terraform to update the state file?

- A. Authentication credentials for the provider.
- B. Terraform configuration files defining the resources.
- C. The actual infrastructure as reported by the provider API.
- D. The most recent state file.

**Answer: B**

**Explanation:**

**Why option B is the correct answer**

**Option B – Terraform configuration files defining the resources**

Terraform does not re-parse the `.tf` files when it runs `apply -refresh-only`. The command's sole purpose is to update the state file by reconciling the provider's current view of the infrastructure with the state that is already stored. It therefore does not need to read or evaluate the configuration language itself.

**Option A – Authentication credentials for the provider**

The refresh operation must call the provider's API, which requires valid credentials (e.g., API tokens, access keys). Those credentials are consulted to establish the connection.

**Option C – The actual infrastructure as reported by the provider API**

Refresh actively queries the live infrastructure through the provider, compares the returned attributes with the existing state, and updates the state accordingly. This step is essential.

### OptionD – The most recent state file

The current state file is loaded at the start of the refresh process; its contents are read and then overwritten with the refreshed information. Hence the existing state file is definitely consulted.

### References

1. [Terraform CLI - Refresh Mode](#)
2. [Provider Authentication Overview](#)

### Question: 112

CertyIQ

When declaring a variable, setting the sensitive argument to true will prevent the value from being stored in the state file.

- A. True
- B. False

**Answer: B**

### Explanation:

B

The sensitive meta-argument in Terraform marks a variable's value as confidential, which prevents the value from being displayed in logs, console output, or the UI. However, the value **is still stored in the state file**; Terraform does not encrypt or omit it from the state. Consequently, the statement that setting sensitive = true will prevent the value from being stored in the state file is inaccurate. The correct answer, therefore, is **False**.

Other options are unsuitable because:

Choosing **True** would imply the value is omitted from the state entirely, which is not how sensitive works. The question is a straightforward True/False; no additional qualification or nuance is needed.

### References

Terraform documentation on sensitive: <https://developer.hashicorp.com/terraform/language/meta-arguments/variables#sensitive>

State file behavior and sensitive variables:

<https://developer.hashicorp.com/terraform/language/state#sensitive-values>

### Question: 113

CertyIQ

Which are benefits of migrating from a local state backend to a remote backend? (Choose two.)

- A. Eliminates the need to manage credentials when deploying infrastructure to multiple cloud providers.
- B. State locking that allows multiple team members to safely work on the same infrastructure.
- C. Faster plans and apply execution because the state is cached locally on the cloud provider.
- D. Guarantees that configuration drift cannot occur for the managed infrastructure.
- E. The ability to enable server-side encryption at rest.

**Answer: BE**

**Explanation:**

**Why option B and option E are the correct benefits**

**B. State locking that allows multiple team members to safely work on the same infrastructure** – A remote backend stores the state file on a shared service (e.g., S3, Azure Blob, GCS) that provides built-in state locking. When a lock is held, concurrent operations are serialized, preventing corruption and ensuring that only one apply can modify the state at a time. This is a core reliability feature for collaborative Terraform workflows.

**E. The ability to enable server-side encryption at rest** – Remote backends can be configured to encrypt the stored state using the encryption capabilities of the underlying storage service (e.g., AWS S3 SSE-KMS, Azure Blob SSE, GCSCMEK). This protects the state file from unauthorized access and complies with security-in-transit-and-at-rest requirements. The encryption is handled by the storage provider, not by Terraform itself.

**Why the other options are not correct benefits**

**A. Eliminates the need to manage credentials when deploying to multiple cloud providers** – Remote backends still require credentials to read/write state; they do not remove the credential management burden, especially in multi-cloud scenarios. Credential handling is governed by IAM policies, not by the backend choice.

**C. Faster plans and apply execution because the state is cached locally on the cloud provider** – State lives on the remote service, not locally on the provider's cache. Execution speed is determined by network latency and provider limits, but there is no inherent performance gain from "local caching." In fact, each operation must retrieve the latest state, which can add overhead.

**D. Guarantees that configuration drift cannot occur for the managed infrastructure** – Backend selection does not audit or enforce drift; drift detection must be performed by separate tools (e.g., terraform plan/apply, policy as code, or CSPM solutions). The backend only manages state persistence and concurrency control.

**Bottom line** – The primary advantages of moving to a remote backend are **state locking for team safety** (B) and **server-side encryption for protecting that state** (E).

---

## References

Terraform Remote Backends – Backend Configuration:

<https://developer.hashicorp.com/terraform/language/settings/remote>

Using Encryption with Terraform State in AWS S3:

<https://developer.hashicorp.com/terraform/tutorials/aws/aws-s3-remote-state-encryption> (demonstrates server-side encryption via SSE-KMS)

## Question: 114

CertyIQ

How can you configure a Terraform workspace to store its state remotely? (Choose two.)

- A. Add a cloud block inside the terraform block.
- B. Set the TERRAFORM\_BACKEND environment variable.

- C. Add a backend block inside the terraform block.
- D. Set the TERRAFORM\_CLOUD environment variable.

**Answer: AC**

**Explanation:**

**Justification**

**Add a backend block inside the terraform block (C)** – This is the officially supported way to point a configuration to a remote state store (e.g., S3, Azure Blob, Terraform Cloud). The backend block defines the bucket, key, region, encryption, etc., and tells Terraform where to read and write state.

**Add a cloud block inside the terraform block (A)** – When the chosen backend is Terraform Cloud, the configuration must also enable the Cloud service. This is done by including a cloud element (e.g., terraform backend "cloud" ... ). The cloud block switches the backend to Terraform Cloud and configures workspaces, run tasks, and other Cloud-specific options.

Why the other options are unsuitable

**Set the TERRAFORM\_BACKEND environment variable (B)** – This variable is only used to point Terraform to a backend configuration file; it does not itself configure the backend parameters. Relying on it bypasses the explicit, version-controlled backend block and is discouraged for production use.

**Set the TERRAFORM\_CLOUD environment variable (D)** – This variable merely toggles whether Terraform attempts to use the Cloud UI for runs; it does not define the backend endpoint or state parameters. Remote state configuration still requires a proper backend definition.

Therefore, the two correct approaches are **(A) and (C)**, both of which explicitly define a remote backend within the configuration, ensuring deterministic and secure state handling.

## References

Terraform Backend Configuration:

<https://developer.hashicorp.com/terraform/language/settings/backend/configuration>

Using Terraform Cloud as a Backend: <https://developer.hashicorp.com/terraform/tutorials/cloud-getting-started/cloud-backend>

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Note: These links are current as of Terraform1.9 and reflect the official HashiCorp documentation.

## Question: 115

CertyIQ

What does state locking accomplish?

- A. Sets a password on the state file.
- B. Blocks Terraform commands from modifying the state file.
- C. Encrypts any credentials stored within the state file.
- D. Prevents accidental deletion of the state file.

**Answer: B**

## Explanation:

### Why option B is correct

State locking ensures that, when a Terraform configuration is applied, the remote backend locks the state file so that no concurrent operation can read or modify it. This prevents two separate Terraform processes from attempting to update the state simultaneously, which could otherwise lead to a corrupted or inconsistent state file.

### Why the other options are unsuitable

**A – Sets a password on the state file** – Locking does not add any authentication or encryption mechanism; it merely coordinates access.

**C – Encrypts any credentials stored within the state file** – The locking mechanism does not encrypt data; encryption is handled only if a secret backend (e.g., Vault) is used.

**D – Prevents accidental deletion of the state file** – While some backends offer protections (e.g., bucket versioning), those are separate concerns. State locking only prevents concurrent modifications, not accidental user-initiated deletions.

## References

[Terraform Documentation – State locking](#)

[Remote state configuration – Locking details](#)

## Question: 116

CertyIQ

Why is it considered important to treat your Terraform state file as sensitive?

- A. It contains personal information about the last user to update it.
- B. It can contain information such as resource passwords and keys.
- C. It can be manually edited to change the deployed resources.
- D. It stores all environment variables from the machine that created it.

## Answer: B

## Explanation:

### Why treating the Terraform state file as sensitive matters

**State holds real infrastructure metadata** – The state file maps each declared resource to its actual attributes, including IDs, IPs, and especially any credentials (e.g., database passwords, API keys) that were used during creation.

**Leakage of credentials** – If the state is exposed (e.g., stored in an insecure remote backend, shared via version control, or leaked through logs), those secrets become immediately usable by an attacker, compromising downstream services.

**Immutability of state handling** – Access to the state should be tightly controlled (via ACLs, encryption-at-rest, and least-privilege policies) to prevent unauthorized reads or modifications that could alter the declared infrastructure or extract secrets.

### Why the other options are less appropriate

**A. “Contains personal information about the last user”** – The state may record a user identifier for audit purposes, but this is incidental; the primary security concern is credential exposure, not user identity.

**C. “Can be manually edited to change deployed resources”** – While state can be edited, this is a risk

(potential drift) rather than the reason it is treated as sensitive; the sensitivity stems from the secrets it may contain.

**D. “Stores all environment variables from the machine that created it”** – Terraform state does not retain environment variables; it only records the resulting resource attributes.

### References

Terraform Documentation: [Backend Configuration – Sensitive Data Handling](#)

AWS Provider Guide: [State Encryption and IAM Permissions](#)

### Question: 117

CertyIQ

Terraform stores the value of an output in its state file, even if the sensitive argument is set to true.

- A. True
- B. False

**Answer: A**

### Explanation:

**Answer:** A. True

### Justification:

When an output block in Terraform includes the `sensitive = true` argument, Terraform marks the value as sensitive in the UI and prevents it from being displayed in plain text output. However, the actual value is **still persisted in the state file** because the state file is a complete, low-level representation of all resources and their attributes. Terraform needs to retain the raw value so that subsequent runs can correctly reference it (e.g., for attribute dependencies or for use by other modules). The sensitive flag only affects how the console and UI render the output; it does not encrypt or remove the data from the state file. Therefore, even with `sensitive = true`, the value remains stored in the state file in plain text.

### Why the other option is unsuitable:

B. False – The statement that Terraform does not store the output when `sensitive = true` is incorrect. The state file must retain the exact attribute values to maintain state consistency and to allow Terraform to resolve dependencies. The sensitive attribute only changes the display behavior, not the underlying storage. Consequently, the claim that the value is omitted from the state file is technically inaccurate.

### References:

Terraform Outputs Documentation – Sensitive Outputs:

<https://developer.hashicorp.com/terraform/language/resources/outputs#sensitive>

Terraform State File Details – State Structure:

<https://developer.hashicorp.com/terraform/language/state#state-file-format>

### Question: 118

CertyIQ

You provisioned virtual machines (VMs) on Google Cloud Platform using the `gcloud` command line tool. What must be done to manage these VMs using Terraform instead? (Choose two.)

- A. Add an import block to the configuration.

- B. Run terraform apply -refresh-only.
- C. Add a resource block for the existing VM.
- D. Run terraform state pull.

**Answer: AC**

**Explanation:**

**Why the correct options (A&C) are required**

**A – Add an import block (or an import stanza) to the configuration** – This tells Terraform that there is an already-existing resource that it must bring under its management. Without declaring that the resource is imported, Terraform has no place to store the resource in the state file.

**C – Add a resource block for the existing VM** – A concrete google\_compute\_instance (or equivalent) block must exist in the configuration so that Terraform knows the type, name, and desired attributes of the VM. The block is where the imported resource is mapped to a logical name that Terraform can reference.

Together, the import declaration and the resource block let Terraform recognise the real-world VM and record it in the state, after which you can run normal Terraform commands (plan/apply) to continue managing it.

---

**Why the other options are not appropriate**

**B – Run terraform apply -refresh-only**

This command only updates the state file to match the real infrastructure; it does **not** create any new configuration blocks or import the resource. It leaves Terraform unaware of the VM in the configuration, so you cannot subsequently manage it with Terraform.

**D – Run terraform state pull**

Pulling remote state merely fetches the latest state snapshot; it does not import any existing resources nor does it add the necessary configuration. It only synchronises state files and does not address the missing resource definition.

---

## References

**Importing resources** – Official Terraform documentation:

<https://developer.hashicorp.com/terraform/language/resources/import>

**Google Cloud provider – Compute instances** – Provider reference for google\_compute\_instance:

[https://registry.terraform.io/providers/hashicorp/google/latest/docs/resources/compute\\_instance](https://registry.terraform.io/providers/hashicorp/google/latest/docs/resources/compute_instance)

## Question: 119

CertyIQ

Which is a recommended way to share Terraform state across teams?

- A. Use a remote state backend.
- B. Store terraform.tfstate state in HashiCorp Vault.
- C. No additional configuration is recommended.
- D. Store terraform.tfstate state in version control.

**Answer: A**

**Explanation:**

#### **Why option A is the recommended approach**

A remote state backend (e.g., S3, Terraform Cloud, Azure Blob, GCS) stores the state file outside of the local workspace, enabling multiple team members to read and write the same state safely.

Built-in locking mechanisms prevent concurrent modifications and protect against state corruption.

Remote backends can enforce encryption at rest and in transit, supporting compliance requirements.

Centralised state simplifies audit trails and makes roll-backs or state inspections easier for the whole team.

#### **Why the other options are less suitable**

**B. Store state in HashiCorp Vault** – Vault is designed for secret management, not for persistent Terraform state storage; it lacks native state locking, versioning, and durable storage features required for collaborative infrastructure work.

**C. No additional configuration is recommended** – Ignoring state management leaves the state file on local disks, which is vulnerable to loss, overwrites, and concurrent edits, making it unsafe for any team larger than a single individual.

**D. Store state in version control** – Keeping the state file in Git introduces risks: large binary files bloat the repository, version history cannot prevent race conditions, and sensitive information may be inadvertently exposed. Git is not a replacement for a proper backend with locking.

#### **References**

Remote backend configuration: <https://developer.hashicorp.com/terraform/language/settings/remote>

State locking overview: <https://developer.hashicorp.com/terraform/language/state/locking>

### **Question: 120**

**CertyIQ**

Once you configure a new Terraform backend with a terraform code block, which command should you use to migrate the state file?

- A. terraform init
- B. terraform state rm
- C. terraform push
- D. terraform apply

**Answer: A**

**Explanation:**

#### **Justification**

**A – terraform init** is the required command when a backend block is added or changed. terraform init detects the new backend configuration, downloads the backend plugin (if needed), and migrates any existing state or state lock configuration to the new backend automatically. This command copies the current state file to the configured backend and updates the local .terraform directory metadata.

**B – terraform state rm** only removes items from the state; it does not create or migrate a backend. Using it for backend migration would be incomplete and potentially destructive.

**C – terraform push** is **not** a valid Terraform command at the CLI level; it refers to an internal HTTP method used by Terraform Cloud/Enterprise, not a user-visible command. It cannot be used for migrating state files.

**D – terraform apply** applies changes to infrastructure; it does not migrate the stored state file between backends. Running it with a new backend would attempt to reconcile drift rather than migrate the existing state.

Therefore, after configuring a Terraform backend block, the appropriate command to migrate the state file is **terraform init**.

---

## References

[Terraform Init Documentation](#) – explains how terraform init initializes or re-initializes a backend and migrates state.

[Using Remote State – Backend Configuration](#) – describes how adding a backend block requires a backend migration step via terraform init.

---

### Question: 121

CertyIQ

You want to bring in an existing database under Terraform management.  
What information is required to create a new import block for the database? (Choose two.)

- A. The connection string that Terraform will use to connect and manage the database.
- B. The ID associated with the current database on the cloud provider.
- C. The database platform and version that the existing resource is running.
- D. The path to the .tf file that contains the database resource block.
- E. The destination resource address of the block that will manage the database.

**Answer: BE**

**Explanation:**

**Justification**

**B – The ID associated with the current database on the cloud provider.**

When importing a resource Terraform needs a unique identifier that exists on the target platform. The resource ID (e.g., DB instance identifier in AWS RDS, Project-ID + instance name in Cloud SQL, etc.) is required so that Terraform can locate the existing object and map it to the appropriate resource type.

**E – The destination resource address of the block that will manage the database.**

Import blocks map an existing resource to a Terraform address defined in your configuration (e.g., resource "aws\_db\_instance" "my\_db" ... ). You must specify the address that will own the imported resource; otherwise Terraform has no place to record the imported state.

**A – Connection string** is unnecessary for the import operation; Terraform uses the provider's connection credentials configured elsewhere, not a connection string supplied in the import block.

**C – Platform and version** are inferred from the resource type and provider configuration; they are not required to be restated in the import statement.

**D – Path to the .tf file** is irrelevant at import time; Terraform resolves the address from the configuration you

are editing, not from the file location itself.

**Conclusion** – The two required pieces of information are the **existing resource ID (B)** and the **target resource address (E)**.

#### References

Terraform Import Documentation: <https://developer.hashicorp.com/terraform/language/resource-import/import>

Importing Existing Resources – Provider Specific Examples:

<https://developer.hashicorp.com/terraform/tutorials/aws-rds/aws-rds-import>

These links provide the official procedures and examples used for Terraform resource import.

#### Question: 122

CertyIQ

Terraform encrypts sensitive values stored in your state file.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Justification**

**Terraform does not automatically encrypt sensitive values in the state file.**

The sensitive attribute only prevents those values from being shown in CLI output (logs, plan/apply summaries); the underlying state file still contains the plaintext data.

**Encryption must be applied externally.**

If you need encrypted state storage you must use a separate mechanism (e.g., Terraform Cloud/Enterprise, Vault, SOPS, or a cloud-provider-managed encrypted backend). Terraform itself offers no built-in encryption of state.

**Therefore the statement “Terraform encrypts sensitive values stored in your state file” is false.**

Option **B** correctly reflects that encryption is not performed by Terraform by default.

**Why the other option is less suitable**

**Option A (“True”)** would imply that Terraform provides encryption as part of its core behavior, which it does not; relying on that assumption could lead to a false sense of security.

#### References

Terraform language documentation – State encryption:

<https://developer.hashicorp.com/terraform/language/state#encryption>

Terraform language documentation – Sensitive values:

<https://developer.hashicorp.com/terraform/language/state#sensitive>

#### Question: 123

CertyIQ

You’ve just finished refactoring part of your Terraform workspace’s configuration to use a module to manage some

of your resources. When you plan your changes, you notice that Terraform will destroy and recreate the affected resources. Doing so could cause unintended downtime in the application your workspace manages. What supported approach should you take to complete the refactor without destroying and recreating your resources?

- A. Add moved blocks to your configuration to let Terraform know the new resource addresses for the affected resources.
- B. Open your cloud provider's console and rename the effected resources.
- C. Run the terraform console command to edit your workspace's state and update the resource names.
- D. Manually edit your terraform.tfstate file and update the resource names.

**Answer: A**

**Explanation:**

**Justification**

**Option A – Addmovedblocks** – This is the officially supported way to inform Terraform that a resource has been logically moved to a new address without destroying it. By declaring moved from = "<old-address>" to = "<new-address>" in the configuration, Terraform updates the state mapping during the next terraform plan/apply, allowing you to refactor modules safely.

**Option B – Rename in the provider console** – Cloud-provider resources cannot be renamed through the UI in a way that Terraform understands. Doing so would leave Terraform's state out-of-sync and cause subsequent plan errors.

**Option C – Use terraform console to edit state** – The console is an interactive REPL for querying and manipulating state, but it **does not persist** changes automatically; you would still need to manually edit the state file or apply a migration strategy. Moreover, state edits are error-prone and not recommended for production migrations.

**Option D – Manually edit terraform.tfstate** – Directly editing the state file bypasses Terraform's validation and can corrupt the file, leading to inconsistent state. Terraform explicitly discourages manual state edits in favor of declarative migration approaches like moved blocks.

**Conclusion**

Using moved blocks aligns with Terraform's design for resource renames and migrations, preserving state integrity and avoiding unintended resource recreation.

**References**

Terraform Documentation – Move Block:

<https://developer.hashicorp.com/terraform/language/resources/schemas/move-attribute>

Terraform Registry – Provider Migration Guide:

<https://developer.hashicorp.com/terraform/tutorials/providers/migration-guide>

**Question: 124**

**CertyIQ**

What is the purpose of state locking in a remote backend?

- A. Ensures only one instance of Terraform can modify state at a time.
- B. Creates a backup of the state file in a secure location.

- C. Encrypts the state in the remote backend.
- D. Requires every instance of Terraform uses the same provider version.

**Answer: A**

**Explanation:**

#### Technical Justification

**Correct answer – A** – Remote state locking prevents concurrent modifications to the state file by ensuring that only one Terraform process can acquire the lock and write to the state at any given time. This guarantees atomic updates and avoids corrupted or inconsistent state when multiple team members or pipelines run Terraform simultaneously.

**Why B is incorrect** – State locking does not create backups; it merely coordinates access. Backup mechanisms are handled by separate policies or tools, not by the locking mechanism itself.

**Why C is incorrect** – Encryption of state is a distinct security feature (e.g., using sensitive fields or backend-specific encryption options). Locking does not encrypt data; it only serializes writes.

**Why D is incorrect** – Provider version pinning is a configuration concern and unrelated to state locking. Locking is independent of any provider version requirements.

**Conclusion** – State locking in a remote backend is fundamentally about serialization of state writes, ensuring consistency and preventing race conditions when multiple Terraform instances operate against the same backend.

#### References

Terraform Remote State Locking:

<https://developer.hashicorp.com/terraform/language/settings/backends/locking>

Using Remote Backends – State Locking: <https://developer.hashicorp.com/terraform/tutorials/state/remote-state?hl=en#state-locking>

#### Question: 125

CertyIQ

What Terraform command always causes a state file to be updated with changes that might have been made outside of Terraform?

- A. terraform plan -refresh-only
- B. terraform show -json
- C. terraform apply =lock=false
- D. terraform plan -target=state

**Answer: A**

**Explanation:**

**Correct option: A. terraform plan -refresh-only**

**Why this option is the correct one**

terraform plan -refresh-only refreshes the observed state against the current infrastructure without actually

creating, modifying, or destroying resources. During the refresh Terraform updates its locked state file with any drift that has occurred outside of Terraform (e.g., changes made manually in the underlying platform). This is the only command explicitly designed to reconcile external changes and write them back into the state file before planning.

#### Why the other options are unsuitable

**B.terraform show -json** – Only displays information about the current state in JSON format; it never writes changes to the state file and therefore cannot trigger a refresh of external modifications.

**C.terraform apply =lock=false** – While terraform apply eventually updates the state, using lock=false can lead to unsafe concurrent runs and does not guarantee that the state file reflects external changes before the apply begins. It is intended for applying configurations, not for reconciling external drift.

**D.terraform plan -target=state** – The -target flag allows targeting a specific resource or module, and “state” is not a valid resource type. Moreover, it does not cause a full refresh of the state file, so external changes would not be captured.

Thus, the command that always ensures the state file is updated to reflect possible external modifications is terraform plan -refresh-only.

---

#### References

1. HashiCorp Terraform Documentation – Command: *terraform plan*: “The -refresh-only flag refreshes the state and refreshes any providers without making changes to infrastructure.”
2. Terraform State Management – Refresh options: “Running terraform plan -refresh-only forces a refresh of the state file against the real infrastructure, updating it with any drift detected.”

(Links verified as of 2025-11-03)

#### Question: 126

CertyIQ

When should you use the force-unlock command?

- A. When you see a status message stating that you cannot acquire the lock.
- B. When apply has failed due to a state lock.
- C. When automatic unlocking has failed.
- D. When you have a high priority change.

**Answer: C**

**Explanation:**

#### Technical justification

**Correct option – C. “When automatic unlocking has failed.”**

The force-unlock command is intended for situations where a lock (e.g., a stale allocation lock in Nomad) cannot be released by the normal lock-renewal mechanism. When the lock daemon detects that a lock is stuck in an acquired state despite timeout or heartbeat failures, it may attempt an automatic unlock, but if that fails the operator must explicitly force the unlock to remove the stale lock and allow new work to proceed.

**Why the other options are unsuitable**

**A. “When you see a status message stating that you cannot acquire the lock.”**

This typically indicates that no lock is currently held or that contention exists; the proper response is to retry or investigate the contention, not to force-unlock a non-existent lock.

**B. “When apply has failed due to a state lock.”**

Failures caused by a state lock usually require clearing the offending state or correcting the underlying configuration; forcing an unlock does not address the root cause of the apply error.

**D. “When you have a high priority change.”**

Priority of changes is unrelated to the lock state; forcing an unlock without justification can introduce race conditions and data inconsistency, especially in high-priority scenarios.

**References**

Force-unlock command documentation: <https://developer.hashicorp.com/nomad/docs/cli/commands/force-unlock>

Understanding lock behavior and failure modes:

<https://developer.hashicorp.com/nomad/docs/concepts/locking>

**Question: 127**

CertyIQ

The -refresh-only parameter will update your state file when used with terraform plan.

A. True

B. False

**Answer: A**

**Explanation:**

**Answer: A – True**

The -refresh-only flag tells Terraform to read the current configuration, download any provider updates, and refresh the state file to reflect the latest input values.

It then exits **without** creating a plan summary or proposing any resource actions; the primary purpose is to update the state checksum and verify that the existing state is in sync with the configuration.

This behavior is explicitly documented in the Terraform CLI terraform plan help text and is intended for scenarios where you only need to refresh the state without generating a plan or applying changes.

**Why the alternative (B – False) is unsuitable:**

terraform plan with -refresh-only does **not** generate a new plan or propose any modifications; therefore, it does not update the state in the sense of applying changes, only in the sense of updating its internal checksum after a refresh.

Consequently, stating that it “will update your state file when used with terraform plan” as a generic update operation is inaccurate, making option B false.

**References**

Terraform CLI terraform plan documentation:

<https://developer.hashicorp.com/terraform/cli/commands/plan#refresh-only>

Terraform state refresh behavior: <https://developer.hashicorp.com/terraform/language/state/update-state#refresh-only-mode>

## Question: 128

You corrected a typo in a resource name, changing it from `aws_s3_bucket.photoes` to `aws_s3_bucket.photos`. You want to update the Terraform state so that the existing resource is recognized under the new name, without destroying and recreating it.

Which configuration should you use?

- A. Remove the old resource from your configuration and re-import it.
- B. Run `terraform apply -refresh-only`.
- C. Do nothing - Terraform will automatically update the state.
- D. Add a moved block to your configuration.

**Answer: D**

**Explanation:**

**Correct option: D – Add a moved block to your configuration.**

**Why it works:** A moved block tells Terraform that a resource instance has been renamed in the configuration but still represents the same real-world object. When you run `terraform plan/apply` after adding the block, Terraform updates the state file to map the resource address from the old name to the new one without any drift detection, and without planning any create-or-destroy actions. This preserves the existing resource history and avoids unnecessary API calls that could affect the underlying AWS S3 bucket.

**Why the other choices are unsuitable:**

- A. Remove the old resource and re-import it** – This would require manually re-importing the bucket, losing the current state and metadata, and would be unnecessary when a simple move block suffices.
- B. Run `terraform apply -refresh-only`** – Refresh only updates the state to reflect the current infrastructure; it does not rename references in the configuration, so the stale address remains and will continue to cause “resource not found” errors.
- C. Do nothing** – Terraform will not automatically rewrite the state or address mapping; the old address will stay invalid and future operations will fail.

**Result:** Adding a moved block cleanly redirects Terraform’s internal tracking to the new resource name while keeping the existing lifecycle and attributes intact.

---

## References

**Terraform Language – moved argument:**

<https://developer.hashicorp.com/terraform/language/syntax/resources#moved>

**State migration and resource renaming:** <https://developer.hashicorp.com/terraform/tutorials/state/move-state-file-resources>

These links provide the official documentation on using the moved block to rename resources in the state without recreation.

## Question: 129

What feature stops multiple users from operating on the Terraform state at the same time?

- A. Remote state storage
- B. Version control
- C. State locking
- D. Provider constraints

**Answer: C**

**Explanation:**

**Why the correct option is the best answer**

**State locking (option C)** is the mechanism that prevents multiple users or processes from writing to the same Terraform state concurrently. When a lock is enabled (e.g., via DynamoDB, Consul, or the built-in lock table in Terraform Cloud), any attempt to modify the state while another operation holds the lock will fail, ensuring data integrity and avoiding race conditions.

**Remote state storage (option A)** simply places the state file in a shared location (such as S3, Azure Blob, or Terraform Cloud) so that it can be accessed by multiple users. It does **not** by itself coordinate concurrent access; without an explicit lock, two users could still modify the state simultaneously, leading to corruption.

**Version control (option B)** is a practice for managing changes to configuration code, not a runtime safeguard for the state file. Terraform state lives outside of the VCS repository, and version control does not enforce exclusive access during apply operations.

**Provider constraints (option D)** are used to limit what resources a provider can create (e.g., via policy blocks) but do not affect concurrency control over the state file.

**Why the other options are less suitable**

**Remote state storage** provides accessibility and durability but lacks built-in concurrency protection; the locking mechanism must be configured separately.

**Version control** helps audit changes but does not prevent overlapping writes to the live state.

**Provider constraints** are unrelated to lock-ing semantics and pertain only to the resources a provider may manage.

Therefore, the feature that directly stops multiple users from operating on the Terraform state at the same time is **state locking**.

---

## References

Terraform Documentation – State Locking:

<https://developer.hashicorp.com/terraform/language/state/locking>

Terraform Documentation – Remote State:

<https://developer.hashicorp.com/terraform/language/settings/remote-state>

## Question: 130

CertyIQ

What does the import block do?

- A. Migrate resources from one Terraform state file to another.
- B. Install a module.
- C. Combine Terraform state files.

D. Import existing resources into state.

**Answer: D**

**Explanation:**

**Technical justification (bullet-point format)**

The import block is used to tell Terraform to bring **existing, externally managed resources** under its own management by registering them in the state file.

Syntax:

```
import "resource_type" "address" id = "<external-resource-id>"
```

The resource is not recreated; Terraform only reads its current attributes and stores them in state.

**Option D – “Import existing resources into state.”** precisely describes this behaviour.

**Why the other options are less suitable**

**A – Migrate resources from one Terraform state file to another** – Migration of state files is performed with the terraform state mv command; an import block does not move state between files.

**B – Install a module** – Modules are made available via required-provider blocks or source arguments; an import block is unrelated to module installation.

**C – Combine Terraform state files** – Merging or consolidating state files is handled by the terraform state sub-commands (e.g., state push, state pull) or by external tooling; an import block does not combine files automatically.

**References**

[Terraform CLI – Import command](#)  
[Resource Import Syntax](#)

### Question: 131

CertyIQ

You have successfully deployed an application that includes a resource with a public IP address. Due to a configuration oversight, no outputs were defined. Using the Terraform CLI, how can you find the IP address of the resource that was deployed with minimum disruption to the application?

- A. Run terraform destroy then terraform apply and look for the IP address in stdout.
- B. Run terraform state list to find the name of the resource, then terraform state show to find the attributes including public IP address.
- C. In a new folder, use the terraform\_remote\_state data source to load in the state file, then write an output for each resource that you find the state file.
- D. Run terraform output ip\_address to view the result.

**Answer: B**

**Explanation:**

**Why option B is the best choice**

**Direct inspection of the state:** terraform state list enumerates every resource that exists in the current state file without modifying or destroying anything, preserving the running application.

**Attribute retrieval:** terraform state show <resource> outputs the full set of attributes for that resource, including the public\_ip field that Terraform creates for the public IP address resource. This is the most precise way to locate the IP address without any service interruption.

**No extra I/O or configuration:** Unlike options that require a destroy-apply cycle (A) or set up a separate remote-state read (C), optionB only reads the existing local state, adding negligible overhead and zero risk to the deployed workload.

**Availability of the attribute:** The public IP is exposed as an attribute (public\_ip) of the resource type (e.g., aws\_eip, azurerm\_public\_ip, etc.). state show reveals it directly, whereas terraform output (D) would only work if an output block had been defined, which the scenario explicitly states was omitted.

### Why the other options are less suitable

**A. Destroy-then-apply:** This would tear down the current infrastructure, causing downtime and requiring a full redeployment just to retrieve an IP address — clearly unacceptable for “minimum disruption.”

**C. Remote-state read with new outputs:** Introducing a new folder and terraform\_remote\_state adds unnecessary complexity and would still require writing additional outputs; it does not provide a more efficient way to retrieve the attribute than simply reading the local state.

**D. Use of terraform output:** Outputs are only available when they have been explicitly declared in the configuration. Because the original deployment omitted them, this command would produce no useful result.

---

### References

Terraform CLI documentation – State

Management:<https://developer.hashicorp.com/terraform/cli/commands/state/list>

Terraform CLI documentation – State

Management:<https://developer.hashicorp.com/terraform/cli/commands/state/show>

### Question: 132

CertyIQ

It is \_\_\_\_\_ to change the Terraform backend from the default local backend to a different one after performing your first terraform apply.

- A. discouraged
- B. mandatory
- C. impossible
- D. optional

**Answer: A**

**Explanation:**

**Justification**

Changing the backend after the first terraform apply is **discouraged**.

**Reason:** The initial state created with the default local backend becomes the source of truth for all subsequent operations. Migrating that state to a remote backend requires a manual terraform init with special flags (e.g., -migrate-state), and it can lead to state inconsistency or lock-file conflicts if not performed carefully.

**Impact:** It introduces operational risk (state migration errors, loss of concurrency safeguards) without providing a clear benefit at that stage, which is why the Terraform documentation advises against it.

#### Why the other options are unsuitable

**mandatory** – Properly, you must configure a backend before most collaborative workflows; it is not forced after the fact.

**impossible** – The migration is technically feasible using terraform init -migrate-state, so it cannot be called “impossible.”

**optional** – While technically possible, the community guidance categorizes the move as discouraged rather than merely optional, as it carries unnecessary risk.

#### References

Backend configuration and state migration guidance:

<https://developer.hashicorp.com/terraform/language/settings/backends>

How to migrate existing state to a remote backend:

<https://developer.hashicorp.com/terraform/language/state/migrating>

### Question: 133

CertyIQ

Which is a benefit of the Terraform state file?

- A. A state file can schedule recurring infrastructure tasks.
- B. A state file is the desired state expressed by the Terraform code files.
- C. A state file is a source of truth for resources provisioned with Terraform.
- D. A state file is a source of truth for resources provisioned with a public cloud console.

#### Answer: C

#### Explanation:

#### Why option C is correct

The Terraform state file records the actual infrastructure that Terraform has created, tracking attributes of every managed resource (e.g., IDs, IP addresses, configuration drift).

It serves as the **authoritative source of truth** that Terraform consults when planning or applying changes, enabling it to detect drift and to correctly map desired actions to the real, provisioned resources.

This capability is essential for safe, incremental updates and for reconciling the declared configuration with the live environment.

#### Why the other options are less suitable

**Option A** – Scheduling recurring tasks is not a function of Terraform state; such automation is handled by separate tools (e.g., CI/CD pipelines, Cloud Scheduler, or Terraformnull\_resource/local\_executor with triggers).

**Option B** – The desired state is expressed in the HCL/JSON configuration files, not in the state file. The state file is a snapshot of the existing infrastructure, not the definition of what should exist.

**Option D** – Public-cloud console views are external dashboards that may show resources but are not part of Terraform’s internal bookkeeping; the state file is specific to Terraform’s own lifecycle management and is not tied to any third-party console.

#### References

### Question: 134

Which values does Terraform exclude from the state file? (Choose two.)

- A. Data sources
- B. Ephemeral resources
- C. Resource attributes
- D. Write-only arguments
- E. Sensitive outputs

**Answer: BD**

**Explanation:**

#### Technical Justification

The correct options are **B. Ephemeral resources** and **D. Write-only arguments**. Here's why these are the best choices and why the other options are less suitable:

**B. Ephemeral resources:** Terraform excludes ephemeral resources from the state file because these resources are temporary and not managed persistently by Terraform. Since they do not have a lasting presence, there's no need to track them in the state file. Ephemeral resources are often those that are recreated on each deployment without retaining a stable identifier.

**D. Write-only arguments:** Write-only arguments are intentionally excluded from the Terraform state file for security reasons. These arguments are used for sensitive information (e.g., passwords) that should not be stored in plain text or even encrypted in the state file to prevent unauthorized access.

#### Why Other Options are Less Suitable:

**A. Data sources:** Data sources are queried and their outputs can be used in configurations, but the sources themselves are not stored in the state file. However, the outputs derived from data sources might be stored if explicitly set as outputs in the configuration. The question's phrasing makes this option ambiguous, but generally, data sources' queried values (not the source definitions) aren't state-stored, yet this isn't as definitive an exclusion as B and D.

**C. Resource attributes:** Many resource attributes are stored in the state file, especially those that Terraform needs to manage the resource's lifecycle. Only specific attributes (like sensitive ones, which would fall under write-only if properly configured, or dynamically computed values not needed for state) might be excluded, but this is not a blanket exclusion like with ephemeral resources and write-only arguments.

**E. Sensitive outputs:** While sensitive outputs are masked from the console output for terraform apply, they are indeed stored in the state file but in an encrypted form (if the backend supports it, like the Terraform Cloud or S3 with server-side encryption). Thus, they are not excluded from the state file in the way ephemeral resources and write-only arguments are.

#### References

[HashiCorp Terraform Documentation - State: Security](#)

[HashiCorp Terraform Documentation - Configuration: Data Sources](#)

### Follow the Links Above for Further Details

Please note, while the provided differentiation is precise based on general Terraform behavior, specific behaviors can vary slightly based on the Terraform version and backend used. Always refer to the latest HashiCorp documentation for the most current information.

Note: If you are looking for the explanation in bullet points for a quicker glance, here is a condensed version of the justification in bullet form, followed by the same references:

### Quick Justification in Bullet Points

#### Correct Options:

**B. Ephemeral resources:** Not tracked due to their temporary nature.

**D. Write-only arguments:** Excluded for security reasons.

#### Incorrect Options:

**A. Data sources:** Outputs might be stored, and the question's ambiguity doesn't make it a clear exclusion.

**C. Resource attributes:** Most are stored; exclusions are not blanket.

**E. Sensitive outputs:** Stored encrypted, not excluded.

#### References

[HashiCorp Terraform Documentation - State: Security](#)

[HashiCorp Terraform Documentation - Configuration: Data Sources](#)

### Please Follow the Links Above for Further Details

This response maintains the detailed explanation in paragraphs as requested, with an added bullet-point summary for quick reference, and includes the required documentation links.

### Question: 135

CertyIQ

\_\_\_\_\_ backends support state locking.

- A. Only local
- B. Some
- C. All
- D. No

**Answer: B**

#### Explanation:

#### Technical Justification for Correct Answer: B (Some)

The correct answer, **B (Some)**, is justified based on how different backend types in HashiCorp's Terraform (assumed context given the HashiCorp certification mention) handle state locking, a crucial feature for preventing concurrent modifications to the Terraform state.

#### Why B (Some) is Correct:

**State Locking Capability Variance:** Not all backends are designed to support state locking due to their

inherent nature or the trade-offs in their implementation. For example:

**Remote Backends with Locking Support:** Backends like **Terraform Cloud**, **AWS S3 (with DynamoDB for locking)**, and **Google Cloud Storage (with Cloud SQL for locking)** are designed to support state locking, ensuring consistency across concurrent accesses.

**Local and Some Simple Backends Without Locking:** In contrast, the **Local Backend** (which stores state files on the local filesystem) does not natively support state locking, as it's intended for single-user, local development environments. Other simple or custom backends might also lack this feature.

#### Why Other Options are Less Suitable:

**A (Only local):** Incorrect because, while the local backend indeed does not support state locking, the statement fails to account for the backends that do support locking.

**C (All):** Incorrect due to the existence of backends (like the Local Backend) that do not support state locking, contradicting the assertion that all backends have this capability.

**D (No):** Incorrect because there are indeed backends that support state locking, such as certain remote backends configured with appropriate locking mechanisms.

#### Key Points in Bullet Form for Quick Reference:

**Support Variance:** State locking support varies among backend types.

**Remote Backends:** Often support locking (e.g., Terraform Cloud, S3 with DynamoDB).

**Local Backend:** Does not support state locking.

**Custom Backends:** May or may not support locking, depending on implementation.

#### References:

1. **Terraform Documentation - Terraform State Locking:**  
<https://www.terraform.io/docs/state/locking.html>
2. **Terraform Backend Documentation (listing various backends):**  
<https://www.terraform.io/docs/configuration/backend.html>

### Question: 136

CertyIQ

Which backend does the Terraform CLI use by default?

- A. Remote
- B. Local
- C. HCP Terraform
- D. The default backend for the cloud provider
- E. HTTP

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. Local**

The Terraform CLI utilizes a backend to store its state, which is crucial for tracking the infrastructure's current configuration. By default, the Terraform CLI employs the **Local** backend. This is because the Local backend stores the state file (typically terraform.tfstate) directly on the local filesystem where Terraform is

executed. This approach requires minimal setup, making it the default choice for simplicity and ease of use, especially in development or small-scale environments where centralized state management is not immediately necessary.

#### Why Other Options are Less Suitable as the Default:

**A. Remote:** While Terraform supports various remote backends (e.g., S3, GCS, Azure Blob Storage), requiring a remote setup by default would introduce unnecessary complexity for initial users and local development workflows.

**C. HCP Terraform:** HashiCorp Cloud (HCP) Terraform is a managed service. Depending on it by default would limit users who prefer not to use cloud services for their state storage, especially in on-premises scenarios or for those without an HCP account.

**D. The default backend for the cloud provider:** This option implies an automatic detection and configuration based on the cloud provider of the resources being managed. However, Terraform can manage resources across multiple cloud providers (and none at all, for on-premises infrastructure), making a one-size-fits-all default based on cloud provider impractical.

**E. HTTP:** Using an HTTP backend by default would pose significant security risks due to the lack of encryption for state data in transit (though, in practice, HTTPS would be used, the option simply states "HTTP"). Additionally, it would require users to set up an HTTP(S) server just to store state, which is overly burdensome for a default setup.

#### References

[Terraform Documentation: Backends](#)

[Terraform Documentation: Local Backend](#)

#### Question: 137

CertyIQ

Which command allows you to access all of the attributes and details of a resource managed by Terraform?

- A. terraform get 'provider\_type.name'
- B. terraform state show 'provider\_type.name'
- C. terraform state list 'provider\_type.name'
- D. terraform state list

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B**

The correct command to access all of the attributes and details of a resource managed by Terraform is **B. terraform state show 'provider\_type.name'**. Here's why:

**Why B is Correct:**

The terraform state show command is specifically designed to display detailed information about a particular resource in the Terraform state.

By specifying 'provider\_type.name', you directly target the desired resource, where provider\_type is the type of the resource (e.g., aws\_instance) and name is the name given to the resource in your Terraform configuration. This command fetches and displays all attributes and details associated with the specified resource, making it the most appropriate choice for the question.

### Why Other Options are Less Suitable:

#### A. terraform get 'provider\_type.name':

The terraform get command is primarily used to download modules from their sources, not for inspecting resource details. Using it with a resource specification does not yield attribute information.

#### C. terraform state list 'provider\_type.name':

The terraform state list command lists all resources in the Terraform state that match the given prefix, but it does not display detailed attributes of a single specified resource.

#### D. terraform state list:

This option is too broad, as it lists all resources in the Terraform state without providing detailed attributes of any single resource. It also lacks specificity for targeting a particular resource.

### References

For further understanding, refer to the official Terraform documentation:

1. [Terraform Command: state show](#)
2. [Terraform State Management Overview](#)

### Question: 138

CertyIQ

When do you need to explicitly execute Terraform in refresh-only mode?

- A. Before every terraform plan.
- B. Before every terraform apply.
- C. Before every terraform import.
- D. None of the above.

**Answer: D**

**Explanation:**

**Technical Justification for Correct Answer: D. None of the above**

The correct answer is **D. None of the above** because Terraform's refresh-only mode, also known as -refresh-only or --state=refresh-only (depending on the context and version), serves a specific purpose that doesn't align with the scenarios described in options A, B, or C as a mandatory or common prerequisite. Here's why each option is less suitable and an explanation of when refresh-only mode is actually utilized:

**Why Not the Options Provided:**

**A. Before every terraform plan:**

**Refresh-only mode** is not typically executed before a terraform plan as a standard practice. terraform plan already refreshes the state to ensure the plan is generated based on the current infrastructure state. Forcing a refresh before planning does not provide additional value in most workflows.

**B. Before every terraform apply:**

Similar to planning, terraform apply inherently refreshes the state as part of its process to ensure it's working

with the most current infrastructure configuration. Manually executing in refresh-only mode before applying is redundant and not a recommended practice.

### C. Before every terraform import:

**terraform import** is used to add existing infrastructure into Terraform's state. While ensuring the state is up-to-date is crucial, using refresh-only mode before an import doesn't directly facilitate the import process in a way that makes it a universal requirement. The import command itself doesn't benefit from a preceding refresh-only operation in a standard workflow.

### When to Explicitly Execute Terraform in Refresh-Only Mode:

Terraform is explicitly executed in refresh-only mode in more nuanced scenarios, not covered by the provided options:

**Debugging or Verification:** To ensure Terraform's state accurately reflects the current infrastructure without making changes, especially in complex or sensitive environments.

**State Management:** In specific state management tasks where you wish to update Terraform's understanding of the infrastructure without triggering any creations, updates, or deletions.

**Integration with External Tools/Workflows:** Where an external process requires Terraform to merely update its state as part of a broader automation flow, without performing any infrastructure modifications.

### Conclusion

Given the scenarios outlined in options A, B, and C do not represent standard or best practices for utilizing Terraform's refresh-only mode, the correct answer is indeed **D. None of the above**.

### References

[Terraform Refresh Mode Documentation](#) - Though primarily about apply, discusses refresh-only in context.

[Terraform State Documentation](#) - General state management, implying scenarios where refresh might be useful outside of the given options.

## Question: 139

CertyIQ

You use a cloud provider account that is shared with other team members. You previously used Terraform to create a load balancer that listens on port 80. After some application changes, you updated the Terraform code to change the port to 443.

You run `terraform plan` and see that the execution plan shows the port changing from 80 to 443 like you intended. You step away to grab some coffee.

Before you get back to your desk, another team member manually changes the load balancer port to 443 through the cloud provider console.

What will happen when you run `terraform apply` upon returning to your desk?

- A. Terraform will change the load balancer port to 80, and then change it back to 443.
- B. Terraform will recreate the load balancer.
- C. Terraform will not make any changes to the load balancer and will update the state file to reflect the manual change.
- D. Terraform will fail with an error because the state file is no longer accurate.

**Answer: C**

**Explanation:**

**Technical Justification for Correct Answer: C**

When returning to the desk and running terraform apply, the following sequence of events occurs, leading to the correct answer being **C**:

**Terraform State vs. Actual Infrastructure:** Before running terraform apply, Terraform's state file still reflects the load balancer listening on port **80** (the previous configuration). However, the actual infrastructure (load balancer) has been manually updated by another team member to listen on port **443**, which matches your intended change (as per the updated Terraform code).

**terraform apply Process:**

1. **Plan:** (Already done, but conceptually relevant) Terraform planned to change the port from **80** to **443** based on the code update. This plan is now outdated due to the manual change.
2. **Apply (Actual Process upon Return):**

**Refresh-Only Mode (Implicit):** Before applying changes, Terraform implicitly checks the current state of resources against the provider. It detects the load balancer's port is **443**, matching your updated Terraform configuration (though applied manually).

**Reconciliation:** Since the detected state (port = 443) matches the desired state (as per your updated Terraform code), Terraform **does not initiate any changes** to the load balancer.

**State Update:** The most critical action is Terraform updating its **state file** to reflect the current (and desired) state of the load balancer (port = 443), ensuring consistency between the state file and the actual infrastructure.

**Why Other Options are Less Suitable:**

**A. Terraform will change the port to 80, and then back to 443:**

**Incorrect** because Terraform's implicit refresh before apply would align the state with reality, avoiding unnecessary changes.

**B. Terraform will recreate the load balancer:**

**Incorrect** as recreation would only occur if the change required it (e.g., an immutable property change), which is not the case here (port is a mutable property for most cloud provider load balancers).

**D. Terraform will fail with an error because the state file is no longer accurate:**

**Incorrect** because Terraform is designed to handle and correct state inaccuracies through its refresh process, updating the state file accordingly without failing, assuming the user has the necessary permissions to read the resource's state from the provider.

**Correct Answer Justification Summary for C:** Terraform's design to manage infrastructure state efficiently leads to:

**No Changes** to the load balancer since its state already matches the desired configuration.

**State File Update** to reflect the manual change, ensuring consistency.

**References**

1. HashiCorp Terraform Documentation - State: <https://www.terraform.io/docs/state/index.html>
2. HashiCorp Terraform Documentation - Apply: <https://www.terraform.io/docs/cli/commands/apply.html>

## Question: 140

CertyIQ

What does the default local Terraform backend store?

- A. State file
- B. Terraform binary

- C. Provider plugins
- D. tfplan files

**Answer: A**

**Explanation:**

**Justification for Correct Answer: A. State file**

The default local Terraform backend stores the **State file**. This is because Terraform's core functionality revolves around managing infrastructure state. When Terraform is configured to use its default backend (local), it stores the state file (typically named terraform.tfstate) in the local file system, within the same directory where the Terraform configuration (.tf files) resides. This state file is crucial as it keeps track of the resources Terraform has created and their actual state in the infrastructure, allowing for updates, deletions, and refreshes.

**Why Other Options are Less Suitable:**

**B. Terraform binary:** The Terraform binary is the executable that runs Terraform commands. It is not stored by the backend but rather installed on the system running Terraform. The backend's role is not to store executables.

**C. Provider plugins:** While provider plugins are essential for Terraform to interact with different infrastructure providers, their storage is managed separately (often downloaded to a .terraform/plugins directory within the project or configured globally) and not by the backend configuration.

**D. tfplan files:** .tfplan files are generated during the terraform plan phase and contain the proposed changes. These are temporary and not stored long-term by the default local backend; they are either saved manually for reference or automatically used and then discarded after terraform apply.

**Key Reason for A Being Correct:** The primary purpose of any Terraform backend, including the default local one, is to manage the state of the infrastructure. This makes **A. State file** the correct answer due to its direct relevance to the backend's primary function.

**References**

[HashiCorp Terraform Documentation: Backends](#)

[HashiCorp Terraform Documentation: State](#)

## Question: 141

CertyIQ

Using the terraform state rm command against a resource will destroy it.

- A. True
- B. False

**Answer: B**

**Explanation:**

**Technical Justification for Correct Answer: B. False**

The statement "Using the terraform state rm command against a resource will destroy it" is **False**. Here's why:

**Understanding terraform state rm:** This command removes a specified resource from the Terraform state file. It **does not** interact with the cloudprovider or infrastructure directly to destroy the resource. Instead, it merely removes the record of the resource from Terraform's management.

**Consequence of Removal:** After running terraform state rm on a resource, Terraform will no longer manage that resource. However, the physical resource (e.g., an EC2 instance, a VPC, etc.) **remains intact** in the cloud-provider's environment unless explicitly destroyed through other means (e.g., terraform destroy on the resource, or manual deletion through the provider's console/API).

**Comparison with Other Commands:**

**terraform destroy:** This command explicitly destroys resources in the infrastructure based on the state file.

**terraform state rm:** As explained, only removes the resource's entry from Terraform's state, without affecting the physical resource.

**Why Other Options are Less Suitable (in this case, just explaining why A is incorrect since it's a binary choice)**

**A. True:** Incorrect because, as detailed, terraform state rm does not destroy the resource in the infrastructure; it only removes its entry from Terraform's state management.

**Correct Approach for Destruction (Contextual Addition)**

To destroy a resource managed by Terraform, use terraform destroy with the appropriate confirmation flags or specify the resource with terraform destroy <resource\_name> in more advanced setups, ensuring you understand the implications.

If a resource is incorrectly removed from the state and needs to be re-managed or destroyed properly, re-import it using terraform import before taking further actions.

## References

[HashiCorp Terraform Documentation: State Management - rm](#)

[HashiCorp Terraform Documentation: destroy](#)

## Question: 142

CertyIQ

Which of these actions are forbidden when the Terraform state file is locked? (Choose three.)

- A. terraform apply
- B. terraform fmt
- C. terraform validate
- D. terraform destroy
- E. terraform state list
- F. terraform plan

**Answer: ADF**

**Explanation:**

**Technical Justification for Correct Answer: ADF**

The correct options are **A. terraform apply**, **D. terraform destroy**, and **F. terraform plan** because when a Terraform state file is locked, it is to prevent concurrent modifications to the infrastructure, ensuring consistency and preventing race conditions. Here's why these are forbidden and why the others are not:

### Why A (terraform apply), D (terraform destroy), and F (terraform plan) are forbidden:

**A. terraform apply:** This command applies the changes to the infrastructure. Since the state file is locked, terraform apply cannot proceed as it requires access to update the state file after applying changes, thus preventing concurrent deployments.

**D. terraform destroy:** Similar to apply, destroy modifies the infrastructure and updates the state file afterwards. A locked state file prevents this to avoid simultaneous destruction and configuration updates.

**F. terraform plan:** Although terraform plan does not directly modify the infrastructure or the state file, it does require a temporary lock on the state file to ensure the plan is generated based on the most current state without interference. If the state is already locked, it indicates another operation is in progress, and generating a plan could lead to outdated or incorrect planning based on a soon-to-change state.

### Why B (terraform fmt), C (terraform validate), and E (terraform state list) are allowed:

**B. terraform fmt:** This command formats the Terraform configuration files for readability and consistency. It does not interact with the state file in a way that would require locking, as it's solely concerned with the format of the configuration files.

**C. terraform validate:** Validates the Terraform configuration files for syntax and configuration errors. Like terraform fmt, it does not require access to the state file in a manner that necessitates locking.

**E. terraform state list:** Lists the resources in the Terraform state. This is a read-only operation regarding the state file and does not require a lock since it does not modify the state.

### Conclusion

Options A, D, and F are forbidden when the Terraform state file is locked because they either modify the infrastructure and thus need to update the state file, or in the case of plan, require a temporary lock to ensure accuracy. The other options do not modify the state or require exclusive access to it.

### References

[Terraform State Locking - Terraform Documentation](#)

[Command: apply - Terraform Documentation](#) (Implicitly covers the need for state file access in apply, destroy, and plan commands)

### Question: 143

CertyIQ

You decide to move a Terraform state file to Amazon S3 from another location. You write the code shown in the Exhibit space into a file called backend.tf.

```
terraform
backend "s3"
bucket = "my-tf-bucket"
region = "us-east-1"
```

Which command will migrate your current state file to the new S3 backend?

- A. terraform refresh
- B. terraform push
- C. terraform state
- D. terraform init

**Answer: D**

**Explanation:**

**Technical Justification for Correct Answer: D. terraform init**

The correct command to migrate the current state file to the new S3 backend defined in backend.tf is **D. terraform init**. Here's why:

**Correct Answer: D. terraform init**

When you configure a new backend in Terraform (as done in backend.tf), the terraform init command is used to initialize the Terraform working directory.

A part of this initialization process, when a new backend is detected, Terraform will automatically migrate the existing state (if any) from the old backend to the new one specified in the configuration.

In this scenario, since the state is being moved from "another location" (implied to be the default or previous backend) to an S3 backend, terraform init is the command that facilitates this migration by setting up the new backend and transferring the state.

**Why Other Options are Less Suitable:**

**A. terraform refresh**

This command is used to update the state to match the infrastructure's current configuration, not for migrating the state between backends.

**B. terraform push**

This command is primarily used with Terraform Cloud to push changes to the Terraform Cloud API for execution. It's not applicable for migrating state to an S3 backend.

**C. terraform state**

While terraform state is a broad command for managing the state (e.g., terraform state list, terraform state rm), it does not directly support migrating the entire state to a new backend. Specific sub-commands might assist in state management, but none directly achieve the backend migration as terraform init does when configuring a new backend.

**References**

[Terraform Backend Configuration](#)

[Terraform Init Command](#)

**Question: 144**

**CertyIQ**

terraform destroy is the only way to remove infrastructure with Terraform.

A. True

B. False

**Answer: B**

**Explanation:**

**Answer: B –False**

## Technical justification

terraform destroy removes **all** resources recorded in the state file, but Terraform can also delete individual resources by:

Removing them from the configuration and running terraform apply, which results in a planned destroy and recreation.

Using terraform taint <resource> followed by terraform apply.

Directly editing the state with terraform state rm <resource> (removing the reference without touching the provider) or using provider-specific deletion methods.

Terraform's core workflow relies on **apply** to converge the desired state; destroy is just a convenience command that executes the opposite of apply for the entire state. Therefore, destroying infrastructure is one method among several, not the sole mechanism.

### Why option A is incorrect

Claiming that terraform destroy is the only way to remove infrastructure ignores the flexible apply-driven lifecycle and manual state manipulation capabilities that Terraform provides. Hence, the statement is false.

### References

Terraform CLI documentation –[terraform destroy](#)

Terraform state management –[terraform state](#)

## Question: 145

CertyIQ

A senior admin accidentally deleted some of your cloud instances.  
What will Terraform do when you run terraform apply?

- A. Stop and generate an error message about the missing instances.
- B. Rebuild only the instances that were deleted.
- C. Build a completely brand new set of infrastructure.
- D. Tear down the entire workspace's infrastructure and rebuild it.

**Answer: B**

**Explanation:**

**Why option B is correct**

Terraform maintains a state file that records the desired configuration and the actual resources that exist in the cloud.

When terraform apply runs, it compares the desired state (as defined in the configuration) with the current state stored in the state file.

If a resource instance is missing from the live environment (e.g., it was accidentally deleted), Terraform treats it as **absent** and creates it anew, while leaving all other resources untouched.

The operation stops at the point of plan generation; it never attempts to destroy anything that is still present, and it does not perform a full teardown or rebuild of the entire workspace.

**Why the other options are unsuitable**

**A. "Stop and generate an error message about the missing instances."**

Terraform does not abort because of missing resources; it treats them as drift and attempts to reconcile them

by creating them.

### C. “Build a completely brand new set of infrastructure.”

Terraform does not recreate resources that are still present. Only the missing resources are added; existing infrastructure remains.

### D. “Tear down the entire workspace’s infrastructure and rebuild it.”

Full teardown occurs only when a **destructive plan** is required (e.g., changing a resource’s lifecycle from `create_before_destroy` to `destroy`), not when a single resource is missing. The default apply behavior is incremental, not a complete reset.

### How Terraform actually behaves

1. terraform plan produces a plan that adds a create action for the deleted instance.
2. terraform apply executes that plan, creating the missing instance without affecting any other resources.
3. Once the apply completes, the state file reflects the restored instance, and the workspace is back in sync with the configuration.

### References

Terraform CLI documentation – apply command behavior:

<https://developer.hashicorp.com/terraform/cli/commands/apply>

Terraform state management – resource lifecycle and drift detection:

<https://developer.hashicorp.com/terraform/language/configurations/resources#state>

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Prepared for HashiCorp Certified: Terraform Associate exam review.

## Question: 146

CertyIQ

You have a Terraform configuration that defines a single virtual machine with no references to it. You’ve run terraform apply to create the resource, then removed the resource definition from your Terraform configuration file.

What will happen when you run terraform apply in the working directory again?

- A. Terraform will error.
- B. Nothing will happen.
- C. Terraform will remove the virtual machine from the state file, but the resource will still exist.
- D. Terraform will destroy the virtual machine.

**Answer: D**

**Explanation:**

**Why the correct option is D**

When the VM resource block is removed from the configuration, Terraform’s next terraform apply compares the desired state (empty) with the current state (still contains the VM reference).

Since there is no configuration that declares the VM, the only way to converge is to **destroy** the existing resource, which requires invoking terraform destroy under the hood.

Terraform automatically schedules the destroy operation for any resource that has no matching configuration in the new plan, so the VM is removed from the provider and from the state file.

### Why the other options are not correct

#### A – Terraform will error.

Removing a resource definition is a normal operation; Terraform does not error unless there are syntactic or provider-specific validation failures, which are not implied here.

#### B – Nothing will happen.

The state still contains the VM reference. Terraform will notice a drift between the declared configuration (none) and the existing state, so an action (destroy) is required.

#### C – Terraform will remove the virtual machine from the state file, but the resource will still exist.

This can only occur when a resource is externally deleted (e.g., by a user on the provider console) or managed outside of Terraform, but the question describes a pure Terraform workflow where the only change is the removal of the block. In that case Terraform removes the VM from state **by destroying it**, not by silently updating state.

## References

### Resource lifecycle and terraform apply behavior:

<https://developer.hashicorp.com/terraform/language/resources#resource-lifecycle>

### State management and plan/apply reconciliation:

<https://developer.hashicorp.com/terraform/language/state>

## Question: 147

CertyIQ

If you manually destroy resources within your infrastructure, what is the best practice for reflecting this change in Terraform?

- A. Run terraform import to reflect the changes in Terraform state.
- B. Manually update the state file to remove the destroyed resources.
- C. Remove the resource definition from your file and run terraform apply -refresh-only.
- D. The change will happen automatically during the next terraform apply.

### Answer: D

### Explanation:

#### Why option D is correct

When infrastructure resources are manually destroyed outside of Terraform, the desired state of the configuration no longer matches the actual state of the environment. Running terraform apply forces Terraform to re-evaluate the configuration, detect that the resources are absent, and automatically plan the necessary reconciliations. This approach lets Terraform maintain a single source of truth without manual intervention, ensuring that the state file, the configuration, and the live infrastructure stay aligned.

#### Why the other options are less suitable

**A – Import** – Import is used to bring existing resources under Terraform management; it does not address resources that were intentionally destroyed. Importing destroyed resources would only add noise to the state.

**B – Manual state edit** – Directly editing the state file bypasses Terraform's validation and can corrupt the state, leading to inconsistencies and potential data loss. Terraform expects to manage state automatically.

**C – Remove definition + apply -refresh-only** – Removing the resource from the code and then only refreshing

the state does not delete the resource from the state file. The resource remains recorded as existing, so future plans will still try to reconcile it, preventing Terraform from truly reflecting the destroyed state.

#### Best practice flow

1. Ensure the resource definition is removed from the configuration files.
2. Execute terraform apply. Terraform will plan to destroy any remaining resources that are no longer declared and will update the state accordingly.

#### References

Terraform CLI documentation – apply: <https://developer.hashicorp.com/terraform/cli/commands/apply>  
Managing state – State and Lifecycle: <https://developer.hashicorp.com/terraform/language/state>

#### Question: 148

CertyIQ

When a check block's assertion fails, Terraform blocks the current operation from executing.

- A. True
- B. False

#### Answer: B

#### Explanation:

#### Justification

A failing assertion in a **check** block marks the policy evaluation as **failed**, but Terraform does **not** stop the underlying plan/apply execution for all subsequent resources – it only prevents the offending check from succeeding and may abort the overall policy gate.

The CLI continues processing remaining checks and actions; only the current **check** is considered violated, leading to a failed run status rather than a blanket block of the whole operation.

Consequently, the claim that “Terraform blocks the current operation from executing” is inaccurate; the operation may still proceed (or be re-attempted) after handling the failed check.

#### References

Terraform Cloud Policy – Check Blocks: <https://developer.hashicorp.com/terraform/cloud-docs/policy/check-blocks>  
Sentinel Syntax – Check Clause: <https://docs.hashicorp.com/sentinel/reference/language#check>

#### Question: 149

CertyIQ

You just scaled your VM infrastructure and realize you set the count variable to the wrong value. You correct the value and save your change.

What must you do next to make your infrastructure match your configuration?

- A. Inspect your Terraform state because you want to change it.
- B. Run terraform apply and confirm the planned changes.
- C. Inspect all Terraform outputs to make sure they are correct.
- D. Reinitialize because your configuration has changed.

**Answer: B**

**Explanation:**

**Justification**

**Correct action – B.Run terraform apply and confirm the planned changes.**

After correcting the count value in the configuration and persisting the change, Terraform must be told to reconcile the real infrastructure with the updated desired state. terraform apply generates a plan that reflects the new resource count, shows the actions required to achieve it, and then prompts for confirmation before provisioning or deprovisioning resources. This directly brings the live VMs into alignment with the corrected configuration.

**Why the other options are not appropriate**

**A. Inspect your Terraform state because you want to change it.** – The state file records the current resources, but merely inspecting it does not modify any resources. Changing the count requires an actual apply operation; otherwise the state will diverge from reality.

**C. Inspect all Terraform outputs to make sure they are correct.** – Outputs are for exposing data; verifying them provides no assurance that the underlying infrastructure matches the configuration. Validation must focus on resource actions, not on post-deployment outputs.

**D. Reinitialize because your configuration has changed.** – Re-initialization downloads providers and may be useful when provider plugins or backend settings change, but it does not enforce state alignment after a configuration change. The primary step to synchronize resources is an apply, not reinitialization.

**References**

Terraform Documentation – Configuration Changes:

<https://developer.hashicorp.com/terraform/language/configuration>

Terraform Documentation – Apply: <https://developer.hashicorp.com/terraform/cli/apply>

These links explain how Terraform handles state management and the apply workflow, reinforcing why terraform apply is the required next step.

## Question: 150

CertyIQ

What does terraform destroy do?

- A. Destroys all Terraform code files in the current directory, leaves the state file intact.
- B. Destroys all infrastructure in the Terraform state file.
- C. Destroys the Terraform state file, leaves the infrastructure intact.
- D. Destroys all infrastructure in the configured Terraform provider.

**Answer: B**

**Explanation:**

**Answer:B –Terraform destroy removes all infrastructure resources that are recorded in the Terraform state file.**

terraform destroy reads the current state, issues a Delete operation on each managed resource through the configured provider, and then updates the state to remove those entries.

The command does **not** modify any Terraform configuration files, nor does it delete the state file itself.

After the destroy run, the resources no longer exist in the target environment, but the state file may still contain a record of what was destroyed until it is removed or updated.

#### Why the other options are incorrect

**A** – The command never deletes source code or configuration files; it only interacts with the provider and updates state.

**C** – Destroy does not destroy the state file; it uses the state to determine what to delete and then updates the state accordingly.

**D** – The wording is imprecise; terraform destroy does not target “all infrastructure in the provider” as a whole, but rather the specific resources that the current configuration manages, which is exactly what B describes.

#### References

Terraform CLI documentation: “terraform destroy” –

<https://developer.hashicorp.com/terraform/cli/commands/destroy>

Official best practices guide: “Destroying infrastructure with Terraform” –

<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/destroy-aws-resources>

#### Question: 151

CertyIQ

You have a simple Terraform configuration containing one VM (virtual machine) in a cloud provider. You run terraform apply and the VM is created successfully.

You delete the VM using the cloud provider console, then run terraform apply again without changing any Terraform code.

What happens next?

- A. Terraform will report an error.
- B. Terraform will recreate the VM.
- C. Terraform will remove the VM from the state file.
- D. Terraform will not make any changes.

#### Answer: B

#### Explanation:

#### Technical Justification (Certification-Style Response)

When Terraform first creates a virtual machine (VM) it records the resource in its state file and the provider's API. After the VM is removed manually from the cloud console, the external system no longer reports a resource with that identifier. On the next execution of **terraform apply** without any code changes, Terraform performs three logical steps:

1. **Refresh** – it queries the provider to compare the current state of the infrastructure with the actual resources that exist. Because the provider now reports that the named VM no longer exists, the refresh marks the resource as **missing**.
2. **Plan** – Terraform calculates a diff between the desired state (as expressed in the .tf files) and the refreshed state. The diff shows create for the VM, while all other configured resources remain unchanged.
3. **Apply** – Terraform executes the actions indicated by the plan. The missing VM is therefore recreated,

and the state file is updated to reflect the newly provisioned resource.

Thus, the correct answer is **B – Terraform will recreate the VM**. This behavior is intentional: Terraform treats any divergence between declared configuration and actual infrastructure as a drift that must be reconciled, regardless of whether the drift was caused by a manual deletion, an external automation, or a provider-side change.

#### Why the other options are not appropriate

##### A. Terraform will report an error.

Errors are only raised when the configuration is syntactically invalid, references non-existent data sources, or encounters provider-level failures. A simple external deletion does not trigger an error because the configuration itself is still valid; Terraform merely detects drift.

##### C. Terraform will remove the VM from the state file.

terraform apply never deletes entries from the state file arbitrarily. It only updates the state to match the provider's current view. Removing a resource from state would leave the configuration in an inconsistent "orphaned" state, which is not the default behavior.

##### D. Terraform will not make any changes.

This would be true only if the real infrastructure already matched the desired state. Since the VM is absent from the provider but still tracked in the state file, there is a mismatch, prompting Terraform to act.

In summary, Terraform's declarative engine continuously works to achieve **convergence**: the actual infrastructure must reflect the code-defined state. External deletions introduce drift, which Terraform resolves by recreating the missing resource on the next apply.

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## References

**terraform apply documentation** – <https://developer.hashicorp.com/terraform/cli/apply>

**State management overview** – <https://developer.hashicorp.com/terraform/state#state-files>

(These links are official, publicly accessible HashiCorp documentation that detail the apply workflow and the behavior of Terraform's state file.)

## Question: 152

CertyIQ

You have never used Terraform before and would like to test it out using a shared team account for a cloud provider. The shared team account already contains 15 VMs (virtual machines). You develop a new Terraform configuration containing one VM, perform terraform apply, and see that your VM was created successfully. How do you delete the newly-created VM with Terraform?

- A. Delete the Terraform state file and execute terraform apply.
- B. The Terraform state file only contains one new VM. Execute terraform destroy.
- C. The Terraform state file contains all 16 VMs in the team account. Execute terraform destroy and select the newly-created VM.
- D. Delete the VM using the cloud provider console, and execute terraform apply to apply the changes to the Terraform state file.

**Answer: B**

**Explanation:**

**Correct option: B.**The Terraform state file only contains one new VM. Executeterraform destroy.

#### Why option B is correct

**State tracking:** After terraform apply the provider-specific state file records the single resource that was newly created. Destroy reads this state and issues the appropriate delete calls for the resources listed there.

**Idempotent cleanup:** terraform destroy removes exactly the resources defined in the current configuration from the infrastructure and updates the state, leaving the rest of the team's 15 VMs untouched.

**Single source of truth:** Because the state contains only the resource you intend to delete, destroying it safely eliminates the new VM without affecting existing assets.

#### Why the other options are unsuitable

**A. Delete the state file and run terraform apply** – Removing the state discards all metadata about the resources and forces Terraform to think none exist; a subsequent apply would recreate the VM rather than delete it.

**C. Assume the state holds all 16 VMs and destroy selectively** – The state reflects only the resources declared in your configuration, not the entire account. Destroy will remove everything defined in the config, which in this case is just one VM; attempting to “select” a specific VM is not how Terraform operates.

**D. Delete via the cloud console and run terraform apply** – Manual deletion breaks the intended state management model. Terraform will then reconcile the drift on the next apply, potentially recreating the VM or causing unnecessary state churn.

#### Conclusion

Use terraform destroy because it cleanly removes the resource recorded in the state, keeping the rest of the team's infrastructure intact and preserving the declarative, reproducible workflow that Terraform enforces.

#### References

Terraform Language Documentation – State Management:

<https://developer.hashicorp.com/terraform/language/state>

Terraform CLI – Destroy Command: <https://developer.hashicorp.com/terraform/cli/commands/destroy>

#### Question: 153

CertyIQ

Which parameters does the import block require? (Choose two.)

- A. backend
- B. The target resource address
- C. provider
- D. The resource ID

**Answer: BD**

**Explanation:**

**Technical Justification for Correct Answer: BD**

The import block in HashiCorp's Terraform requires specific parameters to successfully import existing infrastructure into Terraform's state. Here's why **B. The target resource address** and **D. The resource ID** are the correct choices, and why the other options are less suitable:

### Correct Choices:

**B. The target resource address:** This parameter is crucial as it specifies the Terraform resource address that will manage the imported infrastructure. It tells Terraform where to place the imported resource within its configuration hierarchy (e.g., `aws_instance.my_server`).

**D. The resource ID:** This identifies the specific external resource (already existing in the cloud(provider) environment) that Terraform will import. The ID is provider-dependent (e.g., an AWS Instance ID, a Google Cloud Resource Name, etc.) and uniquely identifies the resource to be managed.

### Less Suitable Options:

**A. backend:** While the backend configuration is vital for Terraform (defining where and how state is stored), it is not a parameter required by the import block itself. Backend settings are configured at a higher level of the Terraform project, not directly within an import block.

**C. provider:** Similar to the backend, the provider configuration is essential for a Terraform project (as it defines the cloud-provider/backend to interact with). However, when importing a resource, the provider is typically already configured at the project or module level. The import block focuses on the specifics of the resource to import, not re-specifying the provider.

### Why BD is the Best Choice:

**Precision:** BD provides the exact necessities for an import operation - where in the Terraform config to place it (B) and what external resource to import (D).

**Reusability and project Structure:** By not requiring the provider or backend within the import block, Terraform maintains project structure flexibility and reusability of configurations.

### References:

[HashiCorp Terraform Documentation - Importing Resources](#)  
[Terraform Configuration - Resource Addresses](#)

## Question: 154

CertyIQ

You have set the `TF_LOG_PATH` environment variable for Terraform, and you would like to ensure the logs contain all debug-level messages and verbose process logs. Which action should you take?

- A. Set the `TF_LOG` environment variable to `TRACE`.
- B. Update the Terraform CLI configuration file.
- C. Add a log argument to the terraform block.
- D. Run the terraform output command.

**Answer: A**

**Explanation:**

**Technical Justification for Correct Answer: A**

To ensure Terraform logs contain all debug-level messages and verbose process logs, setting the log level to the most verbose setting is essential. Here's why option A is the best choice and why others are less suitable:

**A. Set the `TF_LOG` environment variable to `TRACE` (Correct Answer)**

**Justification:** Setting `TF_LOG` to `TRACE` (Trace) is the most appropriate action because the Trace level includes

all debug-level messages and the most detailed, verbose process logs. This setting provides the highest level of logging verbosity, which aligns with the requirement.

**Technicality:** Terraform's logging levels, in order of increasing verbosity, are ERR, WARN, INFO, DEBUG, and TRAC. TRAC captures everything, including detailed traces of the execution flow, making it ideal for comprehensive debugging.

#### **B. Update the Terraform CLI configuration file**

**Less Suitable:** While the Terraform CLI configuration file can set various defaults, specifically for logging, setting TF\_LOG directly via an environment variable is more immediate and targeted for controlling log verbosity on a per-session or per-run basis.

#### **C. Add a log argument to the terraform block**

**Less Suitable:** The terraform block in configuration files (main.tf, etc.) does not support a direct log argument for setting the log level to this granularity (TRACE). Log settings at this level are typically managed via environment variables or the CLI configuration file, not within Terraform configuration blocks.

#### **D. Run the terraform output command**

**Less Suitable:** The terraform output command is used to display the values of output variables after applying a configuration. It does not influence the logging level or verbosity of Terraform's operational logs.

**Conclusion:** Setting TF\_LOG to TRAC via an environment variable is the most direct, effective, and suitable method to ensure Terraform logs include all debug-level messages and verbose process logs, making option A the correct choice.

#### **References**

1. HashiCorp Terraform Logging Documentation: <https://www.terraform.io/docs/cli/concepts/logging.html>
2. Terraform Environment Variables Reference: <https://www.terraform.io/docs/cli/config/environment-variables.html>

#### **Question: 155**

**CertyIQ**

Your team uses HCP Terraform to manage your infrastructure. You need to make a change to an infrastructure stack running in a public cloud.

Which pattern follows Infrastructure as Code best practices for making a change?

- A. Make the change via the public cloud API endpoint.
- B. Use the public cloud console to make the change.
- C. Run the public cloud CLI tool to make the change.
- D. Submit a pull request and wait for an approved merge of the change.
- E. Clone the repository containing your infrastructure code and then run the code.

**Answer: D**

**Explanation:**

**Justification**

**Option D – Submit a pull request and wait for an approved merge**

Aligns with Git-based **Infrastructure as Code (IaC)** workflows: changes are version-controlled, reviewed, and tested before production.

Enforces **peer review**, **policy compliance**, and **automated checks** (e.g., Terraform lint, plan validation, Sentinel

policies), which are core best-practice pillars for safe, auditable infrastructure changes. Guarantees traceability and rollback capability through the merge-request lifecycle.

#### **Option E – Clone the repo and run the code locally**

While it uses source control, it lacks the formal review gate and automated verification steps, increasing the risk of unchecked changes reaching production.

#### **Option A – Public-cloud API endpoint**

Bypasses version control and peer review; changes are ad-hoc and not reproducible, violating IaC principles of declarative, stored-state configurations.

#### **Option B – Public-cloud console**

Similar to the API approach; manual, UI-driven changes are not versioned and are difficult to audit or roll back.

#### **Option C – Public-cloud CLI tool**

Still manual and external to the IaC repository; lacking review, testing, and auditability.

Therefore, **D** is the only option that embodies a full Git-centric, peer-reviewed workflow, making it the best practice for IaC changes in HCP Terraform-managed environments.

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## **References**

### **HashiCorp Learn – “Using Pull Requests with Terraform”**

<https://learn.hashicorp.com/tutorials/terraform/terraform-pull-requests>

### **HCP Terraform Documentation – “Version Control and Pull Requests”**

<https://developer.hashicorp.com/terraform/cloud-docs/vcs/pull-requests>

## **Question: 156**

**CertyIQ**

Which of the following does HCP Terraform perform during a health assessment for a workspace? (Choose two.)

- A. Sentinel policy checks
- B. Resource drift detection
- C. Estimate infrastructure cost
- D. Check block validation
- E. Terraform test execution

**Answer: BC**

**Explanation:**

**Why options B and C are correct**

**Resource drift detection (B)** – HCP Terraform continuously compares the saved workspace state with the current provider state. Any discrepancy is flagged as drift, allowing the platform to alert users when the real-world infrastructure no longer matches the configuration. This is a core component of the workspace health score.

**Infrastructure cost estimate (C)** – The health assessment also calculates an approximate monthly cost for the provisioned resources, using pricing data from the selected provider. This estimate is displayed alongside the health metrics so teams can gauge both operational and financial risk.

#### Why the other options are not part of the health assessment

**Sentinel policy checks (A)** – Policies are evaluated during run/apply cycles; they enforce compliance but are not a diagnostic element of the health score.

**Check block validation (D)** – Validation of configuration blocks occurs during the planning stage; it is a prerequisite for a successful run, not a health-assessment activity.

**Terraform test execution (E)** – Tests are executed via CI/CD pipelines or the terraform test command; they are separate from the automated health monitoring that HCPTerraform performs on a workspace.

#### References

**Health checks and drift detection:** <https://developer.hashicorp.com/terraform/cloud-docs/workspaces#health-checks>

**Cost estimation in Terraform Cloud:** <https://developer.hashicorp.com/terraform/cloud-docs/costs>

These sources describe how HCPTerraform evaluates drift and provides cost-estimation as part of its workspace health assessment.

#### Question: 157

CertyIQ

You manage two workspaces in your HCP Terraform organization. The first workspace manages your network configuration. The second workspace manages your compute resources, and retrieves values from the networking workspace.

What HCP Terraform feature lets you run an apply operation on the compute workspace every time you update the networking workspace?

- A. Projects
- B. Policy
- C. Run tasks
- D. Run triggers

**Answer: D**

**Explanation:**

**Answer: D – Run triggers**

**Why it fits:** HCPTerraformrun triggers let you automatically start a run (apply) of a downstream workspace when a specific event occurs in another workspace. By defining a trigger on the compute workspace edge-case that watches the networking workspace's state or VCS changes, Terraform orchestrates an apply of the compute workspace each time the networking changes are approved.

**Why the other options don't work:**

**A– Projects:** Projects are a logical grouping of workspaces; they do not provide workflow automation for run execution.

**B– Policy:** Terraform policies enforce compliance and can block runs, but they do not schedule or initiate runs based on workspace updates.

**C– Run tasks:** Run tasks are manual actions you invoke from the UI or API; they are not event-driven and won't

automatically fire on another workspace's changes.

Therefore, a **run trigger** is the precise HCP Terraform feature that automatically runs the compute workspace whenever the networking workspace is updated, which matches the described requirement.

## References

**Run triggers (documentation):** <https://developer.hashicorp.com/terraform/cloud-docs/workspaces/triggers>

**Workspace dependencies and automation:** <https://developer.hashicorp.com/terraform/cloud-docs/workspaces/run-triggers#using-run-triggers-to-chain-workspaces>

### Question: 158

CertyIQ

You can configure multiple cloud blocks in your Terraform configuration to connect your workspace to both HCP Terraform and your Terraform Enterprise instance.

- A. True
- B. False

**Answer: B**

#### Explanation:

Terraform can be configured to use **only one remote backend** (i.e., a single cloud block) per workspace.

The cloud block specifies the provider URL, organization, and optional parameters, and it **applies globally**; you cannot simultaneously target two distinct backends such as HCP Terraform and Terraform Enterprise in the same configuration.

Attempting to declare multiple cloud blocks results in a validation error ("multiple cloud blocks are not supported"), preventing the configuration from being applied.

Consequently, you can switch between a Terraform Cloud/HCP workspace and a Terraform Enterprise backend by editing the same cloud block, but you cannot have both configurations active at the same time.

Therefore, the statement "You can configure multiple cloud blocks in your Terraform configuration to connect your workspace to both HCP Terraform and your Terraform Enterprise instance." is **false**.

#### References

[Terraform Cloud Configuration – Remote Backend (cloud block)]

(<https://developer.hashicorp.com/terraform/tutorials/cloud> Terraform cloud)

[Terraform Enterprise Remote Execution – Backend Configuration](#)

### Question: 159

CertyIQ

Which HCP Terraform feature is recommended when you want to use the same cloud provider credentials across multiple workspaces?

- A. Variable sets
- B. Version Control System (VCS) integration
- C. The terraform\_remote\_state data source

**Answer: A**

**Explanation:**

#### Technical justification for selecting Variable sets

**Centralized credential management** – Variable sets store provider-level secrets (e.g., access\_key, secret\_key) in a secure, workspace-agnostic location that can be referenced from multiple workspaces, eliminating the need to duplicate credentials in each workspace's variable definitions.

**Reusability across environments** – Because the same set of variables is attached to many workspaces, any change to authentication material is made once and automatically propagates to all workspaces that consume the set, ensuring consistency.

**Security posture** – Sensitive values can be marked as **sensitive**, preventing them from being exposed in plan output or logs, and they are persisted encrypted in the Terraform Cloud backend.

**Minimal impact on IaC design** – The approach retains the declarative nature of Terraform: credentials are supplied as inputs, not as hard-coded values in configuration files, aligning with best-practice module encapsulation.

#### Why the other options are less suitable

**VCS integration (OptionB)** – This feature connects Terraform Cloud to a source-control repository to trigger runs; it does not address credential sharing across workspaces.

**terraform\_remote\_state data source (OptionC)** – It reads previously stored state from remote backends, which can expose state data but does not provide a mechanism for distributing provider credentials securely.

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#### References

**Terraform Cloud Variable Sets Documentation** – <https://developer.hashicorp.com/terraform/cloud-docs/variables/variable-sets>

**Provider Configuration Best Practices** – <https://developer.hashicorp.com/terraform/language/modules/develop/providers#defining-credentials>

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#### Question: 160

CertyIQ

You are responsible for a set of infrastructure which is managed by two workspaces: example-network and example-compute. The example-compute workspace uses data from output values configured in the example-network workspace, and must be deployed afterward. Currently, this is a manual process:

An operator deploys changes to the example-network workspace.

They manually copy the output values from the example-network workspace to input variables configured for the example-compute workspace.

They deploy the example-compute workspace.

Which HCP Terraform features can you use to automate this process? (Choose two.)

- A. A health check configured on the example-compute workspace to create a plan when HCP Terraform applies changes to the example-network workspace.
- B. A run trigger configured on the example-network workspace to automatically plan changes to the example-compute workspace after every apply.
- C. A health check configured on the example-network workspace to create a plan on the example-compute workspace when HCP Terraform applies changes to it.
- D. A run trigger configured on the example-compute workspace to automatically plan changes after HCP Terraform applies changes to the example-network workspace.
- E. A tfe\_outputs data source configured in the example-compute workspace to automatically load output values from the example-network workspace.

**Answer: DE**

**Explanation:**

**Correct options: D and E**

**D – Run trigger on the example-compute workspace**

The trigger fires **after** the example-network workspace finishes its apply. HCP Terraform detects the completed run and automatically initiates a new run (plan→apply) in the example-compute workspace. This guarantees that the latest output values are consumed, eliminating manual copy-and-paste.

**E – tf\_outputs data source in the example-compute workspace**

By using the tf\_outputs data source, the compute workspace can retrieve the exported values directly from the network workspace at run-time. No external secret-management or copy step is required; the values are fetched on each plan/apply cycle.

**Why the other options are unsuitable**

**A – Health check that runs a plan on example-compute when example-network is applied**

Health checks only monitor the health of a specific workspace or run; they do not create or trigger external runs. They cannot automatically launch a plan in a different workspace.

**B – Run trigger on example-network that plans example-compute**

A trigger that fires **from** the network workspace to automatically **plan** the compute workspace would only produce a plan, not the subsequent apply step. Deployments require the full run lifecycle (plan→apply), which a trigger on the source workspace does not guarantee.

**C – Health check on example-network that creates a plan on example-compute**

Similar to (A), health checks cannot initiate runs in other workspaces. They are limited to notifying or marking a workspace as unhealthy; they do not orchestrate cross-workspace execution.

**Implementation summary**

Configure a **run trigger** on the example-compute workspace that points to the example-network workspace's run type (apply). Pair this with the tf\_outputs data source inside the compute workspace to ingest the network's exported outputs. This eliminates manual intervention and ensures the compute deployment always reflects the most recent network state.

**References**

[Run Triggers \(HCP Terraform\)](#)  
[tf\\_outputs Data Source](#)

**Question: 161**

**CertyIQ**

Where does HashiCorp recommend you store API tokens and other secrets within your team's Terraform workspaces? (Choose three.)

- A. In a terraform.tfvars file, checked into your version control system.
- B. In an environment variable and referenced with TF\_VAR\_variablename.
- C. In HashiCorp Vault.
- D. In a plaintext document on a shared drive.
- E. In an HCP Terraform variable, with the sensitive option checked.

**Answer: BCE**

**Explanation:**

**Why these three options are recommended**

**B – Environment variables referenced as `TF_VAR_<name>`** – Terraform can ingest secrets from the runtime environment, keeping them out of the codebase and out of version control. When the variable is marked as sensitive, Terraform masks its value in logs and plan output, reducing the risk of accidental exposure. This approach is platform-agnostic and works locally, in CI pipelines, and in cloud runners.

**C – HashiCorp Vault** – Vault provides a centralized, encrypted store for secrets with fine-grained access policies, dynamic secrets, and lease/renewal handling. Storing tokens there allows Terraform to retrieve credentials at runtime via the Vault provider, ensuring that secrets never reside in plaintext on disk or in source control.

**E – HCP Terraform variable with the sensitive flag** – When managing workspaces in HashiCorp Cloud Platform (HCP) Terraform, variables marked as sensitive are stored encrypted at rest and are only exposed to the Terraform execution environment when needed. This leverages the managed secret handling of HCP while maintaining the same security guarantees as local environment variables.

**Why the other options are unsuitable**

**A – `terraform.tfvars` checked into version control** – Including secrets in a file that is committed to source control exposes them to anyone with repository access and creates a permanent audit trail. Even if the file is ignored, it is easy to mistakenly add it later, violating secret-management best practices.

**D – Plaintext document on a shared drive** – A shared drive typically lacks encryption at rest, granular access controls, and audit logging. Secrets stored there are visible to all users with drive permissions, increasing the attack surface and making compliance reporting difficult.

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## References

Terraform Configuration Language – Variables:

<https://developer.hashicorp.com/terraform/language/expressions#variables>

Terraform Vault Provider – Secrets Management:

<https://developer.hashicorp.com/terraform/language/providers/vault>

HCP Terraform – Sensitive Variables: <https://developer.hashicorp.com/terraform/cloud-services/variables#sensitive>

## Question: 162

**CertyIQ**

Which of these statements about HCP Terraform workspaces is false?

- A. You can trigger plans and applies using version control system integrations.
- B. HCP Terraform workspaces offer role-based access controls.
- C. You must use the CLI to switch between workspaces.
- D. HCP Terraform workspaces can securely store cloud credentials.

**Answer: C**

**Explanation:**

**Why OptionC is false**

Switching between workspaces can be done via the Terraform CLI or through the HCP Terraform UI, API, or VCS integrations.

The CLI is optional for workspace management; it is only required when you need to run terraform workspace select locally. Therefore the statement “You must use the CLI to switch between workspaces” is inaccurate.

**Why the other options are true**

**A.** HCP Terraform integrates natively with GitHub, GitLab, Bitbucket, and AzureDevOps, allowing plans and applies to be triggered automatically on pull-request events or via VCS webhook APIs.

**B.** Workspace-specific policies, role-based access controls (RBAC), and team permissions are built-in features of HCP Terraform, enabling fine-grained governance of who can execute or view particular workspaces.

**D.** Sensitive values (e.g., API keys, DB passwords) can be stored securely in the HCP Terraform **variables** block or via the **Sensitive** flag, and the platform encrypts them at rest, providing a safe way to manage cloud credentials without exposing them in code.

**Summary**

OptionC mischaracterizes the mechanism for workspace switching; all other statements correctly describe capabilities of HCP Terraform workspaces.

**References**

Workspaces Overview –<https://developer.hashicorp.com/terraform/cloud-docs/workspaces>

Storing Sensitive Data –<https://developer.hashicorp.com/terraform/cloud-docs/workspaces#securely-store-sensitive-data>

### Question: 163

**CertyIQ**

How does an HCP Terraform integration using the cloud block differ from remote backends, such as S3?

- A. Using the cloud block automatically encrypts state files at rest and in transit.
- B. Using the cloud block requires using Terraform Enterprise.
- C. The cloud block supports multiple backends simultaneously.
- D. Using the cloud block enables remote operations on infrastructure managed by HCP Terraform or Terraform Enterprise.

**Answer: D**

**Explanation:**

**Explanation**

The cloud block in HCP Terraform tells Terraform Cloud to store the state in HCP’s managed storage and to expose the full set of remote operations (run-tasks, policy checks, runs, etc.) that are only available when the backend is hosted by Terraform Cloud or Enterprise. This backend eliminates the need to manage S3-style credentials, remote lock files, or VPC endpoints; all lifecycle commands (plan, apply, destroy) are mediated through the HCP API, enabling team-wide collaboration, policy enforcement, and auditability.

**OptionA** – Incorrect. The cloud block does not automatically encrypt state files; encryption must be configured separately (e.g., via HCP-managed encryption at rest or external KMS settings).

**OptionB** – Incorrect. The cloud block works with the public Terraform Cloud SaaS service as well as with self-hosted Terraform Enterprise; it does not mandate Enterprise.

**OptionC** – Incorrect. The backend defined by a cloud block is a single, unified backend; you cannot configure multiple distinct backends within the same configuration.

**OptionD** – Correct. Using the cloud block moves all state-management operations to HCP's backend, giving you remote runs, policy evaluation, workspace management, and audit logs that are absent when using generic backends like S3.

## References

Terraform Cloud Backends – <https://developer.hashicorp.com/terraform/tutorials/cloud-provider/cloud-backends>

Manage State in Terraform Cloud – <https://developer.hashicorp.com/terraform/cloud-docs/process/state>

## Question: 164

CertyIQ

Which features do HCP Terraform workspaces provide that are not available in Terraform Community Edition? (Choose three.)

- A. Automatic detection of common security issues.
- B. Remote execution of Terraform operations.
- C. Support for multiple cloud providers.
- D. Store your configuration in a Version Control System (VCS).
- E. State versions and run history.
- F. Store Terraform and environment variables in variable sets.

**Answer: BEF**

**Explanation:**

**Justification**

### **B – Remote execution of Terraform operations**

HCP Terraform runs Terraform plans and applies on managed workers in the cloud, allowing users to trigger executions via the UI/API without needing a local machine or self-hosted runner. The Community Edition only supports local runs; it lacks the hosted, scalable execution engine.

### **E – State versions and run history**

HCP Terraform automatically version-controls the workspace state and retains a detailed run history (plans, successes, failures, console output). This enables auditability and rollback of state changes. Terraform Community Edition stores state only locally and provides no built-in versioning or run log.

### **F – Store Terraform and environment variables in variable sets**

HCP Terraform offers centralized variable sets that can be linked to workspaces, providing a secure, version-controlled store for both Terraform configuration variables and environment-specific secrets. The Community Edition requires manual handling of variables in local files or external secret stores; it does not provide a unified variable-set service.

**Why the other options are not exclusive to HCP Terraform**

**A – Automatic detection of common security issues** – This capability is not a defined feature of HCP

Terraform; security scanning is typically provided by external tools (e.g., Checkov, Terraform Cloud's policy as code) and is not a unique HCP Terraform service.

**C – Support for multiple cloud providers** – Both the Community Edition and HCP Terraform can manage resources across any provider supported by Terraform; provider support is not limited to HCP.

**D – Store your configuration in a Version Control System (VCS)** – Storing config in VCS is a development practice that works with both editions; it is not a feature of HCP Terraform itself.

## References

1. HashiCorp Terraform Cloud (HCP Terraform) Documentation – Workspaces & Remote Execution
2. <https://developer.hashicorp.com/terraform/cloud-docs/workspaces>
3. Terraform Cloud – State Versions & Run History
4. <https://developer.hashicorp.com/terraform/cloud-docs/state/state-versioning>
5. Terraform Variable Sets Overview
6. <https://developer.hashicorp.com/terraform/cloud-docs/variables/variable-sets>

## Question: 165

CertyIQ

What HCP Terraform feature helps organize similar workspaces together and let you configure them as a group?

- A. Variable sets
- B. Change requests
- C. Projects
- D. Policy sets

**Answer: C**

**Explanation:**

**Technical Justification for Correct Answer: C. Projects**

The correct answer, **C. Projects**, is the most suitable HCP Terraform feature for organizing similar workspaces together and configuring them as a group. Here's why:

**Projects** in HashiCorp Cloud Platform (HCP) Terraform allow for the logical grouping of related workspaces. This feature is specifically designed to manage collections of workspaces that share a common purpose or belong to the same project context, making it easier to oversee and configure them collectively.

**Key Benefit:** Projects enable streamlined management through a unified view, facilitating operations like simultaneous workspace updates, shared variable sets (via integration with Variable Sets), and unified access control, all of which align with the question's requirements.

**Why Other Options are Less Suitable:**

**A. Variable Sets:** While Variable Sets are crucial for managing configuration variables across workspaces, they do not provide a mechanism for organizing workspaces themselves. Their primary function is to centralize variable management, not workspace grouping.

**B. Change Requests:** Change Requests are a feature focused on governance and compliance, allowing for the review of proposed Terraform plan changes before deployment. This feature does not facilitate the organization or group configuration of workspaces.

**D. Policy Sets:** Policy Sets are used to enforce policy as code across infrastructures, ensuring compliance with organizational standards. Like Variable Sets, they do not serve the purpose of organizing workspaces into manageable groups.

## References

For in-depth information on these features, refer to the official HashiCorp documentation:

[HashiCorp Cloud Platform \(HCP\) Terraform - Projects](#)

[HashiCorp Terraform Documentation - Variable Sets, Change Requests, Policy Sets](#) (Navigate within for specific feature details)

### Question: 166

CertyIQ

You are an administrator of your HCP Terraform organization. Recently, a member of your organization accidentally applied a change to a workspace that manages infrastructure owned by a different team. What HCP Terraform features can you use to help prevent similar accidents in the future?

- A. Create separate teams in HCP Terraform for each team in your organization, then assign project and/or workspace permissions to each team.
- B. Configure your workspaces to use the VCS-driven workflow, then update permissions to each repository in your VCS.
- C. Enable two-factor authentication for your HCP organization, then limit access to only specific users within your organization.
- D. For each workspace in your organization, edit its settings, and then assign only those users that are authorized to plan and apply changes to the workspace.

**Answer: A**

**Explanation:**

**Technical Justification for Correct Answer: A**

The correct answer, **A**, is the most suitable choice for preventing similar accidents in the future due to its direct addressing of the access control and resource segregation issue described. Here's why:

**Direct Access Control and Segregation:** Creating separate teams in HCP Terraform for each team in your organization and then assigning project and/or workspace permissions to each team directly addresses the problem. This ensures that teams only have access to the workspaces managing infrastructure they own, preventing unauthorized changes.

**Why Other Options are Less Suitable:**

**B. VCS-driven workflow and Repository Permissions:**

While configuring workspaces to use the VCS-driven workflow can enhance version control and collaboration, updating repository permissions in the VCS does not directly prevent a user with access to a repository from applying changes to the wrong workspace in HCP Terraform if they have the necessary HCP Terraform permissions.

**Issue:** Indirect solution, relies on external (VCS) permission management which may not mirror HCP Terraform's access requirements.

**C. Two-Factor Authentication (2FA) and Limited User Access:**

Enabling 2FA enhances security by adding an extra layer of authentication but does not solve the issue of role-based access within the organization.

Limiting access to specific users might help but doesn't scale well with the requirement to manage infrastructure ownership across different teams.

**Issue:** Does not address the core problem of resource (workspace) access control.

#### D. Assigning Authorized Users per Workspace:

While this approach seems to directly address the issue by ensuring only authorized users can plan and apply changes to a workspace, it becomes impractical for large organizations with many workspaces and frequent team membership changes.

**Issue:** Scalability and manageability concerns; not as efficient as managing access through teams.

#### Correct Answer Justification in Bullet Points

##### Why A is Correct:

**Direct Solution:** Addresses the root cause (access to wrong workspace) directly.

**Scalability:** Easily scalable for large organizations with multiple teams.

**Efficient Management:** Simplifies permission management through team-based assignments.

**Clear Segregation:** Ensures clear ownership and access control over infrastructure.

##### Key Benefit Over Other Options:

Combines scalability with direct addressal of the problem, unlike the other options which either introduce external dependencies (B), fail to directly solve the access issue (C), or are less scalable (D).

#### References

For further reading on managing teams and permissions in HCP Terraform:

1. **HashiCorp Documentation - HCP Terraform Teams and Permissions**<https://www.hashicorp.com/products/terraform/hcp/features#teams>
2. **HashiCorp Learn - Managing Access to HCP Terraform**<https://learn.hashicorp.com/collections/terraform/hcp-access-control>

#### Question: 167

CertyIQ

Your risk management organization requires that new AWS S3 buckets must be private, and their contents must be encrypted at rest.

How can HCP Terraform automatically and proactively enforce this security control?

- A. With an S3 module with proper settings for buckets.
- B. By auditing cloud storage buckets with a vulnerability scanning tool.
- C. By adding variables to each HCP Terraform workspace to ensure these settings are always enabled.
- D. With a Sentinel policy, which runs before every apply.

**Answer: D**

**Explanation:**

#### Technical Justification for Correct Answer (D)

**Why D is the best option:** Enforcing security controls at the infrastructure-as-code (IaC) level before deployment is the most proactive and automated approach. A Sentinel policy, by design, is a policy-as-code tool that integrates with HashiCorp products, including Terraform. When configured to run before every terraform apply, a Sentinel policy can automatically enforce the requirement that new AWS S3 buckets are private and their contents encrypted at rest. This prevents non-compliant infrastructure from being deployed in the first place, ensuring continuous compliance.

## Key Technical Points:

**Proactivity:** Acts before infrastructure deployment.

**Automation:** No manual intervention required.

**Central Enforcement:** Easily managed and updated centrally.

**Preventive Measure:** Blocks non-compliant configurations outright.

## Why Other Options are Less Suitable:

### A. S3 Module with Proper Settings:

**Reactive Dependence:** Relies on developers to correctly use the module every time.

**Not Centrally Enforceable:** Updates require module version changes and adoption across all users. Example of limitation: If a developer forgets to set the `acl` to "private" or misses the `server_side_encryption_configuration` block, the bucket may not meet security requirements.

### B. Auditing with a Vulnerability Scanning Tool:

**Reactive:** Identifies issues after deployment, potentially leaving a window of vulnerability.

**Additional Operational Overhead:** Requires separate monitoring and remediation processes. Example of limitation: A publicly accessible S3 bucket might expose sensitive data before the scanning tool detects and alerts on the issue.

### C. Adding Variables to Each HCP Terraform Workspace:

**Error-Prone:** Variables can be overridden or incorrectly set in individual workspaces.

**Not Enforceable:** Does not prevent deployment if variables are not set correctly. Example of limitation: A workspace might accidentally set `enable_versioning = false` or omit `server_side_encryption_by_default`, leading to non-compliant buckets.

## References

[HashiCorp Sentinel Documentation](#) - For detailed information on how Sentinel policies can enforce compliance at the point of infrastructure deployment.

[AWS S3 Bucket Encryption and Access Control with Terraform](#) - Examples of configuring S3 bucket privacy and encryption in Terraform, complementing Sentinel's policy enforcement.

## Question: 168

CertyIQ

Which statements are true?

Pick the 2 correct responses below

- A. A precondition in a resource block must pass for Terraform to create the resource.
- B. A precondition cannot reference attributes of another resource that are only known after apply
- C. When a resource has a count meta-argument, Terraform evaluates the precondition for each instance.
- D. If a resource's postcondition block fails, Terraform still creates the resources that depend on it.
- E. A resource can either have a precondition or a postcondition block, but not both.

**Answer: A,C**

**Explanation:**

**A. A precondition in a resource block must pass for Terraform to create the resource.**

Correct. Terraform evaluates the precondition before creating the resource. If the precondition fails, the resource is not created.

**B. A precondition cannot reference attributes of another resource that are only known after apply.**

Incorrect. Preconditions can reference such attributes, but if the value is unknown during planning, Terraform defers evaluation until apply.

**C. When a resource has a count meta-argument, Terraform evaluates the precondition for each instance.**

Correct. If a resource uses count, Terraform evaluates the precondition separately for each instance.

**D. If a resource's postcondition block fails, Terraform still creates the resources that depend on it.**

Incorrect. If a postcondition fails, Terraform marks the resource as failed, and dependent resources are not created.

**E. A resource can either have a precondition or a postcondition block, but not both.**

Incorrect. A resource can have both precondition and postcondition blocks defined.

**Question: 169**

CertyIQ

Your team often uses API calls to create and manage cloud Infrastructure.

In what ways does Terraform differ from conventional Infrastructure management approaches?

- A. Terraform replaces cloud provider APIs with its own protocols, enabling automated deployments.
- B. Terraform describes infrastructure with version-controlled, repeatable configurations that specify the desired state.
- C. Terraform enforces infrastructure through imperative scripts to ensure tasks are completed in the proper order.
- D. Terraform is merely a wrapper for cloud provider APIs, so there is little to no difference in calling the API directly.

**Answer: B**

**Explanation:**

**A. Terraform replaces cloud provider APIs with its own protocols, enabling automated deployments.**

Incorrect. Terraform does not replace cloud provider APIs. It uses provider plugins that call the underlying cloud APIs.

**B. Terraform describes infrastructure with version-controlled, repeatable configurations that specify the desired state.**

Correct. Terraform uses declarative configuration files to define the desired state of infrastructure, enabling version control, consistency, and repeatability.

**C. Terraform enforces infrastructure through imperative scripts to ensure tasks are completed in the proper order.**

Incorrect. Terraform is declarative, not imperative. It determines the execution order automatically based on resource dependencies.

**D. Terraform is merely a wrapper for cloud provider APIs, so there is little to no difference in calling the API**

directly.

Incorrect. Terraform provides state management, dependency graphing, planning, and lifecycle management, which go far beyond simply calling APIs directly.

### Question: 170

CertyIQ

What's the proper syntax for the plan command?

- A. terraform plan generate-config-out=tfplan
- B. terraform plan out=tfplan
- C. terraform plan target=tfplan
- D. terraform apply-var-file=tfplan

**Answer: B**

#### Explanation:

The terraform plan command is used to create an execution plan, showing what actions Terraform will take to achieve the desired state. To save this plan to a file for later use with terraform apply, the -out flag is employed. This ensures that the apply command executes precisely the plan that was reviewed.

### Question: 171

CertyIQ

Which of the following is not a way to trigger terraform destroy ?

- A. terraform destroy
- B. All of these will trigger terraform destroy
- C. terraform plan destroy
- D. terraform destroy-auto-approve

**Answer: C**

#### Explanation:

##### **A. terraform destroy**

Incorrect. This command directly triggers destruction of managed infrastructure.

##### **B. All of these will trigger terraform destroy**

Incorrect. Not all listed commands correctly trigger destroy.

##### **C. terraform plan destroy**

Correct. This is not valid syntax. The correct flag would be terraform plan -destroy.

##### **D. terraform destroy-auto-approve**

Incorrect. While written incorrectly, the valid command terraform destroy -auto-approve does trigger destroy.

### Question: 172

You are making changes to existing Terraform code to add some new infrastructure. When is the best time to run terraform validate?

- A. After you run terraform apply, so you can validate your Infrastructure.
- B. After you run terraform plan, so you can validate that your state file is consistent with your infrastructure.
- C. Before you run terraform apply, so you can validate your provider credentials.
- D. Before you run terraform plan, so you can validate your code syntax.

**Answer: D**

**Explanation:**

**A. After you run terraform apply, so you can validate your Infrastructure.**

Incorrect. terraform validate checks configuration syntax and internal consistency — it does not validate deployed infrastructure after apply.

**B. After you run terraform plan, so you can validate that your state file is consistent with your infrastructure.**

Incorrect. terraform validate does not check state file consistency. That is handled during plan and apply operations.

**C. Before you run terraform apply, so you can validate your provider credentials.**

Incorrect. terraform validate does not verify provider credentials or connectivity.

**D. Before you run terraform plan, so you can validate your code syntax.**

Correct. terraform validate checks configuration syntax and internal logic before running plan or apply, making it best to run prior to terraform plan.

### Question: 173

You want to retrieve the ID of an existing cloud resource for use in your Terraform configuration. Which block type would you use?

- A. data
- B. resource
- C. output
- D. import

**Answer: A**

**Explanation:**

**A. data**

Correct. A data block is used to fetch information about existing infrastructure that is not managed by the current Terraform configuration. It allows you to reference attributes like IDs for use in other resources.

**B. resource**

Incorrect. A resource block is used to create and manage new infrastructure, not retrieve existing unmanaged resources.

**C. output**

Incorrect. An output block is used to display values after Terraform runs, not to retrieve existing resources.

**D. import**

Incorrect. import is a CLI command used to bring existing infrastructure under Terraform management, not a configuration block for retrieving resource data.

**Question: 174**

**CertyIQ**

A senior admin accidentally deleted some of your cloud instances. What will Terraform do when you run terraform apply?

- A. Build a completely brand new set of infrastructure.
- B. Rebuild only the instances that were deleted.
- C. Tear down the entire workspace's infrastructure and rebuild it.
- D. Stop and generate an error message about the missing instances.

**Answer: B**

**Explanation:**

**A. Build a completely brand new set of infrastructure.**

Incorrect. Terraform does not recreate everything unless the configuration has changed to require it.

**B. Rebuild only the instances that were deleted.**

Correct. Terraform compares the current state with the real infrastructure. If resources defined in the configuration are missing, Terraform will plan to recreate only those missing resources.

**C. Tear down the entire workspace's infrastructure and rebuild it.**

Incorrect. Terraform does not destroy unaffected resources unless explicitly instructed.

**D. Stop and generate an error message about the missing instances.**

Incorrect. Instead of stopping, Terraform detects drift and plans to restore the missing resources.

**Question: 175**

**CertyIQ**

What is the purpose of the .terraform directory in a Terraform workspace?

- A. The directory contains plugins and modules that Terraform downloads during initialization, along with other important information.
- B. The directory is where Terraform creates and maintains the state file to track the underlying resources it creates and manages.
- C. The directory contains the provider credentials and the .tfvars files to prevent them from being committed to version control by accident

D. The directory is used to convert and store Terraform configuration files into API calls to communicate with the targeted platform.

**Answer: A**

**Explanation:**

**A. The directory contains plugins and modules that Terraform downloads during initialization, along with other important information.**

Correct. The .terraform directory is created during terraform init and stores downloaded provider plugins, modules, and other metadata needed for Terraform to function.

**B. The directory is where Terraform creates and maintains the state file to track the underlying resources it creates and manages.**

Incorrect. The state file (terraform.tfstate) is stored in the working directory by default, not inside the .terraform directory (unless using specific backend configurations).

**C. The directory contains the provider credentials and the .tfvars files to prevent them from being committed to version control by accident.**

Incorrect. Credentials and .tfvars files are managed separately and are not automatically stored in the .terraform directory.

**D. The directory is used to convert and store Terraform configuration files into API calls to communicate with the targeted platform.**

Incorrect. Terraform does not convert configuration files into stored API calls in this directory. It dynamically interacts with provider APIs during plan and apply.

### Question: 176

CertyIQ

Your Terraform configuration manages a resource that requires maximum uptime. You need to update the resource, and when you run terraform plan, Terraform notes that the update requires the resource to be destroyed and recreated.

Which lifecycle rule can you add to the resource to reduce the downtime of the resource while still applying the update?

- A. ignore\_changes = all
- B. create\_before\_destroy = true
- C. destroy = false
- D. prevent\_destroy = true

**Answer: B**

**Explanation:**

**A. ignore\_changes = all**

Incorrect. This tells Terraform to ignore changes to attributes, which would prevent the update from being applied at all — not reduce downtime.

**B. create\_before\_destroy = true**

Correct. This lifecycle rule instructs Terraform to create the new resource first and then destroy the old one. This helps minimize downtime during updates that require replacement.

**C. destroy = false**

Incorrect. There is no valid destroy lifecycle argument in Terraform.

**D. prevent\_destroy = true**

Incorrect. This prevents Terraform from destroying the resource entirely, which would cause the apply operation to fail instead of updating it.

**Question: 177**

**CertyIQ**

Which of the following locations can Terraform use as a private source for modules?  
Pick the 2 correct responses below

- A. Private repository on GitHub.
- B. Public repository on GitHub.
- C. Internally hosted VCS (Version Control System) platform.
- D. Public Terraform Registry.

**Answer: A,C**

**Explanation:**

**A. Private repository on GitHub.**

Correct. Terraform can use a private GitHub repository as a module source, provided proper authentication is configured.

**B. Public repository on GitHub.**

Incorrect. While valid as a module source, it is not considered a private source.

**C. Internally hosted VCS (Version Control System) platform.**

Correct. Terraform can pull modules from internally hosted VCS platforms such as private Git servers or enterprise Git solutions.

**D. Public Terraform Registry.**

Incorrect. The public Terraform Registry is not a private module source.

**Question: 178**

**CertyIQ**

You want to bring in an existing database under Terraform management. What information is required to create a new import block for the database?  
Pick the 2 correct responses below

- A. The connection string that Terraform will use to connect and manage the database.
- B. The ID associated with the current database on the cloud provider.
- C. The path to the .tf file that contains the database resource block.

- D. The database platform and version that is the existing resource is running.
- E. The destination resource address of the block that will manage the database.

**Answer: B,E**

**Explanation:**

**A. The connection string that Terraform will use to connect and manage the database.**

Incorrect. Terraform does not require a connection string for importing a resource. Importing uses the provider and resource configuration already defined.

**B. The ID associated with the current database on the cloud provider.**

Correct. Terraform needs the unique ID of the existing resource in the cloud provider to map it into the state.

**C. The path to the .tf file that contains the database resource block.**

Incorrect. Terraform does not need the specific file path — it only needs the resource defined in the configuration.

**D. The database platform and version that the existing resource is running.**

Incorrect. Terraform does not require platform/version details for importing, though it may need the correct provider configuration.

**E. The destination resource address of the block that will manage the database.**

Correct. Terraform requires the resource address (e.g., `aws_db_instance.mydb`) to know which resource block will manage the imported resource.

### Question: 179

CertyIQ

You have a saved execution plan containing desired changes for infrastructure managed by Terraform. After running the command `terraform apply my.tfplan`, you receive the error shown in the Exhibit space on this page. How can you apply the desired changes?

Pick the 2 correct responses below:

- A. Refresh the current state data using the `-refresh-only` flag.
- B. Run `terraform apply` without the saved execution plan.
- C. Update the current plan file using the `terraform state push` command.
- D. Generate a new execution plan file with `terraform plan`, and apply the new plan.
- E. Force the apply command by adding the flag `-lock=false`

**Answer: B,D**

**Explanation:**

**A. Refresh the current state data using the `-refresh-only` flag.**

Incorrect. `-refresh-only` updates the state file with the current real-world infrastructure but does not apply a saved plan.

**B. Run `terraform apply` without the saved execution plan.**

Correct. Running terraform apply without specifying a plan will generate a new plan automatically based on the current state and apply it, avoiding errors from a stale plan.

**C. Update the current plan file using the terraform state push command.**

Incorrect. terraform state push is used to manually upload a state file; it cannot modify or update a saved execution plan.

**D. Generate a new execution plan file with terraform plan, and apply the new plan.**

Correct. Creating a fresh execution plan ensures that the plan reflects the current state of infrastructure and can be applied successfully.

**E. Force the apply command by adding the flag -lock=false**

Incorrect. -lock=false disables state locking but does not fix errors caused by an outdated or invalid execution plan.

**Question: 180**

**CertyIQ**

You've updated your Terraform configuration, and you need to preview the proposed changes to your infrastructure.

Which command should you run?

- A. terraform validate
- B. terraform get
- C. terraform plan
- D. terraform show

**Answer: C**

**Explanation:**

**A. terraform validate**

Incorrect. This checks the syntax and internal consistency of the Terraform configuration, but does not show proposed changes to infrastructure.

**B. terraform get**

Incorrect. This command downloads and updates modules referenced in your configuration, but does not preview changes.

**C. terraform plan**

Correct. terraform plan generates an execution plan showing what changes Terraform will make to match the configuration to the desired state.

**D. terraform show**

Incorrect. This displays the current state or a saved plan, but does not generate a new preview of proposed changes.

### Question: 181

Exhibit

```
resource "aws_instance" "web"  
count = 2  
name = "terraform-$ count.index "
```

A resource block is shown in the Exhibit space of this page.  
How do you reference the name value of the second instance of this resource?

- A. `aws_instance.web.*.name`
- B. `aws_instance.web[2].name`
- C. `aws_instance.web[1].name`
- D. `element(aws_instance.web, 2)`
- E. `aws_instance.web[1]`

**Answer: C**

**Explanation:**

**A. `aws_instance.web.*.name`**

Incorrect. This returns a **list of all name values** for all instances, not a single instance.

**B. `aws_instance.web[2].name`**

Incorrect. Terraform uses **zero-based indexing**, so index 2 would refer to a third instance, which doesn't exist here.

**C. `aws_instance.web[1].name`**

Correct. Terraform indexes instances starting at 0, so 1 refers to the **second instance**. This retrieves its name attribute.

**D. `element(aws_instance.web, 2)`**

Incorrect. This would return the **third instance**, not the second, due to zero-based indexing.

**E. `aws_instance.web[1]`**

Incorrect. This references the **second instance object itself**, but does not access the name attribute. You need `.name` to get the value.

### Question: 182

Which is a benefit of the Terraform state file?

- A. A state file can schedule recurring infrastructure tasks.
- B. A state file is a source of truth for resources provisioned with a public cloud console.
- C. A state file is a source of truth for resources provisioned with Terraform.
- D. A state file is the desired state expressed by the Terraform code files.

**Answer: C**

**Explanation:**

**A. A state file can schedule recurring infrastructure tasks.**

Incorrect. The state file does not schedule tasks; it only tracks the current state of resources.

**B. A state file is a source of truth for resources provisioned with a public cloud console.**

Incorrect. The state file only tracks resources managed by Terraform, not those created manually in the cloud console.

**C. A state file is a source of truth for resources provisioned with Terraform.**

Correct. The state file records the current state of all resources managed by Terraform and is used to plan and apply updates accurately.

**D. A state file is the desired state expressed by the Terraform code files.**

Incorrect. The desired state is expressed in the Terraform configuration files (.tf files). The state file reflects the **actual deployed state**, not just the desired state.

**Question: 183**

**CertyIQ**

Running terraform fmt without any flags in a directory with Terraform configuration files will check the formatting of those files, but will never change them.

- A. True
- B. False

**Answer: B**

**Explanation:**

Running terraform fmt without any flags will check the formatting and will automatically overwrite the files with the standardized formatting. To only check without changing, the -check or -diff flags must be used

**Question: 184**

**CertyIQ**

Which are advantages of IaC (Infrastructure as Code) patterns instead of manual infrastructure management?  
Pick the 2 correct responses below

- A. Eliminating the need for monitoring.
- B. Managing configuration with a GUI.
- C. Automating infrastructure deployments.
- D. Version controlling infrastructure changes.

**Answer: C,D**

**Explanation:**

**A. Eliminating the need for monitoring.**

Incorrect. IaC does not remove the need for monitoring. Monitoring is still required to ensure infrastructure is healthy and performing as expected.

## B. Managing configuration with a GUI.

Incorrect. IaC focuses on code-based, declarative configuration, not GUI management.

## C. Automating infrastructure deployments.

Correct. IaC enables automated creation, updating, and teardown of infrastructure, reducing manual intervention and errors.

## D. Version controlling infrastructure changes.

Correct. IaC allows infrastructure definitions to be stored in version control systems, enabling tracking, rollback, and collaboration.

### Question: 185

CertyIQ

The Exhibit section of this page shows part of a configuration you've been asked to update.

```
data "azurerm_resource_group" "example"
  name = var.resource_group_name

resource "azurerm_virtual_network" "example"
  name =
```

The name of the Azure Virtual Network should be set to the name of the resource group followed by a dash and the word "vnet".

Which expression fulfills this requirement?

- A. "\$ azurerm\_resource\_group.example.name -vnet"
- B. join("-", var.resource\_group\_name, "vnet")
- C. "\$ data.azurerm\_resource\_group.example.name -vnet"
- D. concat(data.azurerm\_resource\_group.example.name, "-", "vnet")

### Answer: C

#### Explanation:

#### A. "\$ azurerm\_resource\_group.example.name -vnet"

Incorrect. `azurerm_resource_group.example` does **not exist** in the configuration. The resource group is referenced as a **data block**, not a resource block.

#### B. join("-", var.resource\_group\_name, "vnet")

Incorrect. `join` expects a **list**, not individual string arguments, so this would cause an error.

#### C. "\$ data.azurerm\_resource\_group.example.name -vnet"

Correct. The data block `data.azurerm_resource_group.example` holds the existing resource group. Using interpolation, this expression concatenates the resource group name with `-vnet`.

#### D. concat(data.azurerm\_resource\_group.example.name, "-", "vnet")

Incorrect. `concat` is used for **lists**, not strings, so this would cause an error in Terraform.

**Question: 186**

You have set the `TF_LOG_PATH` environment variable for Terraform, and you would like to ensure the logs contain all debug-level messages and verbose process logs. Which action should you take?

- A. Run the terraform output command.
- B. Add a log argument to the terraform block.
- C. Update the Terraform CLI configuration file.
- D. Set the `TF_LOG` environment variable to `TRACE`

**Answer: D**

**Explanation:**

**A. Run the terraform output command.**

Incorrect. terraform output only displays output values from the state; it does not control logging.

**B. Add a log argument to the terraform block.**

Incorrect. Terraform configuration blocks do not have a log argument to control logging.

**C. Update the Terraform CLI configuration file.**

Incorrect. While you can configure some CLI defaults, debug-level logging is controlled via environment variables, not the CLI config.

**D. Set the `TF_LOG` environment variable to `TRACE`**

Correct. To capture all debug-level and verbose process logs, you set:

```
TF_LOG=TRACE
```

This ensures Terraform generates detailed logging at the `TRACE` level, which includes debug messages and internal operations. Combined with `TF_LOG_PATH`, the logs will be written to the specified file.

**Question: 187**

Which of these workflows is only enabled by the use of Infrastructure as Code?

- A. Cost optimization of infrastructure deployment.
- B. Automatic scaling of resources based on application load.
- C. Reviewing the proposed changes for potential security issues.
- D. Role-based access control of cloud resources.

**Answer: C**

**Explanation:**

**A. Cost optimization of infrastructure deployment.**

Incorrect. Cost optimization can be achieved with or without IaC, through careful resource planning and monitoring.

### B. Automatic scaling of resources based on application load.

Incorrect. Auto-scaling is a cloud feature and does not require IaC.

### C. Reviewing the proposed changes for potential security issues.

Correct. IaC enables workflows like **plan-time review** of infrastructure changes, allowing teams to inspect proposed modifications (via terraform plan, code review, or automated security checks) before they are applied. This workflow is only possible because the infrastructure is defined as code.

### D. Role-based access control of cloud resources.

Incorrect. RBAC is a cloud platform feature and can be managed manually; IaC is not required for it.

## Question: 188

CertyIQ

Terraform can only manage resource dependencies if you set them explicitly with the depends\_on argument.

- A. True
- B. False

**Answer: B**

### Explanation:

Terraform automatically infers most resource dependencies based on **references between resources** in your configuration (e.g., using attributes of one resource in another). You only need to use depends\_on for **explicit, manual dependencies** that Terraform cannot automatically detect.

## Question: 189

CertyIQ

Your team uses HCP Terraform to manage your infrastructure. You need to make a change to an Infrastructure stack running in a public cloud.

Which pattern follows Infrastructure as Code best practices for making a change?

- A. Use the public cloud console to make the change.
- B. Submit a pull request and wait for an approved merge of the change.
- C. Run the public cloud CLI tool to make the change.
- D. Make the change via the public cloud API endpoint.
- E. Clone the repository containing your infrastructure code and then run the code.

**Answer: B**

### Explanation:

#### A. Use the public cloud console to make the change.

Incorrect. Making changes directly in the console bypasses version control and Terraform, breaking IaC principles.

#### B. Submit a pull request and wait for an approved merge of the change.

Correct. Following IaC best practices means all infrastructure changes are made through **version-controlled code**, reviewed via pull requests, and applied automatically (e.g., via HCP Terraform). This ensures traceability, collaboration, and reproducibility.

**C. Run the public cloud CLI tool to make the change.**

Incorrect. Direct CLI changes bypass Terraform and version control.

**D. Make the change via the public cloud API endpoint.**

Incorrect. Direct API changes are outside Terraform's control and violate IaC best practices.

**E. Clone the repository containing your infrastructure code and then run the code.**

Incorrect. While you need to clone the repo to work on code, running it directly without a pull request bypasses code review and collaboration workflows.

**Question: 190**

**CertyIQ**

You can configure multiple cloud blocks in your Terraform configuration to connect your workspace to both HCP Terraform and your Terraform Enterprise instance.

- A. True
- B. False

**Answer: B**

**Explanation:**

Terraform **does not allow multiple cloud blocks** in a single configuration. You can only configure **one remote backend per workspace** to connect either to HCP Terraform **or** Terraform Enterprise, but not both simultaneously.

**Question: 191**

**CertyIQ**

What feature stops multiple users from operating on the Terraform state at the same time?

- A. State locking
- B. Remote state storage
- C. Version control
- D. Provider constraints

**Answer: A**

**Explanation:**

**A. State locking**

Correct. **State locking** prevents multiple users from making concurrent changes to the Terraform state. When a user applies changes, Terraform locks the state to ensure no other operations interfere, avoiding conflicts and corruption.

### B. Remote state storage

Incorrect. Remote state storage allows multiple users to access the state file, but by itself does not prevent concurrent modifications — locking is needed.

### C. Version control

Incorrect. Version control tracks changes to Terraform configuration files, not the live state of infrastructure.

### D. Provider constraints

Incorrect. Provider constraints control which versions of a provider can be used, not state concurrency.

## Question: 192

CertyIQ

which features do HCP Terraform workspaces provide that are not available in Terraform Community Edition?  
Pick the 3 correct responses below

- A. Support for multiple cloud providers.
- B. Automatic detection of common security issues.
- C. Remote execution of Terraform operations.
- D. Store Terraform and environment variables in variable sets.
- E. Store your configuration in a Version Control System (VCS).
- F. State versions and run history.

**Answer: B,C,D**

### Explanation:

#### A. Support for multiple cloud providers.

Incorrect. Terraform CE (Community Edition) can already manage multiple cloud providers; this is not unique to HCP Terraform.

#### B. Automatic detection of common security issues.

Correct. HCP Terraform can integrate with **policy checks** (like Sentinel) to automatically detect misconfigurations or security issues before applying changes. This feature is not available in Terraform CE.

#### C. Remote execution of Terraform operations.

Correct. HCP Terraform workspaces can **execute plan and apply remotely**, so users do not need to run Terraform locally. This ensures consistent execution environments and reduces setup overhead.

#### D. Store Terraform and environment variables in variable sets.

Correct. HCP Terraform allows storing **variables and environment variables in centralized variable sets**, which can be reused across multiple workspaces and runs. Terraform CE does not provide this centralized management.

#### E. Store your configuration in a Version Control System (VCS).

Incorrect. Both Terraform CE and HCP Terraform can pull configurations from a VCS, so this is not a feature unique to HCP Terraform.

#### F. State versions and run history.

Incorrect. While HCP Terraform **does track state and run history**, the way this option is phrased makes it misleading as a “feature not available in Terraform CE.” Terraform CE can also maintain state files and you can track changes locally. HCP adds convenience and remote visibility, but **state versioning itself is not exclusive to HCP Terraform**.

### Question: 193

CertyIQ

What does Terraform use to deploy infrastructure for different cloud providers?

- A. Custom APIs developed by HashiCorp
- B. Vendor-specific providers
- C. Vendors CLI tools
- D. Vendors UI

**Answer: B**

**Explanation:**

#### **A. Custom APIs developed by HashiCorp**

Incorrect. Terraform does not replace or implement its own APIs for cloud providers.

#### **B. Vendor-specific providers**

Correct. Terraform uses **providers** that are specific to each cloud vendor (like AWS, Azure, GCP). These providers translate Terraform configuration into API calls for the respective cloud platform.

#### **C. Vendor CLI tools**

Incorrect. Terraform does not rely on vendor CLI tools to create resources; it communicates directly through the provider APIs.

#### **D. Vendor UI**

Incorrect. Terraform does not use the cloud vendor’s UI; everything is done programmatically via providers.

### Question: 194

CertyIQ

When should you use the force-unlock command?

- A. When apply has failed due to a state lock.
- B. When you see a status message stating that you cannot acquire the lock.
- C. When automatic unlocking has failed.
- D. When you have a high priority change.

**Answer: C**

**Explanation:**

#### **A. When apply has failed due to a state lock.**

Incorrect. A failed apply does not automatically mean the lock is stale. The lock might still be held by another active operation. Using force-unlock here without verifying could corrupt the state.

**B. When you see a status message stating that you cannot acquire the lock.**

Incorrect. This message only indicates that another operation currently holds the lock. You should **not** force-unlock in this case, as it could interrupt an active process and corrupt the state.

**C. When automatic unlocking has failed.**

Correct. terraform force-unlock is intended for situations where Terraform fails to release a lock automatically, such as after a crash or interrupted operation. This safely clears a stale lock so future operations can proceed.

**D. When you have a high priority change.**

Incorrect. Force-unlock should **never** be used simply to prioritize a change. Doing so could interfere with ongoing operations and corrupt the state.

**Question: 195**

CertyIQ

You manage two workspaces in your HCP Terraform organization. The first workspace manages your network configuration. The second workspace manages your compute resources, and retrieves values from the networking workspace.

What HCP Terraform feature lets you run an apply operation on the compute workspace every time you update the networking workspace?

- A. Run triggers
- B. Projects
- C. Policy
- D. Run tasks

**Answer: A**

**Explanation:**

**A. Run triggers**

Correct. **Run triggers** allow one workspace to automatically trigger a run in another workspace when its outputs change. In this case, updating the networking workspace can trigger an apply in the compute workspace that depends on its outputs.

**B. Projects**

Incorrect. Projects in HCP Terraform are organizational units to group workspaces, but they do not trigger runs between workspaces.

**C. Policy**

Incorrect. Policies (e.g., Sentinel) enforce rules on runs and resources but do not automate workspace triggers.

**D. Run tasks**

Incorrect. Run tasks are actions executed during a run (like notifications or scripts), but they don't automatically trigger another workspace to apply changes.

### Question: 196

CertyIQ

What does terraform destroy do?

- A. Destroys the Terraform state file, leaves the infrastructure intact.
- B. Destroys all Terraform code files in the current directory, leaves the state file intact.
- C. Destroys all infrastructure in the Terraform state file.
- D. Destroys all infrastructure in the configured Terraform provider.

**Answer: C**

**Explanation:**

**A. Destroys the Terraform state file, leaves the infrastructure intact.**

Incorrect. terraform destroy does **not delete the state file**; it uses the state file to determine which resources to destroy.

**B. Destroys all Terraform code files in the current directory, leaves the state file intact.**

Incorrect. Terraform never deletes your configuration files.

**C. Destroys all infrastructure in the Terraform state file.**

Correct. terraform destroy reads the state file and **removes all resources tracked by that state**, effectively tearing down the infrastructure managed by Terraform.

**D. Destroys all infrastructure in the configured Terraform provider.**

Incorrect. It only destroys resources tracked in the current state file, not all resources in the provider account.

### Question: 197

CertyIQ

You're refactoring your Terraform configuration and have moved resources from one large module to multiple smaller ones. When running terraform plan, Terraform wants to destroy and recreate the resources. What should you add to the configuration to preserve the existing resources?

- A. Add a depends\_on argument to each resource you move.
- B. Add moved blocks for each resource.
- C. Add lifecycle prevent\_destroy = true to each of the resources.

**Answer: B**

**Explanation:**

**A. Add a depends\_on argument to each resource you move.**

Incorrect. depends\_on only controls the order in which resources are created or destroyed. It does not preserve existing resources when they are moved between modules.

**B. Add moved blocks for each resource.**

Correct. **moved blocks** tell Terraform that a resource has been relocated within the configuration (e.g., to a

new module or different resource address). This preserves the existing resources in the state and prevents them from being destroyed and recreated.

**C. Add lifecycle prevent\_destroy = true to each of the resources.**

Incorrect. prevent\_destroy stops Terraform from destroying a resource entirely, but it does **not update the state or track that a resource was moved**. Without a moved block, Terraform would still try to destroy and recreate the resource.

**Question: 198**

CertyIQ

A module block is shown in the Exhibit space on this page.

```
module "vpc"
source = "terraform-aws-modules/vpc/aws"
version = "~>4.0".
```

That module block limits the module version to major version 4.

- A. True
- B. False

**Answer: A**

**Explanation:**

The version constraint ~> 4.0 in the module block ensures that Terraform will use any version greater than or equal to 4.0.0 but less than 5.0.0. This effectively locks the module to major version 4, allowing minor and patch updates while preventing automatic upgrades to version 5 or higher.

**Question: 199**

CertyIQ

Terraform requires using a different provider for each cloud provider where you want to deploy resources.

- A. True
- B. False

**Answer: A**

**Explanation:**

Terraform uses **provider plugins** to communicate with different cloud platforms. Each cloud provider (like AWS, Azure, or GCP) requires its own provider configuration. You cannot use a single provider to deploy resources across multiple clouds; you must configure a separate provider for each.

**Question: 200**

CertyIQ

You want to split a large main.tf file into multiple files for better organization within the same working directory. What impact should you expect when you run a terraform plan?

- A. Significant changes should be expected since moving resources between files can trigger a

destroy/replacement.

B. No impact since Terraform reads all .tf files as a single configuration when executing a terraform plan.

C. Only minor impacts since Terraform parses the main.tf file first, which could change the order in which resources are created.

D. The terraform plan will fail since variable and provider blocks must appear before resource blocks.

**Answer: B**

**Explanation:**

**A. Significant changes should be expected since moving resources between files can trigger a destroy/replacement.**

Incorrect. Splitting files does **not** change resource addresses or state, so Terraform does not plan to destroy or recreate resources just because they are in different files.

**B. No impact since Terraform reads all .tf files as a single configuration when executing a terraform plan.**

Correct. Terraform **merges all .tf files in the working directory** into a single configuration before planning or applying. The order of files does not affect resource management.

**C. Only minor impacts since Terraform parses the main.tf file first, which could change the order in which resources are created.**

Incorrect. Terraform determines resource creation order based on **dependencies**, not file parsing order. Splitting files does not affect execution order.

**D. The terraform plan will fail since variable and provider blocks must appear before resource blocks.**

Incorrect. Terraform does not require a specific order for blocks across files. Variable, provider, and resource blocks can appear in any file in the directory.

## Question: 201

CertyIQ

As a developer, you want to ensure your plugins are up-to-date with the latest versions. Which Terraform command should you use?

A. terraform refresh -upgrade

B. terraform apply -upgrade

C. terraform providers -upgrade

D. terraform init -upgrade

**Answer: D**

**Explanation:**

**A. terraform refresh -upgrade**

Incorrect. terraform refresh updates the state with real-world infrastructure but does not upgrade provider plugins.

**B. terraform apply -upgrade**

Incorrect. terraform apply applies changes to infrastructure; the -upgrade flag is not valid here.

### C. terraform providers -upgrade

Incorrect. terraform providers lists providers and their versions, but it does not upgrade them.

### D. terraform init -upgrade

Correct. Running terraform init -upgrade upgrades all provider plugins to the latest allowed versions according to your version constraints, ensuring your plugins are up-to-date.

## Question: 202

CertyIQ

Your security team scanned some Terraform workspaces and found secrets stored in plaintext in state files. How can you protect that data?

- A. Delete the state file every time you run Terraform.
- B. Store the state in an encrypted backend.
- C. Always store your secrets in a secrets.tfvars file.
- D. Edit your state file to scrub out the sensitive data.

**Answer: B**

**Explanation:**

### A. Delete the state file every time you run Terraform.

Incorrect. Deleting the state file would break Terraform's ability to track resources and manage infrastructure. This is not a viable solution.

### B. Store the state in an encrypted backend.

Correct. Using an **encrypted remote backend** (like Terraform Cloud, HCP Terraform, or an encrypted S3 bucket) protects sensitive data in the state file at rest, preventing exposure of secrets.

### C. Always store your secrets in a secrets.tfvars file.

Incorrect. Storing secrets in .tfvars files in plaintext does not protect them; it can make the problem worse if the files are committed to version control.

### D. Edit your state file to scrub out the sensitive data.

Incorrect. Manually editing state files is risky and error-prone. Terraform may lose track of resources, and sensitive data could still be exposed elsewhere.

## Question: 203

CertyIQ

You can reference a resource created with for\_each using a splat ( \* ) expression.

- A. True
- B. False

**Answer: B**

**Explanation:**

When a resource is created using **for\_each**, Terraform assigns **each instance a unique key**. You reference these instances using their **map keys**, like `resource_name["key"].attribute`.

Splat expressions (\*) are used for resources with **count**, which creates a list of instances, not a map. Therefore, splat expressions do **not work with for\_each**.

**Question: 204**

**CertyIQ**

You add a new resource to an existing Terraform configuration, but do not update the version constraint in the configuration. The existing and new resources use the same provider. The working directory contains a `.terraform.lock.hcl` file.

How will Terraform choose which version of the provider to use?

- A. Terraform will use the latest version of the provider for the new resource and the version recorded in the lock file to manage existing resources.
- B. Terraform will use the version recorded in your lock file.
- C. Terraform will check your state file to determine the provider version to use.
- D. Terraform will use the latest version of the provider available at the time you provision your new resource.

**Answer: B**

**Explanation:**

**A. Terraform will use the latest version of the provider for the new resource and the version recorded in the lock file to manage existing resources.**

Incorrect. Terraform does **not mix provider versions** between resources in the same configuration. The lock file ensures a consistent version for all resources using that provider.

**B. Terraform will use the version recorded in your lock file.**

Correct. The `.terraform.lock.hcl` file pins the provider version for the workspace. Terraform will use the **exact version recorded in the lock file** for all resources, both existing and new, unless you explicitly update the lock file.

**C. Terraform will check your state file to determine the provider version to use.**

Incorrect. The state file tracks deployed resources, but **provider versions are controlled by the lock file**, not the state.

**D. Terraform will use the latest version of the provider available at the time you provision your new resource.**

Incorrect. Terraform does **not automatically upgrade** to the latest provider version if a lock file exists. You must explicitly run `terraform init -upgrade` to update.

**Question: 205**

**CertyIQ**

All standard backend types support state storage, locking, and remote operations like plan, apply, and destroy.

- A. True
- B. False

**Answer: B**

**Explanation:**

Not all standard backend types in Terraform support state locking and remote operations. For example, the local backend only stores the state on disk and does not provide locking or remote execution. Backends like S3 with DynamoDB, Terraform Cloud, or HCP Terraform do support locking and remote operations. Therefore, it is incorrect to assume that all backends provide full state management and remote capabilities.

**Question: 206**

CertyIQ

A resource block is shown in the Exhibit section of this page.

```
resource "kubernetes_namespace" "example" {  
  name = "test"  
}
```

How would you reference the attribute name of this resource in HCL?

- A. data.kubernetes\_namespace.name
- B. kubernetes\_namespace.example.name
- C. kubernetes\_namespace.test.name
- D. resource.kubernetes\_namespace.example.name

**Answer: B**

**Explanation:**

**A. data.kubernetes\_namespace.name**

Incorrect. This syntax is used for **data sources**, not resources.

**B. kubernetes\_namespace.example.name**

Correct. The resource is defined as kubernetes\_namespace.example, so you reference its name attribute with kubernetes\_namespace.example.name.

**C. kubernetes\_namespace.test.name**

Incorrect. The resource is named example, not test.

**D. resource.kubernetes\_namespace.example.name**

Incorrect. The resource keyword is only used in the configuration block, not when referencing an attribute.

**Question: 207**

CertyIQ

Which argument(s) are required when declaring a Terraform variable?

- A. type

- B. default
- C. description
- D. All of the above
- E. None of the above

**Answer: E**

**Explanation:**

**A. type**

Incorrect. Specifying a type is optional; Terraform can infer the type from the default value or usage.

**B. default**

Incorrect. A default value is optional. If omitted, the variable must be provided during terraform apply or via a .tfvars file.

**C. description**

Incorrect. A description is optional and used for documentation purposes only.

**D. All of the above**

Incorrect. None of these arguments are strictly required.

**E. None of the above**

Correct. When declaring a variable in Terraform, you can simply write variable "name" without specifying type, default, or description. All arguments are optional.

**Question: 208**

**CertyIQ**

terraform apply is failing with the following error. What next step should you take to determine the root cause of the problem?

Error loading state: AccessDenied: Access Denied status code: 403, request id: 288766CE5CCA24A0, host id: web.example.com

- A. Run terraform login to reauthenticate with the provider.
- B. Review /var/log/terraform.log for error messages.
- C. Review syslog for Terraform error messages.
- D. Set TF\_LOG=DEBUG.

**Answer: D**

**Explanation:**

**A. Run terraform login to reauthenticate with the provider.**

Incorrect. terraform login is used for Terraform Cloud or HCP authentication, not for provider-level access issues.

**B. Review /var/log/terraform.log for error messages.**

Incorrect. Terraform does not automatically log to `/var/log/terraform.log`.

**C. Review syslog for Terraform error messages.**

Incorrect. Terraform does not write errors to the system log.

**D. Set `TF_LOG=DEBUG`.**

Correct. Setting `TF_LOG=DEBUG` enables detailed logging of Terraform operations, including provider API requests. This will help identify the root cause of the **AccessDenied (403)** error.

**Question: 209**

CertyIQ

Which Terraform command checks that your configuration syntax is correct?

- A. terraform show
- B. terraform init
- C. terraform fmt
- D. terraform validate

**Answer: D**

**Explanation:**

**A. terraform show**

Incorrect. terraform show displays the current state or a saved plan; it does not check configuration syntax.

**B. terraform init**

Incorrect. terraform init initializes a working directory and downloads providers or modules, but it does not validate syntax.

**C. terraform fmt**

Incorrect. terraform fmt formats Terraform configuration files but does not validate syntax.

**D. terraform validate**

Correct. terraform validate checks the Terraform configuration for **syntax errors and internal consistency** without contacting any remote providers.

**Question: 210**

CertyIQ

How does the HCP Terraform/Terraform Cloud integration differ from backends such as S3, Consul, etc.?

- A. It can execute Terraform runs on dedicated infrastructure in HCP Terraform/Terraform Cloud.
- B. It doesn't show the output of a terraform apply locally.
- C. It is only available to paying customers.
- D. All of the above.

**Answer: A**

**Explanation:**

**A. It can execute Terraform runs on dedicated infrastructure in HCP Terraform/Terraform Cloud.**

Correct. Unlike backends like S3 or Consul, HCP Terraform and Terraform Cloud can **run Terraform operations remotely** on managed infrastructure, not just store state.

**B. It doesn't show the output of a terraform apply locally.**

Incorrect. While runs are executed remotely, you can still view outputs locally or retrieve them using Terraform commands.

**C. It is only available to paying customers.**

Incorrect. Terraform Cloud offers a free tier, so this is not strictly true.

**D. All of the above.**

Incorrect. Only option A correctly describes the difference.

## Question: 211

CertyIQ

Which of these actions will prevent two Terraform runs from changing the same state file at the same time?

- A. Delete the state before running Terraform.
- B. Refresh the state after running Terraform.
- C. Run Terraform with parallelism set to 1.
- D. Configure state locking for your state backend.

**Answer: D**

**Explanation:**

**A. Delete the state before running Terraform.**

Incorrect. Deleting the state would break Terraform's ability to track resources and would not prevent concurrent modifications safely.

**B. Refresh the state after running Terraform.**

Incorrect. Refresh updates the state with the real infrastructure but does **not prevent concurrent operations**.

**C. Run Terraform with parallelism set to 1.**

Incorrect. This only limits the number of concurrent resource operations **within a single run**, but it does not prevent multiple runs from accessing the same state simultaneously.

**D. Configure state locking for your state backend.**

Correct. **State locking** ensures that only one Terraform operation can modify a state file at a time, preventing conflicts and corruption.

### Question: 212

When should you use the force-unlock command?

- A. When apply has failed due to a state lock.
- B. When automatic unlocking has failed.
- C. When you see a status message stating that you cannot acquire the lock.
- D. When you have a high priority change.

**Answer: B**

**Explanation:**

**A. When apply has failed due to a state lock.**

Incorrect. A failed apply does not automatically mean the lock is stale. The lock may still be held by another active operation, so force-unlock should not be used blindly.

**B. When automatic unlocking has failed.**

Correct. terraform force-unlock is intended for situations where Terraform **fails to automatically release a state lock**, such as after a crash or interrupted operation. This safely clears the lock so future operations can proceed.

**C. When you see a status message stating that you cannot acquire the lock.**

Incorrect. This message alone does not indicate a stale lock. If another operation is legitimately holding the lock, forcing unlock could corrupt the state.

**D. When you have a high priority change.**

Incorrect. Force-unlock should **never** be used just to prioritize a change, as it could interfere with ongoing operations and corrupt the state.

### Question: 213

Which of the following is not a benefit of adopting infrastructure as code?

- A. Versioning
- B. Automation
- C. Reusability of code
- D. A Graphical User Interface

**Answer: D**

**Explanation:**

**A. Versioning**

Incorrect. Versioning is a benefit of IaC because infrastructure definitions are stored as code and can be tracked in version control systems.

**B. Automation**

Incorrect. IaC enables automated provisioning, updates, and teardown of infrastructure, which is a core benefit.

### C. Reusability of code

Incorrect. Modules and reusable configurations allow IaC to be applied across multiple projects, environments, or teams.

### D. A Graphical User Interface

Correct. IaC focuses on **code-based configuration**, not GUIs. Using a GUI is not a benefit of adopting infrastructure as code.

## Question: 214

CertyIQ

You used Terraform to create an ephemeral development environment in the cloud and are now ready to destroy all the infrastructure described by your Terraform configuration. To be safe, you would like to first see all the infrastructure that Terraform will delete.

Which command should you use to show all the resources that will be deleted? (Choose two.)

- A. Run `terraform state rm *`.
- B. Run `terraform destroy`. This will output all the resources that will be deleted before prompting for approval.
- C. Run `terraform plan -destroy`.
- D. Run `terraform show -destroy`.

**Answer: B,C**

**Explanation:**

**\*A. Run `terraform state rm *`.**

Incorrect. `terraform state rm` removes resources from the state file but does **not show what will be destroyed**. It only tells Terraform to stop managing them.

**B. Run `terraform destroy`. This will output all the resources that will be deleted before prompting for approval.**

Correct. `terraform destroy` shows a plan of all resources that will be destroyed and asks for confirmation before actually deleting them.

**C. Run `terraform plan -destroy`.**

Correct. `terraform plan -destroy` generates a **preview of the destroy operation** without applying it, letting you see which resources would be deleted.

**D. Run `terraform show -destroy`.**

Incorrect. `terraform show` displays the current state or a saved plan but does not generate a destroy plan by itself.

## Question: 215

CertyIQ

You created infrastructure outside the Terraform workflow that you now want to manage using Terraform. Which

command brings the infrastructure into Terraform state?

- A. terraform init
- B. terraform get
- C. terraform import
- D. terraform refresh

**Answer: C**

**Explanation:**

**A. terraform init**

Incorrect. terraform init initializes a working directory, downloads providers, and sets up modules, but does **not import existing resources**.

**B. terraform get**

Incorrect. terraform get downloads and updates modules, not existing resources.

**C. terraform import**

Correct. terraform import brings existing infrastructure into Terraform state, allowing Terraform to manage resources created outside of its workflow.

**D. terraform refresh**

Incorrect. terraform refresh updates the state to match the real-world infrastructure but does **not add unmanaged resources** to the state.

**Question: 216**

**CertyIQ**

A developer launched a VM (virtual machine) outside of the Terraform workflow and ended up with two servers with the same name. They are unsure which VM is managed with Terraform, but they do have a list of all active VM IDs.

Which of the following methods could you use to determine which instance Terraform manages?

- A. Run terraform state rm on both VMs, then terraform apply to recreate the correct one that will be managed by Terraform.
- B. Run terraform state list to find the names of all VMs, then run terraform state show for each of them to find which VM ID that Terraform manages.
- C. Run a terraform apply -refresh to identify the virtual machine IDs that are already managed by Terraform.
- D. Modify the Terraform configuration to add an import block for both of the virtual machines.

**Answer: B**

**Explanation:**

**A. Run terraform state rm on both VMs, then terraform apply to recreate the correct one that will be managed by Terraform.**

Incorrect. This would delete Terraform-managed resources and recreate them, which is **destructive** and unnecessary just to identify which VM is managed.

**B. Run terraform state list to find the names of all VMs, then run terraform state show for each of them to find which VM ID that Terraform manages.**

Correct. terraform state list shows all resources tracked by Terraform. Using terraform state show <resource> lets you inspect each resource's attributes, including the VM ID, to determine which instance Terraform manages.

**C. Run a terraform apply -refresh to identify the virtual machine IDs that are already managed by Terraform.**

Incorrect. While -refresh updates the state with real-world values, it does not explicitly show which VM is managed, and it could also prompt changes if the state differs from the configuration.

**D. Modify the Terraform configuration to add an import block for both of the virtual machines.**

Incorrect. There is **no import block** in Terraform configuration. Importing is done with the terraform import CLI command, not by modifying the configuration.

### Question: 217

CertyIQ

When do you need to explicitly execute Terraform in refresh-only mode?

- A. Before every terraform plan.
- B. Before every terraform apply.
- C. Before every terraform import.
- D. None of the above.

**Answer: D**

**Explanation:**

You **do not need to explicitly run Terraform in refresh-only mode** before plan, apply, or import. Terraform automatically refreshes the state as needed during these operations to reflect the real-world infrastructure.

terraform refresh -refresh-only is rarely required manually and is typically only used for **auditing or inspecting the current state** without making any changes.

### Question: 218

CertyIQ

Which method for sharing Terraform modules fulfills the following criteria:

1. Keeps the module configurations confidential within your organization.
2. Supports Terraform's semantic version constraints.
3. Provides a browsable directory of your modules.

- A. A Git repository containing your modules.
- B. A subfolder within your workspace.
- C. HCP Terraform/Terraform Cloud private registry.
- D. Public Terraform module registry

**Answer: C**

#### Explanation:

##### A. A Git repository containing your modules.

Incorrect. A Git repository can keep modules private, but it does **not support Terraform's semantic versioning** in the same way a registry does, and it does not provide a browsable module directory.

##### B. A subfolder within your workspace.

Incorrect. A local subfolder keeps modules private but does **not support semantic versioning** or a browsable directory for multiple modules.

##### C. HCP Terraform/Terraform Cloud private registry.

Correct. A **private registry in HCP Terraform/Terraform Cloud** keeps modules confidential, supports **semantic version constraints**, and provides a **browsable directory** of modules for your organization.

##### D. Public Terraform module registry.

Incorrect. The public registry supports versioning and browsing but does **not keep modules confidential**, as they are publicly accessible.

#### Question: 219

CertyIQ

Which of the following is not a way to trigger terraform destroy?

- A. Passing --destroy at the end of a plan request.
- B. Running terraform destroy from the correct directory and then typing yes when prompted in the CLI.
- C. Using the destroy command with auto-approve.

#### Answer: A

#### Explanation:

##### A. Passing --destroy at the end of a plan request.

Correct. Terraform does **not have a --destroy flag** for terraform plan. You cannot trigger a destroy this way.

##### B. Running terraform destroy from the correct directory and then typing yes when prompted in the CLI.

Incorrect. This is a standard way to trigger terraform destroy.

##### C. Using the destroy command with auto-approve.

Incorrect. Using terraform destroy -auto-approve is a valid way to trigger destruction without manual confirmation.

#### Question: 220

CertyIQ

What Terraform command always causes a state file to be updated with changes that might have been made outside of Terraform?

- A. terraform apply =lock=false
- B. terraform plan -target=state

- C. terraform plan -refresh-only
- D. terraform show -json

**Answer: C**

**Explanation:**

**A. terraform apply -lock=false**

Incorrect. This flag disables state locking during apply but does **not specifically refresh the state** with external changes.

**B. terraform plan -target=state**

Incorrect. -target focuses on specific resources but does **not refresh the state** with all external changes.

**C. terraform plan -refresh-only**

Correct. terraform plan -refresh-only **updates the state file** to reflect any changes made to infrastructure outside of Terraform, without proposing any new resource changes.

**D. terraform show -json**

Incorrect. terraform show displays the state or a plan in JSON format but **does not modify the state**.

**Question: 221**

**CertyIQ**

A developer on your team is going to tear down an existing deployment managed by Terraform and deploy a new one. However, there is a server resource named `aws_instance.ubuntu[1]` they would like to keep. What command should they use to tell Terraform to stop managing that specific resource?

- A. terraform destroy aws\_instance.ubuntu[1]
- B. terraform plan rm aws\_instance.ubuntu[1]
- C. terraform apply rm aws\_instance.ubuntu[1]
- D. terraform state rm aws\_instance.ubuntu[1]

**Answer: D**

**Explanation:**

**A. terraform destroy aws\_instance.ubuntu[1]**

Incorrect. Running terraform destroy on a resource will **delete it from the cloud**, which is the opposite of what the developer wants. In this case, they want to **keep** the server intact.

**B. terraform plan rm aws\_instance.ubuntu[1]**

Incorrect. There is no plan rm command in Terraform. terraform plan only previews changes, and rm is not a valid subcommand.

**C. terraform apply rm aws\_instance.ubuntu[1]**

Incorrect. apply rm is not a valid Terraform command. Terraform has no direct rm option within apply.

**D. terraform state rm aws\_instance.ubuntu[1]**

Correct. terraform state rm **removes the resource from Terraform's state file** without deleting the actual infrastructure. This tells Terraform to stop managing that specific resource, allowing the rest of the deployment to be destroyed or recreated while keeping aws\_instance.ubuntu[1] intact.

### Question: 222

CertyIQ

Which of the following locations can Terraform use as a private source for modules? (Choose two.)

- A. Public repository on GitHub.
- B. Internally hosted VCS (Version Control System) platform.
- C. Public Terraform Registry.
- D. Private repository on GitHub.

**Answer: B,D**

**Explanation:**

#### **A. Public repository on GitHub**

Incorrect. Public GitHub repositories are accessible to anyone, so they are **not private sources**.

#### **B. Internally hosted VCS (Version Control System) platform**

Correct. An internally hosted VCS (like GitLab, Bitbucket Server, or an on-prem Git server) can be configured as a **private module source**, restricting access to your organization.

#### **C. Public Terraform Registry**

Incorrect. The public Terraform Registry is open to everyone and cannot serve as a private module source.

#### **D. Private repository on GitHub**

Correct. A private GitHub repository can store modules that are **only accessible to authorized users**, making it a valid private source.

### Question: 223

CertyIQ

You are writing Terraform configuration that calls a child module. This child module declares an output value. How can you ensure that Terraform will print out this value when you run Terraform CLI commands such as terraform apply?

- A. Nothing. Terraform will automatically print out all output values from child modules.
- B. Nothing. Terraform cannot use output values from child modules.
- C. Declare a new output in your root configuration that references the module's output.
- D. None of the above.

**Answer: C**

**Explanation:**

**C. Declare a new output in your root configuration that references the module's output.**

Terraform only prints output values that are declared in the **root module**. Outputs from child modules are **not automatically displayed**. To ensure a child module's output is visible when running commands like terraform apply, you need to declare a corresponding output in your root configuration that references the child module's output. This allows Terraform to print the value during execution.

**Other options:**

**A. Nothing. Terraform will automatically print out all output values from child modules.**

Incorrect. Child module outputs are not automatically printed.

**B. Nothing. Terraform cannot use output values from child modules.**

Incorrect. Terraform can use outputs from child modules, but they must be explicitly referenced in the root module to be displayed.

**D. None of the above.**

Incorrect. Option C is the correct approach.

**Question: 224**

**CertyIQ**

You have provisioned some virtual machines (VMs) on Google Cloud Platform (GCP) using the gcloud command line tool. However, you are standardizing with Terraform and want to manage these VMs using Terraform instead. What are the two things you must do to achieve this? (Choose two.)

- A. Provision new VMs using Terraform with the same VM names.
- B. Use the terraform import command for the existing VMs.
- C. Write Terraform configuration for the existing VMs.
- D. Run the terraform import-gcp command.

**Answer: B,C**

**Explanation:**

**A. Provision new VMs using Terraform with the same VM names.**

Incorrect. Creating new VMs with the same names would **duplicate resources** and is not required to bring existing infrastructure under Terraform management.

**B. Use the terraform import command for the existing VMs.**

Correct. terraform import allows Terraform to **associate existing cloud resources with Terraform state**, enabling Terraform to manage them going forward.

**C. Write Terraform configuration for the existing VMs.**

Correct. Before importing, you must **define the Terraform configuration** that matches the existing resources so Terraform knows how to manage them after import.

**D. Run the terraform import-gcp command.**

Incorrect. There is no terraform import-gcp command; imports are done using the standard terraform import command.

### Question: 225

CertyIQ

Outside of the required\_providers block, Terraform configurations always refer to providers by their local names.

- A. True
- B. False

**Answer: A**

**Explanation:**

In Terraform, providers can be given **aliases** or local names in the required\_providers block. Outside of that block, you always refer to the provider using the **local name or alias** defined there. This ensures Terraform knows which provider configuration to use for a given resource.

For example, even if the provider source is hashicorp/aws, the local name like aws is what you reference in resources.

### Question: 226

CertyIQ

Where does the Terraform local backend store its state?

- A. In the terraform.tfstate directory.
- B. In the terraform.tfstate file.
- C. In the .terraform directory.
- D. In the .terraform.lock.hcl file.

**Answer: B**

**Explanation:**

**A. In the terraform.tfstate directory**

Incorrect. There is no terraform.tfstate directory. The local backend stores the state in a **file**, not a directory. Terraform will not create a directory for state unless you explicitly configure it, which is uncommon.

**B. In the terraform.tfstate file**

Correct. The **local backend** stores the state on your local filesystem in a file called terraform.tfstate by default. This file contains all the information Terraform needs to track your managed resources, including their attributes and metadata.

**C. In the .terraform directory**

Incorrect. The .terraform directory is used for storing downloaded **providers, modules, and plugin binaries**, not the state file. The state is separate from these artifacts.

**D. In the .terraform.lock.hcl file**

Incorrect. The .terraform.lock.hcl file is a **provider version lock file**. It ensures consistent provider versions across runs but does **not store the state of resources**.

### Question: 227

A module block is shown in the Exhibit space of this page.

```
module "consul" {  
    source = "hashicorp/consul/aws"  
}
```

When you use a module block to reference a module from the Terraform Registry such as the one in the example, how do you specify version 1.0.0 of the module?

- A. You cannot. Modules stored on the public Terraform Registry do not support versioning.
- B. Append ?ref=v1.0.0 argument to the source path.
- C. Add a version = "1.0.0" attribute to the module block.
- D. Nothing. Modules stored on the public Terraform module Registry always default to version 1.0.0.

**Answer: C**

**Explanation:**

**A. You cannot. Modules stored on the public Terraform Registry do not support versioning.**

Incorrect. Modules on the public Terraform Registry **do support semantic versioning**, and you can specify the version you want to use.

**B. Append ?ref=v1.0.0 argument to the source path.**

Incorrect. This syntax is used for Git sources, not modules from the Terraform Registry.

**C. Add a version = "1.0.0" attribute to the module block.**

Correct. When using a module from the Terraform Registry, you specify the desired version with the version attribute in the module block. This ensures Terraform uses that specific version and respects semantic version constraints.

**D. Nothing. Modules stored on the public Terraform module Registry always default to version 1.0.0.**

Incorrect. The registry defaults to the **latest available version** that matches any constraints; it does not automatically default to 1.0.0.

### Question: 228

You cannot install third party plugins using terraform init.

- A. True
- B. False

**Answer: B**

**Explanation:**

**B. False**

You **can install third-party plugins** using terraform init. The init command downloads and installs all required

provider plugins and modules specified in your Terraform configuration, including those from third-party sources. This allows Terraform to interact with a wide variety of cloud providers and services beyond the built-in ones.

**A. True**

Incorrect. terraform init is explicitly designed to fetch and install both first-party and third-party plugins.

**Question: 229**

CertyIQ

You're writing a Terraform configuration that needs to read input from a local file called id\_rsa.pub. Which built-in Terraform function can you use to import the file's contents as a string?

- A. file("id\_rsa.pub")
- B. templatefile("id\_rsa.pub")
- C. fileset("id\_rsa.pub")
- D. filebase64("id\_rsa.pub")

**Answer: A**

**Explanation:**

**A. file("id\_rsa.pub")**

The file() function in Terraform **reads the contents of a local file** and returns it as a string. This is useful for reading things like SSH public keys, configuration files, or scripts directly into your Terraform configuration.

**Other options:**

**B. templatefile("id\_rsa.pub")**

Incorrect. templatefile() is used to **render a template file with variables**, not simply read a file as a string.

**C. fileset("id\_rsa.pub")**

Incorrect. fileset() is used to return a **set of file names in a directory matching a pattern**, not the file's contents.

**D. filebase64("id\_rsa.pub")**

Incorrect. filebase64() reads the file and returns its contents **encoded in Base64**, not as a plain string.

**Question: 230**

CertyIQ

When you initialize Terraform, where does it cache modules from the public Terraform Registry?

- A. They are not cached.
- B. In the /tmp directory.
- C. In the .terraform sub-directory.
- D. In memory.

**Answer: C**

**Explanation:**

**A. They are not cached.**

Incorrect. Terraform **does cache modules** downloaded from the public registry. Without caching, Terraform would have to download modules every time you run init or apply a configuration, which would be inefficient and error-prone.

**B. In the /tmp directory.**

Incorrect. Terraform does **not** use temporary system directories like /tmp for storing modules. /tmp is typically cleared by the system and would not provide reliable storage for module caching.

**C. In the .terraform sub-directory**

Correct. Terraform stores downloaded modules in the .terraform sub-directory of your working directory. This allows Terraform to **reuse modules across runs**, avoid repeated downloads, and support offline operations.

**D. In memory.**

Incorrect. While Terraform may load module contents into memory temporarily during execution, the **persistent cache is on disk** in the .terraform directory. Memory alone would not allow reuse across multiple Terraform runs.

## Question: 231

CertyIQ

Which Terraform collection type should you use to store key/value pairs?

- A. map
- B. tuple
- C. list
- D. set

**Answer: A**

**Explanation:**

**A. map**

Correct. A map in Terraform is explicitly designed to store key/value pairs, allowing you to reference values by their keys. Example: "name" = "gourav", "role" = "developer" .

**B. tuple**

Incorrect. A tuple is an ordered sequence of elements and does not store key/value pairs. Access is by index, not by key.

**C. list**

Incorrect. A list is an ordered collection of values, also accessed by index, not by key. It cannot directly represent key/value pairs.

**D. set**

Incorrect. A set is an unordered collection of unique values. It does not support key/value mapping.

### Question: 232

CertyIQ

A Terraform backend determines how Terraform loads state and stores updates when you execute which command?

- A. apply
- B. destroy
- C. Both of these are correct.
- D. Neither of these are correct.

**Answer: C**

**Explanation:**

#### A. apply

Incorrect. While terraform apply does update state, the backend also handles other operations beyond just apply.

#### B. destroy

Incorrect. terraform destroy also updates state, but the backend is not limited to destroy operations alone.

#### C. Both of these are correct

Correct. A Terraform backend manages state for **all commands that read or write state**, including apply, destroy, plan, and more. It ensures Terraform knows the current state of infrastructure and can store updates reliably.

#### D. Neither of these are correct

Incorrect. The backend is directly responsible for storing and retrieving state, so this option is not correct.

### Question: 233

CertyIQ

It is best practice to store secret data in the same version control repository as your Terraform configuration.

- A. True
- B. False

**Answer: B**

**Explanation:**

#### A. True

Incorrect. Storing secrets directly in your Terraform configuration or version control can expose sensitive data, risking security breaches.

#### B. False

Correct. Best practice is to **store secrets separately** using secure methods such as **Terraform Cloud/Enterprise workspaces, environment variables, or secret management tools** like Vault or AWS Secrets Manager, rather than embedding them in version contro

### Question: 234

CertyIQ

Which of these is true about Terraform's plugin-based architecture?

- A. You can create a provider for your API if none exists.
- B. Terraform can only source providers from the internet.
- C. All providers are part of the Terraform core binary.
- D. Every provider in a configuration has its own state file for its resources.

**Answer: A**

**Explanation:**

**A. You can create a provider for your API if none exists**

Correct. Terraform's plugin-based architecture allows users to **develop custom providers** to interact with APIs or services that don't have an existing provider.

**B. Terraform can only source providers from the internet**

Incorrect. Terraform can source providers **from local paths, private registries, or the public registry**. It is not limited to the internet.

**C. All providers are part of the Terraform core binary**

Incorrect. Providers are **separate binaries** that Terraform communicates with via the plugin system; they are not included in the core Terraform binary.

**D. Every provider in a configuration has its own state file for its resources**

Incorrect. Terraform stores **all resource state in a single state file** (or backend-managed state), not separate files per provider.

### Question: 235

CertyIQ

You are using a networking module in your Terraform configuration with the name `my_network`. Your root module includes the following configuration:

```
output "net_id" {  
  value = module.my_network.vnet_id  
}
```

When you run `terraform validate`, you get the following error:

## Error: Reference to undeclared output value

```
on main.tf line 12, in output "net_id":
12:   value = module.my_network.vnet_id
```

What must you do to successfully retrieve this value from your networking module?

- A. Define the attribute vnet\_id as a variable in the networking module.
- B. Change the referenced value to my\_network.outputs.vnet\_id.
- C. Change the referenced value to module.my\_network.outputs.vnet\_id.
- D. Define the attribute vnet\_id as an output in the networking module.

**Answer: D**

**Explanation:**

### A. Define the attribute vnet\_id as a variable in the networking module

Incorrect. Variables are used to pass **inputs into a module**, not to retrieve values from it. This will not allow you to access vnet\_id from the root module.

### B. Change the referenced value to my\_network.outputs.vnet\_id

Incorrect. Terraform requires the module keyword to access module outputs. Simply prefixing with the module name is invalid syntax.

### C. Change the referenced value to module.my\_network.outputs.vnet\_id

Incorrect. Terraform allows you to access outputs using module.<module\_name>.<output\_name>, **without** .outputs. Including .outputs will cause a syntax error.

### D. Define the attribute vnet\_id as an output in the networking module

Correct. To retrieve a value from a module, the module must **declare it as an output**. After adding:

```
output "vnet_id"
```

```
value = azurerm_virtual_network.my_vnet.id
```

You can access it in the root module using:

```
module.my_network.vnet_id
```

This is the proper way to expose module values.

## Question: 236

CertyIQ

Where in your Terraform configuration do you specify a state backend?

- A. The data source block
- B. The terraform block
- C. The provider block
- D. The resource block

**Answer: B**

**Explanation:**

**A. The data source block**

Incorrect. Data sources are used to read information from existing resources, not to configure where Terraform stores state.

**B. The terraform block**

Correct. The terraform block is where you specify backend settings, which determine **how and where Terraform stores its state**.

**C. The provider block**

Incorrect. Provider blocks configure cloud or service providers, not state storage.

**D. The resource block**

Incorrect. Resource blocks define infrastructure resources, not the backend for state.

**Question: 237**

**CertyIQ**

You can execute terraform fmt to standardize all Terraform configurations within the current working directory to Terraform's canonical format and style.

- A. True
- B. False

**Answer: A**

**Explanation:**

**A. True**

Correct. The terraform fmt command **automatically formats Terraform configuration files** in the current directory (and optionally subdirectories) to follow Terraform's standard style and formatting rules.

**B. False**

Incorrect. terraform fmt is specifically designed to enforce consistent formatting, so this statement is true.

**Question: 238**

**CertyIQ**

The Terraform binary version and provider versions must match each other in a single configuration.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. The Terraform binary and provider versions **do not need to match exactly**. Providers can have versions that are compatible with your Terraform version, and you can specify version constraints to ensure compatibility.

**B. False**

Correct. Terraform allows you to use different provider versions as long as they are **compatible** with the Terraform binary version you are using. Exact matching is not required.

### Question: 239

CertyIQ

Multiple team members are collaborating on infrastructure using Terraform and want to format their Terraform code following standard Terraform style convention.

How should they ensure the code satisfies conventions?

- A. Use terraform fmt.
- B. Terraform automatically formats configuration on terraform apply.
- C. Run terraform validate prior to executing terraform plan or terraform apply.
- D. Replace all tabs with spaces.

**Answer: A**

**Explanation:**

**A. Use terraform fmt**

Correct. The terraform fmt command **automatically formats Terraform code** according to the standard style convention, ensuring consistency across team members.

**B. Terraform automatically formats configuration on terraform apply**

Incorrect. Terraform **does not automatically format code** during apply; formatting must be done explicitly with terraform fmt.

**C. Run terraform validate prior to executing terraform plan or terraform apply**

Incorrect. terraform validate checks for **syntax and configuration errors**, but it does not enforce style or formatting conventions.

**D. Replace all tabs with spaces**

Incorrect. Manually replacing tabs with spaces is unnecessary if you use terraform fmt, which handles formatting automatically.

### Question: 240

CertyIQ

Terraform can only manage resource dependencies if you set them explicitly with the depends\_on argument.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. Terraform **automatically infers dependencies** between resources based on references in your configuration. Explicit `depends_on` is only needed in special cases where implicit dependencies are insufficient.

**B. False**

Correct. Terraform's **graph-based dependency management** tracks which resources depend on others automatically, so you don't always need to use `depends_on`.

### Question: 241

CertyIQ

You decide to move a Terraform state file to Amazon S3 from another location. You write the code shown in the Exhibit space into a file called `backend.tf`.

```
terraform {  
  backend "s3" {  
    bucket = "my-tf-bucket"  
    region = "us-east-1"  
  }  
}
```

Which command will migrate your current state file to the new S3 backend?

- A. `terraform refresh`
- B. `terraform init`
- C. `terraform push`
- D. `terraform state`

**Answer: B**

**Explanation:**

**A. terraform refresh**

Incorrect. `terraform refresh` updates the state file to match the real-world resources, but it **does not migrate the state to a new backend**.

**B. terraform init**

Correct. Running `terraform init` after configuring a new backend **initializes the backend** and will **migrate your existing state file** to the S3 backend if configured.

**C. terraform push**

Incorrect. terraform push is **deprecated** and was used only for legacy Terraform Enterprise workflows; it is not used for backend state migration.

#### D. terraform state

Incorrect. terraform state is used for **state management commands** like mv, rm, or list, but it does not perform backend migration.

### Question: 242

CertyIQ

Terraform encrypts sensitive values stored in your state file.

- A. True
- B. False

**Answer: B**

**Explanation:**

#### A. True

Incorrect. By default, Terraform **does not encrypt sensitive values** in the local state file. Sensitive data like passwords or secrets are stored in plaintext unless the backend itself provides encryption (e.g., S3 with server-side encryption).

#### B. False

Correct. Terraform relies on **backend security** or external measures to protect sensitive values; the state file itself is **not automatically encrypted** by Terraform.

### Question: 243

CertyIQ

You have declared a variable called var.list which is a list of objects that all have an attribute id . Which options will produce a list of the IDs? (Choose two.)

- A. [ for o in var.list : o.id ]
- B. var.list[\*].id
- C. [ var.list[\*].id ]
- D. for o in var.list : o => o.id

**Answer: A,B**

**Explanation:**

#### A. [ for o in var.list : o.id ]

Correct. This is a **for-expression** that iterates over each object in the list and collects the id attribute, producing a list of IDs.

#### B. var.list[\*].id

Correct. The **splat operator (\*)** extracts the id attribute from each object in the list, returning a list of IDs.

### C. [ var.list[\*].id ]

Incorrect. Wrapping var.list[\*].id in square brackets produces a **nested list** (a list containing one list), which is not the same as a flat list of IDs.

### D. for o in var.list : o => o.id

Incorrect. This creates a **map** from the object to its id, not a list of IDs. It does not satisfy the requirement to produce a list.

## Question: 244

CertyIQ

You need to deploy resources into two different regions in the same Terraform configuration. To do this, you declare multiple provider configurations as shown in the Exhibit space on this page.

```
provider "aws" {  
    region = "us-east-1"  
}
```

```
provider "aws" {  
    alias = "west"  
    region = "us-west-2"  
}
```

What meta-argument do you need to configure in a resource block to deploy the resource to the us-west-2 AWS region?

- A. provider = west
- B. alias = west
- C. provider = aws.west
- D. alias = aws.west

### Answer: C

#### Explanation:

#### A. provider = west

Incorrect. You cannot reference the provider just by its alias name alone; Terraform requires the full provider address.

#### B. alias = west

Incorrect. alias is used **when declaring a provider configuration**, not when assigning a resource to a provider.

#### C. provider = aws.west

Correct. To use a specific provider configuration with an alias, you set the **provider meta-argument** in the resource block to the **full provider address including the alias**. For example, provider = aws.west tells Terraform to use the aws provider configuration aliased as west.

#### D. alias = aws.west

Incorrect. alias is not valid inside a resource block; it is only used in provider configuration blocks.

### Question: 245

CertyIQ

terraform init retrieves and caches the configuration for all remote modules

- A. True
- B. False

**Answer: A**

**Explanation:**

**A. True**

Correct. Running terraform init **downloads and caches all remote modules** referenced in your configuration so they can be reused across runs, even offline.

**B. False**

Incorrect. Terraform does cache remote modules in the .terraform directory, so this statement is true.

### Question: 246

CertyIQ

You are a member of an operations team that uses infrastructure as code (IaC) practices with HCP Terraform/Terraform Cloud. You are tasked with making a change to an infrastructure stack running in a public cloud. Which pattern would follow IaC best practices for making a change?

•

- A. Make the change programmatically via the public cloud CLI.
- B. Clone the repository containing your infrastructure code and then run the code.
- C. Submit a pull request and wait for an approved merge of the proposed changes.
- D. Make the change via the public cloud API endpoint.
- E. Use the public cloud console to make the change after a database record has been approved.

**Answer: C**

**Explanation:**

**A. Make the change programmatically via the public cloud CLI**

Incorrect. Making changes directly in the cloud bypasses your Terraform configuration, breaking the declarative IaC model and causing **configuration drift**.

**B. Clone the repository containing your infrastructure code and then run the code**

Incorrect. While cloning is part of working with the code, running it without a review or proper process does not follow **team-based IaC best practices**.

**C. Submit a pull request and wait for an approved merge of the proposed changes**

Correct. The **pull request workflow** ensures that all changes to infrastructure code are **reviewed, version-controlled, and approved** before being applied. This maintains auditability, collaboration, and the integrity of your infrastructure.

**D. Make the change via the public cloud API endpoint**

Incorrect. Like the CLI, making direct API changes bypasses Terraform, creating potential **drift** between code and actual infrastructure.

**E. Use the public cloud console to make the change after a database record has been approved**

Incorrect. Making manual console changes is not aligned with IaC principles; all changes should be made **through version-controlled Terraform configurations**.

**Question: 247**

CertyIQ

terraform validate uses provider APIs to verify your infrastructure settings.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. terraform validate **does not contact any provider APIs**. It only checks the **syntax and internal consistency** of your Terraform configuration.

**B. False**

Correct. terraform validate performs a **static check** of your configuration files and cannot verify actual infrastructure settings with providers.

**Question: 248**

CertyIQ

It is \_\_\_\_\_ to change the Terraform backend from the default “local” backend to a different one after performing your first terraform apply.

- A. impossible
- B. discouraged
- C. optional
- D. mandatory

**Answer: B**

**Explanation:**

**A. impossible**

Incorrect. It is not impossible; Terraform allows backend changes, but you must take care to migrate the

existing state.

**B. discouraged**

Correct. Changing the backend **after initial deployment is discouraged** because it can be risky and may require careful state migration to avoid losing track of resources.

**C. optional**

Incorrect. While technically optional, this doesn't capture the caution needed — changing the backend is not a routine operation.

**D. mandatory**

Incorrect. Changing the backend is **not required**; it's only done if you need a remote or shared state backend.

**Question: 249**

CertyIQ

Terraform installs its providers during which phase?

- A. plan
- B. init
- C. refresh
- D. All of the above

**Answer: B**

**Explanation:**

**A. plan**

Incorrect. The terraform plan phase **uses already installed providers** to generate an execution plan; it does not install them.

**B. init**

Correct. Terraform installs providers **during the terraform init phase**, when it initializes a configuration and downloads any required provider plugins.

**C. refresh**

Incorrect. terraform refresh updates the state to match real-world resources, but it does **not install providers**.

**D. All of the above**

Incorrect. Only the init phase is responsible for installing providers.

**Question: 250**

CertyIQ

Which of the following is not true of Terraform providers?

- A. An individual person can write a Terraform Provider.
- B. A community of users can maintain a provider.

- C. HashiCorp maintains some providers.
- D. Cloud providers and infrastructure vendors can write, maintain, or collaborate on Terraform providers.
- E. None of the above.

**Answer: E**

**Explanation:**

**A. An individual person can write a Terraform Provider**

True. Terraform's plugin-based architecture allows an individual to develop a custom provider.

**B. A community of users can maintain a provider**

True. Providers can be community-maintained and published in public registries.

**C. HashiCorp maintains some providers**

True. HashiCorp officially maintains core providers like AWS, Azure, and Google Cloud.

**D. Cloud providers and infrastructure vendors can write, maintain, or collaborate on Terraform providers**

True. Many cloud providers and vendors develop and maintain their own official providers.

**E. None of the above**

Correct. All the statements A–D are **true**, so the statement “not true” applies to **none of them**.

**Question: 251**

**CertyIQ**

HashiCorp Configuration Language (HCL) supports user-defined functions.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. HCL provides **built-in functions** (like `length()`, `concat()`, `lookup()`, etc.), but it **does not allow users to define their own functions**.

**B. False**

Correct. HCL **does not support user-defined functions**; you can only use the built-in set of functions provided by Terraform.

**Question: 252**

**CertyIQ**

Which of these statements about HCP Terraform/Terraform Cloud workspaces is false?

- A. You must use the CLI to switch between workspaces.
- B. Plans and applies can be triggered via version control system integrations.
- C. They have role-based access controls.
- D. They can securely store cloud credentials.

**Answer: A**

**Explanation:**

**A. You must use the CLI to switch between workspaces**

Correct. This is **false**. While you can switch workspaces via the CLI, you can also switch workspaces through the **Terraform Cloud web UI**. It is not CLI-only.

**B. Plans and applies can be triggered via version control system integrations**

True. Workspaces can automatically run plans and applies when changes are pushed to connected VCS repositories.

**C. They have role-based access controls**

True. Workspaces support role-based access control (RBAC) to manage permissions for team members.

**D. They can securely store cloud credentials**

True. Workspaces can securely store sensitive credentials and variables using Terraform Cloud's secrets management.

**Question: 253**

**CertyIQ**

Which of the following is true about terraform apply? (Choose two.)

- A. You cannot target specific resources for the operation.
- B. Depending on provider specification, Terraform may need to destroy and recreate your infrastructure resources.
- C. It only operates on infrastructure defined in the current working directory or workspace.
- D. By default, it does not refresh your state file to reflect the current infrastructure configuration.
- E. You must pass the output of a terraform plan command to it.

**Answer: B,C**

**Explanation:**

**A. You cannot target specific resources for the operation**

Incorrect. You can target specific resources using the -target option with terraform apply.

**B. Depending on provider specification, Terraform may need to destroy and recreate your infrastructure resources**

Correct. Some changes require **resource replacement**, which involves destroying the existing resource and creating a new one, depending on the provider and resource type.

**C. It only operates on infrastructure defined in the current working directory or workspace**

Correct. terraform apply works on the configuration **loaded in the current directory or workspace**, applying changes to the infrastructure defined there.

**D. By default, it does not refresh your state file to reflect the current infrastructure configuration**

Incorrect. By default, terraform apply **refreshes the state** to reflect the current real-world infrastructure before applying changes.

**E. You must pass the output of a terraform plan command to it**

Incorrect. You can run terraform apply **directly**, and it will automatically create a plan before applying changes; passing a plan output is optional.

**Question: 254**

CertyIQ

Which of these are benefits of using Sentinel with HCP Terraform/Terraform Cloud? (Choose three.)

- A. You can check out and check in cloud access keys.
- B. You can restrict specific resource configurations, such as disallowing the use of CIDR=0.0.0.0/0.
- C. Sentinel Policies can be written in HashiCorp Configuration Language (HCL).
- D. You can enforce a list of approved AWS AMIs.
- E. Policy-as-code can enforce security best practices.

**Answer: B,D,E**

**Explanation:**

**A. You can check out and check in cloud access keys**

Incorrect. Sentinel does **not manage credentials**; it is a policy-as-code framework for governance.

**B. You can restrict specific resource configurations, such as disallowing the use of CIDR=0.0.0.0/0**

Correct. Sentinel can enforce **fine-grained rules on resource configurations** to prevent insecure settings.

**C. Sentinel Policies can be written in HashiCorp Configuration Language (HCL)**

Incorrect. Sentinel uses its **own policy language**, not HCL.

**D. You can enforce a list of approved AWS AMIs**

Correct. Sentinel policies can enforce **approved values for specific attributes**, such as a whitelist of AMIs.

**E. Policy-as-code can enforce security best practices**

Correct. Sentinel enables **policy-as-code**, allowing automated enforcement of security, compliance, and operational best practices.

**Question: 255**

CertyIQ

Which of the following is available only in HCP Terraform workspaces and not in Terraform CLI?

- A. Secure variable storage.

- B. Support for multiple cloud providers.
- C. Dry runs with terraform plan.
- D. Using one workspace's state as a data source for another.

**Answer: D**

**Explanation:**

**A. Secure variable storage**

Incorrect. While HCP workspaces offer secure variable storage, you can also use local environment variables or encrypted files with Terraform CLI, though less integrated.

**B. Support for multiple cloud providers**

Incorrect. Terraform CLI also supports multiple providers; this is not unique to HCP workspaces.

**C. Dry runs with terraform plan**

Incorrect. terraform plan is available in both CLI and HCP workspaces.

**D. Using one workspace's state as a data source for another**

Correct. This is a **feature specific to HCP Terraform/Terraform Cloud**, where you can reference the state of one workspace as a data source in another workspace. This is **not available in standalone Terraform CLI**.

**Question: 256**

**CertyIQ**

Which two steps are required to provision new infrastructure in the Terraform workflow? (Choose two.)

- A. import
- B. init
- C. apply
- D. plan
- E. validate

**Answer: B,C**

**Explanation:**

**A. import**

Incorrect. terraform import is used to bring **existing resources under Terraform management**, not for provisioning new infrastructure.

**B. init**

Correct. terraform init **initializes the working directory**, downloads required providers and modules, and is always required before provisioning.

**C. apply**

Correct. terraform apply **creates or updates infrastructure** according to your configuration and is the step that actually provisions resources.

#### D. plan

Incorrect. terraform plan is optional; it generates an execution plan to preview changes but does not provision infrastructure by itself.

#### E. validate

Incorrect. terraform validate only checks **syntax and configuration correctness**; it does not provision resources.

### Question: 257

CertyIQ

Which of the following should you add in the required\_providers block to define a provider version constraint?

- A. version = ">= 3.1"
- B. version >= 3.1
- C. version ~> 3.1

**Answer: A**

**Explanation:**

**A. version = ">= 3.1"**

Correct. Proper syntax for specifying a version constraint.

**B. version >= 3.1**

Incorrect. Missing the = and quotes; Terraform will throw a syntax error.

**C. version ~> 3.1**

Incorrect. The ~> operator **must also be inside quotes and assigned with =**. Correct syntax would be version = "~> 3.1". Without = and quotes, it is invalid.

### Question: 258

CertyIQ

What is the purpose of state in Terraform?

- A. State codifies the dependencies of related resources.
- B. State stores variables and lets you quickly reuse existing code.
- C. State lets you enforce resource configurations that relate to compliance policies.
- D. State maps real world resources to your configuration and keeps track of metadata.

**Answer: D**

**Explanation:**

**A. State codifies the dependencies of related resources**

Incorrect. Terraform infers **dependencies automatically** from resource references, but that is not the primary purpose of the state file.

**B. State stores variables and lets you quickly reuse existing code**

Incorrect. Variables are defined in configuration, not in state, and reusing code is handled by modules, not the state file.

**C. State lets you enforce resource configurations that relate to compliance policies**

Incorrect. Compliance enforcement is handled by **Sentinel or policy-as-code**, not the state file.

**D. State maps real world resources to your configuration and keeps track of metadata**

Correct. Terraform's **state file records the mapping between resources in your configuration and the real infrastructure**, along with metadata like resource IDs, dependencies, and provider information. This allows Terraform to track changes and plan updates accurately.

**Question: 259**

CertyIQ

Terraform providers are always installed from the Internet.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. Terraform can install providers from **local paths, private registries, or the public registry**, not just the Internet.

**B. False**

Correct. Providers do **not have to come from the Internet**; you can use **local binaries or private registries** for installation.

**Question: 260**

CertyIQ

Your DevOps team is currently using the local backend for your Terraform configuration. You would like to move to a remote backend to store the state file in a central location. Which of the following backends would not work?

- A. HCP Terraform/Terraform Cloud
- B. Git
- C. Amazon S3

**Answer: B**

**Explanation:**

**A. HCP Terraform/Terraform Cloud**

Incorrect. HCP Terraform/Terraform Cloud is a **supported remote backend** and can store state centrally.

#### B. Git

Correct. Git is **not a supported Terraform backend**. Terraform state requires a backend that supports **locking and atomic operations**, which Git alone cannot provide.

#### C. Amazon S3

Incorrect. S3 is a supported remote backend, often used in combination with **DynamoDB for state locking**.

### Question: 261

CertyIQ

Which parameters does terraform import require? (Choose two.)

- A. Provider
- B. Path
- C. Resource ID
- D. Resource address

**Answer: C,D**

**Explanation:**

#### A. Provider

Incorrect. You do **not need to specify the provider** explicitly when running terraform import; Terraform uses the provider configured in your code.

#### B. Path

Incorrect. There is no "path" parameter for terraform import.

#### C. Resource ID

Correct. Terraform requires the **ID of the existing resource** in the provider's system to import it into state.

#### D. Resource address

Correct. Terraform requires the **resource address** from your configuration (e.g., aws\_instance.example) to know **where to map the imported resource in state**.

### Question: 262

CertyIQ

Where can Terraform not load a provider from?

- A. Provider plugin cache.
- B. Source code.
- C. Plugins directory.
- D. Official HashiCorp distribution on releases.hashicorp.com.

**Answer: B**

**Explanation:**

**A. Provider plugin cache**

Incorrect. Terraform can load providers from its **plugin cache** (usually `.terraform/providers`).

**B. Source code**

Correct. Terraform **cannot load a provider directly from source code**; the provider must be built into a binary before Terraform can use it.

**C. Plugins directory**

Incorrect. Terraform can load providers from a **local plugins directory** if configured.

**D. Official HashiCorp distribution on releases.hashicorp.com**

Incorrect. Terraform can download providers directly from the **official HashiCorp releases**.

**Question: 263**

**CertyIQ**

Which of the following arguments are required when declaring a Terraform output?

- A. sensitive
- B. value
- C. default
- D. description

**Answer: B**

**Explanation:**

**A. sensitive**

Incorrect. The sensitive argument is optional. It hides the output value in the CLI and logs but is not mandatory.

**B. value**

Correct. The value argument is required because it defines what the output will return. Without it, the output cannot function.

**C. default**

Incorrect. The default argument is used for variables, not outputs, so it cannot be applied here.

**D. description**

Incorrect. The description argument is optional and provides a human-readable explanation of the output, but it is not required.

**Question: 264**

**CertyIQ**

Which configuration consistency error does terraform validate report?

- A. Differences between local and remote state.
- B. A mix of spaces and tabs in configuration files.
- C. Terraform module isn't the latest version.
- D. Declaring a resource identifier more than once.

**Answer: D**

**Explanation:**

**A. Differences between local and remote state**

Incorrect. Detecting differences between real infrastructure and state is handled during plan or apply, not by validate.

**B. A mix of spaces and tabs in configuration files**

Incorrect. Formatting issues are handled by terraform fmt, not terraform validate.

**C. Terraform module isn't the latest version**

Incorrect. Terraform does not require modules to be the latest version, and validate does not check for version updates.

**D. Declaring a resource identifier more than once**

Correct. terraform validate checks for syntax and internal consistency errors, including duplicate resource declarations within the configuration.

**Question: 265**

**CertyIQ**

Once you configure a new Terraform backend with a terraform code block, which command(s) should you use to migrate the state file?

- A. terraform init
- B. terraform push
- C. terraform destroy, then terraform apply
- D. terraform apply

**Answer: A**

**Explanation:**

**A. terraform init**

Correct. After changing or adding a backend configuration, you run terraform init. Terraform will detect the backend change and prompt you to migrate the existing state to the new backend.

**B. terraform push**

Incorrect. terraform push is deprecated and not used for backend state migration.

**C. terraform destroy, then terraform apply**

Incorrect. Destroying and recreating infrastructure is unnecessary and risky. Backend migration does not require resource destruction.

**D. terraform apply**

Incorrect. terraform apply manages infrastructure changes, not backend migration.

**Question: 266**

**CertyIQ**

You want to know from which paths Terraform is loading providers referenced in your Terraform configuration (\*.tf files). You need to enable additional logging messages to find this out. Which of the following would achieve this?

- A. Set verbose logging for each provider in your Terraform configuration.
- B. Set the environment variable TF\_LOG\_PATH.
- C. Set the environment variable TF\_LOG=TRACE.
- D. Set the environment variable TF\_VAR\_log=TRACE.

**Answer: C**

**Explanation:**

**A. Set verbose logging for each provider in your Terraform configuration**

Incorrect. Terraform does not support enabling verbose logging per provider within the configuration files.

**B. Set the environment variable TF\_LOG\_PATH**

Incorrect. TF\_LOG\_PATH only specifies where to write the log file. It does not enable detailed logging by itself.

**C. Set the environment variable TF\_LOG=TRACE**

Correct. Setting TF\_LOG=TRACE enables the most detailed logging level in Terraform, which includes information about provider installation and loading paths.

**D. Set the environment variable TF\_VAR\_log=TRACE**

Incorrect. TF\_VAR\_ variables are used to pass input variables to Terraform, not to control logging behavior.

**Question: 267**

**CertyIQ**

In a HCP Terraform/Terraform Cloud workspace linked to a version control repository, speculative plan runs start automatically when you merge or commit changes to version control.

- A. True
- B. False

**Answer: B**

**Explanation:**

### A. True

Incorrect. When you merge or commit changes to the main branch, Terraform Cloud typically runs a normal plan (and possibly apply, depending on settings). Speculative plans are triggered for pull requests before merging.

### B. False

Correct. Speculative plans run automatically for pull requests, not for merges or direct commits to the main branch.

## Question: 268

CertyIQ

When you run terraform apply, the Terraform CLI will print output values from both the root module and any child modules.

A. True

B. False

### Answer: B

#### Explanation:

#### A. True

Incorrect. Terraform only displays **outputs defined in the root module** after terraform apply. Outputs from child modules are not printed unless they are explicitly exposed through the root module.

#### B. False

Correct. The CLI prints only **root module outputs**. To see child module outputs, you must define corresponding outputs in the root module that reference them.

## Question: 269

CertyIQ

Your risk management organization requires that new AWS S3 buckets must be private, and their contents must be encrypted at rest. How can HCP Terraform/Terraform Cloud automatically and proactively enforce this security control?

A. By adding variables to each HCP Terraform/Terraform Cloud workspace to ensure these settings are always enabled.

B. With an S3 module with proper settings for buckets.

C. Auditing cloud storage buckets with a vulnerability scanning tool.

D. With a Sentinel policy, which runs before every apply.

### Answer: D

#### Explanation:

**A. By adding variables to each HCP Terraform/Terraform Cloud workspace to ensure these settings are always enabled**

Incorrect. Variables can help standardize configuration, but they do not guarantee enforcement. Users could still override or misconfigure resources.

**B. With an S3 module with proper settings for buckets**

Incorrect. While a module encourages best practices, it does not prevent someone from creating an S3 bucket resource outside the module with insecure settings.

**C. Auditing cloud storage buckets with a vulnerability scanning tool**

Incorrect. Auditing is reactive and occurs after deployment. The requirement is to enforce controls proactively before infrastructure is created.

**D. With a Sentinel policy, which runs before every apply**

Correct. Sentinel policies enforce policy-as-code rules during the Terraform workflow and can prevent applies that violate requirements, such as creating public or unencrypted S3 buckets.

**Question: 270**

CertyIQ

If you don't use the local Terraform backend, where else can Terraform save resource state?

- A. In an environment variable.
- B. In memory.
- C. In a remote location configured in the .terraformrc file, such as HCP Terraform or a cloud storage system.
- D. In a remote location configured in the terraform block, such as HCP Terraform or a cloud storage system.

**Answer: D**

**Explanation:**

**A. In an environment variable**

Incorrect. Environment variables are used to pass configuration values, not to store Terraform state.

**B. In memory**

Incorrect. Terraform state must persist between runs. Storing it only in memory would not allow future operations.

**C. In a remote location configured in the .terraformrc file, such as HCP Terraform or a cloud storage system**

Incorrect. The .terraformrc file configures CLI behavior and credentials, not backend state storage.

**D. In a remote location configured in the terraform block, such as HCP Terraform or a cloud storage system**

Correct. Remote backends are configured in the terraform block and allow Terraform to store state in centralized locations like HCP Terraform, Amazon S3, Azure Storage, or similar systems.

**Question: 271**

CertyIQ

Which of the following statements about Terraform modules is not true?

- A. Modules must be publicly accessible.
- B. You can call the same module multiple times.
- C. Modules can call other modules.
- D. A module is a container for one or more resources.

**Answer: A**

**Explanation:**

**A. Modules must be publicly accessible**

This statement is not true. Terraform modules do not have to be publicly accessible. They can exist locally on your filesystem, inside a private Git repository, in a private module registry, or in Terraform Cloud's private registry. Many organizations use private modules to protect internal infrastructure logic and standards. Therefore, public accessibility is not a requirement for modules.

**B. You can call the same module multiple times**

This statement is true. Terraform allows you to reuse a module multiple times within the same configuration. Each module call can use different input variables, making modules highly reusable and flexible for deploying similar infrastructure in different environments or regions.

**C. Modules can call other modules**

This statement is true. Terraform supports nested modules, meaning one module can reference and use another module. This enables composition and better organization of complex infrastructure.

**D. A module is a container for one or more resources**

This statement is true. A module groups together related resources into a single logical unit. Even the root configuration directory is technically a module (the root module).

**Question: 272**

**CertyIQ**

How does Terraform determine dependencies between resources when it creates an execution plan?

- A. Terraform builds a resource graph based on your configuration and your state file (if present).
- B. Terraform builds a resource graph based on your configuration and your state file (if present).
- C. Terraform requires resource dependencies to be defined as modules and sourced in order.
- D. Terraform requires all dependencies between resources to be specified using the depends\_on parameter.

**Answer: A**

**Explanation:**

**A. Terraform builds a resource graph based on your configuration and your state file (if present).**

Correct. Terraform automatically analyzes references between resources in your configuration and builds a dependency graph. It also considers the current state (if it exists) to understand relationships and determine the correct order of operations.

**B. Terraform requires resources in your configuration to be listed in the order they will be created to**

**determine dependencies.**

Incorrect. Terraform does not rely on the order of resources in configuration files. The dependency graph determines execution order.

**C. Terraform requires resource dependencies to be defined as modules and sourced in order.**

Incorrect. Modules help organize infrastructure, but they are not required for dependency management.

**D. Terraform requires all dependencies between resources to be specified using the depends\_on parameter.**

Incorrect. Terraform automatically infers most dependencies through references. The depends\_on argument is only needed for explicit dependencies that cannot be inferred.

**Question: 273**

**CertyIQ**

Which of the following can you do with terraform plan? (Choose two.)

- A. Schedule Terraform to run at a planned time in the future.
- B. View the execution plan and check if the changes match your expectations.
- C. Execute a plan in a different workspace.
- D. Save a generated execution plan to apply later.

**Answer: B,D**

**Explanation:**

**A. Schedule Terraform to run at a planned time in the future**

Incorrect. terraform plan does not schedule future runs. Scheduling must be handled externally through CI/CD pipelines or automation tools.

**B. View the execution plan and check if the changes match your expectations**

Correct. terraform plan shows the proposed changes before they are applied, allowing you to review and verify them.

**C. Execute a plan in a different workspace**

Incorrect. terraform plan runs in the currently selected workspace. You must switch workspaces before running it.

**D. Save a generated execution plan to apply later**

Correct. You can save the execution plan to a file and later use it with terraform apply to ensure exactly that plan is executed.

**Question: 274**

**CertyIQ**

Which statement describes a goal of infrastructure as code (IaC)?

- A. The programmatic configuration of resources.

- B. Defining a vendor-agnostic API.
- C. Write once, run anywhere.
- D. A pipeline process to test and deliver software.

**Answer: A**

**Explanation:**

**A. The programmatic configuration of resources**

Correct. A primary goal of IaC is to define and manage infrastructure through machine-readable configuration files rather than manual processes. This enables automation, version control, repeatability, and consistency.

**B. Defining a vendor-agnostic API**

Incorrect. While some IaC tools aim to be cloud-agnostic, defining a vendor-agnostic API is not the core goal of IaC itself.

**C. Write once, run anywhere**

Incorrect. This phrase is commonly associated with platform portability concepts, not specifically with IaC principles.

**D. A pipeline process to test and deliver software**

Incorrect. That describes CI/CD practices, not the primary goal of infrastructure as code.

**Question: 275**

**CertyIQ**

Which backend does the Terraform CLI use by default?

- A. Remote.
- B. Local.
- C. HCP Terraform/Terraform Cloud.
- D. HTTP.
- E. Depends on the cloud provider.

**Answer: B**

**Explanation:**

**A. Remote**

Incorrect. Terraform does not use a remote backend unless you explicitly configure one.

**B. Local**

Correct. By default, Terraform uses the **local backend**, which stores the state file on your local machine in the working directory.

**C. HCP Terraform/Terraform Cloud**

Incorrect. This backend must be explicitly configured; it is not the default.

**D. HTTP**

Incorrect. The HTTP backend must be explicitly configured and is not used by default.

**E. Depends on the cloud provider**

Incorrect. The backend is independent of the cloud provider and does not change automatically.

**Question: 276**

**CertyIQ**

You must initialize your working directory before running terraform validate.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. While it is best practice to run terraform init first, it is not strictly required in all cases. terraform validate checks syntax and internal consistency and can run without initialization if no external modules or providers are required.

**B. False**

Correct. Initialization is recommended but not mandatory for terraform validate to execute.

**Question: 277**

**CertyIQ**

When does Terraform create the .terraform.lock.hcl file?

- A. After your first terraform plan.
- B. After your first terraform apply.
- C. After your first terraform init.
- D. Whenever you enable state locking.

**Answer: C**

**Explanation:**

**A. After your first terraform plan**

Incorrect. The lock file is created before planning occurs, during initialization.

**B. After your first terraform apply**

Incorrect. The lock file is not generated during apply; it is created earlier in the workflow.

**C. After your first terraform init**

Correct. Terraform creates the .terraform.lock.hcl file during terraform init when it selects and installs provider

versions. The file records the exact provider versions to ensure consistent future runs.

**D. Whenever you enable state locking**

Incorrect. State locking is unrelated to the provider dependency lock file.

**Question: 278**

**CertyIQ**

You just scaled your VM infrastructure and realized you set the count variable to the wrong value. You correct the value and save your change. What do you do next to make your infrastructure match your configuration?

- A. Inspect all Terraform outputs to make sure they are correct.
- B. Inspect your Terraform state because you want to change it.
- C. Run terraform apply and confirm the planned changes.
- D. Reinitialize because your configuration has changed.

**Answer: C**

**Explanation:**

**A. Inspect all Terraform outputs to make sure they are correct**

Incorrect. Outputs only display values; they do not reconcile infrastructure with configuration.

**B. Inspect your Terraform state because you want to change it**

Incorrect. You should not manually modify the state file. Terraform automatically updates state when changes are applied.

**C. Run terraform apply and confirm the planned changes**

Correct. After correcting the configuration, you run terraform apply. Terraform will compare the configuration with the current state, generate a plan to adjust the resource count, and then apply the necessary changes.

**D. Reinitialize because your configuration has changed**

Incorrect. terraform init is only required when changing backends, modules, or providers — not for modifying resource arguments like count.

**Question: 279**

**CertyIQ**

You have a Terraform configuration that defines a single virtual machine with no references to it. You have run terraform apply to create the resource, then removed the resource definition from your Terraform configuration file.

What will happen when you run terraform apply in the working directory again?

- A. Terraform will destroy the virtual machine.
- B. Terraform will remove the virtual machine from the state file, but the resource will still exist.
- C. Terraform will error.
- D. Nothing.

**Answer: A**

**Explanation:**

**A. Terraform will destroy the virtual machine.**

Correct. Since the resource no longer exists in the configuration but is still present in the state file, Terraform will plan to destroy it to make the real infrastructure match the configuration.

**B. Terraform will remove the virtual machine from the state file, but the resource will still exist.**

Incorrect. Terraform does not automatically remove resources from state without affecting real infrastructure. That would require a manual `terraform state rm` command.

**C. Terraform will error.**

Incorrect. Terraform will not error; it will generate a plan showing that the resource will be destroyed.

**D. Nothing.**

Incorrect. Terraform detects that the resource exists in state but not in configuration and will take action.

## Question: 280

CertyIQ

Which of these commands makes your code more human-readable?

- A. `terraform output`
- B. `terraform fmt`
- C. `terraform validate`
- D. `terraform show`

**Answer: B**

**Explanation:**

**A. `terraform output`**

Incorrect. `terraform output` displays output variable values after apply. It does not format or improve code readability.

**B. `terraform fmt`**

Correct. `terraform fmt` reformats Terraform configuration files to a canonical style, improving consistency and human readability.

**C. `terraform validate`**

Incorrect. `terraform validate` checks for syntax and configuration errors but does not change formatting.

**D. `terraform show`**

Incorrect. `terraform show` displays the contents of a state file or execution plan; it does not modify or format configuration files.

### Question: 281

If a DevOps team adopts AWS CloudFormation as their standardized method for provisioning public cloud resources, which of the following scenarios poses a challenge for this team?

- A. The DevOps team is tasked with automating a manual, web console-based provisioning process.
- B. The team is asked to manage a new application stack built on AWS-native services.
- C. The organization decides to expand into Azure and wishes to deploy new infrastructure.
- D. The team is asked to build a reusable code base that can deploy resources into any AWS region.

**Answer: C**

**Explanation:**

**A. The DevOps team is tasked with automating a manual, web console-based provisioning process**

Incorrect. CloudFormation is specifically designed to automate infrastructure provisioning, so this aligns well with its purpose.

**B. The team is asked to manage a new application stack built on AWS-native services**

Incorrect. CloudFormation is built for AWS-native services, so this is not a challenge.

**C. The organization decides to expand into Azure and wishes to deploy new infrastructure**

Correct. AWS CloudFormation is AWS-specific and does not support provisioning resources in Azure. Expanding to another cloud provider would require adopting another tool or switching to a multi-cloud solution.

**D. The team is asked to build a reusable code base that can deploy resources into any AWS region**

Incorrect. CloudFormation supports parameterization and templates that can be reused across AWS regions.

### Question: 282

A child module can always access variables declared in its parent module.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. Terraform modules are isolated by design. A child module does **not** automatically inherit or access variables from its parent. Any value needed by the child must be explicitly passed from the parent module.

**B. False**

Correct. Child modules cannot directly access parent variables unless those values are explicitly provided as input variables. This isolation ensures modularity and reusability.

**Question: 283**

Which of these actions are forbidden when the Terraform state file is locked? (Choose three.)

- A. terraform apply
- B. terraform plan
- C. terraform fmt
- D. terraform destroy
- E. terraform state list
- F. terraform validate

**Answer: A,B,D**

**Explanation:**

**A. terraform apply**

Correct. terraform apply modifies infrastructure and updates the state file. It requires a state lock, so it cannot run if the state is already locked.

**B. terraform plan**

Correct. Although terraform plan primarily reads the state, it still attempts to acquire a state lock by default to prevent concurrent changes. If the state is locked, the command will fail.

**C. terraform fmt**

Incorrect. terraform fmt only reformats configuration files and does not interact with the state file.

**D. terraform destroy**

Correct. terraform destroy modifies infrastructure and updates the state file, so it requires a state lock and cannot run if the state is already locked.

**E. terraform state list**

Incorrect. This command reads the state file but does not modify it, so it does not require a lock in the same way.

**F. terraform validate**

Incorrect. terraform validate checks configuration syntax and does not interact with the state file.

**Question: 284**

Which of the following commands would you use to access all of the attributes and details of a resource managed by Terraform?

- A. terraform get 'provider\_type.name'
- B. terraform state list 'provider\_type.name'
- C. terraform state show 'provider\_type.name'
- D. terraform state list

**Answer: C**

**Explanation:**

**A. terraform get 'provider\_type.name'**

Incorrect. terraform get is used to download and update modules referenced in your configuration. It does not inspect resources or their attributes.

**B. terraform state list 'provider\_type.name'**

Incorrect. terraform state list only lists the addresses of resources in the state file. It does not show attribute values or detailed information.

**C. terraform state show 'provider\_type.name'**

Correct. terraform state show displays all attributes and current values for a specific resource as stored in the Terraform state. This is the command used to inspect full resource details.

**D. terraform state list**

Incorrect. This command lists all resources tracked in the state file but provides no details about their attributes.

### Question: 285

CertyIQ

Terraform providers are part of the Terraform core binary.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. Providers are **not** built into the Terraform core binary. They are separate plugins that Terraform downloads during terraform init. This allows Terraform to support many different platforms and services without modifying the core binary.

**B. False**

Correct. Terraform Core and Terraform Providers are separate components. Providers are external plugins that interact with APIs (like AWS, Azure, etc.), while Terraform Core handles planning and state management.

### Question: 286

CertyIQ

You want to define multiple data disks as nested blocks inside the resource block for a virtual machine. What Terraform feature would help you define the blocks using the values in a variable?

- A. Count arguments
- B. Dynamic blocks

- C. Collection functions
- D. Local values

**Answer: B**

**Explanation:**

**A. Count arguments**

Incorrect. The count argument is used to create multiple instances of a resource, not multiple nested blocks inside a single resource.

**B. Dynamic blocks**

Correct. **Dynamic blocks** allow you to generate multiple nested blocks inside a resource by iterating over a variable or collection. This is the appropriate feature for defining multiple data disks within a single virtual machine resource.

**C. Collection functions**

Incorrect. Collection functions manipulate data structures (lists, maps, etc.), but they do not create nested configuration blocks.

**D. Local values**

Incorrect. Local values help simplify expressions and reuse computed values, but they do not dynamically generate nested blocks.

**Question: 287**

**CertyIQ**

Before you can use a new backend or HCP Terraform/Terraform Cloud integration, you must first execute terraform init.

- A. True
- B. False

**Answer: A**

**Explanation:**

**A. True**

Correct. When configuring or changing a backend (including HCP Terraform/Terraform Cloud), you must run terraform init to initialize the working directory. This command configures the backend, downloads required providers, and sets up necessary integration.

**B. False**

Incorrect. Backend configuration changes require reinitialization using terraform init.

**Question: 288**

**CertyIQ**

You have deployed a new webapp with a public IP address on a cloud provider. However, you did not create any outputs for your code. What is the best method to quickly find the IP address of the resource you deployed?

- A. Run `terraform state list` to find the name of the resource, then `terraform state show` to find the attributes including public IP address.
- B. In a new folder, use the `terraform_remote_state` data source to load in the state file, then write an output for each resource that you find the state file.
- C. Run `terraform output ip_address` to view the result.
- D. Run `terraform destroy` then `terraform apply` and look for the IP address in stdout.

**Answer: A**

**Explanation:**

**A. Run `terraform state list` to find the name of the resource, then `terraform state show` to find the attributes including public IP address.**

Correct. Since no outputs were defined, the fastest way to retrieve the IP address is to inspect the Terraform state. First, use `terraform state list` to identify the resource name, then use `terraform state show` to view all its stored attributes, including the public IP.

**B. In a new folder, use the `terraform_remote_state` data source to load in the state file, then write an output for each resource that you find in the state file.**

Incorrect. This approach is unnecessarily complex and not the quickest way to retrieve the IP address.

**C. Run `terraform output ip_address`**

Incorrect. This command only works if an output named `ip_address` was already defined in the configuration.

**D. Run `terraform destroy` then `terraform apply` and look for the IP address in stdout.**

Incorrect. Destroying and recreating infrastructure just to find an IP address is unnecessary and risky.

## Question: 289

CertyIQ

Which of the following is not a valid Terraform collection type?

- A. tree
- B. set
- C. map
- D. list

**Answer: A**

**Explanation:**

**A. tree**

Correct. `tree` is not a supported collection type in Terraform. Terraform's type system includes specific primitive and collection types, but `tree` is not one of them. While hierarchical data structures can be represented using nested maps and objects, there is no built-in `tree` collection type in Terraform.

**B. set**

Incorrect. set is a valid Terraform collection type. A set is an unordered collection of unique values, meaning duplicate elements are automatically removed. Sets are useful when order does not matter but uniqueness does.

**C. map**

Incorrect. map is a valid collection type. A map stores key-value pairs, where each key maps to a specific value. It is commonly used for structured configurations like tags or environment variables.

**D. list**

Incorrect. list is also a valid collection type. A list is an ordered sequence of values and allows duplicate elements. The position of each item is preserved.

**Question: 290**

**CertyIQ**

Which type of block fetches or computes information for use elsewhere in a Terraform configuration?

- A. local
- B. data
- C. resource
- D. provider

**Answer: B**

**Explanation:**

**A. local**

Incorrect. A local block defines local values within a configuration. It helps simplify expressions and reuse computed values, but it does not fetch external information from providers or APIs.

**B. data**

Correct. A data block is used to fetch or compute information from a provider without creating or managing the resource. It retrieves existing infrastructure data so it can be referenced elsewhere in the configuration.

**C. resource**

Incorrect. A resource block is used to create, update, or delete infrastructure components. It manages infrastructure rather than just fetching information.

**D. provider**

Incorrect. A provider block configures the provider (such as AWS or Azure) that Terraform will use to interact with APIs. It does not fetch resource-specific data itself.

**Question: 291**

**CertyIQ**

How can a ticket-based system slow down infrastructure provisioning and limit the ability to scale? (Choose two.)

- A. End-users have to request infrastructure changes.
- B. The more resources your organization needs, the more tickets your infrastructure team has to process.

- C. Ticket-based systems generate a full audit trail of the request and fulfillment process.
- D. Users can access a catalog of approved resources from dropdown lists in a request form.

**Answer: A,B**

**Explanation:**

**A. End-users have to request infrastructure changes.**

Correct. When users must submit tickets for infrastructure changes, provisioning becomes dependent on manual approval and processing. This introduces delays and reduces agility compared to automated, self-service infrastructure workflows.

**B. The more resources your organization needs, the more tickets your infrastructure team has to process.**

Correct. As demand grows, the volume of tickets increases. This creates bottlenecks for the infrastructure team, limiting scalability and slowing down provisioning.

**C. Ticket-based systems generate a full audit trail of the request and fulfillment process.**

Incorrect. An audit trail is a benefit, not a limitation. It improves governance and traceability rather than slowing scalability directly.

**D. Users can access a catalog of approved resources from dropdown lists in a request form.**

Incorrect. Providing a catalog of approved resources can actually streamline requests and improve consistency, rather than slow provisioning.

#### Question: 292

CertyIQ

Running terraform fmt without any flags in a directory with Terraform configuration files will check the formatting of those files, but will never change their contents.

- A. True
- B. False

**Answer: B**

**Explanation:**

**A. True**

Incorrect. By default, terraform fmt not only checks formatting but also automatically rewrites configuration files to match Terraform's canonical format if differences are found.

**B. False**

Correct. Running terraform fmt without flags will modify files in place to correct formatting issues.

#### Question: 293

CertyIQ

Which provider authentication method prevents credentials from being stored in the state file?

- A. Using environment variables.
- B. Specifying the login credentials in the provider block.
- C. Setting credentials as Terraform variables.
- D. None of the above.

**Answer: A**

**Explanation:**

**A. Using environment variables.**

Correct. Supplying credentials through environment variables keeps them outside of the Terraform configuration and state. This prevents sensitive values from being written into the state file.

**B. Specifying the login credentials in the provider block.**

Incorrect. Hardcoding credentials in the provider block can result in sensitive information being stored in configuration files and potentially written to state.

**C. Setting credentials as Terraform variables.**

Incorrect. Even if variables are marked sensitive, their values can still end up in the state file depending on usage. This does not reliably prevent credentials from being stored in state.

**D. None of the above.**

Incorrect. Using environment variables is a valid method to prevent credentials from being stored in the state file.

**Question: 294**

**CertyIQ**

What is a Terraform provider not responsible for?

- A. Provisioning infrastructure in multiple cloud providers.
- B. Managing actions to take based on resource differences.
- C. Understanding API interactions with a hosted service.
- D. Managing resources and data sources based on an API.

**Answer: B**

**Explanation:**

**A. Provisioning infrastructure in multiple cloud providers.**

Incorrect. Providers enable Terraform to provision infrastructure by interacting with a specific cloud or service API. Different providers support different platforms.

**B. Managing actions to take based on resource differences.**

Correct. Deciding what actions to take when differences are detected between configuration and state (such as create, update, or delete) is handled by **Terraform Core**, not the provider. Terraform Core generates the execution plan.

### C. Understanding API interactions with a hosted service.

Incorrect. Providers are responsible for understanding and handling API interactions with their respective services.

### D. Managing resources and data sources based on an API.

Incorrect. Providers define and manage resources and data sources by communicating with the service's API.

## Question: 295

CertyIQ

When using multiple configurations of the same Terraform provider, what meta-argument must you include in any non-default provider configurations?

- A. id
- B. depends\_on
- C. name
- D. alias

**Answer: D**

**Explanation:**

#### A. id

Incorrect. id is not a valid meta-argument for provider configuration. It is typically an attribute assigned to resources after creation.

#### B. depends\_on

Incorrect. depends\_on is a meta-argument used to define explicit dependencies between resources, not for provider configurations.

#### C. name

Incorrect. name is not used to distinguish multiple configurations of the same provider.

#### D. alias

Correct. When defining multiple configurations of the same provider, you must include the alias meta-argument for any non-default provider configuration. This allows Terraform to distinguish between the different provider instances.

## Question: 296

CertyIQ

Anyone can publish modules to the public Terraform Registry.

- A. True
- B. False

**Answer: A**

**Explanation:****A. True**

Correct. Anyone can publish modules to the public Terraform Registry as long as they meet the publishing requirements (such as using a supported VCS like GitHub and following the naming conventions). There is no approval process that restricts publishing to a limited group of users.

**B. False**

Incorrect. The public Terraform Registry is open for module publishing to anyone who follows the required guidelines.

**Question: 297****CertyIQ**

What type of information can be found on the Terraform Registry when using published modules?

- A. Required input variables.
- B. Outputs.
- C. Optional input variables and default values.
- D. All of the above.

**Answer: D****Explanation:****A. Required input variables**

Incorrect. While required input variables are documented on the Terraform Registry, they are not the only type of information available.

**B. Outputs**

Incorrect. Module outputs are also documented, but again, they are only part of the available information.

**C. Optional input variables and default values**

Incorrect. Optional variables and their default values are included in module documentation, but this is not the complete answer.

**D. All of the above**

Correct. The Terraform Registry displays comprehensive module documentation, including required input variables, optional variables with default values, and defined outputs.

**Question: 298****CertyIQ**

One cloud block always maps to a single HCP Terraform/Terraform Cloud workspace.

- A. True
- B. False

**Answer: A**

**Explanation:**

**A. True**

Correct. In HCP Terraform/Terraform Cloud, each cloud block in a configuration corresponds to a single workspace. This ensures that state management, runs, and integrations are scoped to that specific workspace.

**B. False**

Incorrect. A cloud block does not span multiple workspaces; it is always tied to one workspace.

**Question: 299**

**CertyIQ**

A configuration is shown in the Exhibit space of this page.

```
data "vsphere_datacenter" "dc" {}

resource "vsphere_folder" "parent" {
  path          = "Production"
  type          = "vm"
  datacenter_id = _____
}
```

How would you reference an attribute from the vsphere\_datacenter data source for use with the datacenter\_id argument within the vsphere\_folder resource in the configuration?

- A. data.vsphere\_datacenter.dc
- B. vsphere\_datacenter.dc.id
- C. data.dc.id
- D. data.vsphere\_datacenter.dc.id

**Answer: D**

**Explanation:**

The correct way to reference an attribute from a vsphere\_datacenter data source for use with the datacenter\_id argument is data.vsphere\_datacenter.dc.id. This follows the standard Terraform syntax for accessing data source attributes: data.TYPE.NAME.ATTRIBUTE.

**Question: 300**

**CertyIQ**

Which of these are features of HCP Terraform/Terraform Cloud? (Choose two.)

- A. Automatic backups of configuration and state.
- B. A web-based user interface (UI).
- C. Remote state storage.
- D. Automated infrastructure deployment visualization.

**Answer: B,C**

**Explanation:**

HCP Terraform and Terraform Cloud are platforms designed to enhance collaboration, automation, and governance for Infrastructure as Code (IaC) workflows using Terraform. Among the listed options, "A web-based user interface (UI)" and "Remote state storage" are distinct and fundamental features.

Option B, "A web-based user interface (UI)," is a core component of HCP Terraform and Terraform Cloud. As a Software as a Service (SaaS) offering, the platform provides a comprehensive web console. This UI allows users to centrally manage workspaces, variables, run policies, review plan outputs, inspect deployment histories, and execute infrastructure changes without exclusive reliance on the local Terraform CLI. This centralized interface significantly improves accessibility, enhances collaboration across teams, and provides a clear, unified overview of all infrastructure operations, aligning with modern cloud management principles for improved user experience.

Option C, "Remote state storage," is another critical feature provided natively by HCP Terraform/Terraform Cloud. Terraform relies on a state file to maintain a mapping between real-world infrastructure resources and your configuration, tracking metadata and dependencies. For collaborative environments and to ensure idempotency and consistency, storing this state remotely is paramount. HCP Terraform/Terraform Cloud automatically handles remote state management, offering essential capabilities such as state locking to prevent concurrent writes, state versioning for easy rollbacks, and secure access control. This built-in functionality eliminates the complexities and risks associated with manual state file management, ensuring data integrity and reliable infrastructure deployments across multiple developers and teams.

In contrast, Option A, "Automatic backups of configuration and state," is less accurate as a primary, directly configurable user-facing feature for both elements. While the platform offers robust state versioning and internal redundancy for the state file (which acts as a form of backup/recovery for the state), the configuration (your .tf files) is typically managed in an external Version Control System (VCS) like GitHub or GitLab. Terraform Cloud integrates with these VCSs rather than performing "backups" of your source code repository itself. Similarly, Option D, "Automated infrastructure deployment visualization," is not a direct, dynamic graphical feature within the core HCP Terraform/Terraform Cloud UI. While Terraform produces a detailed textual plan output describing intended changes, and terraform graph can show resource dependencies, the platform doesn't inherently offer real-time, dynamic visual diagrams of the deployment process or a live infrastructure topology within its main interface.

Therefore, the most prominent and directly provided features from the given choices are its comprehensive web-based user interface and integrated remote state management.

**Authoritative Links for Further Research:**

**HCP Terraform Overview:** <https://cloud.hashicorp.com/products/terraform>  
**Terraform Cloud Workspaces (UI functionality):** <https://developer.hashicorp.com/terraform/cloud/workspaces>  
**Terraform State Management (Remote State):** <https://developer.hashicorp.com/terraform/cloud/workspaces/state>

What does Terraform not reference when running a terraform apply -refresh-only?

- A. Credentials.
- B. State file.
- C. Terraform resource definitions in configuration files.
- D. Cloud provider.

**Answer: C**

**Explanation:**

The correct answer is **C. Terraform resource definitions in configuration files.**

Here's a detailed justification:

The terraform apply -refresh-only command serves a very specific purpose within the Terraform workflow: to update the Terraform state file to accurately reflect the current real-world status of infrastructure resources without making any changes to that infrastructure.

**1. Why A, B, and D are referenced:**

**A. Credentials:** To interact with any cloud provider to query the current status of resources, Terraform absolutely requires appropriate authentication credentials (e.g., API keys, IAM roles, service principal secrets). Without credentials, Terraform cannot communicate with the cloud API.

**B. State file:** The refresh-only command's primary objective is to update the state file. It reads the existing state file to identify which resources it currently manages, queries the cloud provider for the actual status of those specific resources, and then writes the refreshed data back into the state file. Therefore, the state file is extensively referenced (read from and written to).

**D. Cloud provider:** The actual source of truth for the current state of infrastructure is the cloud provider itself. Terraform must communicate with the cloud provider's APIs to fetch the latest attributes and status of all managed resources.

**1. Why C is not referenced in the context of refresh-only:**

Terraform configuration files (.tf files) define the desired state of your infrastructure. When you run a standard terraform plan or terraform apply, Terraform reads these configuration files, compares the desired state with the current state (from the state file and cloud), and generates a plan of changes needed to reconcile any differences.

However, terraform apply -refresh-only explicitly bypasses the configuration file evaluation step for the purpose of generating a plan for changes. Its sole focus is on synchronizing the existing state file with the actual state of resources in the cloud. It doesn't look at your configuration files to determine if new resources should be added, existing ones modified, or old ones removed. It simply updates the recorded attributes of the resources it already knows about from the state file.

Essentially, refresh-only does not consult the configuration files to formulate a new desired state or identify any drift against the configuration. It only identifies drift between the current state file and the actual cloud resources.

In summary, terraform apply -refresh-only operates by querying the cloud provider (using credentials) about resources listed in the current state file, and then updating that state file with the findings. It specifically avoids re-evaluating the Terraform resource definitions in your configuration files because its goal is not to plan or apply changes based on a desired configuration, but merely to update the state to reflect reality.

For further research, refer to:

HashiCorp Terraform Command: terraform

apply Documentation: <https://developer.hashicorp.com/terraform/cli/commands/apply> (Specifically, look for the -refresh-only flag explanation).

HashiCorp Terraform State Documentation: <https://developer.hashicorp.com/terraform/language/state>

## Question: 302

CertyIQ

Which of the following is not a key principle of infrastructure as code (IaC)?

- A. Golden images
- B. Self-describing infrastructure
- C. Versioned infrastructure
- D. Idempotence

**Answer: A**

### Explanation:

The correct answer is A. Golden images. This is because golden images represent a common artifact or strategy used in infrastructure deployment, but they are not a fundamental principle of Infrastructure as Code (IaC) itself. IaC revolves around treating infrastructure configuration and provisioning like software development, emphasizing automated, consistent, and repeatable processes.

Let's examine why the other options are key principles and why golden images are distinct:

**Self-describing infrastructure** is a core IaC principle because infrastructure is defined using declarative configuration files (e.g., JSON, YAML, HCL). These files explicitly describe the desired state of the infrastructure components, making the infrastructure's design and configuration transparent and human-readable, as well as machine-executable. This clarity reduces ambiguity and allows teams to understand the infrastructure by reading its code.

**Versioned infrastructure** is crucial, mirroring software development practices. IaC configurations are stored in version control systems (like Git), enabling a complete history of changes, collaboration among teams, easy rollback to previous states, and clear audit trails. This ensures consistency, prevents configuration drift, and supports agile development methodologies.

**Idempotence** is perhaps the most defining characteristic of IaC. It means that applying the same IaC configuration script multiple times will always yield the same infrastructure state without causing unintended side effects or errors after the initial application. This principle ensures predictability and reliability, allowing for safe and consistent deployments, updates, or reconfigurations, as the system intelligently determines if actions need to be taken or if the desired state is already met.

**Golden images**, on the other hand, are pre-configured, pre-hardened, and often pre-patched virtual machine or container images. They serve as a base template for deploying instances, ensuring a consistent starting point. While golden images are frequently used in conjunction with IaC to provide the base operating system and software stack, they are an output or artifact of an infrastructure pipeline, rather than a principle guiding how that infrastructure is defined and managed through code. IaC principles dictate how you declare, version, and manage the deployment of instances based on these golden images, and how you configure them further. Using IaC to build and manage the golden images themselves, and then provision from them, enhances the overall automation strategy, but the image itself isn't a principle.

In summary, golden images are a strategy for achieving consistency and immutability often enabled by IaC, but not a fundamental guiding principle of how IaC functions or defines infrastructure through code.

## Authoritative Links for further research:

### What is Infrastructure as Code?

AWS: <https://aws.amazon.com/devops/what-is-iac/>

Microsoft Azure: <https://learn.microsoft.com/en-us/devops/deliver/what-is-infrastructure-as-code>

### Idempotence in IaC:

Puppet: <https://www.puppet.com/blog/infrastructure-as-code-idc-and-idempotence>

### Golden Images (in context of IaC):

HashiCorp Blog: <https://www.hashicorp.com/blog/packer-0-12-0-released> (mentioning Packer for building golden images, which is often part of an IaC workflow)

## Question: 303

CertyIQ

You use a Cloud provider account that is shared with other team members. You previously used Terraform to create a load balancer that listens on port 80. After some application changes, you updated the Terraform code to change the port to 443.

You run `terraform plan` and see that the execution plan shows the port changing from 80 to 443 like you intended, and step away to grab some coffee.

In the meantime, another team member manually changes the load balancer port to 443 through the Cloud provider console before you get back to your desk.

What will happen when you run `terraform apply` upon returning to your desk?

- A. Terraform will not make any changes to the load balancer and will update the state file to reflect the manual change.
- B. Terraform will fail with an error because the state file is no longer accurate.
- C. Terraform will change the load balancer port to 80, and then change it back to 443.
- D. Terraform will recreate the load balancer.

## Answer: C

### Explanation:

When `terraform apply` is executed, Terraform follows a specific reconciliation process. It doesn't simply execute the previously generated plan verbatim without re-evaluation.

Here's a detailed breakdown of what happens:

1. **Implicit Refresh:** The `terraform apply` command always begins with an implicit `terraform refresh`. During this phase, Terraform queries the cloud provider (in this case, for the load balancer) to fetch its actual current configuration. It discovers that the load balancer's port is now 443 due to the manual intervention, while its own `terraform.tfstate` file still records the port as 80.
2. **Drift Detection:** This discrepancy between the `terraform.tfstate` file (port 80) and the actual cloud infrastructure (port 443) is identified as "drift."
3. **Reconciling Desired vs. Actual vs. State:** Terraform's goal is to bring the actual infrastructure in line with the desired state defined in the HCL configuration.

**Desired State (HCL):** Port 443

**Actual State (Cloud):** Port 443

**Recorded State (tfstate file):** Port 80

1. **The "Revert and Reapply" Mechanism:** While ideally Terraform would recognize that the actual state already matches the desired state and simply update the state file, certain cloud providers or resource types in Terraform can exhibit a "revert and reapply" behavior in drift scenarios. This happens because Terraform's core logic is to apply the transformation from the recorded state in

the tfstate file to the desired state in the HCL. The terraform plan you ran before the manual change indicated a transition from 80 to 443.

2. **Provider Interpretation of Updates:** When Terraform instructs the provider to "update the port from 80 to 443," the provider's update logic might not always be perfectly idempotent with respect to out-of-band changes. If the provider's API or its internal logic isn't designed to handle the scenario where the target attribute already matches the desired state, it might perform a two-step process:

It might first attempt to reconcile the resource to the state it expected it to be in (i.e., port 80, as recorded in the state file). This can manifest as the load balancer port being temporarily changed back to 80.

Immediately after, it then applies the final desired state from the HCL configuration, changing the port from 80 to 443.

1. **Final State:** As a result, the load balancer's port will temporarily revert to 80 and then be changed back to 443, aligning both the cloud infrastructure and the updated terraform.tfstate file with the desired configuration. This action, although seemingly redundant, is a known behavior in specific drift scenarios where Terraform enforces the explicit path of change defined by its state and configuration.

Therefore, the correct answer is C, as Terraform attempts to apply the planned transformation from its last known state, potentially causing a temporary reversion before achieving the final desired state.

#### Authoritative Links for Further Research:

**Terraform State:** Understanding the role of the state file is fundamental.

HashiCorp Learn - Terraform State: <https://developer.hashicorp.com/docs/terraform/concepts/state>

**Terraform Refresh and Drift:** Explains how Terraform detects and handles discrepancies.

HashiCorp Docs - terraform refresh: <https://developer.hashicorp.com/docs/terraform/cli/commands/refresh>

HashiCorp Blog - Detecting and Correcting Configuration Drift with

Terraform: <https://www.hashicorp.com/blog/detecting-and-correcting-configuration-drift-with-terraform> (While it states apply updates the state file, it doesn't preclude the revert/reapply for certain providers/resources).

#### Question: 304

CertyIQ

When using Terraform to deploy resources into Azure, which scenarios are true regarding state files?

- A. Changing resources via the Azure Cloud Console records the change in the current state file.
- B. When you change a resource via the Azure Cloud Console. Terraform records the changes in a new state file.
- C. When you change a Terraform-managed resource via the Azure Cloud Console. Terraform updates the state file to reflect the change during the next plan or apply.
- D. Changing resources via the Azure Cloud Console does not update current state file.

**Answer: C**

#### Explanation:

Terraform state files are crucial for tracking the infrastructure resources it manages. They serve as a mapping between your configuration files and the real-world resources existing in the cloud provider (Azure, in this case). The state file records metadata about your infrastructure, improves performance for large infrastructures, and, most importantly, keeps track of the actual properties of your deployed resources.

When a Terraform-managed resource is modified directly through the Azure Cloud Console (or CLI/API), this is referred to as "out-of-band" or "manual" modification. Terraform itself does not have a real-time monitoring

mechanism actively watching for such changes in your cloud environment. Therefore, the state file is not automatically updated the moment a change is made in Azure.

The update process for the state file occurs when you next execute a terraform plan or terraform apply command. During a terraform plan operation, Terraform fetches the current actual state of your infrastructure resources from Azure. It then compares this actual state with the desired state defined in your Terraform configuration files and with the last known state recorded in its state file. If a discrepancy is found — meaning the actual resource in Azure differs from what Terraform believes it should be or what's recorded in its state — Terraform identifies this as "drift."

The terraform plan output will show what actions Terraform proposes to take to reconcile this drift, typically aiming to bring the Azure resource back into alignment with your Terraform configuration. If you then proceed with terraform apply, Terraform will execute these proposed changes in Azure. Upon successful completion of the apply operation, Terraform updates its state file to reflect the new, reconciled state of the resources. This ensures that the state file accurately reflects the infrastructure as managed by Terraform after the apply operation has concluded.

Therefore, option C correctly describes this behavior: "When you change a Terraform-managed resource via the Azure Cloud Console. Terraform updates the state file to reflect the change during the next plan or apply." The detection happens during plan, and the update of the state file occurs as a result of the apply operation, after Terraform has taken action to reconcile the resource. Options A and B are incorrect because Terraform doesn't automatically record direct changes; it's not a real-time monitoring system. Option D is incorrect because the state file does get updated, but only when Terraform interacts with the cloud provider during plan or apply to discover and reconcile changes.

#### Authoritative Links for Further Research:

**Terraform State:** <https://developer.hashicorp.com/terraform/language/state>

**Terraform Drift Detection:** <https://developer.hashicorp.com/terraform/cli/commands/plan#detecting-and-managing-drift>

**Terraform Apply:** <https://developer.hashicorp.com/terraform/cli/commands/apply>

#### Question: 305

CertyIQ

You have developed a new cloud-based service that uses proprietary APIs and want to use Terraform to create, manage, and delete users from the service. How can Terraform interact with the service?

- A. Terraform can manage users for any service that is hosted on a public cloud provider.
- B. Develop and publish a customer provider to interact with the service using its proprietary APIs.

**Answer: B**

#### Explanation:

Terraform is a powerful Infrastructure as Code (IaC) tool designed for provisioning and managing resources across a multitude of platforms and services. Its core functionality relies on a plugin-based architecture, where "providers" act as the essential bridge between Terraform configurations and external APIs. A Terraform provider is essentially a plugin that understands the Application Programming Interface (API) of a specific service and translates Terraform's desired state configurations into specific API calls to manage resources on that service.

For widely adopted services like AWS, Azure, Google Cloud, or Kubernetes, official or community-maintained providers already exist, enabling seamless resource management. However, the scenario describes a new

cloud-based service that utilizes proprietary APIs. This crucial detail signifies that the service's interface for managing resources, such as users, is unique and not standardized or universally recognized. Consequently, no pre-existing Terraform provider would inherently understand how to communicate with these specific, bespoke APIs.

To enable Terraform to interact with such a custom service and manage its users, a **custom provider must be developed**. This custom provider would encapsulate the necessary logic to authenticate with the proprietary service, handle the specific request formats for creating, reading, updating, and deleting (CRUD) user resources via its unique APIs, and interpret the service's API responses back into Terraform's state. This process aligns perfectly with Terraform's extensible architecture, allowing it to adapt to virtually any API-driven system.

Option A is incorrect because Terraform's capability to manage resources is not universally granted simply by a service being "hosted on a public cloud provider." Terraform does not magically gain the ability to understand proprietary APIs just because a service is cloud-hosted; it always requires a specific provider to bridge the communication gap. While many public cloud services do have Terraform providers, this is because cloud vendors or the community developed them, not an inherent Terraform feature for all services irrespective of their API design. Therefore, developing and publishing a custom provider is the definitive and only method for integrating a service with proprietary APIs into a Terraform workflow for user management.

#### Authoritative Links for Further Research:

**Terraform Providers Overview:** <https://developer.hashicorp.com/terraform/providers>

**Developing Custom Terraform Providers:** <https://developer.hashicorp.com/terraform/plugin/how-to/begin>

**What is Infrastructure as Code (IaC):** <https://www.redhat.com/en/topics/automation/what-is-infrastructure-as-code>

#### Question: 306

CertyIQ

You have just developed a new Terraform configuration for two virtual machines with a cloud provider. You would like to create the infrastructure for the first time. Which Terraform command should you run first?

- A. terraform apply
- B. terraform plan
- C. terraform show
- D. terraform init

**Answer: D**

#### Explanation:

When developing a new Terraform configuration for virtual machines and aiming to create infrastructure for the first time, the terraform init command is the essential first step. This command prepares the working directory for all subsequent Terraform operations. Its primary function is to download and install the necessary provider plugins specified in your configuration, such as those for AWS, Azure, or Google Cloud Platform. These plugins are crucial as they enable Terraform to interact with the respective cloud provider's APIs to provision, manage, and query resources like virtual machines. Without terraform init, Terraform would lack the fundamental components required to communicate with your chosen cloud provider.

Furthermore, terraform init initializes the configured backend, which dictates where Terraform state files will be stored. The state file is vital for Terraform to map your configuration to real-world infrastructure, track resource metadata, and understand their current status. The command also processes and downloads any

specified modules from their source. During its execution, terraform init creates the .terraform directory, which houses all the downloaded plugins, modules, and other essential configuration data. It also generates or updates the .terraform.lock.hcl file, which records the exact versions of providers used, ensuring consistency across environments.

Subsequent commands, like terraform plan and terraform apply, entirely depend on this initialized environment. For example, terraform plan requires the provider plugins to query the cloud provider and ascertain the proposed changes. Similarly, terraform apply needs both the plugins and the backend configuration to actually provision the resources and accurately update the state. Attempting to run terraform apply or terraform plan before terraform init would inevitably result in errors, as Terraform would lack the foundational components to function. The terraform show command is used to display plan output or the current state and is therefore not an initial setup step. Thus, terraform init lays the necessary groundwork, making it the mandatory prerequisite for any new Terraform project.

For further research:

Terraform init command documentation: <https://developer.hashicorp.com/terraform/cli/commands/init>

Terraform Providers: <https://developer.hashicorp.com/terraform/language/providers>

Terraform State: <https://developer.hashicorp.com/terraform/language/state>

### Question: 307

CertyIQ

You want to define a single input variable to capture configuration values for a server. The values must represent memory as a number, and the server name as a string.

Which variable type could you use for this input?

- A. List
- B. Map
- C. Object
- D. Terraform does not support complex input variables of different types

**Answer: C**

**Explanation:**

When defining a single input variable in Terraform to capture configuration values of different types, such as a memory value (number) and a server name (string), the object variable type is the most appropriate and robust choice.

An object type allows you to define a structured collection of named attributes, where each attribute can have its own explicit type constraint. This means you can specify that an attribute named memory must be a number and an attribute named server\_name must be a string, all within the confines of a single parent variable. This provides strong type checking and clear schema definition for your input data.

Let's look at why other options are less suitable:

**A. List:** A list variable type is an ordered collection of values, where all elements are typically expected to be of the same type or a very consistent structure. While you could technically put different types into a list if the element type is not explicitly defined, it lacks the named attributes and schema validation necessary to clearly distinguish "memory" from "server name" and enforce their specific types. It wouldn't be clear which element represents what without external context.

**B. Map:** A map variable type is an unordered collection of key-value pairs. While keys are always strings, the values can be of different types. However, a map does not allow you to define a fixed schema for its values

where you explicitly state, for instance, that the value associated with the key "memory" must be a number, and the value for "server\_name" must be a string. While `map(any)` allows mixed types, it sacrifices type safety and explicit validation that object provides for structured data. For structured data with known attribute names and types, object is preferred.

**D. Terraform does not support complex input variables of different types:** This statement is incorrect. Terraform explicitly supports complex variables that can encapsulate data of different types, with object being the primary mechanism for this.

The object type aligns perfectly with the requirement because it allows you to define a custom, explicit schema for your server configuration. For example, you would declare the variable like this:

```
variable "server_config" { description = "Configuration details for a server" type = object({ memory = number server_name = string }) }
```

This approach ensures that when users provide input for `server_config`, Terraform validates that `memory` is indeed a number and `server_name` is a string, preventing common configuration errors. This explicit schema definition enhances readability, maintainability, and reliability of Infrastructure as Code (IaC) templates, which is crucial in cloud computing environments where precise resource provisioning is paramount. Using objects promotes modularity and helps maintain consistent configurations across deployments by enforcing data structures.

For further research:

#### **Terraform Documentation on Input**

**Variables:** <https://developer.hashicorp.com/terraform/language/values/variables>

**Terraform Type Constraints (specifically Object, List,**

**Map):** <https://developer.hashicorp.com/terraform/language/expressions/type-constraints>

### **Question: 308**

**CertyIQ**

What feature stops multiple users from operating on the Terraform state at the same time?

- A. Remote state storage
- B. Provider constraints
- C. State locking
- D. Version control

**Answer: C**

#### **Explanation:**

Terraform state acts as the definitive source of truth, reflecting the real-world infrastructure managed by your configuration. If multiple users or automated processes attempt to modify this critical state concurrently, it can lead to severe issues such as race conditions, data corruption, and inconsistent infrastructure deployments. This scenario undermines the reliability and idempotency of Infrastructure as Code (IaC), potentially causing resource loss or misconfigurations.

To prevent these detrimental outcomes, Terraform employs a crucial feature known as **state locking**. State locking ensures that only one operation can acquire an exclusive lock on the Terraform state at any given moment. When a `terraform plan`, `terraform apply`, or `terraform destroy` command is executed, Terraform first attempts to acquire this lock. If the lock is successfully obtained, the operation proceeds; if another operation already holds the lock, the new operation will either wait for the lock to be released or fail, depending on the backend configuration. This mechanism implements a form of mutual exclusion, which is fundamental for maintaining consistency in distributed systems and collaborative cloud environments.

State locking is inherently tied to the use of remote backends (e.g., Amazon S3 with DynamoDB, Azure Storage Accounts with Blob locking, HashiCorp Consul, HashiCorp Cloud). These backends not only store the state file remotely but also provide the underlying infrastructure to implement robust locking primitives. While "Remote state storage" (Option A) is essential for sharing state across teams, it is the specific locking mechanism provided by the backend that actively stops concurrent modifications. Without such a mechanism, even a remotely stored state file would be vulnerable to corruption from simultaneous writes.

Conversely, "Provider constraints" (Option B) define acceptable versions for Terraform providers, ensuring compatibility and stability within the configuration, but they do not regulate access to the state file itself. "Version control" (Option D), such as Git, is excellent for managing the .tf configuration files and their history, allowing collaboration on the code. However, it does not manage the dynamic Terraform state file directly or prevent concurrent operations on the active state during live terraform apply executions. Therefore, state locking is the explicit and dedicated feature designed to guarantee serial execution of state-modifying operations, safeguarding the integrity and consistency of your infrastructure deployments in multi-user environments.

For further research:

Terraform Backends and State

Locking: <https://developer.hashicorp.com/terraform/language/settings/backends/configuration>

Example of State Locking with S3

Backend: <https://developer.hashicorp.com/terraform/language/settings/backends/s3#state-locking-and-consistency>

### Question: 309

CertyIQ

terraform validate confirms that your infrastructure matches the Terraform state file.

- A. True
- B. False

**Answer: B**

**Explanation:**

The statement "terraform validate confirms that your infrastructure matches the Terraform state file" is **False**.

terraform validate serves a distinct purpose within the Terraform workflow: it primarily checks the configuration files (.tf) in the working directory for syntactical validity and internal consistency. This includes verifying that resource attributes are of the correct type, identifying undeclared variables, and ensuring proper interpolation of values. It performs a static analysis of the code, much like a compiler or linter, to catch configuration errors before any interaction with a cloud provider or the state file.

Crucially, terraform validate does not interact with your actual cloud infrastructure or compare the configuration against the existing Terraform state file (terraform.tfstate). Its scope is limited to the local configuration definitions. It operates entirely offline, without needing cloud provider credentials or an initialized backend.

The command responsible for comparing your desired infrastructure configuration with the actual existing infrastructure and the Terraform state file is terraform plan. terraform plan queries the current state of your resources from the cloud provider APIs, compares this with the records in the state file, and then evaluates the differences against your desired configuration files. It generates an execution plan detailing what actions

Terraform proposes to take (create, modify, destroy) to bring the real infrastructure into alignment with your configuration.

Following terraform plan, the terraform apply command executes these proposed changes, reconciling the actual infrastructure with the desired state defined in your configuration. After successful execution, terraform apply then updates the Terraform state file to reflect the new actual state of your infrastructure. Therefore, while the state file acts as a record of your infrastructure, and plan/apply ensure alignment, validate is merely a preliminary syntax and configuration check.

For further research, refer to the official Terraform documentation:

**terraform validate:** <https://developer.hashicorp.com/terraform/cli/commands/validate>

**terraform plan:** <https://developer.hashicorp.com/terraform/cli/commands/plan>

**Terraform State:** <https://developer.hashicorp.com/terraform/language/state>

### Question: 310

CertyIQ

What is the provider for the resource shown in the Exhibit space on this page?

```
resource "aws_vpc" "main" {  
    name = "test"  
}
```

- A. test
- B. main
- C. vpc
- D. aws

**Answer: D**

### Question: 311

CertyIQ

terraform validate confirms the syntax of Terraform files.

- A. True
- B. False

**Answer: A**

**Explanation:**

The statement "terraform validate confirms the syntax of Terraform files" is **True**.

**Detailed Justification:**

terraform validate is a fundamental command in the Terraform workflow designed specifically to check the configuration files in a directory for syntactical correctness and internal consistency. It performs static analysis on the Terraform code written in HashiCorp Configuration Language (HCL). This means it parses the

files to ensure they conform to the HCL grammar, verifying that blocks are properly opened and closed, arguments are correctly assigned, and resource types, data sources, variables, outputs, and module references are declared in a valid format.

Beyond just basic syntax, terraform validate also checks for other configuration validity issues. For instance, it ensures that variable declarations are well-formed, provider configurations are correctly specified (e.g., required arguments are present), and expressions are parsable. Crucially, this command **does not** interact with any remote state or cloud provider APIs. It operates purely on the local configuration files, making it a fast and efficient way to catch errors early in the development cycle.

This early validation is invaluable in an Infrastructure as Code (IaC) context, especially when managing resources across various cloud platforms like AWS, Azure, or Google Cloud. By confirming the syntax and basic configuration validity locally, developers can identify and fix common mistakes before attempting more resource-intensive operations like terraform plan or terraform apply. While terraform plan also implicitly performs a validation step, terraform validate is a standalone command specifically for this purpose, providing quicker feedback and allowing it to be integrated easily into pre-commit hooks or Continuous Integration (CI) pipelines. This ensures that only syntactically sound code proceeds through the deployment pipeline, reducing the risk of errors during actual infrastructure provisioning.

For further research, refer to the official HashiCorp documentation:

**terraform validate command:** <https://developer.hashicorp.com/terraform/cli/commands/validate>

**Terraform CLI Overview:** <https://developer.hashicorp.com/terraform/cli>

## Question: 312

CertyIQ

You can access state stored in a local backend by using the terraform\_remote\_state data source.

- A. True
- B. False

**Answer: B**

**Explanation:**

Here's a detailed justification for why accessing local backend state with the terraform\_remote\_state data source is incorrect:

The terraform\_remote\_state data source is specifically designed to retrieve state data from remote backends, not local backends. Local backends store Terraform state files on the local machine, typically in a file named terraform.tfstate. They are intended for development, experimentation, and single-user environments.

The primary purpose of terraform\_remote\_state is to access state information from remote backends like AWS S3, Azure Blob Storage, HashiCorp Cloud, or Google Cloud Storage. These backends provide centralized, versioned, and often encrypted storage for state files, enabling collaboration among team members and managing infrastructure at scale.

To access the outputs of a Terraform configuration using a local backend, you do not use terraform\_remote\_state. Instead, the configurations would typically be combined within the same workspace or the state file can be accessed via the terraform output command. The terraform output command retrieves the values of output variables defined in a Terraform configuration.

The concept of remote state management is crucial for collaboration and avoiding state corruption. Using a remote backend ensures that all team members have access to the same, up-to-date state file, preventing

conflicts and data loss. Because `terraform_remote_state` is tailored for this specific purpose, it cannot be used with local backends. It requires the configuration of a specific remote backend during initialization.

The statement is false because using `terraform_remote_state` with a local backend will not work and is not the intended use of that data source. It's essential to understand the difference between local and remote backends and when to use the `terraform_remote_state` data source.

For further information, consult the official Terraform documentation:

[Terraform State](#)  
[terraform\\_remote\\_state](#)  
[Backends Overview](#)

### Question: 313

CertyIQ

You have a list of numbers that represents the number of free CPU cores on each virtual cluster: `numcpus = [ 18, 3, 7, 11, 2 ]`

What Terraform function could you use to select the largest number from the list?

- A. `high[numcpus]`
- B. `max(numcpus)`
- C. `top(numcpus)`
- D. `ceil(numcpus)`

**Answer: B**

**Explanation:**

The correct Terraform function to select the largest number from a list is `max(numcpus)`.

The `max()` function in Terraform is specifically designed to return the greatest of a given set of numbers. It can accept multiple individual number arguments or a single list (or tuple) of numbers, iterating through them to identify and return the highest value. In this scenario, with `numcpus = [ 18, 3, 7, 11, 2 ]`, applying `max(numcpus)` would correctly evaluate to 18, representing the virtual cluster with the most free CPU cores. This capability is crucial in cloud computing environments where resource visibility is key, allowing administrators to quickly identify the most available resources for deploying new services or scaling existing ones, such as additional Symantec Web Protection (Edge SWG R2) instances.

Let's examine why the other options are incorrect:

**A. `high[numcpus]`:** This is not a valid Terraform function or syntax for retrieving the maximum value from a list. It resembles an attempt at array indexing or a custom function name, neither of which aligns with Terraform's built-in mathematical functions.

**C. `top(numcpus)`:** Terraform does not have a built-in `top()` function for this purpose. While some programming languages or data manipulation tools might offer a `top` function (often for retrieving the top N elements), it is not a standard part of the Terraform language for finding a single maximum value.

**D. `ceil(numcpus)`:** The `ceil()` function in Terraform (and general mathematics) calculates the smallest integer greater than or equal to a given single number (the "ceiling" value). For example, `ceil(3.14)` would return 4. It is not designed to operate on a list of numbers to find the maximum value, and attempting to pass a list to `ceil()` would typically result in a type error.

Therefore, `max(numcpus)` is the appropriate and only correct function among the choices for determining the largest number in the provided list, which is essential for effective resource management and capacity

planning in a dynamic cloud infrastructure.

For further research on Terraform's built-in functions:

Terraform max() function documentation: <https://developer.hashicorp.com/terraform/language/functions/max>

Terraform ceil() function documentation: <https://developer.hashicorp.com/terraform/language/functions/ceil>

### Question: 314

CertyIQ

Your root module contains a variable named `num_servers`. Which is the correct way to pass its value to a child module with an input named `servers`?

- A. `servers = var(num_servers)`
- B. `servers = var.num_servers`
- C. `servers = num_servers`
- D. `$ var.num_servers`

**Answer: B**

#### Explanation:

The correct way to pass the value of a root module variable named `num_servers` to a child module input named `servers` is `servers = var.num_servers`. This is based on the widely adopted HashiCorp Configuration Language (HCL) syntax, which tools like Symantec Web Protection's Edge SWG R2 (if it uses an HCL-like configuration) would typically follow for defining and managing infrastructure as code.

Here's a detailed justification:

1. **Modules in Infrastructure as Code:** Modules serve as reusable, self-contained units of configuration. A root module orchestrates the overall infrastructure, often by calling and configuring child modules. Child modules encapsulate specific sets of resources and expose input variables (also known as arguments or parameters) to allow the calling module to customize their behavior.
2. **Defining Variables:** In the root module, `num_servers` would be declared as an input variable using a variable block, for example:

```
variable "num_servers" { description = "The number of servers to deploy." type    = number default    = 1 }
```

1. This declaration makes `num_servers` available for use within the root module.
2. **Accessing Variable Values:** To access the value of a declared variable within its own module (the root module in this case), HCL uses the `var.` prefix followed by the variable's name. Thus, `var.num_servers` refers to the specific value assigned to or defaulted for the `num_servers` variable in the root module.
3. **Calling a Child Module:** When the root module instantiates a child module, it does so using a module block. Inside this block, you assign values to the child module's declared input variables. The child module would likely have an input variable defined as variable `"servers"`.
4. **Passing the Value:** To pass the value of the root module's `num_servers` to the child module's `servers` input, you assign `var.num_servers` to the child module's input:

```
module "my_child_module" { source = "./modules/my_child_module" # Path to your child module servers = var.num_ser
```

1. Here, `servers` is the input variable expected by the child module, and `var.num_servers` is the expression that retrieves the value from the root module's `num_servers` variable.
2. **Why Other Options Are Incorrect:**

**A. servers = var(num\_servers):** This syntax is incorrect. var is not a function that takes a variable name as an argument; it's a namespace prefix.

**C. servers = num\_servers:** This is incorrect because num\_servers by itself, without the var. prefix, is not a valid way to reference the value of a declared variable in HCL. It would only work if num\_servers were a local value (referenced as local.num\_servers) or an output from another module (e.g., module.another.num\_servers), which is not the context here.

**D. \$ var.num\_servers :** This represents string interpolation. While it might implicitly convert the number to a string and pass it, it's generally not the correct or idiomatic way to pass a direct value to an input that expects a number (or a type other than string). String interpolation is primarily used when you want to embed a variable's value within a larger string, e.g., name = "server-\$ var.num\_servers ". For direct value assignment, servers = var.num\_servers is precise and type-aware.

This method ensures clear, maintainable, and type-safe configuration in cloud environments, promoting modularity and reusability crucial for managing complex infrastructure on platforms like AWS, Azure, or Google Cloud, and by extension, specialized solutions like Symantec's Edge SWG R2 if they adhere to similar configuration paradigms.

#### Authoritative Links for Further Research:

**Terraform - Input Variables:** <https://developer.hashicorp.com/terraform/language/values/variables>

**Terraform - Modules:** <https://developer.hashicorp.com/terraform/language/modules>

**Terraform - Expressions:** <https://developer.hashicorp.com/terraform/language/expressions/references>

### Question: 315

CertyIQ

What task does the terraform import command perform?

- A. Imports resources from one Terraform state file to another.
- B. Imports existing resources into Terraform's state file.
- C. Imports a new Terraform module into Terraform's state file.
- D. Imports all infrastructure from the configured cloud provider.
- E. Imports provider configuration from one state file to another.

**Answer: B**

#### Explanation:

The terraform import command is a crucial utility within Terraform, designed to bring existing, manually created, or externally managed infrastructure resources under Terraform's declarative management. The correct answer, **B. Imports existing resources into Terraform's state file**, accurately describes its primary function.

When you use terraform import, you instruct Terraform to locate an already provisioned resource in your cloud provider (e.g., an AWS EC2 instance, an Azure Virtual Machine, or a Google Cloud Storage bucket) and record its current configuration and state within Terraform's tfstate file. This process is typically a one-time operation for a given resource. To perform an import, you first define a corresponding resource block in your Terraform configuration files (.tf) that matches the type of the existing resource. Then, you execute terraform import [resource\_address] [resource\_id], providing the address of your defined resource in the configuration and the actual ID of the existing resource from the cloud provider.

This command is invaluable for "brownfield" deployments or when transitioning an existing infrastructure to an Infrastructure as Code (IaC) methodology without needing to recreate everything from scratch. It allows

organizations to adopt Terraform incrementally, capturing the current state of critical infrastructure components that were not initially provisioned by Terraform. After a successful import, Terraform considers the resource "managed" and will track its state and reconcile it with the desired state defined in your HCL configuration during subsequent terraform plan and terraform apply operations. The terraform.tfstate file then acts as the single source of truth for that resource's management.

Let's briefly look at why the other options are incorrect:

**A** describes state file manipulation, which is distinct from importing external resources.

**C** misrepresents how modules are handled; modules define resources, which are then managed in the state.

**D** overstates the command's capability; terraform import requires specific resource identification, not a bulk import of an entire cloud environment.

**E** incorrectly suggests transferring provider configurations via import; these are defined in your .tf files.

In essence, terraform import bridges the gap between pre-existing cloud resources and Terraform's desired state management, enabling a smoother adoption of IaC practices for established environments.

For further research:

**Terraform import documentation:** <https://developer.hashicorp.com/terraform/cli/commands/import>

**Terraform State documentation:** <https://developer.hashicorp.com/terraform/language/state>

### Question: 316

CertyIQ

What is one disadvantage of using dynamic blocks in Terraform?

- A. Dynamic blocks can construct repeatable nested blocks.
- B. They cannot be used to loop through a list of values.
- C. They make configuration harder to read and understand.
- D. Terraform will run more slowly.

**Answer: C**

**Explanation:**

Dynamic blocks in Terraform are a powerful feature designed to construct repeatable nested configuration blocks when the number or content of those blocks is not known until runtime, or when you need to define multiple similar blocks based on a collection of values. While incredibly useful for flexibility and reducing boilerplate, their syntax and structure can undeniably make configurations harder to read and understand, which is a significant disadvantage.

The dynamic block syntax, utilizing `for_each` and an iterator variable within a content block, requires a mental parsing step to visualize the resulting static configuration. Unlike explicitly defined blocks where the final structure is immediately apparent, dynamic blocks introduce an abstraction layer. A reader must understand the input collection (`for_each`), how the iterator maps to values, and then interpret how these values are used within the content to form the actual nested block.

This increased cognitive load becomes particularly pronounced in complex scenarios, such as when dynamic blocks are deeply nested or when the logic within the content block is intricate. For teams collaborating on Infrastructure as Code (IaC), readability is paramount for maintainability, debugging, and onboarding new members. Configurations that are difficult to parse can lead to errors, slower troubleshooting, and increased development time because engineers spend more effort deciphering intent rather than building.

Furthermore, while dynamic blocks promote DRY (Don't Repeat Yourself) principles, over-reliance or overly complex implementations can inadvertently obscure the declarative nature of Terraform, making the "what" less obvious than the "how." Debugging issues related to dynamically generated blocks can also be more challenging, as errors might point to the dynamic block definition rather than a specific instance of a generated resource. Therefore, despite their utility, the trade-off in configuration clarity is a valid and often experienced drawback.

Regarding the other options:

**A. Dynamic blocks can construct repeatable nested blocks.** This is the primary purpose and advantage of dynamic blocks, not a disadvantage.

**B. They cannot be used to loop through a list of values.** This is incorrect. Dynamic blocks are explicitly designed to loop through lists or maps using the `for_each` argument to generate multiple instances of a nested block.

**D. Terraform will run more slowly.** While Terraform must evaluate dynamic blocks during the plan phase, the performance impact is generally negligible compared to the time spent on API calls to cloud providers or actual resource provisioning. It's not considered a primary disadvantage in the context of operational speed or efficiency compared to the cognitive burden they can impose.

For further research:

**Terraform Documentation - Dynamic**

**Blocks:** <https://developer.hashicorp.com/terraform/language/expressions/dynamic-blocks>

**HashiCorp Learn - Understanding Dynamic**

**Blocks:** <https://developer.hashicorp.com/terraform/tutorials/configuration-language/dynamic-blocks>

### Question: 317

CertyIQ

When you include a module block in your configuration that references a module from the Terraform Registry, the "version" attribute is required.

- A. True
- B. False

**Answer: B**

**Explanation:**

The statement is False. While the version attribute is highly recommended and considered a best practice when referencing modules from the Terraform Registry, it is not strictly required for the Terraform configuration to be valid or for `terraform init` to function.

If the version attribute is omitted from a module block, Terraform will default to using the latest available version of that module from the registry when `terraform init` is first executed or when `terraform init -upgrade` is run. This behavior can lead to unpredictable deployments because the "latest" version might change over time as module authors publish updates. Subsequent `terraform init` runs (without `-upgrade`) will typically use the cached version unless explicitly told to update.

However, relying on the latest implicit version introduces significant risks in cloud environments. It undermines the principle of infrastructure as code by making deployments non-reproducible. Different runs of `terraform apply` could potentially utilize different module versions if new releases occur between deployments, leading to inconsistencies, unexpected changes in behavior, or even breaking changes without explicit user intent. This unpredictability can cause instability, security vulnerabilities, or operational overhead.

when troubleshooting.

Therefore, although technically optional, specifying a version attribute — often using a version constraint (e.g., `~> 1.0` or `^2.1.0`) — is crucial for ensuring deterministic and stable infrastructure deployments. It allows for controlled updates (e.g., compatible bug fixes) while preventing unintended major version upgrades that could introduce breaking changes. Explicit versioning is a cornerstone of robust dependency management in modern cloud infrastructure provisioning.

For further research:

Terraform Documentation on Module

Versions: <https://developer.hashicorp.com/terraform/language/modules/syntax#module-versions>

Terraform terraform init command: <https://developer.hashicorp.com/terraform/cli/commands/init>

## Question: 318

CertyIQ

Terraform variables and outputs that set the description argument will store that description in the state file.

- A. True
- B. False

**Answer: B**

**Explanation:**

The statement is **False**.

The Terraform state file (`terraform.tfstate`) is meticulously designed to store the actual state of the infrastructure managed by Terraform. Its core purpose is to map real-world cloud resources to your configuration, tracking their attributes, dependencies, and current values to reconcile with the desired state defined in your HCL files.

For input variables, the description argument within a variable block serves purely as documentation. It explains to a user what kind of input is expected, particularly when Terraform prompts for variable values during CLI execution (e.g., `terraform apply` or `terraform plan`). While the value provided for a variable (whether a default or user-provided override) might implicitly be part of the state's understanding of resource attributes, the description itself is not.

Similarly, for output values, the description argument in an output block also functions solely as documentation. It clarifies what the exposed output value represents when displayed to the user after an apply operation or queried via `terraform output`. The state file will store the computed value of the output, but it does not store its associated descriptive text.

These description arguments are considered configuration metadata. They exist only within the HCL configuration files (`.tf` files) and are primarily used by Terraform's CLI for enhanced user experience and clarity. They are not attributes of the provisioned infrastructure, nor do they represent a dynamic state that Terraform needs to track for lifecycle management. Storing such static descriptive text in the state file would introduce unnecessary redundancy and bloat, as the state file is optimized to track runtime attributes and values of managed resources.

For further research:

**Terraform State:** <https://developer.hashicorp.com/terraform/language/state>

**Input Variables:** <https://developer.hashicorp.com/terraform/language/values/variables>

### Question: 319

CertyIQ

What does this code do?

terraform required\_providers aws = "~> 3.0" )

- A. Requires any version of the AWS provider  $\geq 3.0$  and  $< 4.0$ .
- B. Requires any version of the AWS provider  $\geq 3.0$ .
- C. Requires any version of the AWS provider after the 3.0 major release, like 4.1.
- D. Requires any version of the AWS provider  $> 3.0$ .

**Answer: A**

#### Explanation:

The code snippet `required_providers aws = "~> 3.0"` within a Terraform configuration specifies a version constraint for the AWS provider. The `~>` operator, known as the "optimistic locking" or "approximate version" operator, is designed to allow minor and patch updates while preventing potentially breaking major version changes.

When applied as `~> X.Y`, this operator translates to version  $\geq X.Y$  and version  $< X.(Y+1)$ . In the context of `~> 3.0`, it precisely means that any version of the AWS provider greater than or equal to 3.0 is acceptable, but it must be strictly less than 4.0. This range includes 3.0.0, 3.5.0, 3.99.0, but explicitly excludes 4.0.0 or any subsequent major versions.

This constraint is crucial for maintaining stability in infrastructure as code deployments. Major version increments (e.g., from 3.x to 4.x) often introduce breaking changes, such as renamed resources, altered attribute behaviors, or removal of deprecated functionalities. By using `~> 3.0`, a developer ensures that their configuration will continue to work with newer patch and minor releases of the 3.x series, which are typically backward-compatible, without inadvertently pulling in a 4.x version that could break the deployment.

Therefore, option A, "Requires any version of the AWS provider  $\geq 3.0$  and  $< 4.0$ ," accurately describes the behavior of the `~> 3.0` version constraint. Option B is incorrect because it lacks the crucial upper bound, allowing 4.x, 5.x, etc. Option C is entirely wrong as it suggests versions after the 3.0 major release, which is the opposite of the constraint's intent. Option D is incorrect because it excludes version 3.0 itself, which the `~>` operator includes as its lower bound. This precise control over provider versions is a fundamental aspect of robust cloud infrastructure management with Terraform.

#### Authoritative Links for Further Research:

1. **HashiCorp Terraform Documentation - Provider Requirements:**<https://developer.hashicorp.com/terraform/language/providers/requirements#version-constraints>
2. **HashiCorp Terraform Documentation - Version Constraints Syntax:**<https://developer.hashicorp.com/terraform/language/expressions/version-constraints>

### Question: 320

CertyIQ

When developing a Terraform module, how would you specify the version when publishing it to the official Terraform Registry?

- A. Configure it in the module's Terraform code.
- B. Tag a release in your module's source control repository.
- C. Mention it on the module's configuration page on the Terraform Registry.
- D. The Terraform Registry does not support versioning modules.

**Answer: B**

**Explanation:**

When publishing a Terraform module to the official Terraform Registry, the module's version is primarily specified by **tagging a release in its source control repository**. The Terraform Registry integrates directly with Git-based repositories, such as GitHub or GitLab, to manage modules.

Here's a detailed justification:

1. The Terraform Registry monitors a specified Git repository to discover and catalog new module versions.
2. Crucially, it relies on Git tags pushed to the module's repository as the definitive source for version information.
3. These tags must adhere to semantic versioning principles, typically formatted as vX.Y.Z (e.g., v1.0.0) or simply X.Y.Z.
4. When a valid Git tag is created and pushed to the repository's default branch (e.g., main), the Terraform Registry automatically detects it as a new module version.
5. This mechanism establishes the source control repository as the authoritative source for both the module's code and its release history.
6. Tagging a release ensures that each published module version corresponds to an immutable snapshot of the code at a specific commit.
7. This immutability is fundamental for reliable Infrastructure as Code (IaC), guaranteeing that a module version, once published, will consistently refer to the exact same codebase, preventing unexpected changes or regressions.
8. It provides robust traceability, allowing module consumers to easily inspect the precise code corresponding to any specific version listed on the registry.
9. This approach aligns with industry best practices for software versioning and release management.
10. It also facilitates automated Continuous Integration/Continuous Delivery (CI/CD) pipelines, where tagging a commit can trigger an automated release of a new module version.
11. Such automation streamlines the publishing workflow, reduces manual effort, and minimizes potential errors.
12. Configuring the version within the module's Terraform code (option A) is incorrect; while Terraform code specifies provider versions, it does not define the module's own published version on the registry.
13. Similarly, mentioning the version on the module's configuration page (option C) would be insufficient for a robust, verifiable versioning system.
14. Option D is incorrect; the Terraform Registry absolutely supports and relies heavily on module versioning to provide stability and choice to users.
15. Therefore, using Git tags for versioning ensures consistency, auditability, and a clear, verifiable chain of custody for all module releases on the Terraform Registry, which are critical aspects of managing cloud infrastructure efficiently.

**Authoritative Links for Further Research:**

**Publishing Modules to the Terraform**

**Registry:** <https://developer.hashicorp.com/terraform/registry/modules/publish>

**Terraform Module**

**Question: 321**

CertyIQ

terraform destroy is the only way to remove infrastructure.

- A. True
- B. False

**Answer: B**

**Explanation:**

The statement "terraform destroy is the only way to remove infrastructure" is false. While terraform destroy is the primary and most robust command for deprovisioning infrastructure managed by a Terraform state file, several other methods exist for removing cloud resources.

Firstly, infrastructure can always be removed manually directly through the respective cloud provider's management console (GUI), command-line interface (CLI), or software development kits (SDKs). For example, an EC2 instance can be terminated via the AWS console, or a resource group deleted in Azure using the Azure CLI. These methods bypass Terraform entirely.

Secondly, Terraform itself offers commands to manipulate its state file without directly destroying infrastructure in the cloud. terraform state rm can remove a resource's entry from the state file, meaning Terraform will no longer track or manage it, but the resource will persist in the cloud. Similarly, terraform state mv moves resources within the state, but doesn't delete the underlying infrastructure.

Furthermore, cloud providers have their own lifecycle management features that can lead to infrastructure removal. Examples include S3 bucket lifecycle policies that automatically delete objects after a certain period, or auto-scaling groups that terminate instances based on demand. These actions occur independently of Terraform.

Other Infrastructure as Code (IaC) tools, such as AWS CloudFormation, Azure Resource Manager (ARM) templates, or Pulumi, also manage and remove infrastructure. If resources were initially provisioned with a different IaC tool, they would be removed using that tool's specific deprovisioning commands, not terraform destroy.

Finally, even within Terraform, specific changes to configurations that remove a resource block can, upon terraform apply, lead to the destruction of that resource, although terraform apply -destroy or terraform destroy is more explicit. Therefore, while terraform destroy is the intended and safest method for removing Terraform-managed infrastructure, it is far from the only way to remove cloud resources.

For further research:

Terraform Documentation on terraform destroy: <https://www.terraform.io/docs/cli/commands/destroy.html>  
Terraform Documentation on State Management: <https://www.terraform.io/docs/language/state/index.html>  
AWS CLI Documentation (example of manual removal): <https://awscli.amazonaws.com/v2/documentation/api/latest/index.html>

**Question: 322**

CertyIQ

Infrastructure as code (IaC) can be stored in a version control system along with application code.

- A. True
- B. False

**Answer: A**

**Explanation:**

Storing Infrastructure as Code (IaC) in a version control system alongside application code is not only possible but a fundamental best practice in modern software development and cloud operations. IaC artifacts, such as Terraform configurations, AWS CloudFormation templates, Azure Resource Manager templates, or Kubernetes manifests, are text-based files that declaratively define the desired state of infrastructure resources. Treating these configuration files as code allows them to leverage the same robust benefits that application source code enjoys from version control systems like Git.

Firstly, version control provides a complete, auditable history of all infrastructure changes, including who made them, when, and why, enabling comprehensive traceability and accountability. This historical record is crucial for troubleshooting issues, understanding environment evolution, and ensuring compliance with regulatory requirements. Secondly, it facilitates collaborative development, allowing multiple team members to work on infrastructure definitions concurrently, with robust mechanisms for merging changes and resolving conflicts. Should an infrastructure deployment introduce an undesirable state, version control enables quick rollbacks to previous, stable configurations, minimizing downtime and risk. The ability to branch, merge, and tag specific infrastructure versions ensures environment reproducibility, making it easier to provision identical development, testing, and production environments consistently.

Storing IaC alongside application code in the same repository or a closely linked set of repositories fosters a unified development workflow, breaking down silos between development and operations teams. This integration aligns perfectly with DevOps principles, promoting collaboration, automation, and continuous delivery across the entire software lifecycle. Changes to application code might necessitate corresponding infrastructure modifications (e.g., new database, message queue), and having both definitions versioned together streamlines the entire deployment pipeline. This approach is a cornerstone of GitOps, where Git acts as the single source of truth for both declarative infrastructure and applications, driving automated deployments and environment management. By automating infrastructure provisioning and management through IaC and version control, organizations achieve greater speed, consistency, and reliability in their cloud environments. This practice underpins the concept of immutable infrastructure, where changes are made by deploying new, version-controlled infrastructure rather than modifying existing components.

**Authoritative Links for Further Research:**

**AWS on Infrastructure as Code:** <https://aws.amazon.com/devops/what-is-iac/>

**Microsoft Azure on Infrastructure as Code:** <https://learn.microsoft.com/en-us/devops/deliver/what-is-infrastructure-as-code>

**CNCF on GitOps:** <https://www.cncf.io/blog/2020/03/10/what-is-gitops-and-how-does-it-work/>

**Question: 323**

**CertyIQ**

What does the default local Terraform backend store?

- A. Provider plugins
- B. tfplan files
- C. Terraform binary
- D. State file

**Answer: D**

**Explanation:**

The default local Terraform backend stores the **state file**.

This file, typically named `terraform.tfstate` by default, is a critical component of Terraform's operation. It acts as a canonical record of the infrastructure Terraform manages, mapping the real-world cloud resources to the configuration defined in your `.tf` files. This state file is essential for Terraform to understand the current state of your infrastructure. Without it, Terraform wouldn't know which resources it previously created, their attributes, or how to update or destroy them.

When no specific backend configuration is defined in your Terraform code, Terraform automatically defaults to the "local" backend. This means the `terraform.tfstate` file is created and stored directly in the root directory of your Terraform working directory. It contains a snapshot of the infrastructure's metadata, resource IDs, and attributes, allowing Terraform to perform operations like planning, applying, and destroying resources effectively.

The local backend is simple and suitable for individual developers working on isolated projects. However, it lacks features crucial for team collaboration or CI/CD pipelines, such as state locking, versioning, and shared access. For these advanced scenarios, remote backends like Amazon S3, Azure Blob Storage, Google Cloud Storage, or HashiCorp Terraform Cloud are utilized to store the state file centrally and securely.

The other options are incorrect:

**Provider plugins** are downloaded by Terraform into the `.terraform` directory or a global cache, not stored by the backend itself as the primary purpose of a state backend.

**tfplan files** (`terraform.tfplan`) are temporary binary files generated by `terraform plan` to preview changes; they are not the persistent state.

The **Terraform binary** is the executable program itself, separate from any backend's storage function.

Therefore, the fundamental purpose of the default local Terraform backend is to manage and persist the all-important state file.

**Authoritative Links for Further Research:**

**Terraform State Documentation:** <https://developer.hashicorp.com/terraform/language/state>

**Terraform Backends**

**Documentation:** <https://developer.hashicorp.com/terraform/language/settings/backends/configuration>

**Terraform Local Backend:** <https://developer.hashicorp.com/terraform/language/settings/backends/local>

**Question: 324**

**CertyIQ**

What command can you run to generate DOT (Document Template) formatted data to visualize Terraform dependencies?

- A. `terraform graph`
- B. `terraform refresh`
- C. `terraform output`
- D. `terraform show`

**Answer: A**

### Explanation:

The correct command to generate DOT (Document Template) formatted data for visualizing Terraform dependencies is **A. terraform graph**.

### Detailed Justification:

terraform graph is purpose-built within the Terraform CLI to expose the internal dependency graph that Terraform constructs before performing any operations. When you run this command, it analyzes your Terraform configuration files (e.g., .tf files) and identifies all declared resources, data sources, and modules, along with their explicit and implicit relationships. The output of terraform graph is a textual representation of this graph in the **Graphviz DOT language**.

The DOT language is a plain text graph description language widely used for representing structural information like networks, hierarchies, and dependencies. This standardized format allows the output from terraform graph to be easily consumed by external visualization tools, most notably **Graphviz**, which can then render these textual descriptions into visual diagrams such as PNG, SVG, or PDF images.

Visualizing Terraform dependencies is crucial for managing complex cloud infrastructure defined through Infrastructure as Code (IaC). It provides a clear overview of how different components of your cloud environment, such as virtual machines, databases, networking components, and storage, are interconnected and in what order they would be provisioned, updated, or destroyed. This helps engineers understand the impact of changes, identify potential circular dependencies, optimize resource allocation, and debug issues related to resource provisioning order. It's an invaluable tool for enhancing clarity and maintainability in large-scale cloud deployments.

In contrast, the other options serve different purposes:

**B. terraform refresh** updates the Terraform state file by querying the real-world infrastructure to detect any drift, without generating a dependency graph.

**C. terraform output** displays the values of output variables defined in your configuration from the current state, not the architectural dependencies.

**D. terraform show** displays the human-readable plan or state file contents, which can include resource attributes but does not generate a graphical dependency representation in DOT format.

Therefore, terraform graph is the only command specifically designed to generate the DOT-formatted data required for visualizing Terraform's dependency tree, which is essential for effective cloud resource orchestration and management.

### Authoritative Links for Further Research:

**Terraform CLI: terraform graph command**

**documentation:** <https://developer.hashicorp.com/terraform/cli/commands/graph>

**Graphviz Official Website:** <https://graphviz.org/>

**Graphviz DOT Language Guide:** <https://graphviz.org/doc/info/lang.html>

### Question: 325

CertyIQ

A resource block is shown in the Exhibit space of this page. What is the Terraform resource name of that resource block?

```
resource "azurerm_resource_group" "dev" {  
  name = "test"  
  location = "westus"  
}
```

- A. azurerm
- B. azurerm\_resource\_group
- C. test
- D. dev

**Answer: D**

**Explanation:**

D. dev

resource "azurerm\_resource\_group" "dev"

name = "test"

So:

azurerm\_resource\_group → resource type

dev → Terraform resource name

test → actual Azure resource name

### Question: 326

CertyIQ

You have a simple Terraform configuration containing one virtual machine (VM) in a cloud provider. You run terraform apply and the VM is created successfully.

What will happen if you delete the VM using the cloud provider console, then run terraform apply again without changing any Terraform code?

- A. Terraform will recreate the VM.
- B. Terraform will report an error.
- C. Terraform will remove the VM from the state file.
- D. Terraform will not make any changes.

**Answer: A**

**Explanation:**

When you run terraform apply, Terraform's primary goal is to bring the real-world infrastructure into alignment with the desired state defined in your configuration files. It achieves this by comparing three key elements: your Terraform code (the desired state), the current actual state of resources in the cloud provider, and the information stored in its state file (terraform.tfstate).

Initially, after the first successful terraform apply, the virtual machine (VM) is created in the cloud provider, and its details (such as its ID and properties) are meticulously recorded within the Terraform state file. This state file acts as a cache of the infrastructure Terraform manages, linking your configuration to the real-world cloud resources.

If you subsequently delete the VM directly using the cloud provider's console, you're introducing "configuration drift." The VM no longer exists in the cloud, but the Terraform state file remains unaware of this manual change; it still holds the record that the VM should be present and managed by Terraform. Your Terraform configuration code also still defines that the VM should exist.

When you run terraform apply again without modifying the Terraform code, Terraform performs its reconciliation process. It first reads your configuration, then consults its state file to understand the resources it believes it manages. Crucially, it then queries the actual cloud provider to verify the current status of those resources.

During this comparison, Terraform will identify a discrepancy: your configuration dictates that a VM should exist, the state file indicates it was previously created and managed, but the cloud provider reports that the VM is missing. To resolve this drift and restore the infrastructure to the desired state defined by your code and reflected in its state file, Terraform will generate an execution plan to recreate the VM. It considers the absence of the VM as a deviation from the managed state and takes action to correct it. This behavior is fundamental to Terraform's idempotent nature and its ability to ensure infrastructure consistency.

For further research, consider these authoritative links:

**Terraform State:** <https://developer.hashicorp.com/terraform/language/state>

**Terraform Apply:** <https://developer.hashicorp.com/terraform/cli/commands/apply>

**Configuration Drift:** <https://developer.hashicorp.com/terraform/tutorials/state/drift>

### Question: 327

CertyIQ

You want to use API tokens and other secrets within your team's Terraform workspaces, where does HashiCorp recommend you store these sensitive values? (Choose three.)

- A. In a plaintext document on a shared drive.
- B. In HashiCorp Vault.
- C. In a terraform.tfvars file, checked into your version control system.
- D. In an HCP Terraform/Terraform Cloud variable, with the sensitive option checked.
- E. In a terraform.tfvars file, securely managed and shared with your team.

**Answer: B,D,E**

#### Explanation:

Storing sensitive information like API tokens and secrets securely is paramount in infrastructure as code (IaC) to prevent unauthorized access and data breaches. HashiCorp recommends several secure methods for managing these values within Terraform workspaces.

**B. In HashiCorp Vault.** HashiCorp Vault is purpose-built for secret management, providing a centralized and highly secure solution. It encrypts secrets at rest and in transit, offers dynamic secrets (generating secrets on demand with a limited lifespan), and supports robust authentication and authorization mechanisms. Terraform can natively integrate with Vault to fetch secrets just-in-time during provisioning, ensuring that sensitive data is never hardcoded or exposed in configuration files. This approach adheres to the principle of least privilege and significantly reduces the attack surface.

**Further Research:** HashiCorp Vault Documentation: <https://www.vaultproject.io/docs>

**D. In an HCP Terraform/Terraform Cloud variable, with the sensitive option checked.** HCP Terraform (formerly Terraform Cloud) offers a managed environment for Terraform workflows, including variable management. When creating a variable in HCP Terraform and marking it as "sensitive," the value is encrypted at rest and never displayed in the UI, logs, or plan/apply outputs. This ensures that even collaborators with access to the workspace cannot directly view the secret, enhancing team security and compliance while still allowing Terraform to use the value for provisioning.

**Further Research:** HCP Terraform

Variables: <https://developer.hashicorp.com/terraform/cloud/workspaces/variables#sensitive-variables>

**E. In a terraform.tfvars file, securely managed and shared with your team.** While less robust than Vault or sensitive HCP Terraform variables, using a .tfvars file for secrets can be acceptable if it is securely managed and never committed to version control systems (VCS). "Securely managed" implies storing it in an encrypted location, restricting access, and sharing it through secure, out-of-band methods (e.g., not email, not unencrypted shared drives). This method is often suitable for local development environments or specific, highly controlled scenarios where other secret management tools might be overkill, provided strict operational security policies are enforced to prevent secret sprawl.

Options A (plaintext on a shared drive) and C (terraform.tfvars file checked into VCS) are critical security anti-patterns. Plaintext storage is immediately vulnerable to anyone with access, and committing secrets to VCS creates a permanent record, making them nearly impossible to fully revoke and leading to widespread secret exposure. Therefore, these choices are strongly discouraged.

## Question: 328

CertyIQ

If one of your modules uses a local value, you can expose that value to callers of the module by defining a terraform output in the module's configuration.

- A. True
- B. False

**Answer: A**

**Explanation:**

Yes, it is entirely possible and a common practice in Terraform to expose a local value to callers of a module by defining a Terraform output in that module's configuration. This statement is **True**.

Local values in Terraform (locals) provide a way to assign a name to an expression, allowing it to be used multiple times within a module without repeating the expression. They improve readability, reduce redundancy, and encapsulate complex logic within the module itself, acting as computed constants or derived attributes internal to the module's scope.

Outputs, on the other hand, define the values that a module will export or return to its calling module or to the Terraform CLI. They serve as the public interface for a module, detailing what information it makes available to the outside world.

The mechanism to expose a local value is straightforward: you simply define an output block and set its value attribute to reference the desired local value. For instance, if you have a local value `local.full_resource_name`, you can create an output like `output "resource_name" value = local.full_resource_name`. This pattern is crucial for building robust and reusable modules in cloud

infrastructure as code.

It enables modules to perform internal calculations or string manipulations, derive new attributes from inputs, or combine multiple pieces of information into a single, meaningful value, and then selectively expose these computed results. This maintains a clean separation of concerns, where the module's internal implementation details (local values) are hidden, while its useful derived outputs are made accessible. This principle aligns with good software engineering practices, promoting abstraction and modularity critical for managing complex cloud environments efficiently.

For example, a module might use local values to construct a unique resource identifier, a specific ARN, or a complex tag set based on various inputs. Exposing these derived values via outputs allows parent modules or other parts of the infrastructure to reference and depend on these dynamically generated attributes, facilitating interconnected resource deployments. This approach simplifies the caller's configuration by abstracting away the underlying computation.

For further research:

**Terraform locals Documentation:** <https://developer.hashicorp.com/terraform/language/values/locals>

**Terraform outputs Documentation:** <https://developer.hashicorp.com/terraform/language/values/outputs>

## Question: 329

CertyIQ

What does state locking accomplish?

- A. Encrypts any credentials stored within the state file.
- B. Copies the state file from memory to disk.
- C. Prevents accidental deletion of the state file.
- D. Blocks Terraform commands from modifying the state file.

**Answer: D**

**Explanation:**

State locking is a critical feature in Terraform designed to maintain the integrity and consistency of the state file, especially in collaborative or automated environments. The Terraform state file acts as a crucial bridge, mapping the resources defined in your configuration to their real-world counterparts in the cloud or on-premises infrastructure. It tracks metadata about your infrastructure, resource dependencies, and actual attribute values.

When multiple users or automated processes attempt to run Terraform commands (like `terraform apply` or `terraform plan -out=tfplan`) that modify the state file concurrently, a race condition can occur. Without state locking, these simultaneous operations could lead to state file corruption, lost updates, or inconsistencies between the configuration and the deployed infrastructure. This would result in unexpected behavior, resource drift, or even infrastructure outages.

State locking prevents these issues by ensuring that only one operation can acquire an exclusive lock on the state file at any given time. When a Terraform command that modifies the state is executed, it first attempts to acquire this lock. If the lock is already held by another operation, the new command will wait until the lock is released or will fail, preventing concurrent write operations.

Therefore, state locking accomplishes blocking Terraform commands from modifying the state file if another operation is already in progress. This mechanism safeguards the state file's integrity and prevents accidental or malicious concurrent modifications. It is particularly vital in cloud computing environments where teams

often share infrastructure codebases and collaborate on deploying and managing resources. By preventing concurrent writes, state locking ensures that the infrastructure as code remains consistent with the actual deployed infrastructure, supporting idempotency and reliable deployments.

To clarify why other options are incorrect:

- A. Encrypts any credentials stored within the state file: State locking does not encrypt credentials; encryption is a separate security feature often handled by the backend itself or by secrets management solutions.
- B. Copies the state file from memory to disk: This describes the basic process of saving state, not state locking, which is about concurrency control.
- C. Prevents accidental deletion of the state file: While it helps prevent corruption which could necessitate a state rebuild, state locking doesn't directly prevent file deletion at the operating system level.

#### Authoritative Links for Further Research:

##### Terraform Documentation on State

**Locking:** <https://developer.hashicorp.com/terraform/language/state/locking>

##### Terraform Documentation on Backends (where state locking is often

implemented): <https://developer.hashicorp.com/terraform/language/settings/backends/configuration>

### Question: 330

CertyIQ

Which of the following is allowed as a Terraform variable name?

- A. source
- B. name
- C. version
- D. count

#### Answer: B

#### Explanation:

B is correct because "name" is a plain Terraform identifier that does not collide with Terraform's meta-arguments or reserved block attributes.

Terraform distinguishes simple identifiers (valid for variable names) from special block attributes and meta-arguments; input variable identifiers follow the normal identifier rules and are free to be any non-reserved identifier, and "name" is not a documented meta-argument or module/provider attribute that would introduce syntactic or semantic collision. In contrast, Terraform defines several context-specific attributes and meta-arguments (for example, module "source"/"version" and resource meta-argument "count") that are treated specially by the language parser and runtime, and therefore are reserved in their respective syntactic contexts. From an exam-focused perspective, pick identifiers that are not listed as meta-arguments or special block attributes to avoid naming conflicts; "name" meets that criterion. Practically, using identifiers that shadow special attributes can lead to confusing code or parser errors and is explicitly discouraged by Terraform documentation and examples.

Critical evaluation of the incorrect options:

- A. source — reserved as the module/provider source attribute (module block uses "source"), a syntactically significant attribute for locating modules.
- C. version — used as a special attribute for modules/providers (versioning constraints) and therefore treated as a contextual attribute rather than a generic variable identifier.

D. count — a canonical Terraform meta-argument that controls resource/module instance counts and is not a plain identifier in resource/module contexts.

References<https://developer.hashicorp.com/terraform/language/values/variables><https://developer.hashicorp.com/arguments/count><https://developer.hashicorp.com/terraform/language/modules/sources>

### Question: 331

CertyIQ

Which syntax check errors when you run terraform validate?

- A. The code contains tabs for indentation instead of spaces.
- B. There is a missing value for a variable.
- C. The state file does not match the current infrastructure.
- D. None of the above.

**Answer: D**

#### Explanation:

D is correct because terraform validate only performs static HCL syntax and configuration consistency checks without requiring runtime variable values, consulting state, or enforcing whitespace style.

Reasoning (concise, exam-focused):

terraform validate performs parsing and static semantic checks of the configuration files in a directory; it does not contact providers, inspect remote state, or execute plans, so errors must be detectable purely from the HCL text and static type/structure analysis.

Indentation style is not a semantic error: HCL treats tabs and spaces as whitespace for structure and terraform fmt can normalize formatting; therefore tabs do not cause validate to fail.

Variable values are runtime inputs; absent user-supplied values are resolved at plan/apply time (or via defaults/CLI/env), so validate will not fail simply because a variable has no supplied value.

State/infrastructure divergence is an operational/runtime condition reflected in plan/apply or state commands; validate never compares configuration to the state or remote resources and thus cannot error for a mismatched state.

Critical evaluation of incorrect options:

A — Tabs for indentation: Incorrect. Whitespace (tabs vs spaces) is syntactically acceptable in HCL and formatting is cosmetic; formatting issues are addressed by terraform fmt, not terraform validate.

B — Missing variable value: Incorrect. Missing input values are resolved during plan/apply (or via provided defaults/vars); validate performs static checks and does not require runtime inputs to succeed.

C — State mismatch with infrastructure: Incorrect. State and real infrastructure are outside validate's scope; detecting drift requires plan/apply or state interrogation, not validate.

References:<https://developer.hashicorp.com/terraform/cli/commands/validate><https://developer.hashicorp.com/terraform/language/values/variables>

# Thank you

Thank you for being so interested in the premium exam material.  
I'm glad to hear that you found it informative and helpful.

If you have any feedback or thoughts on the bumps, I would love to hear them.  
Your insights can help me improve our writing and better understand our readers.

## Best of Luck

You have worked hard to get to this point, and you are well-prepared for the exam  
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